

Basin 01013500: Fish River near Fort Kent, Maine

Dominant forcing: Tair Peak TE: 0.0049 Peak lag: 14 day(s)
Summary: Tair-dominant at lag 14 day(s); top drivers consistent with low high_prec_freq, high gauge_lat, low low_prec_freq.

Top 10 driver attributes

1. high_prec_freq

z = -1.76 r = -0.49 score = 0.864

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. gauge_lat

z = +1.53 r = +0.51 score = 0.776

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. low_prec_freq

z = -1.49 r = -0.50 score = 0.753

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. pet_mean

z = -1.47 r = -0.33 score = 0.487

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lon

z = +1.68 r = -0.25 score = 0.427

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. runoff_ratio

z = +0.67 r = +0.56 score = 0.372

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. frac_snow

z = +0.67 r = +0.46 score = 0.308

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. frac_forest

z = +0.72 r = +0.35 score = 0.248

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. slope_mean

z = -0.52 r = +0.41 score = 0.212

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. aridity

z = -0.69 r = -0.30 score = 0.210

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01022500: Narraguagus River at Cherryfield, Maine

Dominant forcing: PRCP Peak TE: 0.0167 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high runoff_ratio.

Top 10 driver attributes

1. gauge_lon

z = +1.72 r = -0.27 score = 0.467

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.06 r = +0.20 score = 0.213

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. runoff_ratio

z = +0.92 r = +0.23 score = 0.213

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. aridity

z = -0.76 r = -0.28 score = 0.210

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
5. sand_frac

z = +1.47 r = +0.12 score = 0.172

high sand fraction promotes fast infiltration and subsurface drainage.
6. high_prec_dur

z = -0.76 r = +0.23 score = 0.171

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. elev_mean

z = -0.85 r = -0.20 score = 0.166

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. pet_mean

z = -1.20 r = -0.12 score = 0.147

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. glim_1st_class_frac

z = -0.96 r = -0.15 score = 0.140

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. root_depth_99

z = +1.35 r = +0.10 score = 0.137

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01030500: Mattawamkeag River near Mattawamkeag, Maine

Dominant forcing: PRCP Peak TE: 0.0134 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high area_gages2, high area_geospa_fabric.

Top 10 driver attributes

1. gauge_lon z = +1.70 r = -0.27 score = 0.460

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

2. area_gages2 z = +1.69 r = +0.18 score = 0.306

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.

3. area_geospa_fabric z = +1.68 r = +0.17 score = 0.286

a large geospatial drainage area integrates many sub-basin responses.

4. slope_fdc z = +1.25 r = +0.20 score = 0.251

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

5. low_prec_freq z = -1.11 r = -0.22 score = 0.249

few dry days indicate steady moisture supply, which supports continuous streamflow.

6. aridity z = -0.70 r = -0.28 score = 0.194

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

7. high_prec_dur z = -0.74 r = +0.23 score = 0.168

short heavy-rainfall events produce sharper but briefer runoff peaks.

8. runoff_ratio z = +0.72 r = +0.23 score = 0.167

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

9. pet_mean z = -1.34 r = -0.12 score = 0.163

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

10. high_prec_freq z = -0.83 r = -0.19 score = 0.157

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01031500: Piscataquis River near Dover-Foxcroft, Maine

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high glim_2nd_class_frac, low glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.63 r = -0.27 score = 0.443

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = +2.01 r = +0.15 score = 0.308

a large secondary geological class adds heterogeneity to the subsurface flow response.
3. glim_1st_class_frac

z = -1.67 r = -0.15 score = 0.242

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
4. high_prec_dur

z = -1.05 r = +0.23 score = 0.238

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. aridity

z = -0.76 r = -0.28 score = 0.210

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. runoff_ratio

z = +0.81 r = +0.23 score = 0.187

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. q95

z = +0.60 r = +0.31 score = 0.183

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. low_prec_freq

z = -0.78 r = -0.22 score = 0.174

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. pet_mean

z = -1.29 r = -0.12 score = 0.157

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01047000: Carrabassett River near North Anson, Maine

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low glim_1st_class_frac, high runoff_ratio.

Top 10 driver attributes

1. gauge_lon

z = +1.59 r = -0.27 score = 0.433

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_1st_class_frac

z = -2.36 r = -0.15 score = 0.343

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
3. runoff_ratio

z = +1.16 r = +0.23 score = 0.267

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. high_prec_dur

z = -0.97 r = +0.23 score = 0.219

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. gvf_diff

z = +0.83 r = -0.24 score = 0.201

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. aridity

z = -0.69 r = -0.28 score = 0.192

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. q95

z = +0.62 r = +0.31 score = 0.188

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. root_depth_99

z = +1.69 r = +0.10 score = 0.171

very deep maximum rooting allows vegetation to tap deep storage in droughts.
9. frac_forest

z = +0.94 r = +0.17 score = 0.162

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. pet_mean

z = -1.26 r = -0.12 score = 0.153

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01052500: Diamond River near Wentworth Location, NH

Dominant forcing: Tair Peak TE: 0.0066 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.74 r = -0.50 score = 0.878

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.63 r = -0.49 score = 0.804

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +1.10 r = +0.56 score = 0.613

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. gauge_lat

z = +1.08 r = +0.51 score = 0.547

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. stream_elas

z = -1.03 r = -0.43 score = 0.443

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. pet_mean

z = -1.25 r = -0.33 score = 0.412

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. frac_snow

z = +0.86 r = +0.46 score = 0.397

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. gauge_lon

z = +1.53 r = -0.25 score = 0.388

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. q95

z = +0.73 r = +0.51 score = 0.374

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01054200: Wild River at Gilead, Maine

Dominant forcing: Tair Peak TE: 0.0037 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with high runoff_ratio, high q95, high gauge_lat.

Top 10 driver attributes

1. runoff_ratio

z = +1.22 r = +0.56 score = 0.677

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. q95

z = +1.02 r = +0.51 score = 0.522

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. gauge_lat

z = +0.99 r = +0.51 score = 0.500

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. low_prec_freq

z = -0.98 r = -0.50 score = 0.493

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. pet_mean

z = -1.19 r = -0.33 score = 0.393

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lon

z = +1.53 r = -0.25 score = 0.390

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. slope_mean

z = +0.93 r = +0.41 score = 0.380

steep terrain promotes fast surface runoff and a quick streamflow response.
8. high_prec_freq

z = -0.75 r = -0.49 score = 0.371

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. q_mean

z = +0.80 r = +0.45 score = 0.359

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
10. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01055000: Swift River near Roxbury, Maine

Dominant forcing: SRAD Peak TE: 0.0035 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high gauge_lon, high runoff_ratio, high gvf_diff.

Top 10 driver attributes

1. gauge_lon z = +1.55 r = -0.43 score = 0.672

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. runoff_ratio z = +1.14 r = +0.43 score = 0.487

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. gvf_diff z = +1.00 r = -0.33 score = 0.328

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. gauge_lat z = +1.04 r = +0.29 score = 0.304

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. q95 z = +0.69 r = +0.43 score = 0.296

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. high_prec_dur z = -0.95 r = +0.29 score = 0.274

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. low_prec_freq z = -0.85 r = -0.31 score = 0.265

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. clay_frac z = -1.02 r = -0.22 score = 0.226

low clay fraction allows easier infiltration and reduces surface runoff.
9. frac_forest z = +0.94 r = +0.23 score = 0.222

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. slope_mean z = +0.56 r = +0.37 score = 0.208

steep terrain promotes fast surface runoff and a quick streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01057000: Little Androscoggin River near South Paris, Maine

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high glim_2nd_class_frac, high sand_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.56 r = -0.27 score = 0.423

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = +1.77 r = +0.15 score = 0.271

a large secondary geological class adds heterogeneity to the subsurface flow response.
3. sand_frac

z = +2.08 r = +0.12 score = 0.244

high sand fraction promotes fast infiltration and subsurface drainage.
4. glim_1st_class_frac

z = -1.65 r = -0.15 score = 0.241

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gvf_diff

z = +0.92 r = -0.24 score = 0.222

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. soil_conductivity

z = +1.50 r = +0.15 score = 0.222

high soil conductivity allows fast infiltration and subsurface flow.
7. high_prec_dur

z = -0.94 r = +0.23 score = 0.214

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. aridity

z = -0.75 r = -0.28 score = 0.206

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. pet_mean

z = -1.18 r = -0.12 score = 0.144

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. frac_forest

z = +0.81 r = +0.17 score = 0.140

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01073000: OYSTER RIVER NEAR DURHAM, NH

Dominant forcing: PRCP Peak TE: 0.0057 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, low aridity.

Top 10 driver attributes

1. gauge_lon

z = +1.53 r = -0.27 score = 0.416

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.07 r = +0.20 score = 0.215

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. aridity

z = -0.69 r = -0.28 score = 0.191

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. elev_mean

z = -0.92 r = -0.20 score = 0.179

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. sand_frac

z = +1.29 r = +0.12 score = 0.151

high sand fraction promotes fast infiltration and subsurface drainage.
6. high_prec_dur

z = -0.65 r = +0.23 score = 0.146

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. pet_mean

z = -1.04 r = -0.12 score = 0.127

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. frac_forest

z = +0.70 r = +0.17 score = 0.121

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. soil_conductivity

z = +0.79 r = +0.15 score = 0.117

high soil conductivity allows fast infiltration and subsurface flow.
10. glim_1st_class_frac

z = +0.78 r = -0.15 score = 0.114

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01078000: SMITH RIVER NEAR BRISTOL, NH

Dominant forcing: PRCP Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high glim_2nd_class_frac, high sand_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.48 r = -0.27 score = 0.403

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = +2.11 r = +0.15 score = 0.324

a large secondary geological class adds heterogeneity to the subsurface flow response.
3. sand_frac

z = +2.09 r = +0.12 score = 0.245

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_conductivity

z = +1.50 r = +0.15 score = 0.221

high soil conductivity allows fast infiltration and subsurface flow.
5. high_prec_dur

z = -0.93 r = +0.23 score = 0.211

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. aridity

z = -0.72 r = -0.28 score = 0.199

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. glim_1st_class_frac

z = -1.24 r = -0.15 score = 0.180

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. gvf_diff

z = +0.69 r = -0.24 score = 0.167

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. root_depth_99

z = +1.63 r = +0.10 score = 0.165

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. frac_forest

z = +0.93 r = +0.17 score = 0.161

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01118300: PENDLETON HILL BROOK NEAR CLARKS FALLS, CT.

Dominant forcing: PRCP Peak TE: 0.0041 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +1.48

r = -0.27

score = 0.401

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.93

r = +0.20

score = 0.388

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gvf_diff

z = +0.99

r = -0.24

score = 0.240

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. stream_elas

z = -1.86

r = -0.12

score = 0.229

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. aridity

z = -0.75

r = -0.28

score = 0.208

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. hfd_mean

z = -0.81

r = -0.24

score = 0.192

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. sand_frac

z = +1.54

r = +0.12

score = 0.181

high sand fraction promotes fast infiltration and subsurface drainage.
8. elev_mean

z = -0.92

r = -0.20

score = 0.180

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. runoff_ratio

z = +0.68

r = +0.23

score = 0.158

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. p_mean

z = +0.45

r = +0.34

score = 0.156

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01121000: MOUNT HOPE RIVER NEAR WARRENVILLE, CT.

Dominant forcing: Tair Peak TE: 0.0030 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with high gauge_lon, low stream_elas, low clay_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.46 r = -0.25 score = 0.371

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. stream_elas

z = -0.85 r = -0.43 score = 0.366

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. clay_frac

z = -1.15 r = -0.30 score = 0.340

low clay fraction allows easier infiltration and reduces surface runoff.
4. slope_fdc

z = +1.39 r = +0.24 score = 0.335

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. runoff_ratio

z = +0.60 r = +0.56 score = 0.332

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. frac_forest

z = +0.87 r = +0.35 score = 0.302

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. pet_mean

z = -0.88 r = -0.33 score = 0.290

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. slope_mean

z = -0.62 r = +0.41 score = 0.252

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. gauge_lat

z = +0.50 r = +0.51 score = 0.252

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. elev_mean

z = -0.79 r = +0.30 score = 0.234

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01123000: LITTLE RIVER NEAR HANOVER, CT.

Dominant forcing: PRCP Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high slope_fdc.

Top 10 driver attributes

1. gauge_lon

z = +1.46 r = -0.27 score = 0.398

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.24 r = -0.24 score = 0.301

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. slope_fdc

z = +1.13 r = +0.20 score = 0.226

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. aridity

z = -0.74 r = -0.28 score = 0.203

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
5. hfd_mean

z = -0.70 r = -0.24 score = 0.166

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. elev_mean

z = -0.81 r = -0.20 score = 0.159

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. p_mean

z = +0.40 r = +0.34 score = 0.138

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. high_prec_dur

z = -0.61 r = +0.23 score = 0.138

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01134500: MOOSE RIVER AT VICTORY, VT

Dominant forcing: PRCP Peak TE: 0.0077 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low low_prec_freq, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +1.48 r = -0.27 score = 0.401

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. low_prec_freq

z = -1.52 r = -0.22 score = 0.341

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gvf_diff

z = +1.04 r = -0.24 score = 0.252

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. high_prec_dur

z = -1.03 r = +0.23 score = 0.234

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. glim_2nd_class_frac

z = +1.49 r = +0.15 score = 0.228

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. high_prec_freq

z = -1.12 r = -0.19 score = 0.210

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. runoff_ratio

z = +0.91 r = +0.23 score = 0.210

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. aridity

z = -0.74 r = -0.28 score = 0.203

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_forest

z = +0.96 r = +0.17 score = 0.165

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. pet_mean

z = -1.25 r = -0.12 score = 0.153

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01137500: AMMONOOSUC RIVER AT BETHLEHEM JUNCTION, NH

Dominant forcing: SRAD Peak TE: 0.0052 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high gauge_lon, high slope_mean, low low_prec_freq.

Top 10 driver attributes

1. gauge_lon

z = +1.49 r = -0.43 score = 0.644

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_mean

z = +1.36 r = +0.37 score = 0.499

steep terrain promotes fast surface runoff and a quick streamflow response.
3. low_prec_freq

z = -1.43 r = -0.31 score = 0.447

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. high_prec_freq

z = -1.19 r = -0.33 score = 0.390

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. runoff_ratio

z = +0.90 r = +0.43 score = 0.385

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. q5

z = +1.15 r = +0.31 score = 0.359

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. gauge_lat

z = +0.96 r = +0.29 score = 0.283

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. root_depth_50

z = +2.13 r = -0.13 score = 0.267

deep median rooting allows vegetation to access stored soil water during dry periods.
9. clay_frac

z = -1.20 r = -0.22 score = 0.264

low clay fraction allows easier infiltration and reduces surface runoff.
10. frac_forest

z = +0.96 r = +0.23 score = 0.226

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01139000: WELLS RIVER AT WELLS RIVER, VT

Dominant forcing: SRAD Peak TE: 0.0076 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low low_prec_freq.

Top 10 driver attributes

1. gauge_lon z = +1.46 r = -0.43 score = 0.632

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +0.97 r = -0.33 score = 0.319

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. low_prec_freq z = -1.00 r = -0.31 score = 0.311

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. runoff_ratio z = +0.66 r = +0.43 score = 0.283

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. glim_1st_class_frac z = -1.31 r = -0.21 score = 0.280

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. gauge_lat z = +0.94 r = +0.29 score = 0.276

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. high_prec_dur z = -0.94 r = +0.29 score = 0.272

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. p_seasonality z = +0.52 r = -0.46 score = 0.237

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. frac_forest z = +0.90 r = +0.23 score = 0.211

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. lai_diff z = +1.46 r = -0.14 score = 0.203

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01139800: EAST ORANGE BRANCH AT EAST ORANGE, VT

Dominant forcing: SRAD Peak TE: 0.0085 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high carbonate_rocks_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.45 r = -0.43 score = 0.625

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.37 r = -0.33 score = 0.448

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. carbonate_rocks_frac

z = +3.38 r = -0.11 score = 0.380

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. runoff_ratio

z = +0.69 r = +0.43 score = 0.296

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. high_prec_dur

z = -0.97 r = +0.29 score = 0.279

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. gauge_lat

z = +0.93 r = +0.29 score = 0.273

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. low_prec_freq

z = -0.83 r = -0.31 score = 0.258

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. p_seasonality

z = +0.51 r = -0.46 score = 0.231

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. lai_diff

z = +1.64 r = -0.14 score = 0.229

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
10. frac_forest

z = +0.97 r = +0.23 score = 0.227

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01142500: AYERS BROOK AT RANDOLPH, VT

Dominant forcing: Tair Peak TE: 0.0030 Peak lag: 13 day(s)
Summary: Tair-dominant at lag 13 day(s); top drivers consistent with high gauge_lat, low low_prec_freq, low pet_mean.

Top 10 driver attributes

1. gauge_lat z = +0.90 r = +0.51 score = 0.455

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
2. low_prec_freq z = -0.85 r = -0.50 score = 0.429

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. pet_mean z = -1.15 r = -0.33 score = 0.382

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. gauge_lon z = +1.43 r = -0.25 score = 0.363

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. frac_forest z = +0.91 r = +0.35 score = 0.316

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. runoff_ratio z = +0.57 r = +0.56 score = 0.316

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. aridity z = -0.66 r = -0.30 score = 0.201

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. stream_elas z = -0.42 r = -0.43 score = 0.179

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. low_prec_dur z = -0.86 r = -0.20 score = 0.175

short dry spells limit basin drying and keep storage close to full.
10. clay_frac z = -0.58 r = -0.30 score = 0.173

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01144000: WHITE RIVER AT WEST HARTFORD, VT

Dominant forcing: Tair Peak TE: 0.0043 Peak lag: 13 day(s)
Summary: Tair-dominant at lag 13 day(s); top drivers consistent with high gauge_lat, low low_prec_freq, low pet_mean.

Top 10 driver attributes

1. gauge_lat z = +0.86 r = +0.51 score = 0.434

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
2. low_prec_freq z = -0.84 r = -0.50 score = 0.424

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. pet_mean z = -1.13 r = -0.33 score = 0.376

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. runoff_ratio z = +0.67 r = +0.56 score = 0.372

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon z = +1.44 r = -0.25 score = 0.367

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. frac_forest z = +0.92 r = +0.35 score = 0.317

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. clay_frac z = -0.73 r = -0.30 score = 0.216

low clay fraction allows easier infiltration and reduces surface runoff.
8. aridity z = -0.69 r = -0.30 score = 0.210

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. high_prec_freq z = -0.41 r = -0.49 score = 0.204

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. geol_porostiy z = -1.42 r = -0.13 score = 0.182

low geological porosity reduces subsurface storage and produces a flashier response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01162500: PRIEST BROOK NEAR WINCHENDON, MA

Dominant forcing: PRCP Peak TE: 0.0083 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon z = +1.46 r = -0.27 score = 0.397

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc z = +1.24 r = +0.20 score = 0.250

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. high_prec_dur z = -1.06 r = +0.23 score = 0.241

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. sand_frac z = +1.86 r = +0.12 score = 0.217

high sand fraction promotes fast infiltration and subsurface drainage.
5. aridity z = -0.75 r = -0.28 score = 0.206

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. root_depth_99 z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. soil_conductivity z = +1.26 r = +0.15 score = 0.186

high soil conductivity allows fast infiltration and subsurface flow.
8. frac_forest z = +0.88 r = +0.17 score = 0.151

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. p_mean z = +0.37 r = +0.34 score = 0.128

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. pet_mean z = -0.95 r = -0.12 score = 0.116

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01169000: NORTH RIVER AT SHATTUCKVILLE, MA

Dominant forcing: Tair Peak TE: 0.0022 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with high runoff_ratio, low stream_elas, low low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = +0.79 r = +0.56 score = 0.437

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. stream_elas

z = -0.96 r = -0.43 score = 0.412

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. low_prec_freq

z = -0.75 r = -0.50 score = 0.380

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. gauge_lon

z = +1.42 r = -0.25 score = 0.362

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. frac_forest

z = +0.96 r = +0.35 score = 0.332

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. gauge_lat

z = +0.65 r = +0.51 score = 0.329

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. pet_mean

z = -0.96 r = -0.33 score = 0.319

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. q95

z = +0.48 r = +0.51 score = 0.247

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. aridity

z = -0.78 r = -0.30 score = 0.237

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. clay_frac

z = -0.77 r = -0.30 score = 0.228

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01170100: GREEN RIVER NEAR COLRAIN, MA

Dominant forcing: Tair Peak TE: 0.0032 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with low low_prec_freq, high runoff_ratio, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq

z = -0.81 r = -0.50 score = 0.406

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. runoff_ratio

z = +0.73 r = +0.56 score = 0.406

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. gauge_lon

z = +1.43 r = -0.25 score = 0.363

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. gauge_lat

z = +0.66 r = +0.51 score = 0.336

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. frac_forest

z = +0.96 r = +0.35 score = 0.332

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. pet_mean

z = -0.98 r = -0.33 score = 0.325

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. clay_frac

z = -0.88 r = -0.30 score = 0.259

low clay fraction allows easier infiltration and reduces surface runoff.
8. aridity

z = -0.78 r = -0.30 score = 0.237

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. q95

z = +0.43 r = +0.51 score = 0.222

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. stream_elas

z = -0.51 r = -0.43 score = 0.219

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01181000: WEST BRANCH WESTFIELD RIVER AT HUNTINGTON, MA

Dominant forcing: Tair Peak TE: 0.0016 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with high runoff_ratio, high gauge_lon, low low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = +0.76 r = +0.56 score = 0.421

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. gauge_lon

z = +1.41 r = -0.25 score = 0.360

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. low_prec_freq

z = -0.67 r = -0.50 score = 0.338

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. frac_forest

z = +0.95 r = +0.35 score = 0.329

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. stream_elas

z = -0.73 r = -0.43 score = 0.312

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. pet_mean

z = -0.94 r = -0.33 score = 0.311

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. gauge_lat

z = +0.57 r = +0.51 score = 0.291

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. clay_frac

z = -0.83 r = -0.30 score = 0.244

low clay fraction allows easier infiltration and reduces surface runoff.
9. q95

z = +0.45 r = +0.51 score = 0.231

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. aridity

z = -0.75 r = -0.30 score = 0.228

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01187300: HUBBARD RIVER NR. WEST HARTLAND, CT.

Dominant forcing: Tair Peak TE: 0.0018 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low stream_elas, high gauge_lon, high frac_forest.

Top 10 driver attributes

1. stream_elas

z = -1.49 r = -0.43 score = 0.640

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. gauge_lon

z = +1.41 r = -0.25 score = 0.359

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. frac_forest

z = +0.95 r = +0.35 score = 0.328

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. pet_mean

z = -0.91 r = -0.33 score = 0.302

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. runoff_ratio

z = +0.51 r = +0.56 score = 0.285

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. gauge_lat

z = +0.54 r = +0.51 score = 0.271

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. clay_frac

z = -0.90 r = -0.30 score = 0.266

low clay fraction allows easier infiltration and reduces surface runoff.
8. low_prec_freq

z = -0.52 r = -0.50 score = 0.260

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. aridity

z = -0.78 r = -0.30 score = 0.236

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. q95

z = +0.36 r = +0.51 score = 0.186

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01195100: INDIAN RIVER NEAR CLINTON, CT.

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +1.43 r = -0.27 score = 0.390

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.75 r = +0.20 score = 0.351

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gvf_diff

z = +1.17 r = -0.24 score = 0.285

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. hfd_mean

z = -0.81 r = -0.24 score = 0.193

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. aridity

z = -0.68 r = -0.28 score = 0.187

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. elev_mean

z = -0.94 r = -0.20 score = 0.184

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. sand_frac

z = +1.43 r = +0.12 score = 0.168

high sand fraction promotes fast infiltration and subsurface drainage.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. soil_conductivity

z = +0.89 r = +0.15 score = 0.131

high soil conductivity allows fast infiltration and subsurface flow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01333000: GREEN RIVER AT WILLIAMSTOWN, MA

Dominant forcing: Tair Peak TE: 0.0068 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with high runoff_ratio, low low_prec_freq, high gauge_lon.

Top 10 driver attributes

1. runoff_ratio

z = +0.73 r = +0.56 score = 0.407

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. low_prec_freq

z = -0.76 r = -0.50 score = 0.384

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gauge_lon

z = +1.39 r = -0.25 score = 0.355

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. pet_mean

z = -1.04 r = -0.33 score = 0.343

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lat

z = +0.67 r = +0.51 score = 0.336

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. frac_forest

z = +0.93 r = +0.35 score = 0.320

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. slope_mean

z = +0.61 r = +0.41 score = 0.247

steep terrain promotes fast surface runoff and a quick streamflow response.
8. aridity

z = -0.75 r = -0.30 score = 0.226

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. high_prec_freq

z = -0.42 r = -0.49 score = 0.209

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. clay_frac

z = -0.60 r = -0.30 score = 0.177

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01350000: SCHOHARIE CREEK AT PRATTSVILLE NY

Dominant forcing: Tair Peak TE: 0.0015 Peak lag: 24 day(s)
Summary: Tair-dominant at lag 24 day(s); top drivers consistent with high runoff_ratio, high slope_mean, high gauge_lon.

Top 10 driver attributes

1. runoff_ratio

z = +0.68 r = +0.56 score = 0.375

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. slope_mean

z = +0.89 r = +0.41 score = 0.364

steep terrain promotes fast surface runoff and a quick streamflow response.
3. gauge_lon

z = +1.32 r = -0.25 score = 0.335

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. frac_forest

z = +0.96 r = +0.35 score = 0.331

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. pet_mean

z = -0.98 r = -0.33 score = 0.323

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lat

z = +0.59 r = +0.51 score = 0.299

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. low_prec_freq

z = -0.53 r = -0.50 score = 0.269

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. aridity

z = -0.77 r = -0.30 score = 0.234

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. clay_frac

z = -0.76 r = -0.30 score = 0.225

low clay fraction allows easier infiltration and reduces surface runoff.
10. q95

z = +0.43 r = +0.51 score = 0.221

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01350080: MANOR KILL AT WEST CONESVILLE NEAR GILBOA NY

Dominant forcing: SRAD Peak TE: 0.0026 Peak lag: 26 day(s)
Summary: SRAD-dominant at lag 26 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, high gvf_diff.

Top 10 driver attributes

1. gauge_lon z = +1.32 r = -0.43 score = 0.570

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur z = -0.96 r = +0.29 score = 0.277

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gvf_diff z = +0.72 r = -0.33 score = 0.235

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. glim_1st_class_frac z = +1.05 r = -0.21 score = 0.225

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
5. frac_forest z = +0.95 r = +0.23 score = 0.223

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. glim_2nd_class_frac z = -1.06 r = +0.18 score = 0.187

a small secondary geological class means the subsurface is more uniform.
7. lai_diff z = +1.27 r = -0.14 score = 0.178

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
8. gauge_lat z = +0.60 r = +0.29 score = 0.176

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. low_prec_freq z = -0.54 r = -0.31 score = 0.168

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. stream_elas z = -0.63 r = -0.26 score = 0.164

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01350140: MINE KILL NEAR NORTH BLENHEIM NY

Dominant forcing: Tair Peak TE: 0.0045 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low low_prec_freq, low stream_elas, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq

z = -0.78 r = -0.50 score = 0.391

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.84 r = -0.43 score = 0.359

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. gauge_lon

z = +1.31 r = -0.25 score = 0.335

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. frac_forest

z = +0.94 r = +0.35 score = 0.326

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. pet_mean

z = -0.99 r = -0.33 score = 0.326

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lat

z = +0.61 r = +0.51 score = 0.309

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. slope_fdc

z = +1.19 r = +0.24 score = 0.288

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. high_prec_freq

z = -0.56 r = -0.49 score = 0.274

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. aridity

z = -0.68 r = -0.30 score = 0.205

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. clay_frac

z = -0.68 r = -0.30 score = 0.200

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01365000: RONDOUT CREEK NEAR LOWES CORNERS NY

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low soil_depth_statsgo, high gvf_diff.

Top 10 driver attributes

1. gauge_lon z = +1.31 r = -0.27 score = 0.357

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. soil_depth_statsgo z = -2.22 r = +0.13 score = 0.298

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
3. gvf_diff z = +1.21 r = -0.24 score = 0.293

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. runoff_ratio z = +1.11 r = +0.23 score = 0.256

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. p_mean z = +0.68 r = +0.34 score = 0.233

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. aridity z = -0.83 r = -0.28 score = 0.229

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. q_mean z = +0.79 r = +0.28 score = 0.220

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
8. high_prec_dur z = -0.93 r = +0.23 score = 0.211

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. q95 z = +0.60 r = +0.31 score = 0.182

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. frac_forest z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01411300: TUCKAHOE RIVER AT HEAD OF RIVER NJ

Dominant forcing: PRCP Peak TE: 0.0095 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, high gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.378

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.29 r = -0.27 score = 0.351

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. gvf_diff

z = +0.79 r = -0.24 score = 0.193

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. elev_mean

z = -0.95 r = -0.20 score = 0.185

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. root_depth_99

z = +1.79 r = +0.10 score = 0.181

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. soil_conductivity

z = +1.13 r = +0.15 score = 0.166

high soil conductivity allows fast infiltration and subsurface flow.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. q95

z = -0.41 r = +0.31 score = 0.126

low peak-flow values reflect a damped streamflow response with limited peaks.
10. stream_elas

z = -1.00 r = -0.12 score = 0.124

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01413500: EAST BR DELAWARE R AT MARGARETVILLE NY

Dominant forcing: Tair Peak TE: 0.0022 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with low low_prec_freq, low stream_elas, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.86 r = -0.50 score = 0.431

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.79 r = -0.43 score = 0.340

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. frac_forest

z = +0.96 r = +0.35 score = 0.332

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. gauge_lon

z = +1.30 r = -0.25 score = 0.332

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. pet_mean

z = -1.00 r = -0.33 score = 0.330

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. runoff_ratio

z = +0.53 r = +0.56 score = 0.295

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gauge_lat

z = +0.56 r = +0.51 score = 0.282

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. slope_fdc

z = +1.09 r = +0.24 score = 0.263

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. high_prec_freq

z = -0.49 r = -0.49 score = 0.241

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. aridity

z = -0.77 r = -0.30 score = 0.233

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01414500: MILL BROOK NEAR DUNRAVEN NY

Dominant forcing: Tair Peak TE: 0.0029 Peak lag: 24 day(s)
Summary: Tair-dominant at lag 24 day(s); top drivers consistent with low low_prec_freq, high runoff_ratio, low stream_elas.

Top 10 driver attributes

1. low_prec_freq

z = -0.88 r = -0.50 score = 0.443

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. runoff_ratio

z = +0.75 r = +0.56 score = 0.418

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. stream_elas

z = -0.86 r = -0.43 score = 0.369

low streamflow elasticity means streamflow is buffered against rainfall variability.
4. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. pet_mean

z = -1.00 r = -0.33 score = 0.333

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lon

z = +1.30 r = -0.25 score = 0.331

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. gauge_lat

z = +0.55 r = +0.51 score = 0.278

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. aridity

z = -0.81 r = -0.30 score = 0.244

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. slope_fdc

z = +1.00 r = +0.24 score = 0.242

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. clay_frac

z = -0.76 r = -0.30 score = 0.225

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01415000: TREMPER KILL NEAR ANDES NY

Dominant forcing: Tair Peak TE: 0.0024 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with low low_prec_freq, high frac_forest, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq

z = -1.11 r = -0.50 score = 0.561

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
3. gauge_lon

z = +1.29 r = -0.25 score = 0.329

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. pet_mean

z = -0.99 r = -0.33 score = 0.328

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. stream_elas

z = -0.73 r = -0.43 score = 0.313

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. high_prec_freq

z = -0.58 r = -0.49 score = 0.285

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. gauge_lat

z = +0.55 r = +0.51 score = 0.279

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. slope_fdc

z = +1.05 r = +0.24 score = 0.254

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. aridity

z = -0.75 r = -0.30 score = 0.228

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. runoff_ratio

z = +0.40 r = +0.56 score = 0.220

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01423000: WEST BRANCH DELAWARE RIVER AT WALTON NY

Dominant forcing: Tair Peak TE: 0.0033 Peak lag: 11 day(s)
Summary: Tair-dominant at lag 11 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, low pet_mean.

Top 10 driver attributes

1. low_prec_freq z = -1.23 r = -0.50 score = 0.622

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq z = -1.10 r = -0.49 score = 0.539

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. pet_mean z = -1.01 r = -0.33 score = 0.334

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. gauge_lon z = +1.27 r = -0.25 score = 0.324

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. frac_forest z = +0.93 r = +0.35 score = 0.321

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. gauge_lat z = +0.56 r = +0.51 score = 0.284

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. slope_fdc z = +1.12 r = +0.24 score = 0.271

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. runoff_ratio z = +0.47 r = +0.56 score = 0.262

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. aridity z = -0.74 r = -0.30 score = 0.224

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. stream_elas z = -0.45 r = -0.43 score = 0.194

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01434025: BISCUIT BK ABOVE PIGEON BK AT FROST VALLEY NY

Dominant forcing: Tair Peak TE: 0.0028 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low stream_elas, high runoff_ratio, low low_prec_freq.

Top 10 driver attributes

1. stream_elas

z = -1.40 r = -0.43 score = 0.601

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. runoff_ratio

z = +1.06 r = +0.56 score = 0.590

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. low_prec_freq

z = -0.85 r = -0.50 score = 0.429

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. q95

z = +0.81 r = +0.51 score = 0.416

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. q_mean

z = +0.86 r = +0.45 score = 0.383

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lon

z = +1.31 r = -0.25 score = 0.334

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. pet_mean

z = -0.93 r = -0.33 score = 0.307

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. aridity

z = -0.88 r = -0.30 score = 0.268

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gauge_lat

z = +0.53 r = +0.51 score = 0.267

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01435000: NEVERSINK RIVER NEAR CLARYVILLE NY

Dominant forcing: Tair Peak TE: 0.0027 Peak lag: 24 day(s)
Summary: Tair-dominant at lag 24 day(s); top drivers consistent with high runoff_ratio, high q_mean, low low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = +1.29 r = +0.56 score = 0.715

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. q_mean

z = +0.95 r = +0.45 score = 0.427

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. low_prec_freq

z = -0.80 r = -0.50 score = 0.402

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. q95

z = +0.74 r = +0.51 score = 0.378

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. gauge_lon

z = +1.31 r = -0.25 score = 0.333

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. stream_elas

z = -0.75 r = -0.43 score = 0.322

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. pet_mean

z = -0.88 r = -0.33 score = 0.292

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. q5

z = +0.78 r = +0.36 score = 0.286

high low-flow values reflect strong baseflow and steady year-round streamflow.
10. aridity

z = -0.85 r = -0.30 score = 0.258

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01439500: Bush Kill at Shoemakers, PA

Dominant forcing: PRCP Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low aridity.

Top 10 driver attributes

1. gauge_lon

z = +1.28 r = -0.27 score = 0.348

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.42 r = -0.24 score = 0.344

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. aridity

z = -0.68 r = -0.28 score = 0.188

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. hfd_mean

z = -0.79 r = -0.24 score = 0.187

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. high_prec_dur

z = -0.81 r = +0.23 score = 0.185

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. slope_fdc

z = +0.86 r = +0.20 score = 0.173

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. frac_forest

z = +0.96 r = +0.17 score = 0.165

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. glim_2nd_class_frac

z = -1.07 r = +0.15 score = 0.165

a small secondary geological class means the subsurface is more uniform.
9. runoff_ratio

z = +0.70 r = +0.23 score = 0.162

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. glim_1st_class_frac

z = +1.05 r = -0.15 score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01440000: FLAT BROOK NEAR FLATBROOKVILLE NJ

Dominant forcing: PRCP Peak TE: 0.0046 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low aridity.

Top 10 driver attributes

1. gauge_lon

z = +1.29 r = -0.27 score = 0.349

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.43 r = -0.24 score = 0.347

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. aridity

z = -0.64 r = -0.28 score = 0.178

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. slope_fdc

z = +0.81 r = +0.20 score = 0.163

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. hfd_mean

z = -0.68 r = -0.24 score = 0.161

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. stream_elas

z = -1.13 r = -0.12 score = 0.139

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. high_prec_dur

z = -0.59 r = +0.23 score = 0.133

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. elev_mean

z = -0.67 r = -0.20 score = 0.131

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. frac_forest

z = +0.72 r = +0.17 score = 0.124

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. gvf_max

z = +1.04 r = +0.12 score = 0.120

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01440400: Brodhead Creek near Analomink, PA

Dominant forcing: PRCP Peak TE: 0.0026 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon

z = +1.27 r = -0.27 score = 0.345

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.30 r = -0.24 score = 0.317

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur

z = -0.89 r = +0.23 score = 0.202

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. aridity

z = -0.71 r = -0.28 score = 0.195

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
5. hfd_mean

z = -0.79 r = -0.24 score = 0.187

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. frac_forest

z = +0.95 r = +0.17 score = 0.164

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. runoff_ratio

z = +0.70 r = +0.23 score = 0.162

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. slope_fdc

z = +0.69 r = +0.20 score = 0.139

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01451800: Jordan Creek near Schnecksville, PA

Dominant forcing: VP Peak TE: 0.0051 Peak lag: 24 day(s)
Summary: VP-dominant at lag 24 day(s); top drivers consistent with high high_prec_freq, high runoff_ratio, low aridity.

Top 10 driver attributes

1. high_prec_freq

z = +0.63 r = -0.47 score = 0.297

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. runoff_ratio

z = +0.49 r = +0.50 score = 0.246

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. aridity

z = -0.59 r = -0.31 score = 0.187

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. slope_mean

z = -0.66 r = +0.27 score = 0.180

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. pet_mean

z = -0.66 r = -0.25 score = 0.169

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. silt_frac

z = +1.25 r = +0.13 score = 0.168

high silt fraction holds water and slows infiltration.
7. gauge_lon

z = +1.24 r = -0.13 score = 0.164

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. slope_fdc

z = +0.68 r = +0.24 score = 0.162

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. low_prec_dur

z = -0.55 r = -0.27 score = 0.147

short dry spells limit basin drying and keep storage close to full.
10. stream_elas

z = -0.36 r = -0.36 score = 0.132

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01466500: MCDONALDS BRANCH IN LEBANON STATE FOREST NJ

Dominant forcing: PRCP Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high soil_depth_pelletier, high gauge_lon.

Top 10 driver attributes

1. soil_conductivity

z = +4.08 r = +0.15 score = 0.601

high soil conductivity allows fast infiltration and subsurface flow.
2. soil_depth_pelletier

z = +2.39 r = +0.16 score = 0.381

deep soils provide more storage and slower release of water to streamflow.
3. gauge_lon

z = +1.31 r = -0.27 score = 0.357

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. sand_frac

z = +3.00 r = +0.12 score = 0.351

high sand fraction promotes fast infiltration and subsurface drainage.
5. stream_elas

z = -2.06 r = -0.12 score = 0.254

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. q95

z = -0.69 r = +0.31 score = 0.211

low peak-flow values reflect a damped streamflow response with limited peaks.
7. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
8. elev_mean

z = -0.92 r = -0.20 score = 0.179

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. frac_forest

z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01484100: BEAVERDAM BRANCH AT HOUSTON, DE

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, low elev_mean.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.38 r = +0.16 score = 0.381

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.25 r = -0.27 score = 0.340

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. elev_mean

z = -0.94 r = -0.20 score = 0.184

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
4. high_prec_dur

z = -0.74 r = +0.23 score = 0.167

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. sand_frac

z = +1.42 r = +0.12 score = 0.166

high sand fraction promotes fast infiltration and subsurface drainage.
6. q95

z = -0.55 r = +0.31 score = 0.166

low peak-flow values reflect a damped streamflow response with limited peaks.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. soil_conductivity

z = +0.96 r = +0.15 score = 0.141

high soil conductivity allows fast infiltration and subsurface flow.
10. frac_forest

z = -0.75 r = +0.17 score = 0.129

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01485500: NASSAWANGO CREEK NEAR SNOW HILL, MD

Dominant forcing: PRCP Peak TE: 0.0106 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, high soil_conductivity.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.28 r = +0.16 score = 0.364

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.25 r = -0.27 score = 0.340

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. soil_conductivity

z = +1.65 r = +0.15 score = 0.244

high soil conductivity allows fast infiltration and subsurface flow.
4. slope_fdc

z = +1.16 r = +0.20 score = 0.233

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. elev_mean

z = -0.95 r = -0.20 score = 0.185

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. high_prec_dur

z = -0.79 r = +0.23 score = 0.178

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. hfd_mean

z = -0.67 r = -0.24 score = 0.159

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. root_depth_99

z = +1.46 r = +0.10 score = 0.148

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01486000: MANOKIN BRANCH NEAR PRINCESS ANNE, MD

Dominant forcing: PRCP Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, high slope_fdc.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.19 r = +0.16 score = 0.350

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.24 r = -0.27 score = 0.337

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_fdc

z = +1.07 r = +0.20 score = 0.214

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. elev_mean

z = -0.95 r = -0.20 score = 0.186

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. root_depth_99

z = +1.77 r = +0.10 score = 0.179

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. high_prec_dur

z = -0.67 r = +0.23 score = 0.151

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. hfd_mean

z = -0.62 r = -0.24 score = 0.148

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. gvf_diff

z = -0.58 r = -0.24 score = 0.141

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01487000: NANTICOKE RIVER NEAR BRIDGEVILLE, DE

Dominant forcing: PRCP Peak TE: 0.0105 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, low elev_mean.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.36 r = +0.16 score = 0.377

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.25 r = -0.27 score = 0.339

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. elev_mean

z = -0.95 r = -0.20 score = 0.185

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
4. slope_fdc

z = +0.85 r = +0.20 score = 0.170

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
6. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. aridity

z = -0.45 r = -0.28 score = 0.124

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. high_prec_dur

z = -0.54 r = +0.23 score = 0.123

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. q95

z = -0.40 r = +0.31 score = 0.123

low peak-flow values reflect a damped streamflow response with limited peaks.
10. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01491000: CHOPTANK RIVER NEAR GREENSBORO, MD

Dominant forcing: PRCP Peak TE: 0.0057 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, low frac_forest.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.32 r = +0.16 score = 0.370

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.23 r = -0.27 score = 0.335

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. frac_forest

z = -1.09 r = +0.17 score = 0.189

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. elev_mean

z = -0.94 r = -0.20 score = 0.184

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. slope_fdc

z = +0.90 r = +0.20 score = 0.181

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. gvf_diff

z = +0.69 r = -0.24 score = 0.168

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. high_prec_dur

z = -0.71 r = +0.23 score = 0.161

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. aridity

z = -0.44 r = -0.28 score = 0.122

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01510000: OTSELIC RIVER AT CINCINNATUS NY

Dominant forcing: Tair Peak TE: 0.0037 Peak lag: 14 day(s)
Summary: Tair-dominant at lag 14 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.53 r = -0.50 score = 0.772

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.04 r = -0.49 score = 0.512

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.69 r = +0.56 score = 0.385

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. pet_mean

z = -0.98 r = -0.33 score = 0.326

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lat

z = +0.63 r = +0.51 score = 0.320

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. frac_forest

z = +0.91 r = +0.35 score = 0.316

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lon

z = +1.23 r = -0.25 score = 0.312

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. stream_elas

z = -0.65 r = -0.43 score = 0.280

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. slope_fdc

z = +1.04 r = +0.24 score = 0.250

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. geol_permeability

z = -1.82 r = -0.11 score = 0.206

low geological permeability slows subsurface flow and reduces groundwater contribution.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01516500: Corey Creek near Mainesburg, PA

Dominant forcing: Tair Peak TE: 0.0024 Peak lag: 19 day(s)
Summary: Tair-dominant at lag 19 day(s); top drivers consistent with low stream_elas, low low_prec_freq, low pet_mean.

Top 10 driver attributes

1. stream_elas

z = -0.81 r = -0.43 score = 0.347

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. low_prec_freq

z = -0.62 r = -0.50 score = 0.314

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. pet_mean

z = -0.94 r = -0.33 score = 0.311

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. frac_forest

z = +0.88 r = +0.35 score = 0.303

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. gauge_lon

z = +1.16 r = -0.25 score = 0.295

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. high_prec_freq

z = -0.59 r = -0.49 score = 0.290

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. gauge_lat

z = +0.49 r = +0.51 score = 0.247

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. silt_frac

z = +1.57 r = +0.13 score = 0.203

high silt fraction holds water and slows infiltration.
9. q5

z = -0.55 r = +0.36 score = 0.201

low low-flow values reflect basins that nearly dry out during dry periods.
10. slope_fdc

z = +0.76 r = +0.24 score = 0.184

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01518862: Cowanesque River at Westfield, PA

Dominant forcing: Tair Peak TE: 0.0024 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with low low_prec_freq, high frac_forest, low pet_mean.

Top 10 driver attributes

1. low_prec_freq

z = -0.95 r = -0.50 score = 0.480

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. frac_forest

z = +0.92 r = +0.35 score = 0.319

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
3. pet_mean

z = -0.94 r = -0.33 score = 0.310

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. gauge_lon

z = +1.13 r = -0.25 score = 0.287

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. high_prec_freq

z = -0.53 r = -0.49 score = 0.263

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. gauge_lat

z = +0.51 r = +0.51 score = 0.260

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. slope_fdc

z = +1.00 r = +0.24 score = 0.242

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. silt_frac

z = +1.49 r = +0.13 score = 0.192

high silt fraction holds water and slows infiltration.
9. p_seasonality

z = +0.50 r = -0.38 score = 0.190

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. q5

z = -0.51 r = +0.36 score = 0.186

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01532000: Towanda Creek near Monroeton, PA

Dominant forcing: Tair Peak TE: 0.0014 Peak lag: 30 day(s)
Summary: Tair-dominant at lag 30 day(s); top drivers consistent with low low_prec_freq, high gauge_lon, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.68 r = -0.50 score = 0.343

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gauge_lon

z = +1.19 r = -0.25 score = 0.303

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. frac_forest

z = +0.85 r = +0.35 score = 0.293

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. pet_mean

z = -0.88 r = -0.33 score = 0.293

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lat

z = +0.47 r = +0.51 score = 0.239

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. high_prec_freq

z = -0.47 r = -0.49 score = 0.231

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. silt_frac

z = +1.42 r = +0.13 score = 0.183

high silt fraction holds water and slows infiltration.
8. q5

z = -0.49 r = +0.36 score = 0.180

low low-flow values reflect basins that nearly dry out during dry periods.
9. glim_1st_class_frac

z = +1.06 r = -0.16 score = 0.171

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. low_prec_dur

z = -0.83 r = -0.20 score = 0.168

short dry spells limit basin drying and keep storage close to full.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01539000: Fishing Creek near Bloomsburg, PA

Dominant forcing: SRAD Peak TE: 0.0015 Peak lag: 8 day(s)
Summary: SRAD-dominant at lag 8 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +1.19 r = -0.43 score = 0.516

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -1.12 r = +0.29 score = 0.321

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gvf_diff

z = +0.94 r = -0.33 score = 0.308

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. runoff_ratio

z = +0.57 r = +0.43 score = 0.243

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
7. low_prec_freq

z = -0.53 r = -0.31 score = 0.166

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. lai_diff

z = +1.17 r = -0.14 score = 0.164

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
9. p_seasonality

z = +0.34 r = -0.46 score = 0.156

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. stream_elas

z = -0.55 r = -0.26 score = 0.141

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01542810: Waldy Run near Emporium, PA

Dominant forcing: VP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -1.35 r = -0.50 score = 0.674

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.12 r = -0.47 score = 0.526

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.97 r = +0.29 score = 0.281

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. low_prec_dur

z = -0.93 r = -0.27 score = 0.248

short dry spells limit basin drying and keep storage close to full.
5. pet_mean

z = -0.93 r = -0.25 score = 0.238

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. slope_fdc

z = +0.85 r = +0.24 score = 0.203

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. aridity

z = -0.64 r = -0.31 score = 0.201

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. stream_elas

z = -0.53 r = -0.36 score = 0.191

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. silt_frac

z = +1.40 r = +0.13 score = 0.189

high silt fraction holds water and slows infiltration.
10. runoff_ratio

z = +0.34 r = +0.50 score = 0.171

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01543000: Driftwood Br Sinnemahoning Cr at Sterling Run, PA

Dominant forcing: VP Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -1.24 r = -0.50 score = 0.621

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.99 r = -0.47 score = 0.464

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.95 r = +0.29 score = 0.275

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. low_prec_dur

z = -0.92 r = -0.27 score = 0.245

short dry spells limit basin drying and keep storage close to full.
5. slope_fdc

z = +0.98 r = +0.24 score = 0.235

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. pet_mean

z = -0.92 r = -0.25 score = 0.234

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. runoff_ratio

z = +0.42 r = +0.50 score = 0.212

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. aridity

z = -0.63 r = -0.31 score = 0.196

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. silt_frac

z = +1.17 r = +0.13 score = 0.158

high silt fraction holds water and slows infiltration.
10. gvf_max

z = +1.03 r = +0.14 score = 0.147

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01543500: Sinnemahoning Creek at Sinnemahoning, PA

Dominant forcing: VP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -1.25 r = -0.50 score = 0.626

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.84 r = -0.47 score = 0.396

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.93 r = +0.29 score = 0.272

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. slope_fdc

z = +1.10 r = +0.24 score = 0.264

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. low_prec_dur

z = -0.92 r = -0.27 score = 0.245

short dry spells limit basin drying and keep storage close to full.
6. pet_mean

z = -0.91 r = -0.25 score = 0.231

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. aridity

z = -0.63 r = -0.31 score = 0.197

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. runoff_ratio

z = +0.39 r = +0.50 score = 0.195

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. gvff_max

z = +1.02 r = +0.14 score = 0.146

high maximum green vegetation fraction reflects dense canopy cover.
10. gauge_lon

z = +1.09 r = -0.13 score = 0.144

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01544500: Kettle Creek at Cross Fork, PA

Dominant forcing: VP Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low stream_elas, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq

z = -1.06 r = -0.50 score = 0.527

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -1.02 r = -0.36 score = 0.367

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. high_prec_freq

z = -0.75 r = -0.47 score = 0.355

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. frac_forest

z = +0.97 r = +0.29 score = 0.281

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. slope_fdc

z = +1.04 r = +0.24 score = 0.248

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. low_prec_dur

z = -0.90 r = -0.27 score = 0.241

short dry spells limit basin drying and keep storage close to full.
7. pet_mean

z = -0.89 r = -0.25 score = 0.227

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. runoff_ratio

z = +0.42 r = +0.50 score = 0.211

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. aridity

z = -0.61 r = -0.31 score = 0.191

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gauge_lat

z = +0.43 r = +0.35 score = 0.151

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01545600: Young Womans Creek near Renovo, PA

Dominant forcing: Tair Peak TE: 0.0024 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low low_prec_freq, low stream_elas, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq

z = -0.99 r = -0.50 score = 0.498

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.98 r = -0.43 score = 0.419

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. high_prec_freq

z = -0.74 r = -0.49 score = 0.366

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. pet_mean

z = -0.88 r = -0.33 score = 0.292

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lon

z = +1.12 r = -0.25 score = 0.284

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. slope_fdc

z = +1.01 r = +0.24 score = 0.244

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. gauge_lat

z = +0.41 r = +0.51 score = 0.208

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. aridity

z = -0.61 r = -0.30 score = 0.186

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. low_prec_dur

z = -0.89 r = -0.20 score = 0.181

short dry spells limit basin drying and keep storage close to full.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01547700: Marsh Creek at Blanchard, PA

Dominant forcing: SRAD Peak TE: 0.0014 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon

z = +1.12 r = -0.43 score = 0.485

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.20 r = -0.33 score = 0.392

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur

z = -1.13 r = +0.29 score = 0.326

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. glim_1st_class_frac

z = +1.05 r = -0.21 score = 0.224

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
5. low_prec_freq

z = -0.71 r = -0.31 score = 0.222

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. lai_diff

z = +1.43 r = -0.14 score = 0.200

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
7. glim_2nd_class_frac

z = -1.06 r = +0.18 score = 0.186

a small secondary geological class means the subsurface is more uniform.
8. stream_elas

z = +0.71 r = -0.26 score = 0.183

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. frac_forest

z = +0.71 r = +0.23 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. p_seasonality

z = +0.34 r = -0.46 score = 0.156

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01548500: Pine Creek at Cedar Run, PA

Dominant forcing: Tair Peak TE: 0.0029 Peak lag: 24 day(s)
Summary: Tair-dominant at lag 24 day(s); top drivers consistent with low low_prec_freq, low stream_elas, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.96 r = -0.50 score = 0.483

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.93 r = -0.43 score = 0.398

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. frac_forest

z = +0.93 r = +0.35 score = 0.322

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. pet_mean

z = -0.93 r = -0.33 score = 0.308

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. high_prec_freq

z = -0.62 r = -0.49 score = 0.306

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. gauge_lon

z = +1.13 r = -0.25 score = 0.288

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. slope_fdc

z = +0.97 r = +0.24 score = 0.234

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. gauge_lat

z = +0.44 r = +0.51 score = 0.221

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. low_prec_dur

z = -0.87 r = -0.20 score = 0.176

short dry spells limit basin drying and keep storage close to full.
10. glim_1st_class_frac

z = +1.06 r = -0.16 score = 0.171

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01549500: Blockhouse Creek near English Center, PA

Dominant forcing: Tair Peak TE: 0.0035 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low stream_elas, high frac_forest, low pet_mean.

Top 10 driver attributes

1. stream_elas

z = -1.06 r = -0.43 score = 0.453

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. frac_forest

z = +0.94 r = +0.35 score = 0.324

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
3. pet_mean

z = -0.89 r = -0.33 score = 0.294

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. gauge_lon

z = +1.14 r = -0.25 score = 0.291

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. low_prec_freq

z = -0.57 r = -0.50 score = 0.289

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. gauge_lat

z = +0.43 r = +0.51 score = 0.216

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. high_prec_freq

z = -0.41 r = -0.49 score = 0.204

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. slope_fdc

z = +0.83 r = +0.24 score = 0.202

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. silt_frac

z = +1.41 r = +0.13 score = 0.182

high silt fraction holds water and slows infiltration.
10. aridity

z = -0.60 r = -0.30 score = 0.181

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01550000: Lycoming Creek near Trout Run, PA

Dominant forcing: Tair Peak TE: 0.0023 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.68 r = -0.50 score = 0.342

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.69 r = -0.49 score = 0.339

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.94 r = +0.35 score = 0.323

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. pet_mean

z = -0.89 r = -0.33 score = 0.295

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lon

z = +1.16 r = -0.25 score = 0.295

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. runoff_ratio

z = +0.47 r = +0.56 score = 0.260

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. slope_fdc

z = +1.00 r = +0.24 score = 0.241

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. gauge_lat

z = +0.42 r = +0.51 score = 0.211

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. aridity

z = -0.61 r = -0.30 score = 0.185

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. clay_frac

z = -0.61 r = -0.30 score = 0.179

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01552000: Loyalsock Creek at Loyalsockville, PA

Dominant forcing: Tair Peak TE: 0.0021 Peak lag: 30 day(s)
Summary: Tair-dominant at lag 30 day(s); top drivers consistent with low low_prec_freq, low stream_elas, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.85 r = -0.50 score = 0.427

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.79 r = -0.43 score = 0.339

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. frac_forest

z = +0.94 r = +0.35 score = 0.326

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. gauge_lon

z = +1.16 r = -0.25 score = 0.296

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. pet_mean

z = -0.86 r = -0.33 score = 0.284

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. high_prec_freq

z = -0.52 r = -0.49 score = 0.258

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. runoff_ratio

z = +0.46 r = +0.56 score = 0.256

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. slope_fdc

z = +0.85 r = +0.24 score = 0.206

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. gauge_lat

z = +0.40 r = +0.51 score = 0.202

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. aridity

z = -0.64 r = -0.30 score = 0.193

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01552500: Muncy Creek near Sonestown, PA

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 11 day(s)
Summary: Tair-dominant at lag 11 day(s); top drivers consistent with high runoff_ratio, low stream_elas, low low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = +0.69 r = +0.56 score = 0.381

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. stream_elas

z = -0.83 r = -0.43 score = 0.357

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. low_prec_freq

z = -0.67 r = -0.50 score = 0.340

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. frac_forest

z = +0.96 r = +0.35 score = 0.330

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. gauge_lon

z = +1.19 r = -0.25 score = 0.302

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. pet_mean

z = -0.89 r = -0.33 score = 0.295

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. aridity

z = -0.69 r = -0.30 score = 0.209

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. gauge_lat

z = +0.41 r = +0.51 score = 0.205

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. slope_fdc

z = +0.72 r = +0.24 score = 0.174

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. glim_1st_class_frac

z = +1.06 r = -0.16 score = 0.171

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01557500: Bald Eagle Creek at Tyrone, PA

Dominant forcing: Tair Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, high runoff_ratio, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq

z = -0.71 r = -0.50 score = 0.358

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. runoff_ratio

z = +0.51 r = +0.56 score = 0.286

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. gauge_lon

z = +1.08 r = -0.25 score = 0.276

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. pet_mean

z = -0.77 r = -0.33 score = 0.255

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. slope_fdc

z = +1.05 r = +0.24 score = 0.253

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. frac_forest

z = +0.70 r = +0.35 score = 0.243

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. high_prec_freq

z = -0.44 r = -0.49 score = 0.214

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. low_prec_dur

z = -0.80 r = -0.20 score = 0.162

short dry spells limit basin drying and keep storage close to full.
9. soil_depth_pelletier

z = -0.60 r = -0.26 score = 0.154

shallow soils limit storage and produce a quicker streamflow response.
10. aridity

z = -0.50 r = -0.30 score = 0.152

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01567500: Bixler Run near Loysville, PA

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 24 day(s)
Summary: SRAD-dominant at lag 24 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon z = +1.13 r = -0.43 score = 0.490

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +0.87 r = -0.33 score = 0.286

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_1st_class_frac z = -1.16 r = -0.21 score = 0.248

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
4. high_prec_dur z = -0.71 r = +0.29 score = 0.205

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. soil_depth_pelletier z = -0.61 r = -0.24 score = 0.147

shallow soils limit storage and produce a quicker streamflow response.
6. soil_conductivity z = -0.54 r = +0.25 score = 0.138

low soil conductivity restricts infiltration, often increasing surface runoff.
7. elev_mean z = -0.57 r = +0.24 score = 0.137

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. p_seasonality z = +0.29 r = -0.46 score = 0.133

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. soil_porosity z = +0.69 r = -0.17 score = 0.119

high soil porosity increases water storage capacity and dampens streamflow response.
10. sand_frac z = -0.64 r = +0.18 score = 0.113

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01568000: Sherman Creek at Shermans Dale, PA

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.15 r = -0.27 score = 0.312

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.11 r = -0.24 score = 0.270

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_1st_class_frac

z = -1.25 r = -0.15 score = 0.182

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
4. slope_fdc

z = +0.88 r = +0.20 score = 0.178

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. hfd_mean

z = -0.62 r = -0.24 score = 0.147

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. aridity

z = -0.53 r = -0.28 score = 0.147

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. high_prec_dur

z = -0.61 r = +0.23 score = 0.139

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. elev_mean

z = -0.63 r = -0.20 score = 0.123

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
10. gvf_max

z = +0.72 r = +0.12 score = 0.083

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01580000: DEER CREEK AT ROCKS, MD

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with low stream_elas, high baseflow_index, high gauge_lon.

Top 10 driver attributes

1. stream_elas

z = -0.93 r = -0.43 score = 0.399

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. baseflow_index

z = +1.38 r = +0.22 score = 0.310

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. gauge_lon

z = +1.20 r = -0.25 score = 0.304

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. slope_mean

z = -0.72 r = +0.41 score = 0.292

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. q5

z = +0.66 r = +0.36 score = 0.242

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. elev_mean

z = -0.74 r = +0.30 score = 0.221

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. frac_snow

z = -0.47 r = +0.46 score = 0.218

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. geol_porostiy

z = -1.67 r = -0.13 score = 0.213

low geological porosity reduces subsurface storage and produces a flashier response.
9. q95

z = -0.41 r = +0.51 score = 0.210

low peak-flow values reflect a damped streamflow response with limited peaks.
10. frac_forest

z = -0.58 r = +0.35 score = 0.200

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01583500: WESTERN RUN AT WESTERN RUN, MD

Dominant forcing: Tair Peak TE: 0.0020 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with low stream_elas, high baseflow_index, high gauge_lon.

Top 10 driver attributes

1. stream_elas

z = -1.00 r = -0.43 score = 0.430

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. baseflow_index

z = +1.47 r = +0.22 score = 0.330

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. gauge_lon

z = +1.18 r = -0.25 score = 0.300

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. slope_mean

z = -0.69 r = +0.41 score = 0.281

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. q95

z = -0.52 r = +0.51 score = 0.265

low peak-flow values reflect a damped streamflow response with limited peaks.
6. elev_mean

z = -0.77 r = +0.30 score = 0.228

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. frac_snow

z = -0.48 r = +0.46 score = 0.220

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. geol_porostiy

z = -1.54 r = -0.13 score = 0.197

low geological porosity reduces subsurface storage and produces a flashier response.
9. q5

z = +0.47 r = +0.36 score = 0.171

high low-flow values reflect strong baseflow and steady year-round streamflow.
10. low_q_freq

z = -1.20 r = -0.14 score = 0.164

infrequent low-flow events indicate continuous streamflow throughout the year.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01586610: MORGAN RUN NEAR LOUISVILLE, MD

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with low stream_elas, low slope_mean, high gauge_lon.

Top 10 driver attributes

1. stream_elas

z = -0.78 r = -0.43 score = 0.333

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. slope_mean

z = -0.73 r = +0.41 score = 0.297

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. gauge_lon

z = +1.16 r = -0.25 score = 0.296

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. frac_forest

z = -0.84 r = +0.35 score = 0.292

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. baseflow_index

z = +1.05 r = +0.22 score = 0.235

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
6. elev_mean

z = -0.72 r = +0.30 score = 0.213

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. q95

z = -0.41 r = +0.51 score = 0.210

low peak-flow values reflect a damped streamflow response with limited peaks.
8. geol_porostiy

z = -1.63 r = -0.13 score = 0.208

low geological porosity reduces subsurface storage and produces a flashier response.
9. frac_snow

z = -0.44 r = +0.46 score = 0.202

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. high_prec_freq

z = +0.32 r = -0.49 score = 0.159

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01591400: CATTAIL CREEK NEAR GLENWOOD, MD

Dominant forcing: PRCP Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low glim_2nd_class_frac, low q95.

Top 10 driver attributes

1. gauge_lon

z = +1.16 r = -0.27 score = 0.314

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = -1.04 r = +0.15 score = 0.159

a small secondary geological class means the subsurface is more uniform.
3. q95

z = -0.52 r = +0.31 score = 0.159

low peak-flow values reflect a damped streamflow response with limited peaks.
4. elev_mean

z = -0.78 r = -0.20 score = 0.152

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. glim_1st_class_frac

z = +1.03 r = -0.15 score = 0.150

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. high_prec_dur

z = -0.65 r = +0.23 score = 0.147

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. frac_forest

z = -0.83 r = +0.17 score = 0.143

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. aridity

z = -0.42 r = -0.28 score = 0.116

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.
10. soil_depth_statsgo

z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01594950: MCMILLAN F NEAR FORT PENDLETON, MD

Dominant forcing: Tair Peak TE: 0.0044 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low_low_prec_freq, low_high_prec_freq, high_slope_fdc.

Top 10 driver attributes

1. low_prec_freq z = -1.72 r = -0.50 score = 0.867

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq z = -1.25 r = -0.49 score = 0.615

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. slope_fdc z = +1.97 r = +0.24 score = 0.476

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. frac_forest z = +0.90 r = +0.35 score = 0.310

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. gauge_lon z = +1.01 r = -0.25 score = 0.258

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. geol_permeability z = -2.02 r = -0.11 score = 0.229

low geological permeability slows subsurface flow and reduces groundwater contribution.
7. aridity z = -0.70 r = -0.30 score = 0.212

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. q5 z = -0.56 r = +0.36 score = 0.205

low low-flow values reflect basins that nearly dry out during dry periods.
9. pet_mean z = -0.60 r = -0.33 score = 0.199

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. low_prec_dur z = -0.97 r = -0.20 score = 0.198

short dry spells limit basin drying and keep storage close to full.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01596500: SAVAGE RIVER NEAR BARTON, MD

Dominant forcing: Tair Peak TE: 0.0052 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low low_prec_freq, high slope_fdc, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq

z = -0.90 r = -0.50 score = 0.452

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. slope_fdc

z = +1.15 r = +0.24 score = 0.277

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gauge_lon

z = +1.03 r = -0.25 score = 0.262

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. frac_forest

z = +0.63 r = +0.35 score = 0.219

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. clay_frac

z = -0.71 r = -0.30 score = 0.210

low clay fraction allows easier infiltration and reduces surface runoff.
6. pet_mean

z = -0.63 r = -0.33 score = 0.209

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. high_prec_freq

z = -0.39 r = -0.49 score = 0.193

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. runoff_ratio

z = +0.33 r = +0.56 score = 0.182

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. low_prec_dur

z = -0.84 r = -0.20 score = 0.171

short dry spells limit basin drying and keep storage close to full.
10. glim_1st_class_frac

z = +1.06 r = -0.16 score = 0.171

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01605500: SOUTH BRANCH POTOMAC RIVER AT FRANKLIN, WV

Dominant forcing: Tair Peak TE: 0.0006 Peak lag: 22 day(s)
Summary: Tair-dominant at lag 22 day(s); top drivers consistent with high frac_forest, low low_prec_freq, high gauge_lon.

Top 10 driver attributes

1. frac_forest

z = +0.94 r = +0.35 score = 0.326

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. low_prec_freq

z = -0.63 r = -0.50 score = 0.320

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gauge_lon

z = +1.01 r = -0.25 score = 0.258

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. slope_mean

z = +0.51 r = +0.41 score = 0.210

steep terrain promotes fast surface runoff and a quick streamflow response.
5. q95

z = -0.38 r = +0.51 score = 0.194

low peak-flow values reflect a damped streamflow response with limited peaks.
6. high_prec_freq

z = -0.39 r = -0.49 score = 0.193

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. runoff_ratio

z = -0.31 r = +0.56 score = 0.173

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. pet_mean

z = -0.49 r = -0.33 score = 0.162

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.156

shallow soils limit storage and produce a quicker streamflow response.
10. q_mean

z = -0.31 r = +0.45 score = 0.140

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01606500: SO. BRANCH POTOMAC RIVER NR PETERSBURG, WV

Dominant forcing: PRCP Peak TE: 0.0010 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low low_prec_freq.

Top 10 driver attributes

1. gauge_lon z = +1.02 r = -0.27 score = 0.278

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +1.12 r = -0.24 score = 0.273

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. low_prec_freq z = -1.13 r = -0.22 score = 0.253

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. high_prec_dur z = -0.82 r = +0.23 score = 0.186

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. slope_fdc z = +0.92 r = +0.20 score = 0.185

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. soil_depth_statsgo z = -1.29 r = +0.13 score = 0.173

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. high_prec_freq z = -0.86 r = -0.19 score = 0.163

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. aridity z = -0.51 r = -0.28 score = 0.141

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_forest z = +0.79 r = +0.17 score = 0.136

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. gvf_max z = +0.87 r = +0.12 score = 0.101

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01613050: Tonoloway Creek near Needmore, PA

Dominant forcing: PRCP Peak TE: 0.0017 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon

z = +1.09 r = -0.27 score = 0.296

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.00 r = -0.24 score = 0.243

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur

z = -1.02 r = +0.23 score = 0.232

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. slope_fdc

z = +0.90 r = +0.20 score = 0.181

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
6. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. frac_forest

z = +0.79 r = +0.17 score = 0.137

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. hfd_mean

z = -0.56 r = -0.24 score = 0.133

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. elev_mean

z = -0.63 r = -0.20 score = 0.124

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. aridity

z = -0.43 r = -0.28 score = 0.119

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01620500: NORTH RIVER NEAR STOKESVILLE, VA

Dominant forcing: SRAD Peak TE: 0.0012 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon z = +1.02 r = -0.43 score = 0.441

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +0.75 r = -0.33 score = 0.245

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_1st_class_frac z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
4. slope_mean z = +0.61 r = +0.37 score = 0.224

steep terrain promotes fast surface runoff and a quick streamflow response.
5. frac_forest z = +0.89 r = +0.23 score = 0.210

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. glim_2nd_class_frac z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
7. p_seasonality z = +0.41 r = -0.46 score = 0.188

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. lai_diff z = +1.30 r = -0.14 score = 0.182

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
9. runoff_ratio z = +0.40 r = +0.43 score = 0.171

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. q5 z = -0.49 r = +0.31 score = 0.153

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01632000: N F SHENANDOAH RIVER AT COOTES STORE, VA

Dominant forcing: Tair Peak TE: 0.0013 Peak lag: 1 day(s)

Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low q5, high p_seasonality.

Top 10 driver attributes

1. gauge_lon z = +1.04 r = -0.25 score = 0.266

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

2. q5 z = -0.57 r = +0.36 score = 0.209

low low-flow values reflect basins that nearly dry out during dry periods.

3. p_seasonality z = +0.49 r = -0.38 score = 0.189

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

4. q95 z = -0.32 r = +0.51 score = 0.163

low peak-flow values reflect a damped streamflow response with limited peaks.

5. q_mean z = -0.35 r = +0.45 score = 0.159

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

6. geol_permeability z = -1.40 r = -0.11 score = 0.158

low geological permeability slows subsurface flow and reduces groundwater contribution.

7. soil_depth_pelletier z = -0.60 r = -0.26 score = 0.154

shallow soils limit storage and produce a quicker streamflow response.

8. frac_snow z = -0.33 r = +0.46 score = 0.153

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

9. runoff_ratio z = -0.28 r = +0.56 score = 0.153

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

10. pet_mean z = -0.43 r = -0.33 score = 0.142

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01632900: SMITH CREEK NEAR NEW MARKET, VA

Dominant forcing: Tair Peak TE: 0.0020 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with high clay_frac, high stream_elas, low q95.

Top 10 driver attributes

1. clay_frac

z = +1.74 r = -0.30 score = 0.513

high clay fraction reduces infiltration and increases surface runoff.
2. stream_elas

z = +1.06 r = -0.43 score = 0.454

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. q95

z = -0.53 r = +0.51 score = 0.273

low peak-flow values reflect a damped streamflow response with limited peaks.
4. gauge_lon

z = +1.06 r = -0.25 score = 0.269

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. runoff_ratio

z = -0.48 r = +0.56 score = 0.267

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q_mean

z = -0.45 r = +0.45 score = 0.201

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. glim_2nd_class_frac

z = +1.64 r = +0.12 score = 0.199

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. p_seasonality

z = +0.49 r = -0.38 score = 0.186

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. frac_snow

z = -0.39 r = +0.46 score = 0.180

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. soil_depth_pelletier

z = -0.60 r = -0.26 score = 0.155

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01634500: CEDAR CREEK NEAR WINCHESTER, VA

Dominant forcing: Tair Peak TE: 0.0026 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with high stream_elas, high gauge_lon, high p_seasonality.

Top 10 driver attributes

1. stream_elas

z = +1.35 r = -0.43 score = 0.577

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. gauge_lon

z = +1.08 r = -0.25 score = 0.274

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. p_seasonality

z = +0.49 r = -0.38 score = 0.188

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. q95

z = -0.34 r = +0.51 score = 0.172

low peak-flow values reflect a damped streamflow response with limited peaks.
5. pet_mean

z = -0.48 r = -0.33 score = 0.157

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.157

shallow soils limit storage and produce a quicker streamflow response.
7. elev_mean

z = -0.53 r = +0.30 score = 0.157

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. q_mean

z = -0.33 r = +0.45 score = 0.146

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. frac_snow

z = -0.31 r = +0.46 score = 0.142

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. gvf_max

z = +0.96 r = +0.14 score = 0.132

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01638480: CATOCTIN CREEK AT TAYLORSTOWN, VA

Dominant forcing: VP Peak TE: 0.0023 Peak lag: 19 day(s)

Summary: VP-dominant at lag 19 day(s); top drivers consistent with high slope_fdc, low q5, low q95.

Top 10 driver attributes

1. slope_fdc

z = +0.91 r = +0.24 score = 0.218

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. q5

z = -0.54 r = +0.34 score = 0.183

low low-flow values reflect basins that nearly dry out during dry periods.
3. q95

z = -0.36 r = +0.49 score = 0.175

low peak-flow values reflect a damped streamflow response with limited peaks.
4. stream_elas

z = +0.48 r = -0.36 score = 0.174

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. slope_mean

z = -0.62 r = +0.27 score = 0.169

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. gauge_lon

z = +1.12 r = -0.13 score = 0.148

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. frac_snow

z = -0.46 r = +0.31 score = 0.143

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. q_mean

z = -0.32 r = +0.43 score = 0.138

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. frac_forest

z = -0.47 r = +0.29 score = 0.135

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. runoff_ratio

z = -0.25 r = +0.50 score = 0.126

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01639500: BIG PIPE CREEK AT BRUCEVILLE, MD

Dominant forcing: Tair Peak TE: 0.0033 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with low frac_forest, low slope_mean, high gauge_lon.

Top 10 driver attributes

1. frac_forest

z = -1.40 r = +0.35 score = 0.485

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. slope_mean

z = -0.73 r = +0.41 score = 0.299

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. gauge_lon

z = +1.14 r = -0.25 score = 0.291

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. elev_mean

z = -0.71 r = +0.30 score = 0.212

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. clay_frac

z = -0.63 r = -0.30 score = 0.188

low clay fraction allows easier infiltration and reduces surface runoff.
6. frac_snow

z = -0.40 r = +0.46 score = 0.185

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
7. silt_frac

z = +1.36 r = +0.13 score = 0.176

high silt fraction holds water and slows infiltration.
8. soil_depth_pelletier

z = -0.60 r = -0.26 score = 0.154

shallow soils limit storage and produce a quicker streamflow response.
9. q95

z = -0.30 r = +0.51 score = 0.151

low peak-flow values reflect a damped streamflow response with limited peaks.
10. pet_mean

z = -0.45 r = -0.33 score = 0.150

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01644000: GOOSE CREEK NEAR LEESBURG, VA

Dominant forcing: VP Peak TE: 0.0024 Peak lag: 14 day(s)
Summary: VP-dominant at lag 14 day(s); top drivers consistent with high stream_elas, high slope_fdc, low q5.

Top 10 driver attributes

1. stream_elas

z = +1.13 r = -0.36 score = 0.407

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. slope_fdc

z = +1.14 r = +0.24 score = 0.272

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. q5

z = -0.57 r = +0.34 score = 0.191

low low-flow values reflect basins that nearly dry out during dry periods.
4. q95

z = -0.33 r = +0.49 score = 0.161

low peak-flow values reflect a damped streamflow response with limited peaks.
5. gauge_lon

z = +1.12 r = -0.13 score = 0.148

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. frac_snow

z = -0.46 r = +0.31 score = 0.143

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
7. q_mean

z = -0.32 r = +0.43 score = 0.136

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. slope_mean

z = -0.47 r = +0.27 score = 0.128

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. runoff_ratio

z = -0.25 r = +0.50 score = 0.124

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. low_prec_dur

z = -0.46 r = -0.27 score = 0.123

short dry spells limit basin drying and keep storage close to full.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01658500: S F QUANTICO CREEK NEAR INDEPENDENT HILL, VA

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 29 day(s)
Summary: Tair-dominant at lag 29 day(s); top drivers consistent with low slope_mean, high frac_forest, high gauge_lon.

Top 10 driver attributes

1. slope_mean

z = -0.83 r = +0.41 score = 0.340

gentle terrain slows surface runoff and produces a smoother streamflow signal.
2. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
3. gauge_lon

z = +1.13 r = -0.25 score = 0.288

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. elev_mean

z = -0.84 r = +0.30 score = 0.250

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. frac_snow

z = -0.53 r = +0.46 score = 0.246

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
6. glim_2nd_class_frac

z = +1.91 r = +0.12 score = 0.231

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. runoff_ratio

z = -0.39 r = +0.56 score = 0.214

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q5

z = -0.59 r = +0.36 score = 0.214

low low-flow values reflect basins that nearly dry out during dry periods.
9. q95

z = -0.36 r = +0.51 score = 0.183

low peak-flow values reflect a damped streamflow response with limited peaks.
10. glim_1st_class_frac

z = -1.06 r = -0.16 score = 0.172

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01664000: RAPPAHANNOCK RIVER AT REMINGTON, VA

Dominant forcing: PRCP Peak TE: 0.0045 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high slope_fdc.

Top 10 driver attributes

1. gauge_lon z = +1.11 r = -0.27 score = 0.301

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +1.02 r = -0.24 score = 0.247

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. slope_fdc z = +1.00 r = +0.20 score = 0.201

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. glim_1st_class_frac z = -1.28 r = -0.15 score = 0.186

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. high_prec_dur z = -0.77 r = +0.23 score = 0.174

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. elev_mean z = -0.69 r = -0.20 score = 0.135

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. q95 z = -0.36 r = +0.31 score = 0.109

low peak-flow values reflect a damped streamflow response with limited peaks.
8. aridity z = -0.39 r = -0.28 score = 0.108

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. p_seasonality z = +0.41 r = -0.24 score = 0.100

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. soil_depth_pelletier z = -0.60 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01666500: ROBINSON RIVER NEAR LOCUST DALE, VA

Dominant forcing: VP Peak TE: 0.0017 Peak lag: 16 day(s)
Summary: VP-dominant at lag 16 day(s); top drivers consistent with low q95, high high_prec_freq, low frac_snow.

Top 10 driver attributes

1. q95 z = -0.36 r = +0.49 score = 0.179

low peak-flow values reflect a damped streamflow response with limited peaks.
2. high_prec_freq z = +0.32 r = -0.47 score = 0.152

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. frac_snow z = -0.47 r = +0.31 score = 0.147

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
4. gauge_lon z = +1.09 r = -0.13 score = 0.144

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. frac_forest z = +0.47 r = +0.29 score = 0.136

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. low_prec_dur z = -0.47 r = -0.27 score = 0.125

short dry spells limit basin drying and keep storage close to full.
7. slope_fdc z = +0.51 r = +0.24 score = 0.123

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. slope_mean z = -0.45 r = +0.27 score = 0.122

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. aridity z = -0.37 r = -0.31 score = 0.115

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality z = +0.43 r = -0.27 score = 0.115

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01667500: RAPIDAN RIVER NEAR CULPEPER, VA

Dominant forcing: PRCP Peak TE: 0.0021 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low glim_1st_class_frac, high gauge_lon, high gvf_diff.

Top 10 driver attributes

1. glim_1st_class_frac z = -2.19 r = -0.15 score = 0.318

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
2. gauge_lon z = +1.10 r = -0.27 score = 0.298

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. gvf_diff z = +0.81 r = -0.24 score = 0.198

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. glim_2nd_class_frac z = +1.07 r = +0.15 score = 0.163

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. high_prec_dur z = -0.62 r = +0.23 score = 0.142

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. elev_mean z = -0.72 r = -0.20 score = 0.141

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. slope_fdc z = +0.66 r = +0.20 score = 0.132

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. hfd_mean z = -0.43 r = -0.24 score = 0.103

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. q95 z = -0.34 r = +0.31 score = 0.103

low peak-flow values reflect a damped streamflow response with limited peaks.
10. p_seasonality z = +0.42 r = -0.24 score = 0.102

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01669000: PISCATAWAY CREEK NEAR TAPPAHANNOCK, VA

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high soil_depth_pelletier.

Top 10 driver attributes

1. gauge_lon

z = +1.17

r = -0.27

score = 0.316

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.30

r = +0.20

score = 0.262

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. soil_depth_pelletier

z = +1.52

r = +0.16

score = 0.243

deep soils provide more storage and slower release of water to streamflow.
4. high_prec_dur

z = -0.97

r = +0.23

score = 0.220

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. sand_frac

z = +1.84

r = +0.12

score = 0.216

high sand fraction promotes fast infiltration and subsurface drainage.
6. elev_mean

z = -0.91

r = -0.20

score = 0.178

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. q95

z = -0.52

r = +0.31

score = 0.157

low peak-flow values reflect a damped streamflow response with limited peaks.
9. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. root_depth_99

z = +1.49

r = +0.10

score = 0.150

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 01669520: DRAGON SWAMP AT MASCOT, VA

Dominant forcing: PRCP Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high soil_depth_pelletier.

Top 10 driver attributes

1. gauge_lon

z = +1.18 r = -0.27 score = 0.320

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.51 r = +0.20 score = 0.304

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. soil_depth_pelletier

z = +1.68 r = +0.16 score = 0.269

deep soils provide more storage and slower release of water to streamflow.
4. high_prec_dur

z = -0.92 r = +0.23 score = 0.209

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. sand_frac

z = +1.57 r = +0.12 score = 0.183

high sand fraction promotes fast infiltration and subsurface drainage.
6. elev_mean

z = -0.93 r = -0.20 score = 0.182

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. root_depth_99

z = +1.72 r = +0.10 score = 0.174

very deep maximum rooting allows vegetation to tap deep storage in droughts.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. frac_forest

z = +0.78 r = +0.17 score = 0.134

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02011400: JACKSON RIVER NEAR BACOVA, VA

Dominant forcing: Tair Peak TE: 0.0017 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with high frac_forest, low low_prec_freq, high gauge_lon.

Top 10 driver attributes

1. frac_forest

z = +0.95 r = +0.35 score = 0.330

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. low_prec_freq

z = -0.51 r = -0.50 score = 0.258

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gauge_lon

z = +0.98 r = -0.25 score = 0.250

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. slope_mean

z = +0.59 r = +0.41 score = 0.240

steep terrain promotes fast surface runoff and a quick streamflow response.
5. slope_fdc

z = +0.92 r = +0.24 score = 0.222

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. runoff_ratio

z = -0.37 r = +0.56 score = 0.203

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. glim_2nd_class_frac

z = +1.62 r = +0.12 score = 0.197

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. q95

z = -0.37 r = +0.51 score = 0.189

low peak-flow values reflect a damped streamflow response with limited peaks.
9. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.156

shallow soils limit storage and produce a quicker streamflow response.
10. glim_1st_class_frac

z = -0.90 r = -0.16 score = 0.146

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02011460: BACK CREEK NEAR SUNRISE, VA

Dominant forcing: Tair Peak TE: 0.0018 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with high frac_forest, low low_prec_freq, high slope_mean.

Top 10 driver attributes

1. frac_forest

z = +0.95 r = +0.35 score = 0.330

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. low_prec_freq

z = -0.65 r = -0.50 score = 0.328

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. slope_mean

z = +0.66 r = +0.41 score = 0.267

steep terrain promotes fast surface runoff and a quick streamflow response.
4. slope_fdc

z = +1.09 r = +0.24 score = 0.263

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. gauge_lon

z = +0.99 r = -0.25 score = 0.252

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. high_prec_freq

z = -0.39 r = -0.49 score = 0.193

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. pet_mean

z = -0.49 r = -0.33 score = 0.161

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.157

shallow soils limit storage and produce a quicker streamflow response.
9. geol_permeability

z = -1.38 r = -0.11 score = 0.157

low geological permeability slows subsurface flow and reduces groundwater contribution.
10. aridity

z = -0.52 r = -0.30 score = 0.156

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02013000: DUNLAP CREEK NEAR COVINGTON, VA

Dominant forcing: Tair Peak TE: 0.0032 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with high frac_forest, high gauge_lon, high slope_mean.

Top 10 driver attributes

1. frac_forest

z = +0.93 r = +0.35 score = 0.321

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. gauge_lon

z = +0.97 r = -0.25 score = 0.247

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_mean

z = +0.52 r = +0.41 score = 0.213

steep terrain promotes fast surface runoff and a quick streamflow response.
4. frac_snow

z = -0.39 r = +0.46 score = 0.181

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
5. low_prec_freq

z = -0.34 r = -0.50 score = 0.173

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. q95

z = -0.33 r = +0.51 score = 0.167

low peak-flow values reflect a damped streamflow response with limited peaks.
7. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.156

shallow soils limit storage and produce a quicker streamflow response.
8. runoff_ratio

z = -0.28 r = +0.56 score = 0.156

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. q_mean

z = -0.34 r = +0.45 score = 0.150

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. slope_fdc

z = +0.59 r = +0.24 score = 0.142

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02014000: POTTS CREEK NEAR COVINGTON, VA

Dominant forcing: PRCP Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high gauge_lon, high slope_fdc.

Top 10 driver attributes

1. gvf_diff

z = +1.20 r = -0.24 score = 0.293

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. gauge_lon

z = +0.97 r = -0.27 score = 0.264

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_fdc

z = +0.98 r = +0.20 score = 0.197

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. high_prec_dur

z = -0.86 r = +0.23 score = 0.196

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. soil_depth_statsgo

z = -1.04 r = +0.13 score = 0.139

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
6. frac_forest

z = +0.72 r = +0.17 score = 0.125

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gvf_max

z = +0.98 r = +0.12 score = 0.114

high maximum green vegetation fraction reflects dense canopy cover.
8. aridity

z = -0.39 r = -0.28 score = 0.108

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. q95

z = -0.34 r = +0.31 score = 0.104

low peak-flow values reflect a damped streamflow response with limited peaks.
10. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.098

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02015700: BULLPASTURE RIVER AT WILLIAMSVILLE, VA

Dominant forcing: Tair Peak TE: 0.0017 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with high frac_forest, high gauge_lon, high glim_2nd_class_frac.

Top 10 driver attributes

1. frac_forest

z = +0.94 r = +0.35 score = 0.326

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. gauge_lon

z = +1.00 r = -0.25 score = 0.255

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. glim_2nd_class_frac

z = +2.02 r = +0.12 score = 0.245

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. slope_mean

z = +0.48 r = +0.41 score = 0.195

steep terrain promotes fast surface runoff and a quick streamflow response.
5. glim_1st_class_frac

z = -1.15 r = -0.16 score = 0.186

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. low_prec_freq

z = -0.34 r = -0.50 score = 0.172

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. low_q_freq

z = -1.16 r = -0.14 score = 0.158

infrequent low-flow events indicate continuous streamflow throughout the year.
8. soil_depth_pelletier

z = -0.61 r = -0.26 score = 0.155

shallow soils limit storage and produce a quicker streamflow response.
9. q5

z = +0.41 r = +0.36 score = 0.150

high low-flow values reflect strong baseflow and steady year-round streamflow.
10. pet_mean

z = -0.45 r = -0.33 score = 0.149

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02016000: COWPASTURE RIVER NEAR CLIFTON FORGE, VA

Dominant forcing: PRCP Peak TE: 0.0035 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low soil_depth_statsgo.

Top 10 driver attributes

1. gauge_lon

z = +0.99 r = -0.27 score = 0.269

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +0.92 r = -0.24 score = 0.223

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. soil_depth_statsgo

z = -1.51 r = +0.13 score = 0.202

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
4. high_prec_dur

z = -0.73 r = +0.23 score = 0.165

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. frac_forest

z = +0.78 r = +0.17 score = 0.135

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. gvf_max

z = +0.96 r = +0.12 score = 0.111

high maximum green vegetation fraction reflects dense canopy cover.
7. aridity

z = -0.39 r = -0.28 score = 0.107

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. slope_fdc

z = +0.52 r = +0.20 score = 0.105

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. glim_2nd_class_frac

z = +0.67 r = +0.15 score = 0.103

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02017500: JOHNS CREEK AT NEW CASTLE, VA

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high slope_fdc, high gvf_diff, high gauge_lon.

Top 10 driver attributes

1. slope_fdc

z = +1.36

r = +0.20

score = 0.274

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. gvf_diff

z = +1.09

r = -0.24

score = 0.264

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. gauge_lon

z = +0.97

r = -0.27

score = 0.263

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. high_prec_dur

z = -0.83

r = +0.23

score = 0.188

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. glim_2nd_class_frac

z = +0.97

r = +0.15

score = 0.148

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. soil_depth_statsgo

z = -1.09

r = +0.13

score = 0.145

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. frac_forest

z = +0.79

r = +0.17

score = 0.137

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. gvf_max

z = +0.91

r = +0.12

score = 0.105

high maximum green vegetation fraction reflects dense canopy cover.
9. soil_depth_pelletier

z = -0.61

r = +0.16

score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
10. aridity

z = -0.32

r = -0.28

score = 0.089

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02018000: CRAIG CREEK AT PARR, VA

Dominant forcing: PRCP Peak TE: 0.0073 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high slope_fdc.

Top 10 driver attributes

1. gauge_lon

z = +0.98 r = -0.27 score = 0.266

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.05 r = -0.24 score = 0.256

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. slope_fdc

z = +1.01 r = +0.20 score = 0.203

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. soil_depth_statsgo

z = -1.19 r = +0.13 score = 0.159

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
5. high_prec_dur

z = -0.60 r = +0.23 score = 0.136

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. frac_forest

z = +0.66 r = +0.17 score = 0.114

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gvf_max

z = +0.93 r = +0.12 score = 0.108

high maximum green vegetation fraction reflects dense canopy cover.
8. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
9. aridity

z = -0.32 r = -0.28 score = 0.089

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. hfd_mean

z = -0.37 r = -0.24 score = 0.088

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02027000: TYE RIVER NEAR LOVINGSTON, VA

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 14 day(s)
Summary: SRAD-dominant at lag 14 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high glim_2nd_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.04 r = -0.43 score = 0.448

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.07 r = -0.33 score = 0.349

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_2nd_class_frac

z = +1.85 r = +0.18 score = 0.324

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. glim_1st_class_frac

z = -1.31 r = -0.21 score = 0.280

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. frac_forest

z = +0.81 r = +0.23 score = 0.190

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. high_prec_dur

z = -0.61 r = +0.29 score = 0.177

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. lai_diff

z = +1.22 r = -0.14 score = 0.170

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
8. runoff_ratio

z = +0.37 r = +0.43 score = 0.159

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. soil_depth_pelletier

z = -0.61 r = -0.24 score = 0.147

shallow soils limit storage and produce a quicker streamflow response.
10. p_seasonality

z = +0.31 r = -0.46 score = 0.142

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02027500: PINEY RIVER AT PINEY RIVER, VA

Dominant forcing: PRCP Peak TE: 0.0012 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high slope_fdc.

Top 10 driver attributes

1. gauge_lon

z = +1.03 r = -0.27 score = 0.281

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.07 r = -0.24 score = 0.260

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. slope_fdc

z = +0.96 r = +0.20 score = 0.194

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. glim_2nd_class_frac

z = +1.18 r = +0.15 score = 0.180

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. stream_elas

z = -1.38 r = -0.12 score = 0.170

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. aridity

z = -0.55 r = -0.28 score = 0.151

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. high_prec_dur

z = -0.66 r = +0.23 score = 0.149

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. hfd_mean

z = -0.60 r = -0.24 score = 0.143

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. frac_forest

z = +0.81 r = +0.17 score = 0.139

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. glim_1st_class_frac

z = -0.81 r = -0.15 score = 0.118

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02028500: ROCKFISH RIVER NEAR GREENFIELD, VA

Dominant forcing: SRAD Peak TE: 0.0024 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high gauge_lon, high gvf_diff, high frac_forest.

Top 10 driver attributes

1. gauge_lon

z = +1.05 r = -0.43 score = 0.452

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = +1.14 r = -0.33 score = 0.375

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. frac_forest

z = +0.87 r = +0.23 score = 0.205

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. glim_1st_class_frac

z = -0.90 r = -0.21 score = 0.192

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. lai_diff

z = +1.29 r = -0.14 score = 0.181

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
6. p_seasonality

z = +0.36 r = -0.46 score = 0.166

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. slope_mean

z = +0.44 r = +0.37 score = 0.161

steep terrain promotes fast surface runoff and a quick streamflow response.
8. high_prec_dur

z = -0.54 r = +0.29 score = 0.155

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. soil_depth_pelletier

z = -0.61 r = -0.24 score = 0.146

shallow soils limit storage and produce a quicker streamflow response.
10. frac_snow

z = -0.48 r = +0.26 score = 0.122

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02038850: HOLIDAY CREEK NEAR ANDERSONVILLE, VA

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, low q95.

Top 10 driver attributes

1. gauge_lon

z = +1.06 r = -0.27 score = 0.287

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -0.82 r = +0.23 score = 0.185

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. q95

z = -0.51 r = +0.31 score = 0.154

low peak-flow values reflect a damped streamflow response with limited peaks.
4. root_depth_99

z = +1.44 r = +0.10 score = 0.145

very deep maximum rooting allows vegetation to tap deep storage in droughts.
5. elev_mean

z = -0.74 r = -0.20 score = 0.145

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. frac_forest

z = +0.78 r = +0.17 score = 0.134

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. hfd_mean

z = -0.51 r = -0.24 score = 0.122

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. high_prec_freq

z = +0.62 r = -0.19 score = 0.117

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. soil_conductivity

z = -0.76 r = +0.15 score = 0.112

low soil conductivity restricts infiltration, often increasing surface runoff.
10. gvf_max

z = +0.96 r = +0.12 score = 0.112

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02046000: STONY CREEK NEAR DINWIDDIE, VA

Dominant forcing: PRCP Peak TE: 0.0044 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon

z = +1.12 r = -0.27 score = 0.305

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +0.87 r = +0.20 score = 0.175

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. high_prec_dur

z = -0.77 r = +0.23 score = 0.174

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. elev_mean

z = -0.86 r = -0.20 score = 0.167

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. root_depth_99

z = +1.58 r = +0.10 score = 0.160

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. frac_forest

z = +0.80 r = +0.17 score = 0.137

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. hfd_mean

z = -0.53 r = -0.24 score = 0.125

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. runoff_ratio

z = -0.54 r = +0.23 score = 0.124

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. q95

z = -0.40 r = +0.31 score = 0.123

low peak-flow values reflect a damped streamflow response with limited peaks.
10. q_mean

z = -0.41 r = +0.28 score = 0.114

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02051000: NORTH MEHERRIN RIVER NEAR LUNENBURG, VA

Dominant forcing: Tair Peak TE: 0.0023 Peak lag: 5 day(s)
Summary: Tair-dominant at lag 5 day(s); top drivers consistent with high clay_frac, low runoff_ratio, low slope_mean.

Top 10 driver attributes

1. clay_frac

z = +2.81 r = -0.30 score = 0.831

high clay fraction reduces infiltration and increases surface runoff.
2. runoff_ratio

z = -0.66 r = +0.56 score = 0.369

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. slope_mean

z = -0.82 r = +0.41 score = 0.335

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. high_prec_freq

z = +0.66 r = -0.49 score = 0.327

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. stream_elas

z = +0.67 r = -0.43 score = 0.287

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. frac_forest

z = +0.80 r = +0.35 score = 0.278

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lon

z = +1.08 r = -0.25 score = 0.274

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. frac_snow

z = -0.59 r = +0.46 score = 0.274

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. q95

z = -0.51 r = +0.51 score = 0.261

low peak-flow values reflect a damped streamflow response with limited peaks.
10. elev_mean

z = -0.77 r = +0.30 score = 0.230

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02051500: MEHERRIN RIVER NEAR LAWRENCEVILLE, VA

Dominant forcing: PRCP Peak TE: 0.0205 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, high glim_2nd_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.11 r = -0.27 score = 0.301

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -0.99 r = +0.23 score = 0.225

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. glim_2nd_class_frac

z = +1.33 r = +0.15 score = 0.204

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. elev_mean

z = -0.82 r = -0.20 score = 0.161

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. glim_1st_class_frac

z = -1.01 r = -0.15 score = 0.147

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. q95

z = -0.46 r = +0.31 score = 0.141

low peak-flow values reflect a damped streamflow response with limited peaks.
7. hfd_mean

z = -0.54 r = -0.24 score = 0.129

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. runoff_ratio

z = -0.56 r = +0.23 score = 0.129

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. frac_forest

z = +0.72 r = +0.17 score = 0.124

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. q_mean

z = -0.44 r = +0.28 score = 0.121

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02053200: POTECASTI CREEK NEAR UNION, NC

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, high slope_fdc.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.27 r = +0.16 score = 0.362

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.16 r = -0.27 score = 0.314

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_fdc

z = +1.38 r = +0.20 score = 0.278

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. elev_mean

z = -0.93 r = -0.20 score = 0.182

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. glim_2nd_class_frac

z = -0.97 r = +0.15 score = 0.149

a small secondary geological class means the subsurface is more uniform.
6. high_prec_dur

z = -0.65 r = +0.23 score = 0.147

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. glim_1st_class_frac

z = +0.98 r = -0.15 score = 0.143

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. runoff_ratio

z = -0.56 r = +0.23 score = 0.128

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. q_mean

z = -0.41 r = +0.28 score = 0.114

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. p_seasonality

z = +0.46 r = -0.24 score = 0.113

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02053800: S F ROANOKE RIVER NEAR SHAWSVILLE, VA

Dominant forcing: PRCP Peak TE: 0.0035 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low glim_1st_class_frac, high gvf_diff.

Top 10 driver attributes

1. gauge_lon z = +0.96 r = -0.27 score = 0.260

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_1st_class_frac z = -1.51 r = -0.15 score = 0.220

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
3. gvf_diff z = +0.86 r = -0.24 score = 0.209

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. glim_2nd_class_frac z = +1.19 r = +0.15 score = 0.183

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. high_prec_dur z = -0.80 r = +0.23 score = 0.182

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. q95 z = -0.50 r = +0.31 score = 0.151

low peak-flow values reflect a damped streamflow response with limited peaks.
7. slope_fdc z = +0.63 r = +0.20 score = 0.126

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. frac_forest z = +0.67 r = +0.17 score = 0.115

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. aridity z = -0.37 r = -0.28 score = 0.103

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. soil_depth_pelletier z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02055100: TINKER CREEK NEAR DALEVILLE, VA

Dominant forcing: PRCP Peak TE: 0.0008 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high gauge_lon, high slope_fdc.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.27 r = -0.12 score = 0.273

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gauge_lon

z = +0.98 r = -0.27 score = 0.266

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_fdc

z = +1.17 r = +0.20 score = 0.235

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. q95

z = -0.47 r = +0.31 score = 0.144

low peak-flow values reflect a damped streamflow response with limited peaks.
5. gvf_diff

z = +0.58 r = -0.24 score = 0.141

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. soil_conductivity

z = -0.77 r = +0.15 score = 0.113

low soil conductivity restricts infiltration, often increasing surface runoff.
7. soil_depth_statsgo

z = -0.80 r = +0.13 score = 0.107

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. sand_frac

z = -0.91 r = +0.12 score = 0.107

low sand fraction reduces infiltration and may favor surface runoff.
9. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.098

shallow soils limit storage and produce a quicker streamflow response.
10. q_mean

z = -0.33 r = +0.28 score = 0.092

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02056900: BLACKWATER RIVER NEAR ROCKY MOUNT, VA

Dominant forcing: SRAD Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high clay_frac, low q95.

Top 10 driver attributes

1. gauge_lon

z = +0.98 r = -0.43 score = 0.425

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. clay_frac

z = +1.72 r = -0.22 score = 0.380

high clay fraction reduces infiltration and increases surface runoff.
3. q95

z = -0.49 r = +0.43 score = 0.213

low peak-flow values reflect a damped streamflow response with limited peaks.
4. gvf_diff

z = +0.61 r = -0.33 score = 0.201

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. baseflow_index

z = +0.87 r = +0.22 score = 0.191

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
6. frac_forest

z = +0.73 r = +0.23 score = 0.172

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. p_seasonality

z = +0.33 r = -0.46 score = 0.152

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. soil_depth_pelletier

z = -0.61 r = -0.24 score = 0.145

shallow soils limit storage and produce a quicker streamflow response.
9. frac_snow

z = -0.56 r = +0.26 score = 0.142

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. soil_conductivity

z = -0.52 r = +0.25 score = 0.131

low soil conductivity restricts infiltration, often increasing surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02059500: GOOSE CREEK NEAR HUDDLESTON, VA

Dominant forcing: PRCP Peak TE: 0.0018 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low q95, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +1.00 r = -0.27 score = 0.273

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. q95

z = -0.54 r = +0.31 score = 0.165

low peak-flow values reflect a damped streamflow response with limited peaks.
3. gvf_diff

z = +0.66 r = -0.24 score = 0.161

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. q_mean

z = -0.41 r = +0.28 score = 0.114

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
5. elev_mean

z = -0.56 r = -0.20 score = 0.109

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. runoff_ratio

z = -0.47 r = +0.23 score = 0.109

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. stream_elas

z = -0.83 r = -0.12 score = 0.102

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
9. soil_depth_statsgo

z = +0.71 r = +0.13 score = 0.095

deep soils (STATSGO) increase basin storage and delay streamflow response.
10. frac_forest

z = +0.54 r = +0.17 score = 0.094

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02064000: FALLING RIVER NEAR NARUNA, VA

Dominant forcing: SRAD Peak TE: 0.0007 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high clay_frac, high gauge_lon, low slope_mean.

Top 10 driver attributes

1. clay_frac

z = +2.56 r = -0.22 score = 0.564

high clay fraction reduces infiltration and increases surface runoff.
2. gauge_lon

z = +1.04 r = -0.43 score = 0.449

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_mean

z = -0.77 r = +0.37 score = 0.283

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. q95

z = -0.50 r = +0.43 score = 0.215

low peak-flow values reflect a damped streamflow response with limited peaks.
5. glim_2nd_class_frac

z = +1.22 r = +0.18 score = 0.214

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. soil_conductivity

z = -0.71 r = +0.25 score = 0.180

low soil conductivity restricts infiltration, often increasing surface runoff.
7. high_prec_freq

z = +0.54 r = -0.33 score = 0.177

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. elev_mean

z = -0.72 r = +0.24 score = 0.174

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. frac_forest

z = +0.72 r = +0.23 score = 0.170

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. runoff_ratio

z = -0.37 r = +0.43 score = 0.160

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02065500: CUB CREEK AT PHENIX, VA

Dominant forcing: PRCP Peak TE: 0.0052 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low glim_1st_class_frac, high gauge_lon, high pet_mean.

Top 10 driver attributes

1. glim_1st_class_frac z = -2.20 r = -0.15 score = 0.320

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
2. gauge_lon z = +1.05 r = -0.27 score = 0.285

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. pet_mean z = +1.64 r = -0.12 score = 0.199

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. high_prec_dur z = -0.77 r = +0.23 score = 0.175

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. glim_2nd_class_frac z = +1.12 r = +0.15 score = 0.171

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. q95 z = -0.55 r = +0.31 score = 0.166

low peak-flow values reflect a damped streamflow response with limited peaks.
7. elev_mean z = -0.75 r = -0.20 score = 0.146

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. frac_forest z = +0.71 r = +0.17 score = 0.123

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. hfd_mean z = -0.44 r = -0.24 score = 0.104

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. root_depth_99 z = +1.02 r = +0.10 score = 0.103

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02069700: SOUTH MAYO RIVER NEAR NETTLERIDGE, VA

Dominant forcing: SRAD Peak TE: 0.0012 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high clay_frac, high baseflow_index.

Top 10 driver attributes

1. gauge_lon

z = +0.97 r = -0.43 score = 0.417

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. clay_frac

z = +1.83 r = -0.22 score = 0.404

high clay fraction reduces infiltration and increases surface runoff.
3. baseflow_index

z = +1.38 r = +0.22 score = 0.302

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. gvf_diff

z = +0.88 r = -0.33 score = 0.288

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. q5

z = +0.72 r = +0.31 score = 0.224

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. high_prec_dur

z = -0.77 r = +0.29 score = 0.222

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. frac_forest

z = +0.87 r = +0.23 score = 0.205

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. slope_mean

z = -0.49 r = +0.37 score = 0.181

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. q95

z = -0.40 r = +0.43 score = 0.171

low peak-flow values reflect a damped streamflow response with limited peaks.
10. soil_conductivity

z = -0.65 r = +0.25 score = 0.165

low soil conductivity restricts infiltration, often increasing surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02070000: NORTH MAYO RIVER NEAR SPENCER, VA

Dominant forcing: PRCP Peak TE: 0.0019 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high pet_mean, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +0.97 r = -0.27 score = 0.265

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. pet_mean

z = +1.76 r = -0.12 score = 0.214

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
3. gvf_diff

z = +0.79 r = -0.24 score = 0.192

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. frac_forest

z = +0.87 r = +0.17 score = 0.150

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. q95

z = -0.48 r = +0.31 score = 0.146

low peak-flow values reflect a damped streamflow response with limited peaks.
6. elev_mean

z = -0.57 r = -0.20 score = 0.112

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. q5

z = +0.60 r = +0.18 score = 0.109

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. high_prec_dur

z = -0.46 r = +0.23 score = 0.104

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. gvf_max

z = +0.90 r = +0.12 score = 0.104

high maximum green vegetation fraction reflects dense canopy cover.
10. soil_depth_pelletier

z = -0.60 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02074500: SANDY RIVER NEAR DANVILLE, VA

Dominant forcing: VP Peak TE: 0.0017 Peak lag: 8 day(s)
Summary: VP-dominant at lag 8 day(s); top drivers consistent with high stream_elas, high clay_frac, high pet_mean.

Top 10 driver attributes

1. stream_elas

z = +2.22 r = -0.36 score = 0.803

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. clay_frac

z = +2.69 r = -0.22 score = 0.591

high clay fraction reduces infiltration and increases surface runoff.
3. pet_mean

z = +1.14 r = -0.25 score = 0.291

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. q95

z = -0.57 r = +0.49 score = 0.282

low peak-flow values reflect a damped streamflow response with limited peaks.
5. slope_mean

z = -0.75 r = +0.27 score = 0.204

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. runoff_ratio

z = -0.39 r = +0.50 score = 0.196

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. frac_snow

z = -0.62 r = +0.31 score = 0.192

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. gauge_lat

z = -0.50 r = +0.35 score = 0.177

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. frac_forest

z = +0.59 r = +0.29 score = 0.170

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. high_prec_freq

z = +0.32 r = -0.47 score = 0.152

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02077200: HYCO CREEK NEAR LEASBURG, NC

Dominant forcing: SRAD Peak TE: 0.0013 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low stream_elas, low slope_mean.

Top 10 driver attributes

1. gauge_lon z = +1.02 r = -0.43 score = 0.442

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. stream_elas z = -1.24 r = -0.26 score = 0.320

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. slope_mean z = -0.83 r = +0.37 score = 0.305

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. sand_frac z = -1.38 r = +0.18 score = 0.243

low sand fraction reduces infiltration and may favor surface runoff.
5. glim_1st_class_frac z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. soil_porosity z = +1.25 r = -0.17 score = 0.215

high soil porosity increases water storage capacity and dampens streamflow response.
7. gvff_diff z = +0.61 r = -0.33 score = 0.201

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. baseflow_index z = -0.91 r = +0.22 score = 0.200

a low baseflow index means streamflow responds quickly to recent rainfall.
9. q5 z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.
10. runoff_ratio z = -0.46 r = +0.43 score = 0.198

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02081500: TAR RIVER NEAR TAR RIVER, NC

Dominant forcing: PRCP Peak TE: 0.0062 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, low hfd_mean.

Top 10 driver attributes

1. gauge_lon

z = +1.06 r = -0.27 score = 0.288

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -1.00 r = +0.23 score = 0.227

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. hfd_mean

z = -0.68 r = -0.24 score = 0.162

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. elev_mean

z = -0.80 r = -0.20 score = 0.156

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. runoff_ratio

z = -0.55 r = +0.23 score = 0.128

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q_mean

z = -0.42 r = +0.28 score = 0.117

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. q95

z = -0.37 r = +0.31 score = 0.113

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q5

z = -0.62 r = +0.18 score = 0.112

low low-flow values reflect basins that nearly dry out during dry periods.
9. soil_depth_statsgo

z = +0.76 r = +0.13 score = 0.101

deep soils (STATSGO) increase basin storage and delay streamflow response.
10. frac_forest

z = +0.56 r = +0.17 score = 0.096

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02082950: LITTLE FISHING CREEK NEAR WHITE OAK, NC

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, high root_depth_99.

Top 10 driver attributes

1. gauge_lon

z = +1.10 r = -0.27 score = 0.300

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -0.96 r = +0.23 score = 0.219

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. root_depth_99

z = +1.76 r = +0.10 score = 0.178

very deep maximum rooting allows vegetation to tap deep storage in droughts.
4. elev_mean

z = -0.86 r = -0.20 score = 0.169

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. slope_fdc

z = +0.69 r = +0.20 score = 0.139

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. q95

z = -0.43 r = +0.31 score = 0.131

low peak-flow values reflect a damped streamflow response with limited peaks.
7. runoff_ratio

z = -0.57 r = +0.23 score = 0.131

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q_mean

z = -0.43 r = +0.28 score = 0.119

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. glim_2nd_class_frac

z = +0.69 r = +0.15 score = 0.106

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. frac_snow

z = -0.71 r = -0.15 score = 0.104

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02092500: TRENT RIVER NEAR TRENTON, NC

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gauge_lon, high glim_2nd_class_frac.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.05 r = +0.16 score = 0.328

deep soils provide more storage and slower release of water to streamflow.
2. gauge_lon

z = +1.13 r = -0.27 score = 0.307

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. glim_2nd_class_frac

z = +1.58 r = +0.15 score = 0.243

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. elev_mean

z = -0.94 r = -0.20 score = 0.183

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
5. organic_frac

z = +1.07 r = +0.17 score = 0.183

high organic soil fraction increases water retention and supports baseflow.
6. sand_frac

z = +1.50 r = +0.12 score = 0.175

high sand fraction promotes fast infiltration and subsurface drainage.
7. slope_fdc

z = +0.81 r = +0.20 score = 0.163

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. p_seasonality

z = +0.56 r = -0.24 score = 0.136

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. p_mean

z = +0.37 r = +0.34 score = 0.128

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. soil_conductivity

z = +0.85 r = +0.15 score = 0.125

high soil conductivity allows fast infiltration and subsurface flow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02096846: CANE CREEK NEAR ORANGE GROVE, NC

Dominant forcing: VP Peak TE: 0.0030 Peak lag: 28 day(s)
Summary: VP-dominant at lag 28 day(s); top drivers consistent with low runoff_ratio, high clay_frac, low q95.

Top 10 driver attributes

1. runoff_ratio

z = -0.70 r = +0.50 score = 0.350

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. clay_frac

z = +1.21 r = -0.22 score = 0.265

high clay fraction reduces infiltration and increases surface runoff.
3. q95

z = -0.49 r = +0.49 score = 0.243

low peak-flow values reflect a damped streamflow response with limited peaks.
4. high_prec_freq

z = +0.50 r = -0.47 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. slope_mean

z = -0.82 r = +0.27 score = 0.221

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. gauge_lat

z = -0.63 r = +0.35 score = 0.220

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. frac_forest

z = +0.74 r = +0.29 score = 0.215

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. q5

z = -0.64 r = +0.34 score = 0.214

low low-flow values reflect basins that nearly dry out during dry periods.
9. frac_snow

z = -0.68 r = +0.31 score = 0.213

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. q_mean

z = -0.48 r = +0.43 score = 0.207

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02102908: FLAT CREEK NEAR INVERNESS, NC

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low gvf_diff, high sand_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.02 r = -0.27 score = 0.278

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff

z = -1.09 r = -0.24 score = 0.266

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. sand_frac

z = +2.21 r = +0.12 score = 0.259

high sand fraction promotes fast infiltration and subsurface drainage.
4. stream_elas

z = -1.71 r = -0.12 score = 0.212

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. soil_conductivity

z = +1.42 r = +0.15 score = 0.209

high soil conductivity allows fast infiltration and subsurface flow.
6. q5

z = +1.02 r = +0.18 score = 0.184

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. elev_mean

z = -0.83 r = -0.20 score = 0.162

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -0.97 r = +0.15 score = 0.150

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +0.99 r = -0.15 score = 0.143

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. q95

z = -0.42 r = +0.31 score = 0.129

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02108000: NORTHEAST CAPE FEAR RIVER NEAR CHINQUAPIN, NC

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high sand_frac.

Top 10 driver attributes

1. gauge_lon

z = +1.11 r = -0.27 score = 0.301

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.05 r = +0.20 score = 0.211

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. sand_frac

z = +1.64 r = +0.12 score = 0.192

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_depth_pelletier

z = +1.14 r = +0.16 score = 0.183

deep soils provide more storage and slower release of water to streamflow.
5. elev_mean

z = -0.93 r = -0.20 score = 0.181

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. p_seasonality

z = +0.60 r = -0.24 score = 0.147

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. p_mean

z = +0.39 r = +0.34 score = 0.135

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. frac_snow

z = -0.83 r = -0.15 score = 0.121

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. soil_conductivity

z = +0.81 r = +0.15 score = 0.119

high soil conductivity allows fast infiltration and subsurface flow.
10. gauge_lat

z = -0.85 r = +0.13 score = 0.107

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02111180: ELK CREEK AT ELKVILLE, NC

Dominant forcing: PRCP Peak TE: 0.0018 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low soil_depth_statsgo, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon z = +0.89 r = -0.27 score = 0.241

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. soil_depth_statsgo z = -1.43 r = +0.13 score = 0.192

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
3. high_prec_dur z = -0.84 r = +0.23 score = 0.190

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gvf_diff z = +0.76 r = -0.24 score = 0.185

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. frac_forest z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. aridity z = -0.53 r = -0.28 score = 0.145

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. root_depth_99 z = +1.43 r = +0.10 score = 0.145

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. gvf_max z = +1.01 r = +0.12 score = 0.117

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02111500: REDDIES RIVER AT NORTH WILKESBORO, NC

Dominant forcing: SRAD Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high baseflow_index, high glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +0.90 r = -0.43 score = 0.390

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. baseflow_index

z = +1.44 r = +0.22 score = 0.315

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
4. q5

z = +0.72 r = +0.31 score = 0.226

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. high_prec_dur

z = -0.76 r = +0.29 score = 0.218

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. frac_forest

z = +0.92 r = +0.23 score = 0.217

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
8. gauge_lat

z = -0.59 r = +0.29 score = 0.173

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. q95

z = -0.40 r = +0.43 score = 0.172

low peak-flow values reflect a damped streamflow response with limited peaks.
10. gvf_diff

z = +0.52 r = -0.33 score = 0.169

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02112120: ROARING RIVER NEAR ROARING RIVER, NC

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, low q95.

Top 10 driver attributes

1. gauge_lon z = +0.91 r = -0.27 score = 0.247

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur z = -0.78 r = +0.23 score = 0.177

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. q95 z = -0.40 r = +0.31 score = 0.121

low peak-flow values reflect a damped streamflow response with limited peaks.
4. aridity z = -0.44 r = -0.28 score = 0.121

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
5. frac_forest z = +0.70 r = +0.17 score = 0.121

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. gvf_diff z = +0.48 r = -0.24 score = 0.117

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. q5 z = +0.60 r = +0.18 score = 0.109

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. gvf_max z = +0.92 r = +0.12 score = 0.107

high maximum green vegetation fraction reflects dense canopy cover.
9. soil_depth_pelletier z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
10. frac_snow z = -0.62 r = -0.15 score = 0.090

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02112360: MITCHELL RIVER NEAR STATE ROAD, NC

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, high q5.

Top 10 driver attributes

1. gauge_lon z = +0.92 r = -0.27 score = 0.251

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur z = -0.85 r = +0.23 score = 0.193

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. q5 z = +1.00 r = +0.18 score = 0.180

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. gvf_diff z = +0.56 r = -0.24 score = 0.136

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. q95 z = -0.40 r = +0.31 score = 0.123

low peak-flow values reflect a damped streamflow response with limited peaks.
8. frac_forest z = +0.65 r = +0.17 score = 0.112

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. aridity z = -0.37 r = -0.28 score = 0.101

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gvf_max z = +0.85 r = +0.12 score = 0.098

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02118500: HUNTING CREEK NEAR HARMONY, NC

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low glim_2nd_class_frac, high glim_1st_class_frac.

Top 10 driver attributes

1. gauge_lon

z = +0.93 r = -0.27 score = 0.252

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
3. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
4. gvf_diff

z = +0.61 r = -0.24 score = 0.149

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. q95

z = -0.45 r = +0.31 score = 0.138

low peak-flow values reflect a damped streamflow response with limited peaks.
6. high_prec_dur

z = -0.58 r = +0.23 score = 0.132

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. aridity

z = -0.39 r = -0.28 score = 0.108

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. elev_mean

z = -0.54 r = -0.20 score = 0.105

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. soil_depth_statsgo

z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.
10. stream_elas

z = +0.80 r = -0.12 score = 0.098

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02125000: BIG BEAR CR NR RICHFIELD, NC

Dominant forcing: SRAD Peak TE: 0.0041 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high stream_elas, high gauge_lon, low baseflow_index.

Top 10 driver attributes

1. stream_elas

z = +2.19 r = -0.26 score = 0.567

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. gauge_lon

z = +0.95 r = -0.43 score = 0.412

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. baseflow_index

z = -1.35 r = +0.22 score = 0.296

a low baseflow index means streamflow responds quickly to recent rainfall.
4. slope_mean

z = -0.78 r = +0.37 score = 0.288

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. glim_2nd_class_frac

z = +1.59 r = +0.18 score = 0.280

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. clay_frac

z = +1.08 r = -0.22 score = 0.238

high clay fraction reduces infiltration and increases surface runoff.
7. high_prec_freq

z = +0.72 r = -0.33 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. sand_frac

z = -1.32 r = +0.18 score = 0.232

low sand fraction reduces infiltration and may favor surface runoff.
9. gauge_lat

z = -0.75 r = +0.29 score = 0.220

southern latitude is associated with warmer climates and rainfall-driven streamflow.
10. q5

z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02128000: LITTLE RIVER NEAR STAR, NC

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low hfd_mean, low elev_mean.

Top 10 driver attributes

1. gauge_lon z = +0.98 r = -0.27 score = 0.267

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. hfd_mean z = -0.63 r = -0.24 score = 0.151

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. elev_mean z = -0.72 r = -0.20 score = 0.140

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
4. q95 z = -0.40 r = +0.31 score = 0.122

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_snow z = -0.75 r = -0.15 score = 0.109

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
6. gvf_diff z = +0.44 r = -0.24 score = 0.108

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. frac_forest z = +0.59 r = +0.17 score = 0.101

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. q5 z = -0.55 r = +0.18 score = 0.100

low low-flow values reflect basins that nearly dry out during dry periods.
9. sand_frac z = -0.84 r = +0.12 score = 0.099

low sand fraction reduces infiltration and may favor surface runoff.
10. q_mean z = -0.36 r = +0.28 score = 0.099

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02137727: CATAWBA R NR PLEASANT GARDENS, NC

Dominant forcing: SRAD Peak TE: 0.0014 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high q5, high gvf_diff.

Top 10 driver attributes

1. gauge_lon z = +0.85 r = -0.43 score = 0.366

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. q5 z = +0.91 r = +0.31 score = 0.283

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. gvf_diff z = +0.78 r = -0.33 score = 0.255

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. baseflow_index z = +1.16 r = +0.22 score = 0.255

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
5. root_depth_50 z = +1.98 r = -0.13 score = 0.248

deep median rooting allows vegetation to access stored soil water during dry periods.
6. glim_1st_class_frac z = +1.05 r = -0.21 score = 0.225

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. frac_forest z = +0.93 r = +0.23 score = 0.219

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. gauge_lat z = -0.68 r = +0.29 score = 0.200

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. high_prec_freq z = -0.61 r = -0.33 score = 0.200

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. glim_2nd_class_frac z = -1.07 r = +0.18 score = 0.187

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02140991: JOHNS RIVER AT ARNEYS STORE, NC

Dominant forcing: SRAD Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high q5, high root_depth_50.

Top 10 driver attributes

1. gauge_lon

z = +0.87 r = -0.43 score = 0.375

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. q5

z = +0.89 r = +0.31 score = 0.279

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. root_depth_50

z = +1.84 r = -0.13 score = 0.230

deep median rooting allows vegetation to access stored soil water during dry periods.
4. baseflow_index

z = +1.00 r = +0.22 score = 0.220

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
5. frac_forest

z = +0.92 r = +0.23 score = 0.216

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. high_prec_dur

z = -0.74 r = +0.29 score = 0.212

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. gauge_lat

z = -0.66 r = +0.29 score = 0.192

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. lai_diff

z = +1.35 r = -0.14 score = 0.188

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
9. glim_1st_class_frac

z = +0.85 r = -0.21 score = 0.183

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. low_prec_freq

z = -0.56 r = -0.31 score = 0.175

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02143000: HENRY FORK NEAR HENRY RIVER, NC

Dominant forcing: PRCP Peak TE: 0.0048 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high root_depth_99, low glim_2nd_class_frac.

Top 10 driver attributes

1. gauge_lon z = +0.89 r = -0.27 score = 0.241

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. root_depth_99 z = +1.72 r = +0.10 score = 0.174

very deep maximum rooting allows vegetation to tap deep storage in droughts.
3. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
4. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
5. aridity z = -0.44 r = -0.28 score = 0.121

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. hfd_mean z = -0.47 r = -0.24 score = 0.112

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. p_mean z = +0.32 r = +0.34 score = 0.109

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. frac_snow z = -0.72 r = -0.15 score = 0.105

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. soil_depth_statsgo z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.
10. q95 z = -0.33 r = +0.31 score = 0.102

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02143040: JACOB FORK AT RAMSEY, NC

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high p_mean, high root_depth_99.

Top 10 driver attributes

1. gauge_lon

z = +0.88 r = -0.27 score = 0.238

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. p_mean

z = +0.56 r = +0.34 score = 0.192

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
4. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. aridity

z = -0.59 r = -0.28 score = 0.162

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. frac_forest

z = +0.82 r = +0.17 score = 0.142

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. gvf_max

z = +1.00 r = +0.12 score = 0.116

high maximum green vegetation fraction reflects dense canopy cover.
9. gvf_diff

z = +0.47 r = -0.24 score = 0.114

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
10. hfd_mean

z = -0.47 r = -0.24 score = 0.112

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02149000: COVE CREEK NEAR LAKE LURE, NC

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low glim_2nd_class_frac.

Top 10 driver attributes

1. gauge_lon z = +0.84 r = -0.27 score = 0.229

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +0.86 r = -0.24 score = 0.209

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
4. frac_forest z = +0.90 r = +0.17 score = 0.155

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. root_depth_99 z = +1.52 r = +0.10 score = 0.154

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. p_mean z = +0.40 r = +0.34 score = 0.138

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. q5 z = +0.70 r = +0.18 score = 0.126

high low-flow values reflect strong baseflow and steady year-round streamflow.
9. aridity z = -0.44 r = -0.28 score = 0.122

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gvf_max z = +0.99 r = +0.12 score = 0.114

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02152100: FIRST BROAD RIVER NEAR CASAR, NC

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high root_depth_50, high stream_elas.

Top 10 driver attributes

1. gauge_lon

z = +0.87 r = -0.43 score = 0.376

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. root_depth_50

z = +2.25 r = -0.13 score = 0.282

deep median rooting allows vegetation to access stored soil water during dry periods.
3. stream_elas

z = +1.01 r = -0.26 score = 0.261

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. gvf_diff

z = +0.77 r = -0.33 score = 0.251

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. baseflow_index

z = +1.01 r = +0.22 score = 0.220

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
7. gauge_lat

z = -0.72 r = +0.29 score = 0.211

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. lai_diff

z = +1.43 r = -0.14 score = 0.199

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
9. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
10. frac_snow

z = -0.73 r = +0.26 score = 0.185

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02177000: CHATTOOGA RIVER NEAR CLAYTON, GA

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high gauge_lon, high baseflow_index.

Top 10 driver attributes

1. q5

z = +1.93

r = +0.31

score = 0.604

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lon

z = +0.77

r = -0.43

score = 0.333

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. baseflow_index

z = +1.49

r = +0.22

score = 0.326

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. q_mean

z = +0.92

r = +0.35

score = 0.326

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
5. p_mean

z = +1.50

r = +0.20

score = 0.301

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. root_depth_50

z = +2.30

r = -0.13

score = 0.288

deep median rooting allows vegetation to access stored soil water during dry periods.
7. runoff_ratio

z = +0.67

r = +0.43

score = 0.285

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. gauge_lat

z = -0.85

r = +0.29

score = 0.249

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. frac_forest

z = +0.90

r = +0.23

score = 0.210

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. low_prec_freq

z = -0.65

r = -0.31

score = 0.204

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02178400: TALLULAH RIVER NEAR CLAYTON, GA

Dominant forcing: SRAD Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high gauge_lon, high q_mean.

Top 10 driver attributes

1. q5

z = +2.00

r = +0.31

score = 0.625

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lon

z = +0.76

r = -0.43

score = 0.327

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. q_mean

z = +0.89

r = +0.35

score = 0.314

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. baseflow_index

z = +1.42

r = +0.22

score = 0.311

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
5. runoff_ratio

z = +0.70

r = +0.43

score = 0.298

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. p_mean

z = +1.39

r = +0.20

score = 0.278

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
7. root_depth_50

z = +2.05

r = -0.13

score = 0.257

deep median rooting allows vegetation to access stored soil water during dry periods.
8. gvf_diff

z = +0.76

r = -0.33

score = 0.248

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. gauge_lat

z = -0.84

r = +0.29

score = 0.245

southern latitude is associated with warmer climates and rainfall-driven streamflow.
10. frac_forest

z = +0.91

r = +0.23

score = 0.213

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02193340: KETTLE CREEK NEAR WASHINGTON, GA

Dominant forcing: Tair Peak TE: 0.0036 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with high clay_frac, low gauge_lat, low runoff_ratio.

Top 10 driver attributes

1. clay_frac

z = +2.11 r = -0.30 score = 0.622

high clay fraction reduces infiltration and increases surface runoff.
2. gauge_lat

z = -1.07 r = +0.51 score = 0.540

southern latitude is associated with warmer climates and rainfall-driven streamflow.
3. runoff_ratio

z = -0.72 r = +0.56 score = 0.400

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. high_prec_freq

z = +0.81 r = -0.49 score = 0.397

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. frac_snow

z = -0.81 r = +0.46 score = 0.374

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
6. stream_elas

z = +0.84 r = -0.43 score = 0.359

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. slope_mean

z = -0.82 r = +0.41 score = 0.335

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. q95

z = -0.60 r = +0.51 score = 0.304

low peak-flow values reflect a damped streamflow response with limited peaks.
9. low_prec_freq

z = +0.53 r = -0.50 score = 0.265

many dry days indicate a drier climate where streamflow depends on stored water.
10. q_mean

z = -0.50 r = +0.45 score = 0.225

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02196000: STEVENS CREEK NEAR MODOC, SC

Dominant forcing: PRCP Peak TE: 0.0081 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, high stream_elas, high gauge_lon.

Top 10 driver attributes

1. pet_mean z = +2.02 r = -0.12 score = 0.246

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. stream_elas z = +1.96 r = -0.12 score = 0.242

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. gauge_lon z = +0.84 r = -0.27 score = 0.228

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. hfd_mean z = -0.89 r = -0.24 score = 0.212

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. runoff_ratio z = -0.92 r = +0.23 score = 0.212

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. high_prec_dur z = -0.80 r = +0.23 score = 0.181

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. q95 z = -0.57 r = +0.31 score = 0.175

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean z = -0.60 r = +0.28 score = 0.165

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. gvf_diff z = -0.67 r = -0.24 score = 0.163

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. elev_mean z = -0.79 r = -0.20 score = 0.153

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02198100: BEAVERDAM CREEK NEAR SARDIS, GA

Dominant forcing: PRCP Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high soil_depth_pelletier, high sand_frac.

Top 10 driver attributes

1. gauge_lon z = +0.86 r = -0.27 score = 0.234

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. soil_depth_pelletier z = +1.29 r = +0.16 score = 0.206

deep soils provide more storage and slower release of water to streamflow.
3. sand_frac z = +1.73 r = +0.12 score = 0.202

high sand fraction promotes fast infiltration and subsurface drainage.
4. hfd_mean z = -0.84 r = -0.24 score = 0.199

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. high_prec_freq z = +0.98 r = -0.19 score = 0.185

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. pet_mean z = +1.47 r = -0.12 score = 0.179

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. elev_mean z = -0.86 r = -0.20 score = 0.168

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. low_prec_freq z = +0.75 r = -0.22 score = 0.167

many dry days indicate a drier climate where streamflow depends on stored water.
9. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02202600: BLACK CREEK NEAR BLITCHTON, GA

Dominant forcing: PRCP Peak TE: 0.0085 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high soil_depth_pelletier, high sand_frac, high stream_elas.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.22 r = +0.16 score = 0.355

deep soils provide more storage and slower release of water to streamflow.
2. sand_frac

z = +2.18 r = +0.12 score = 0.255

high sand fraction promotes fast infiltration and subsurface drainage.
3. stream_elas

z = +1.98 r = -0.12 score = 0.244

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. gauge_lon

z = +0.88 r = -0.27 score = 0.240

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. elev_mean

z = -0.92 r = -0.20 score = 0.180

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. soil_conductivity

z = +1.21 r = +0.15 score = 0.178

high soil conductivity allows fast infiltration and subsurface flow.
7. gauge_lat

z = -1.36 r = +0.13 score = 0.171

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. runoff_ratio

z = -0.74 r = +0.23 score = 0.171

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. high_prec_freq

z = +0.85 r = -0.19 score = 0.160

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q_mean

z = -0.50 r = +0.28 score = 0.139

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02212600: FALLING CREEK NEAR JULIETTE, GA

Dominant forcing: SRAD Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high clay_frac, low gauge_lat, high gauge_lon.

Top 10 driver attributes

1. clay_frac

z = +1.69

r = -0.22

score = 0.372

high clay fraction reduces infiltration and increases surface runoff.
2. gauge_lat

z = -1.18

r = +0.29

score = 0.346

southern latitude is associated with warmer climates and rainfall-driven streamflow.
3. gauge_lon

z = +0.74

r = -0.43

score = 0.322

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. runoff_ratio

z = -0.72

r = +0.43

score = 0.308

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. slope_mean

z = -0.83

r = +0.37

score = 0.304

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. root_depth_50

z = +2.12

r = -0.13

score = 0.265

deep median rooting allows vegetation to access stored soil water during dry periods.
7. high_prec_freq

z = +0.81

r = -0.33

score = 0.263

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. frac_snow

z = -0.83

r = +0.26

score = 0.212

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. q5

z = -0.62

r = +0.31

score = 0.195

low low-flow values reflect basins that nearly dry out during dry periods.
10. q95

z = -0.44

r = +0.43

score = 0.192

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02215100: TUCSAWHATCHEE CREEK NEAR HAWKINSVILLE, GA

Dominant forcing: PRCP Peak TE: 0.0109 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, low frac_forest, high gauge_lon.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.30 r = -0.12 score = 0.275

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. frac_forest

z = -1.31 r = +0.17 score = 0.226

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. gauge_lon

z = +0.76 r = -0.27 score = 0.206

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. hfd_mean

z = -0.80 r = -0.24 score = 0.189

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. gauge_lat

z = -1.35 r = +0.13 score = 0.170

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. sand_frac

z = +1.44 r = +0.12 score = 0.168

high sand fraction promotes fast infiltration and subsurface drainage.
7. elev_mean

z = -0.82 r = -0.20 score = 0.161

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. high_prec_dur

z = -0.63 r = +0.23 score = 0.143

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. frac_snow

z = -0.87 r = -0.15 score = 0.127

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. runoff_ratio

z = -0.53 r = +0.23 score = 0.123

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02216180: TURNPIKE CREEK NEAR MCRAE, GA

Dominant forcing: PRCP Peak TE: 0.0063 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, high sand_frac, high gauge_lon.

Top 10 driver attributes

1. zero_q_freq

z = +2.51 r = -0.13 score = 0.335

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. sand_frac

z = +1.96 r = +0.12 score = 0.229

high sand fraction promotes fast infiltration and subsurface drainage.
3. gauge_lon

z = +0.79 r = -0.27 score = 0.216

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. soil_depth_pelletier

z = +1.28 r = +0.16 score = 0.205

deep soils provide more storage and slower release of water to streamflow.
5. stream_elas

z = +1.52 r = -0.12 score = 0.188

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. gauge_lat

z = -1.39 r = +0.13 score = 0.176

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. elev_mean

z = -0.86 r = -0.20 score = 0.168

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. soil_conductivity

z = +1.10 r = +0.15 score = 0.163

high soil conductivity allows fast infiltration and subsurface flow.
10. hfd_mean

z = -0.68 r = -0.24 score = 0.162

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02221525: MURDER CREEK BELOW EATONTON, GA

Dominant forcing: PRCP Peak TE: 0.0092 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low glim_2nd_class_frac, low runoff_ratio.

Top 10 driver attributes

1. gauge_lon

z = +0.76 r = -0.27 score = 0.206

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. glim_2nd_class_frac

z = -1.07 r = +0.15 score = 0.165

a small secondary geological class means the subsurface is more uniform.
3. runoff_ratio

z = -0.69 r = +0.23 score = 0.159

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. q95

z = -0.51 r = +0.31 score = 0.156

low peak-flow values reflect a damped streamflow response with limited peaks.
5. glim_1st_class_frac

z = +1.05 r = -0.15 score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. hfd_mean

z = -0.64 r = -0.24 score = 0.153

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. high_prec_freq

z = +0.79 r = -0.19 score = 0.148

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. gauge_lat

z = -1.15 r = +0.13 score = 0.145

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. elev_mean

z = -0.74 r = -0.20 score = 0.145

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q_mean

z = -0.47 r = +0.28 score = 0.131

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02231000: ST. MARYS RIVER NR MACCLENNY, FLA.

Dominant forcing: PRCP Peak TE: 0.0055 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high soil_conductivity, low gvf_diff.

Top 10 driver attributes

1. organic_frac

z = +2.70 r = +0.17 score = 0.460

high organic soil fraction increases water retention and supports baseflow.
2. soil_conductivity

z = +2.93 r = +0.15 score = 0.432

high soil conductivity allows fast infiltration and subsurface flow.
3. gvf_diff

z = -1.55 r = -0.24 score = 0.378

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +2.34 r = +0.16 score = 0.374

deep soils provide more storage and slower release of water to streamflow.
5. sand_frac

z = +2.31 r = +0.12 score = 0.270

high sand fraction promotes fast infiltration and subsurface drainage.
6. gauge_lon

z = +0.85 r = -0.27 score = 0.230

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. gauge_lat

z = -1.71 r = +0.13 score = 0.215

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. p_seasonality

z = +0.86 r = -0.24 score = 0.211

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. runoff_ratio

z = -0.78 r = +0.23 score = 0.181

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. elev_mean

z = -0.92 r = -0.20 score = 0.179

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02231342: FT DRUM CREEK AT SUNSHINE ST PKY NEAR FT DRUM, FL

Dominant forcing: PRCP Peak TE: 0.0073 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high sand_frac, high hfd_mean.

Top 10 driver attributes

1. soil_conductivity

z = +4.26 r = +0.15 score = 0.627

high soil conductivity allows fast infiltration and subsurface flow.
2. sand_frac

z = +3.44 r = +0.12 score = 0.403

high sand fraction promotes fast infiltration and subsurface drainage.
3. hfd_mean

z = +1.60 r = -0.24 score = 0.381

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. soil_depth_pelletier

z = +2.38 r = +0.16 score = 0.380

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +1.40 r = -0.24 score = 0.342

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gauge_lat

z = -2.24 r = +0.13 score = 0.283

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. gvff_diff

z = -1.09 r = -0.24 score = 0.264

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. gauge_lon

z = +0.92 r = -0.27 score = 0.251

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. frac_forest

z = -1.34 r = +0.17 score = 0.232

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. elev_mean

z = -0.94 r = -0.20 score = 0.183

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02235200: BLACKWATER CREEK NEAR CASSIA, FL

Dominant forcing: PRCP Peak TE: 0.0086 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high organic_frac, low gvf_diff.

Top 10 driver attributes

1. soil_conductivity

z = +5.20 r = +0.15 score = 0.766

high soil conductivity allows fast infiltration and subsurface flow.
2. organic_frac

z = +3.63 r = +0.17 score = 0.620

high organic soil fraction increases water retention and supports baseflow.
3. gvf_diff

z = -1.28 r = -0.24 score = 0.311

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. sand_frac

z = +2.62 r = +0.12 score = 0.306

high sand fraction promotes fast infiltration and subsurface drainage.
5. p_seasonality

z = +1.22 r = -0.24 score = 0.297

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. pet_mean

z = +2.41 r = -0.12 score = 0.294

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. soil_depth_pelletier

z = +1.73 r = +0.16 score = 0.276

deep soils provide more storage and slower release of water to streamflow.
8. runoff_ratio

z = -1.10 r = +0.23 score = 0.253

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. gauge_lat

z = -1.99 r = +0.13 score = 0.251

southern latitude is associated with warmer climates and rainfall-driven streamflow.
10. gauge_lon

z = +0.88 r = -0.27 score = 0.240

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02245500: SOUTH FORK BLACK CREEK NR PENNEY FARMS, FLA.

Dominant forcing: PRCP Peak TE: 0.0112 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high sand_frac, low gvf_diff.

Top 10 driver attributes

1. soil_conductivity

z = +4.69 r = +0.15 score = 0.691

high soil conductivity allows fast infiltration and subsurface flow.
2. sand_frac

z = +3.17 r = +0.12 score = 0.371

high sand fraction promotes fast infiltration and subsurface drainage.
3. gvf_diff

z = -1.52 r = -0.24 score = 0.370

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +1.87 r = +0.16 score = 0.299

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +0.99 r = -0.24 score = 0.242

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gauge_lon

z = +0.86 r = -0.27 score = 0.233

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. pet_mean

z = +1.84 r = -0.12 score = 0.225

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
8. gauge_lat

z = -1.78 r = +0.13 score = 0.225

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. organic_frac

z = +1.29 r = +0.17 score = 0.220

high organic soil fraction increases water retention and supports baseflow.
10. hfd_mean

z = +0.83 r = -0.24 score = 0.196

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02246000: NORTH FORK BLACK CREEK NR MIDDLEBURG, FLA.

Dominant forcing: PRCP Peak TE: 0.0098 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high sand_frac, low gvf_diff.

Top 10 driver attributes

1. soil_conductivity

z = +3.24 r = +0.15 score = 0.477

high soil conductivity allows fast infiltration and subsurface flow.
2. sand_frac

z = +3.20 r = +0.12 score = 0.374

high sand fraction promotes fast infiltration and subsurface drainage.
3. gvf_diff

z = -1.44 r = -0.24 score = 0.350

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +2.02 r = +0.16 score = 0.323

deep soils provide more storage and slower release of water to streamflow.
5. gauge_lon

z = +0.86 r = -0.27 score = 0.233

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. p_seasonality

z = +0.94 r = -0.24 score = 0.229

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. gauge_lat

z = -1.75 r = +0.13 score = 0.221

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. elev_mean

z = -0.92 r = -0.20 score = 0.180

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02296500: CHARLIE CREEK NEAR GARDNER FL

Dominant forcing: PRCP Peak TE: 0.0066 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +3.94 r = +0.15 score = 0.580

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +2.07 r = -0.24 score = 0.493

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. sand_frac

z = +3.18 r = +0.12 score = 0.372

high sand fraction promotes fast infiltration and subsurface drainage.
4. p_seasonality

z = +1.52 r = -0.24 score = 0.371

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. soil_depth_pelletier

z = +2.32 r = +0.16 score = 0.370

deep soils provide more storage and slower release of water to streamflow.
6. stream_elas

z = +2.46 r = -0.12 score = 0.304

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. gauge_lat

z = -2.28 r = +0.13 score = 0.288

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. gvf_diff

z = -1.12 r = -0.24 score = 0.272

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. gauge_lon

z = +0.86 r = -0.27 score = 0.234

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. frac_forest

z = -1.27 r = +0.17 score = 0.218

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02297155: HORSE CREEK NEAR MYAKKA HEAD FL

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +4.43 r = +0.15 score = 0.653

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +2.20 r = -0.24 score = 0.522

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. sand_frac

z = +3.50 r = +0.12 score = 0.409

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_depth_pelletier

z = +2.40 r = +0.16 score = 0.383

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +1.52 r = -0.24 score = 0.371

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gvf_diff

z = -1.19 r = -0.24 score = 0.288

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lat

z = -2.26 r = +0.13 score = 0.285

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. frac_forest

z = -1.38 r = +0.17 score = 0.237

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. pet_mean

z = +1.95 r = -0.12 score = 0.237

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
10. gauge_lon

z = +0.85 r = -0.27 score = 0.231

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02297310: HORSE CREEK NEAR ARCADIA FL

Dominant forcing: PRCP Peak TE: 0.0047 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high hfd_mean, high soil_conductivity, high sand_frac.

Top 10 driver attributes

1. hfd_mean

z = +2.55 r = -0.24 score = 0.606

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
2. soil_conductivity

z = +4.02 r = +0.15 score = 0.593

high soil conductivity allows fast infiltration and subsurface flow.
3. sand_frac

z = +3.36 r = +0.12 score = 0.393

high sand fraction promotes fast infiltration and subsurface drainage.
4. p_seasonality

z = +1.57 r = -0.24 score = 0.383

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. soil_depth_pelletier

z = +2.39 r = +0.16 score = 0.381

deep soils provide more storage and slower release of water to streamflow.
6. gauge_lat

z = -2.31 r = +0.13 score = 0.292

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. pet_mean

z = +2.26 r = -0.12 score = 0.276

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
8. gvf_diff

z = -1.12 r = -0.24 score = 0.272

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. glim_2nd_class_frac

z = +1.72 r = +0.15 score = 0.264

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. gauge_lon

z = +0.85 r = -0.27 score = 0.231

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02298123: PRAIRIE CREEK NEAR FORT OGDEN FL

Dominant forcing: PRCP Peak TE: 0.0104 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high p_seasonality.

Top 10 driver attributes

1. soil_conductivity

z = +4.29 r = +0.15 score = 0.633

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +2.34 r = -0.24 score = 0.555

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. p_seasonality

z = +1.64 r = -0.24 score = 0.401

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. soil_depth_pelletier

z = +2.40 r = +0.16 score = 0.384

deep soils provide more storage and slower release of water to streamflow.
5. sand_frac

z = +3.09 r = +0.12 score = 0.361

high sand fraction promotes fast infiltration and subsurface drainage.
6. gauge_lat

z = -2.34 r = +0.13 score = 0.295

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. gvff_diff

z = -1.17 r = -0.24 score = 0.284

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. gauge_lon

z = +0.86 r = -0.27 score = 0.235

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. frac_forest

z = -1.35 r = +0.17 score = 0.232

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. carbonate_rocks_frac

z = +1.90 r = -0.12 score = 0.228

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02298608: MYAKKA RIVER AT MYAKKA CITY FL

Dominant forcing: PRCP Peak TE: 0.0086 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +4.17 r = +0.15 score = 0.615

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +2.58 r = -0.24 score = 0.614

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. sand_frac

z = +3.29 r = +0.12 score = 0.385

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.379

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +1.55 r = -0.24 score = 0.377

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gvf_diff

z = -1.20 r = -0.24 score = 0.290

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lat

z = -2.29 r = +0.13 score = 0.288

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. frac_forest

z = -1.51 r = +0.17 score = 0.261

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. gauge_lon

z = +0.84 r = -0.27 score = 0.228

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. pet_mean

z = +1.50 r = -0.12 score = 0.183

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02299950: MANATEE RIVER NEAR MYAKKA HEAD FL

Dominant forcing: PRCP Peak TE: 0.0074 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +4.64 r = +0.15 score = 0.683

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +2.42 r = -0.24 score = 0.576

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. sand_frac

z = +3.55 r = +0.12 score = 0.416

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.379

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +1.50 r = -0.24 score = 0.366

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gvf_diff

z = -1.36 r = -0.24 score = 0.331

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lat

z = -2.26 r = +0.13 score = 0.285

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. frac_forest

z = -1.56 r = +0.17 score = 0.269

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. gauge_lon

z = +0.84 r = -0.27 score = 0.227

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. elev_mean

z = -0.92 r = -0.20 score = 0.180

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02300700: BULLFROG CREEK NEAR WIMAUMA FL

Dominant forcing: PRCP Peak TE: 0.0070 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high hfd_mean, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +4.80 r = +0.15 score = 0.707

high soil conductivity allows fast infiltration and subsurface flow.
2. hfd_mean

z = +1.89 r = -0.24 score = 0.449

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. sand_frac

z = +3.24 r = +0.12 score = 0.379

high sand fraction promotes fast infiltration and subsurface drainage.
4. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.379

deep soils provide more storage and slower release of water to streamflow.
5. p_seasonality

z = +1.45 r = -0.24 score = 0.354

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. pet_mean

z = +2.82 r = -0.12 score = 0.344

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. gvf_diff

z = -1.27 r = -0.24 score = 0.308

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. gauge_lat

z = -2.20 r = +0.13 score = 0.278

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. frac_forest

z = -1.49 r = +0.17 score = 0.257

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. gauge_lon

z = +0.83 r = -0.27 score = 0.225

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02310947: WITHLACOOCHEE RIVER NEAR CUMPRESSCO, FLA.

Dominant forcing: PRCP Peak TE: 0.0064 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high stream_elas, high organic_frac.

Top 10 driver attributes

1. soil_conductivity

z = +3.77 r = +0.15 score = 0.555

high soil conductivity allows fast infiltration and subsurface flow.
2. stream_elas

z = +4.14 r = -0.12 score = 0.511

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. organic_frac

z = +2.19 r = +0.17 score = 0.373

high organic soil fraction increases water retention and supports baseflow.
4. p_seasonality

z = +1.36 r = -0.24 score = 0.333

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. sand_frac

z = +2.70 r = +0.12 score = 0.316

high sand fraction promotes fast infiltration and subsurface drainage.
6. hfd_mean

z = +1.19 r = -0.24 score = 0.283

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
7. gauge_lat

z = -2.10 r = +0.13 score = 0.265

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. runoff_ratio

z = -1.15 r = +0.23 score = 0.264

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. gvf_diff

z = -1.06 r = -0.24 score = 0.257

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. soil_depth_pelletier

z = +1.57 r = +0.16 score = 0.250

deep soils provide more storage and slower release of water to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02312200: LITTLE WITHLACOOCHEE RIVER AT RERDELL, FLA.

Dominant forcing: PRCP Peak TE: 0.0067 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high stream_elas, high soil_conductivity, high sand_frac.

Top 10 driver attributes

1. stream_elas

z = +5.65 r = -0.12 score = 0.698

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. soil_conductivity

z = +3.12 r = +0.15 score = 0.460

high soil conductivity allows fast infiltration and subsurface flow.
3. sand_frac

z = +3.10 r = +0.12 score = 0.363

high sand fraction promotes fast infiltration and subsurface drainage.
4. glim_2nd_class_frac

z = +2.33 r = +0.15 score = 0.357

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. p_seasonality

z = +1.28 r = -0.24 score = 0.313

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. runoff_ratio

z = -1.19 r = +0.23 score = 0.276

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. gauge_lat

z = -2.05 r = +0.13 score = 0.259

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. gvf_diff

z = -1.01 r = -0.24 score = 0.245

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. gauge_lon

z = +0.84 r = -0.27 score = 0.228

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. pet_mean

z = +1.79 r = -0.12 score = 0.218

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02314500: SUWANNEE RIVER AT US 441, AT FARGO, GA

Dominant forcing: PRCP Peak TE: 0.0055 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high organic_frac, high soil_conductivity, high soil_depth_pelletier.

Top 10 driver attributes

1. organic_frac

z = +3.55 r = +0.17 score = 0.605

high organic soil fraction increases water retention and supports baseflow.
2. soil_conductivity

z = +2.96 r = +0.15 score = 0.436

high soil conductivity allows fast infiltration and subsurface flow.
3. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.379

deep soils provide more storage and slower release of water to streamflow.
4. gvf_diff

z = -1.54 r = -0.24 score = 0.374

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. slope_fdc

z = +1.40 r = +0.20 score = 0.282

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. area_gages2

z = +1.49 r = +0.18 score = 0.268

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
7. sand_frac

z = +2.09 r = +0.12 score = 0.244

high sand fraction promotes fast infiltration and subsurface drainage.
8. gauge_lon

z = +0.82 r = -0.27 score = 0.222

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. runoff_ratio

z = -0.96 r = +0.23 score = 0.221

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. gauge_lat

z = -1.64 r = +0.13 score = 0.208

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02315500: SUWANNEE RIVER AT WHITE SPRINGS, FLA.

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, high organic_frac.

Top 10 driver attributes

1. area_gages2

z = +3.14 r = +0.18 score = 0.567

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +2.99 r = +0.17 score = 0.509

a large geospatial drainage area integrates many sub-basin responses.
3. organic_frac

z = +2.75 r = +0.17 score = 0.469

high organic soil fraction increases water retention and supports baseflow.
4. soil_conductivity

z = +2.75 r = +0.15 score = 0.405

high soil conductivity allows fast infiltration and subsurface flow.
5. gvf_diff

z = -1.58 r = -0.24 score = 0.383

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. soil_depth_pelletier

z = +2.38 r = +0.16 score = 0.380

deep soils provide more storage and slower release of water to streamflow.
7. sand_frac

z = +2.22 r = +0.12 score = 0.260

high sand fraction promotes fast infiltration and subsurface drainage.
8. runoff_ratio

z = -0.97 r = +0.23 score = 0.223

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. gauge_lon

z = +0.80 r = -0.27 score = 0.219

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. gauge_lat

z = -1.71 r = +0.13 score = 0.216

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02324400: FENHOLLOWAY RIVER NEAR FOLEY, FLA.

Dominant forcing: PRCP Peak TE: 0.0064 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high soil_conductivity, high soil_depth_pelletier.

Top 10 driver attributes

1. organic_frac

z = +14.92 r = +0.17 score = 2.543

high organic soil fraction increases water retention and supports baseflow.
2. soil_conductivity

z = +8.02 r = +0.15 score = 1.182

high soil conductivity allows fast infiltration and subsurface flow.
3. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.378

deep soils provide more storage and slower release of water to streamflow.
4. gvf_diff

z = -1.51 r = -0.24 score = 0.367

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. runoff_ratio

z = -1.15 r = +0.23 score = 0.266

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. pet_mean

z = +1.85 r = -0.12 score = 0.226

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. gauge_lat

z = -1.76 r = +0.13 score = 0.222

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. slope_fdc

z = +1.09 r = +0.20 score = 0.219

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. gauge_lon

z = +0.76 r = -0.27 score = 0.206

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. stream_elas

z = +1.62 r = -0.12 score = 0.200

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02327100: SOPCHOPPY RIVER NR SOPCHOPPY, FLA.

Dominant forcing: PRCP Peak TE: 0.0096 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high soil_conductivity, low gvf_diff.

Top 10 driver attributes

1. organic_frac

z = +7.49 r = +0.17 score = 1.277

high organic soil fraction increases water retention and supports baseflow.
2. soil_conductivity

z = +5.77 r = +0.15 score = 0.850

high soil conductivity allows fast infiltration and subsurface flow.
3. gvf_diff

z = -1.65 r = -0.24 score = 0.401

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +2.37 r = +0.16 score = 0.378

deep soils provide more storage and slower release of water to streamflow.
5. root_depth_99

z = +3.49 r = +0.10 score = 0.353

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. p_mean

z = +0.65 r = +0.34 score = 0.222

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
7. gauge_lat

z = -1.75 r = +0.13 score = 0.221

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. slope_fdc

z = +1.04 r = +0.20 score = 0.209

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. sand_frac

z = +1.72 r = +0.12 score = 0.201

high sand fraction promotes fast infiltration and subsurface drainage.
10. gauge_lon

z = +0.70 r = -0.27 score = 0.189

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02342933: SOUTH FORK COWIKEE CREEK NEAR BATESVILLE AL

Dominant forcing: Tair Peak TE: 0.0012 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low gauge_lat, low frac_snow, low slope_mean.

Top 10 driver attributes

1. gauge_lat z = -1.39 r = +0.51 score = 0.702

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. frac_snow z = -0.86 r = +0.46 score = 0.397

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
3. slope_mean z = -0.81 r = +0.41 score = 0.330

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. low_prec_freq z = +0.60 r = -0.50 score = 0.304

many dry days indicate a drier climate where streamflow depends on stored water.
5. runoff_ratio z = -0.54 r = +0.56 score = 0.298

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. pet_mean z = +0.88 r = -0.33 score = 0.291

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. frac_forest z = +0.72 r = +0.35 score = 0.249

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. elev_mean z = -0.80 r = +0.30 score = 0.239

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. high_prec_freq z = +0.48 r = -0.49 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q5 z = -0.59 r = +0.36 score = 0.215

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02349900: TURKEY CREEK AT BYROMVILLE, GA

Dominant forcing: SRAD Peak TE: 0.0026 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, low frac_forest, low slope_mean.

Top 10 driver attributes

1. gauge_lat

z = -1.35 r = +0.29 score = 0.397

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. frac_forest

z = -1.45 r = +0.23 score = 0.341

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. slope_mean

z = -0.91 r = +0.37 score = 0.334

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. carbonate_rocks_frac

z = +2.87 r = -0.11 score = 0.322

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
5. gauge_lon

z = +0.73 r = -0.43 score = 0.317

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. runoff_ratio

z = -0.62 r = +0.43 score = 0.266

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. sand_frac

z = +1.48 r = +0.18 score = 0.261

high sand fraction promotes fast infiltration and subsurface drainage.
8. soil_porosity

z = -1.32 r = -0.17 score = 0.226

low soil porosity reduces water storage and produces a flashier response.
9. frac_snow

z = -0.87 r = +0.26 score = 0.222

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. high_prec_dur

z = -0.73 r = +0.29 score = 0.210

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02350900: KINCHAFOONEE CREEK NEAR DAWSON, GA

Dominant forcing: PRCP Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high sand_frac, high gauge_lon, low gauge_lat.

Top 10 driver attributes

1. sand_frac

z = +1.80

r = +0.12

score = 0.210

high sand fraction promotes fast infiltration and subsurface drainage.
2. gauge_lon

z = +0.71

r = -0.27

score = 0.193

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. gauge_lat

z = -1.44

r = +0.13

score = 0.181

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. slope_fdc

z = +0.85

r = +0.20

score = 0.171

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. elev_mean

z = -0.79

r = -0.20

score = 0.154

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. hfd_mean

z = -0.60

r = -0.24

score = 0.143

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. q95

z = -0.45

r = +0.31

score = 0.139

low peak-flow values reflect a damped streamflow response with limited peaks.
8. high_prec_dur

z = -0.58

r = +0.23

score = 0.132

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. frac_snow

z = -0.86

r = -0.15

score = 0.127

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. runoff_ratio

z = -0.54

r = +0.23

score = 0.124

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02361000: CHOCTAWHATCHEE RIVER NEAR NEWTON, AL.

Dominant forcing: PRCP Peak TE: 0.0015 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high sand_frac, low gauge_lat, high p_mean.

Top 10 driver attributes

1. sand_frac

z = +1.72

r = +0.12

score = 0.201

high sand fraction promotes fast infiltration and subsurface drainage.
2. gauge_lat

z = -1.52

r = +0.13

score = 0.191

southern latitude is associated with warmer climates and rainfall-driven streamflow.
3. p_mean

z = +0.50

r = +0.34

score = 0.171

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. gauge_lon

z = +0.63

r = -0.27

score = 0.171

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. elev_mean

z = -0.82

r = -0.20

score = 0.160

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. high_prec_dur

z = -0.66

r = +0.23

score = 0.150

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. hfd_mean

z = -0.54

r = -0.24

score = 0.129

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. frac_snow

z = -0.87

r = -0.15

score = 0.127

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. soil_conductivity

z = +0.81

r = +0.15

score = 0.119

high soil conductivity allows fast infiltration and subsurface flow.
10. slope_fdc

z = +0.56

r = +0.20

score = 0.113

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02363000: PEA RIVER NEAR ARITON AL

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high pet_mean, high slope_fdc, low gauge_lat.

Top 10 driver attributes

1. pet_mean

z = +2.49

r = -0.12

score = 0.304

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. slope_fdc

z = +1.46

r = +0.20

score = 0.293

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gauge_lat

z = -1.47

r = +0.13

score = 0.185

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. high_prec_dur

z = -0.79

r = +0.23

score = 0.178

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. p_mean

z = +0.50

r = +0.34

score = 0.172

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. gauge_lon

z = +0.62

r = -0.27

score = 0.168

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. glim_2nd_class_frac

z = -1.07

r = +0.15

score = 0.164

a small secondary geological class means the subsurface is more uniform.
8. sand_frac

z = +1.38

r = +0.12

score = 0.162

high sand fraction promotes fast infiltration and subsurface drainage.
9. elev_mean

z = -0.79

r = -0.20

score = 0.155

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. glim_1st_class_frac

z = +1.05

r = -0.15

score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02369800: BLACKWATER RIVER NEAR BRADLEY AL

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low gvf_diff, high soil_depth_pelletier.

Top 10 driver attributes

1. p_mean

z = +0.97

r = +0.34

score = 0.333

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. gvf_diff

z = -1.36

r = -0.24

score = 0.332

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. soil_depth_pelletier

z = +1.62

r = +0.16

score = 0.259

deep soils provide more storage and slower release of water to streamflow.
4. root_depth_99

z = +2.05

r = +0.10

score = 0.207

very deep maximum rooting allows vegetation to tap deep storage in droughts.
5. pet_mean

z = +1.68

r = -0.12

score = 0.204

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
6. gauge_lat

z = -1.58

r = +0.13

score = 0.199

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. organic_frac

z = +1.04

r = +0.17

score = 0.178

high organic soil fraction increases water retention and supports baseflow.
8. elev_mean

z = -0.87

r = -0.20

score = 0.171

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. sand_frac

z = +1.33

r = +0.12

score = 0.156

high sand fraction promotes fast infiltration and subsurface drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02371500: CONECUH RIVER AT BRANTLEY AL

Dominant forcing: PRCP Peak TE: 0.0060 Peak lag: 3 day(s)
Summary: PRCP-dominant at lag 3 day(s); top drivers consistent with high slope_fdc, low high_prec_dur, low gauge_lat.

Top 10 driver attributes

1. slope_fdc

z = +1.18 r = +0.20 score = 0.237

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. high_prec_dur

z = -0.84 r = +0.23 score = 0.191

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gauge_lat

z = -1.47 r = +0.13 score = 0.186

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. p_mean

z = +0.50 r = +0.34 score = 0.174

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gauge_lon

z = +0.59 r = -0.27 score = 0.160

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. elev_mean

z = -0.81 r = -0.20 score = 0.157

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. glim_2nd_class_frac

z = -1.00 r = +0.15 score = 0.153

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +0.96 r = -0.15 score = 0.140

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. hfd_mean

z = -0.53 r = -0.24 score = 0.126

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. frac_snow

z = -0.85 r = -0.15 score = 0.125

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02372250: PATSALIGA CREEK NEAR BRANTLEY AL

Dominant forcing: PRCP Peak TE: 0.0034 Peak lag: 3 day(s)
Summary: PRCP-dominant at lag 3 day(s); top drivers consistent with low gauge_lat, high p_mean, high slope_fdc.

Top 10 driver attributes

1. gauge_lat z = -1.47 r = +0.13 score = 0.185

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. p_mean z = +0.51 r = +0.34 score = 0.177

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. slope_fdc z = +0.85 r = +0.20 score = 0.171

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. hfd_mean z = -0.68 r = -0.24 score = 0.162

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. elev_mean z = -0.81 r = -0.20 score = 0.158

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. gauge_lon z = +0.58 r = -0.27 score = 0.157

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. high_prec_dur z = -0.69 r = +0.23 score = 0.157

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. sand_frac z = +1.29 r = +0.12 score = 0.151

high sand fraction promotes fast infiltration and subsurface drainage.
9. glim_2nd_class_frac z = -0.97 r = +0.15 score = 0.149

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac z = +0.98 r = -0.15 score = 0.143

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02374500: MURDER CREEK NEAR EVERGREEN AL

Dominant forcing: PRCP Peak TE: 0.0087 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, high p_mean, low high_prec_dur.

Top 10 driver attributes

1. pet_mean

z = +2.81 r = -0.12 score = 0.343

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. p_mean

z = +0.82 r = +0.34 score = 0.284

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. high_prec_dur

z = -0.90 r = +0.23 score = 0.204

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gauge_lat

z = -1.50 r = +0.13 score = 0.190

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. gvf_diff

z = -0.71 r = -0.24 score = 0.173

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. elev_mean

z = -0.83 r = -0.20 score = 0.161

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. hfd_mean

z = -0.63 r = -0.24 score = 0.150

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. gauge_lon

z = +0.54 r = -0.27 score = 0.147

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. frac_forest

z = +0.83 r = +0.17 score = 0.143

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. frac_snow

z = -0.86 r = -0.15 score = 0.126

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02381600: FAUSETT CREEK NEAR TALKING ROCK, GA

Dominant forcing: VP Peak TE: 0.0024 Peak lag: 24 day(s)

Summary: VP-dominant at lag 24 day(s); top drivers consistent with high runoff_ratio, high q_mean, high q95.

Top 10 driver attributes

1. runoff_ratio

z = +2.05 r = +0.50 score = 1.022

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. q_mean

z = +1.43 r = +0.43 score = 0.611

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. q95

z = +0.94 r = +0.49 score = 0.466

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. clay_frac

z = -1.49 r = -0.22 score = 0.328

low clay fraction allows easier infiltration and reduces surface runoff.
5. q5

z = +0.95 r = +0.34 score = 0.318

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. gauge_lat

z = -0.90 r = +0.35 score = 0.316

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. frac_snow

z = -0.79 r = +0.31 score = 0.247

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. silt_frac

z = -1.80 r = +0.13 score = 0.242

low silt fraction limits water retention in the soil column.
9. frac_forest

z = +0.79 r = +0.29 score = 0.229

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. aridity

z = -0.57 r = -0.31 score = 0.179

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02384540: MILL CREEK NEAR CRANDALL, GA

Dominant forcing: PRCP Peak TE: 0.0035 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low aridity, low hfd_mean.

Top 10 driver attributes

1. p_mean

z = +1.18

r = +0.34

score = 0.408

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. aridity

z = -0.80

r = -0.28

score = 0.220

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
3. hfd_mean

z = -0.81

r = -0.24

score = 0.193

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. root_depth_99

z = +1.89

r = +0.10

score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
5. gauge_lon

z = +0.68

r = -0.27

score = 0.185

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. high_prec_dur

z = -0.70

r = +0.23

score = 0.158

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. frac_forest

z = +0.86

r = +0.17

score = 0.148

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. slope_fdc

z = +0.68

r = +0.20

score = 0.137

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02395120: TWO RUN CREEK NEAR KINGSTON, GA

Dominant forcing: Tair Peak TE: 0.0037 Peak lag: 29 day(s)
Summary: Tair-dominant at lag 29 day(s); top drivers consistent with low gauge_lat, high clay_frac, low frac_snow.

Top 10 driver attributes

1. gauge_lat

z = -0.96 r = +0.51 score = 0.486

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. clay_frac

z = +1.57 r = -0.30 score = 0.464

high clay fraction reduces infiltration and increases surface runoff.
3. frac_snow

z = -0.81 r = +0.46 score = 0.375

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
4. slope_mean

z = -0.78 r = +0.41 score = 0.316

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. high_prec_freq

z = +0.48 r = -0.49 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. glim_1st_class_frac

z = -1.27 r = -0.16 score = 0.205

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. elev_mean

z = -0.65 r = +0.30 score = 0.192

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. stream_elas

z = +0.45 r = -0.43 score = 0.191

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. gauge_lon

z = +0.67 r = -0.25 score = 0.171

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. low_prec_freq

z = +0.33 r = -0.50 score = 0.167

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02408540: HATCHET CREEK BELOW ROCKFORD AL

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low high_prec_dur, high gauge_lon.

Top 10 driver attributes

1. p_mean

z = +0.57

r = +0.34

score = 0.197

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. high_prec_dur

z = -0.85

r = +0.23

score = 0.192

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gauge_lon

z = +0.59

r = -0.27

score = 0.159

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. root_depth_99

z = +1.54

r = +0.10

score = 0.156

very deep maximum rooting allows vegetation to tap deep storage in droughts.
5. gauge_lat

z = -1.22

r = +0.13

score = 0.153

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. hfd_mean

z = -0.63

r = -0.24

score = 0.150

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. elev_mean

z = -0.66

r = -0.20

score = 0.129

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. aridity

z = -0.46

r = -0.28

score = 0.128

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_snow

z = -0.82

r = -0.15

score = 0.120

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. slope_fdc

z = +0.58

r = +0.20

score = 0.116

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02415000: HILLABEE CREEK NEAR HACKNEYVILLE AL

Dominant forcing: Tair Peak TE: 0.0041 Peak lag: 14 day(s)
Summary: Tair-dominant at lag 14 day(s); top drivers consistent with low gauge_lat, low frac_snow, low slope_mean.

Top 10 driver attributes

1. gauge_lat

z = -1.19

r = +0.51

score = 0.600

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. frac_snow

z = -0.81

r = +0.46

score = 0.375

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
3. slope_mean

z = -0.72

r = +0.41

score = 0.293

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. frac_forest

z = +0.73

r = +0.35

score = 0.252

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. geol_porostiy

z = -1.67

r = -0.13

score = 0.213

low geological porosity reduces subsurface storage and produces a flashier response.
6. elev_mean

z = -0.62

r = +0.30

score = 0.185

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. aridity

z = -0.51

r = -0.30

score = 0.156

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. gauge_lon

z = +0.61

r = -0.25

score = 0.156

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. soil_depth_pelletier

z = -0.60

r = -0.26

score = 0.154

shallow soils limit storage and produce a quicker streamflow response.
10. slope_fdc

z = +0.59

r = +0.24

score = 0.142

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02422500: MULBERRY CREEK AT JONES AL

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low high_prec_dur, high p_mean, low gauge_lat.

Top 10 driver attributes

1. high_prec_dur

z = -0.89 r = +0.23 score = 0.202

short heavy-rainfall events produce sharper but briefer runoff peaks.
2. p_mean

z = +0.52 r = +0.34 score = 0.181

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. gauge_lat

z = -1.28 r = +0.13 score = 0.161

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. hfd_mean

z = -0.67 r = -0.24 score = 0.159

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. elev_mean

z = -0.78 r = -0.20 score = 0.153

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. gauge_lon

z = +0.55 r = -0.27 score = 0.149

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. glim_2nd_class_frac

z = -0.90 r = +0.15 score = 0.138

a small secondary geological class means the subsurface is more uniform.
8. aridity

z = -0.46 r = -0.28 score = 0.128

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_snow

z = -0.84 r = -0.15 score = 0.123

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. glim_1st_class_frac

z = +0.84 r = -0.15 score = 0.122

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02427250: PINE BARREN CREEK NEAR SNOW HILL, AL.

Dominant forcing: PRCP Peak TE: 0.0110 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, low high_prec_dur, low hfd_mean.

Top 10 driver attributes

1. pet_mean

z = +2.13

r = -0.12

score = 0.260

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. high_prec_dur

z = -1.03

r = +0.23

score = 0.234

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. hfd_mean

z = -0.86

r = -0.24

score = 0.204

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. p_mean

z = +0.56

r = +0.34

score = 0.192

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gauge_lat

z = -1.39

r = +0.13

score = 0.176

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. glim_2nd_class_frac

z = -1.07

r = +0.15

score = 0.164

a small secondary geological class means the subsurface is more uniform.
7. elev_mean

z = -0.84

r = -0.20

score = 0.163

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. gvfdiff

z = -0.64

r = -0.24

score = 0.156

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. glim_1st_class_frac

z = +1.05

r = -0.15

score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. gauge_lon

z = +0.54

r = -0.27

score = 0.146

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02430085: RED BUD CREEK NR MOORES MILL, MS

Dominant forcing: SRAD Peak TE: 0.0033 Peak lag: 5 day(s)
Summary: SRAD-dominant at lag 5 day(s); top drivers consistent with low stream_elas, low slope_mean, low gauge_lat.

Top 10 driver attributes

1. stream_elas

z = -1.28 r = -0.26 score = 0.331

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. slope_mean

z = -0.84 r = +0.37 score = 0.308

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. gauge_lat

z = -0.92 r = +0.29 score = 0.269

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
5. frac_snow

z = -0.81 r = +0.26 score = 0.207

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
6. gauge_lon

z = +0.46 r = -0.43 score = 0.200

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
8. gvf_diff

z = +0.57 r = -0.33 score = 0.188

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. elev_mean

z = -0.78 r = +0.24 score = 0.187

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. high_prec_dur

z = -0.53 r = +0.29 score = 0.153

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02430615: MUD CREEK NR FAIRVIEW, MS

Dominant forcing: SRAD Peak TE: 0.0036 Peak lag: 5 day(s)
Summary: SRAD-dominant at lag 5 day(s); top drivers consistent with low stream_elas, low slope_mean, low gauge_lat.

Top 10 driver attributes

1. stream_elas

z = -1.23 r = -0.26 score = 0.319

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. slope_mean

z = -0.85 r = +0.37 score = 0.313

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. gauge_lat

z = -0.93 r = +0.29 score = 0.273

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
5. frac_forest

z = +0.94 r = +0.23 score = 0.220

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. root_depth_50

z = +1.72 r = -0.13 score = 0.215

deep median rooting allows vegetation to access stored soil water during dry periods.
7. frac_snow

z = -0.82 r = +0.26 score = 0.210

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. gauge_lon

z = +0.46 r = -0.43 score = 0.198

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. elev_mean

z = -0.80 r = +0.24 score = 0.192

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02450250: SIPSEY FORK NEAR GRAYSON AL

Dominant forcing: PRCP Peak TE: 0.0047 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low soil_depth_statsgo, high p_mean, low hfd_mean.

Top 10 driver attributes

1. soil_depth_statsgo

z = -2.56 r = +0.13 score = 0.342

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
2. p_mean

z = +0.82 r = +0.34 score = 0.281

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. hfd_mean

z = -1.16 r = -0.24 score = 0.275

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. gvf_diff

z = +0.86 r = -0.24 score = 0.209

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. aridity

z = -0.59 r = -0.28 score = 0.162

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. gauge_lon

z = +0.52 r = -0.27 score = 0.141

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. slope_fdc

z = +0.64 r = +0.20 score = 0.129

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. high_prec_dur

z = -0.57 r = +0.23 score = 0.129

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. elev_mean

z = -0.64 r = -0.20 score = 0.125

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. gauge_lat

z = -0.95 r = +0.13 score = 0.120

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02464000: NORTH RIVER NEAR SAMANTHA AL

Dominant forcing: PRCP Peak TE: 0.0071 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, high p_mean, low high_prec_dur.

Top 10 driver attributes

1. hfd_mean

z = -1.11

r = -0.24

score = 0.263

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. p_mean

z = +0.72

r = +0.34

score = 0.248

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. high_prec_dur

z = -0.88

r = +0.23

score = 0.200

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. frac_forest

z = +0.93

r = +0.17

score = 0.160

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. aridity

z = -0.57

r = -0.28

score = 0.159

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. root_depth_99

z = +1.52

r = +0.10

score = 0.154

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. elev_mean

z = -0.78

r = -0.20

score = 0.153

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. gauge_lat

z = -1.11

r = +0.13

score = 0.140

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. gauge_lon

z = +0.51

r = -0.27

score = 0.137

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. sand_frac

z = -1.14

r = +0.12

score = 0.133

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02464146: TURKEY CREEK NEAR TUSCALOOSA

Dominant forcing: Tair Peak TE: 0.0013 Peak lag: 20 day(s)
Summary: Tair-dominant at lag 20 day(s); top drivers consistent with low gauge_lat, low frac_snow, low slope_mean.

Top 10 driver attributes

1. gauge_lat z = -1.12 r = +0.51 score = 0.566

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. frac_snow z = -0.82 r = +0.46 score = 0.380

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
3. slope_mean z = -0.79 r = +0.41 score = 0.321

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. high_prec_freq z = +0.64 r = -0.49 score = 0.316

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. frac_forest z = +0.85 r = +0.35 score = 0.295

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. elev_mean z = -0.79 r = +0.30 score = 0.234

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. glim_2nd_class_frac z = +1.46 r = +0.12 score = 0.178

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. aridity z = -0.53 r = -0.30 score = 0.161

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. soil_depth_pelletier z = -0.61 r = -0.26 score = 0.155

shallow soils limit storage and produce a quicker streamflow response.
10. pet_mean z = +0.44 r = -0.33 score = 0.145

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02464360: BINION CREEK BELOW GIN CREEK NEAR SAMANTHA AL

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, high p_mean, high root_depth_99.

Top 10 driver attributes

1. hfd_mean

z = -0.98 r = -0.24 score = 0.233

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. p_mean

z = +0.60 r = +0.34 score = 0.205

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. root_depth_99

z = +1.61 r = +0.10 score = 0.163

very deep maximum rooting allows vegetation to tap deep storage in droughts.
4. stream_elas

z = -1.30 r = -0.12 score = 0.161

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. elev_mean

z = -0.82 r = -0.20 score = 0.160

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. frac_forest

z = +0.89 r = +0.17 score = 0.154

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lat

z = -1.12 r = +0.13 score = 0.141

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. gauge_lon

z = +0.50 r = -0.27 score = 0.136

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. high_prec_dur

z = -0.57 r = +0.23 score = 0.128

short heavy-rainfall events produce sharper but briefer runoff peaks.
10. aridity

z = -0.46 r = -0.28 score = 0.128

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02465493: ELLIOTTS CREEK AT MOUNDVILLE AL

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low hfd_mean, high pet_mean.

Top 10 driver attributes

1. p_mean

z = +0.56

r = +0.34

score = 0.192

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. hfd_mean

z = -0.79

r = -0.24

score = 0.187

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. pet_mean

z = +1.45

r = -0.12

score = 0.176

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. elev_mean

z = -0.84

r = -0.20

score = 0.164

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. gauge_lat

z = -1.20

r = +0.13

score = 0.151

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. gauge_lon

z = +0.50

r = -0.27

score = 0.137

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. q95

z = -0.41

r = +0.31

score = 0.126

low peak-flow values reflect a damped streamflow response with limited peaks.
10. frac_snow

z = -0.84

r = -0.15

score = 0.123

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02469800: SATILPA CREEK NEAR COFFEEVILLE AL

Dominant forcing: PRCP Peak TE: 0.0132 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high root_depth_99, low hfd_mean.

Top 10 driver attributes

1. p_mean

z = +0.59

r = +0.34

score = 0.205

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. root_depth_99

z = +1.89

r = +0.10

score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
3. hfd_mean

z = -0.79

r = -0.24

score = 0.187

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. gvf_diff

z = -0.75

r = -0.24

score = 0.182

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. gauge_lat

z = -1.44

r = +0.13

score = 0.182

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. elev_mean

z = -0.87

r = -0.20

score = 0.169

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. high_prec_dur

z = -0.73

r = +0.23

score = 0.166

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. frac_forest

z = +0.92

r = +0.17

score = 0.159

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. gauge_lon

z = +0.48

r = -0.27

score = 0.130

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. stream_elas

z = +1.03

r = -0.12

score = 0.127

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02472000: LEAF RIVER NR COLLINS, MS

Dominant forcing: PRCP Peak TE: 0.0098 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high p_mean, low hfd_mean, low gauge_lat.

Top 10 driver attributes

1. p_mean

z = +0.64

r = +0.34

score = 0.220

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. hfd_mean

z = -0.88

r = -0.24

score = 0.210

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. gauge_lat

z = -1.45

r = +0.13

score = 0.183

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. gvf_diff

z = -0.73

r = -0.24

score = 0.177

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. high_prec_dur

z = -0.72

r = +0.23

score = 0.164

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. elev_mean

z = -0.81

r = -0.20

score = 0.158

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. stream_elas

z = +1.10

r = -0.12

score = 0.136

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. aridity

z = -0.48

r = -0.28

score = 0.132

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_snow

z = -0.84

r = -0.15

score = 0.124

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. area_gages2

z = +0.67

r = +0.18

score = 0.120

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02472500: BOUIE CREEK NR HATTIESBURG, MS

Dominant forcing: PRCP Peak TE: 0.0063 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low high_prec_dur, low gvf_diff.

Top 10 driver attributes

1. p_mean

z = +0.71

r = +0.34

score = 0.244

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. high_prec_dur

z = -1.01

r = +0.23

score = 0.228

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gvf_diff

z = -0.89

r = -0.24

score = 0.216

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +1.26

r = +0.16

score = 0.201

deep soils provide more storage and slower release of water to streamflow.
5. gauge_lat

z = -1.50

r = +0.13

score = 0.189

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. pet_mean

z = +1.50

r = -0.12

score = 0.183

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. slope_fdc

z = -0.86

r = +0.20

score = 0.173

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. hfd_mean

z = -0.69

r = -0.24

score = 0.163

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. elev_mean

z = -0.82

r = -0.20

score = 0.160

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02479155: CYPRESS CREEK NR JANICE, MS

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high root_depth_99, high p_mean, low gvf_diff.

Top 10 driver attributes

1. root_depth_99

z = +3.44 r = +0.10 score = 0.348

very deep maximum rooting allows vegetation to tap deep storage in droughts.
2. p_mean

z = +0.83 r = +0.34 score = 0.286

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. gvf_diff

z = -1.16 r = -0.24 score = 0.283

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lat

z = -1.58 r = +0.13 score = 0.199

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. hfd_mean

z = -0.77 r = -0.24 score = 0.183

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. soil_depth_pelletier

z = +1.14 r = +0.16 score = 0.182

deep soils provide more storage and slower release of water to streamflow.
7. elev_mean

z = -0.88 r = -0.20 score = 0.171

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. high_prec_dur

z = -0.75 r = +0.23 score = 0.170

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. sand_frac

z = +1.42 r = +0.12 score = 0.167

high sand fraction promotes fast infiltration and subsurface drainage.
10. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02479300: RED CREEK AT VESTRY, MS

Dominant forcing: PRCP Peak TE: 0.0153 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low gvf_diff, high soil_depth_pelletier.

Top 10 driver attributes

1. p_mean

z = +0.96 r = +0.34 score = 0.332

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. gvf_diff

z = -1.17 r = -0.24 score = 0.284

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. soil_depth_pelletier

z = +1.48 r = +0.16 score = 0.237

deep soils provide more storage and slower release of water to streamflow.
4. gauge_lat

z = -1.63 r = +0.13 score = 0.206

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. high_prec_dur

z = -0.79 r = +0.23 score = 0.179

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. elev_mean

z = -0.88 r = -0.20 score = 0.173

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. aridity

z = -0.59 r = -0.28 score = 0.162

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. hfd_mean

z = -0.60 r = -0.24 score = 0.143

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02479560: ESCATAWPA RIVER NEAR AGRICOLA MS

Dominant forcing: PRCP Peak TE: 0.0178 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low gvf_diff, low gauge_lat.

Top 10 driver attributes

1. p_mean

z = +0.99

r = +0.34

score = 0.340

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. gvf_diff

z = -1.12

r = -0.24

score = 0.272

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. gauge_lat

z = -1.62

r = +0.13

score = 0.204

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. soil_depth_pelletier

z = +1.21

r = +0.16

score = 0.193

deep soils provide more storage and slower release of water to streamflow.
5. elev_mean

z = -0.88

r = -0.20

score = 0.173

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. aridity

z = -0.60

r = -0.28

score = 0.167

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. hfd_mean

z = -0.70

r = -0.24

score = 0.166

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. slope_fdc

z = +0.70

r = +0.20

score = 0.141

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02481000: BILOXI RIVER AT WORTHAM, MS

Dominant forcing: PRCP Peak TE: 0.0105 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low gvf_diff, high soil_depth_pelletier.

Top 10 driver attributes

1. p_mean

z = +1.09

r = +0.34

score = 0.375

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. gvf_diff

z = -1.26

r = -0.24

score = 0.307

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. soil_depth_pelletier

z = +1.48

r = +0.16

score = 0.236

deep soils provide more storage and slower release of water to streamflow.
4. gauge_lat

z = -1.67

r = +0.13

score = 0.211

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. q95

z = +0.59

r = +0.31

score = 0.180

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. elev_mean

z = -0.90

r = -0.20

score = 0.176

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. aridity

z = -0.63

r = -0.28

score = 0.173

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. root_depth_99

z = +1.38

r = +0.10

score = 0.139

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 02481510: WOLF RIVER NR LANDON, MS

Dominant forcing: PRCP Peak TE: 0.0111 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low gvf_diff, low gauge_lat.

Top 10 driver attributes

1. **p_mean**

z = +0.97

r = +0.34

score = 0.333

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **gvf_diff**

z = -1.19

r = -0.24

score = 0.289

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. **gauge_lat**

z = -1.68

r = +0.13

score = 0.212

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. **soil_depth_pelletier**

z = +1.27

r = +0.16

score = 0.203

deep soils provide more storage and slower release of water to streamflow.
5. **elev_mean**

z = -0.88

r = -0.20

score = 0.173

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. **glim_2nd_class_frac**

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. **aridity**

z = -0.59

r = -0.28

score = 0.162

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. **glim_1st_class_frac**

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. **high_prec_dur**

z = -0.61

r = +0.23

score = 0.139

short heavy-rainfall events produce sharper but briefer runoff peaks.
10. **hfd_mean**

z = -0.54

r = -0.24

score = 0.129

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03010655: Oswayo Creek at Shinglehouse, PA

Dominant forcing: PRCP Peak TE: 0.0056 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, low low_prec_freq, high gauge_lon.

Top 10 driver attributes

1. gvf_diff z = +1.36 r = -0.24 score = 0.332

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. low_prec_freq z = -1.34 r = -0.22 score = 0.300

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gauge_lon z = +1.09 r = -0.27 score = 0.295

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. slope_fdc z = +1.36 r = +0.20 score = 0.273

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. high_prec_dur z = -0.96 r = +0.23 score = 0.218

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. hfd_mean z = -0.70 r = -0.24 score = 0.166

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. aridity z = -0.60 r = -0.28 score = 0.165

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. frac_forest z = +0.95 r = +0.17 score = 0.163

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. high_prec_freq z = -0.82 r = -0.19 score = 0.154

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03011800: Kinzua Creek near Guffey, PA

Dominant forcing: SRAD Peak TE: 0.0026 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low low_prec_freq, high gauge_lon, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq

z = -1.57 r = -0.31 score = 0.491

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gauge_lon

z = +1.05 r = -0.43 score = 0.455

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. high_prec_freq

z = -1.38 r = -0.33 score = 0.451

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. gvf_diff

z = +1.37 r = -0.33 score = 0.449

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. runoff_ratio

z = +0.65 r = +0.43 score = 0.277

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. high_prec_dur

z = -0.88 r = +0.29 score = 0.254

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. frac_forest

z = +0.97 r = +0.23 score = 0.227

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. lai_diff

z = +1.51 r = -0.14 score = 0.211

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
10. slope_mean

z = -0.56 r = +0.37 score = 0.206

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03015500: Brokenstraw Creek at Youngsville, PA

Dominant forcing: Tair Peak TE: 0.0043 Peak lag: 17 day(s)
Summary: Tair-dominant at lag 17 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.62 r = -0.50 score = 0.816

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.22 r = -0.49 score = 0.599

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.78 r = +0.56 score = 0.434

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. frac_forest

z = +0.92 r = +0.35 score = 0.319

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. pet_mean

z = -0.95 r = -0.33 score = 0.314

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lon

z = +1.02 r = -0.25 score = 0.259

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. gauge_lat

z = +0.50 r = +0.51 score = 0.253

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. slope_fdc

z = +0.97 r = +0.24 score = 0.234

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. q95

z = +0.43 r = +0.51 score = 0.219

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. aridity

z = -0.70 r = -0.30 score = 0.211

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03021350: French Creek near Wattsburg, PA

Dominant forcing: Tair Peak TE: 0.0098 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.55 r = -0.50 score = 0.783

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.21 r = -0.49 score = 0.593

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +1.01 r = +0.56 score = 0.558

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. q95

z = +0.81 r = +0.51 score = 0.414

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. pet_mean

z = -0.94 r = -0.33 score = 0.311

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. frac_forest

z = +0.90 r = +0.35 score = 0.311

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lat

z = +0.53 r = +0.51 score = 0.269

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. silt_frac

z = +2.07 r = +0.13 score = 0.268

high silt fraction holds water and slows infiltration.
9. gauge_lon

z = +0.99 r = -0.25 score = 0.251

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. geol_permeability

z = -2.09 r = -0.11 score = 0.237

low geological permeability slows subsurface flow and reduces groundwater contribution.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03026500: Sevenmile Run near Rasselas, PA

Dominant forcing: Tair Peak TE: 0.0032 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -1.51 r = -0.50 score = 0.760

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.10 r = -0.49 score = 0.539

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. slope_fdc

z = +1.31 r = +0.24 score = 0.317

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. pet_mean

z = -0.95 r = -0.33 score = 0.313

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. stream_elas

z = -0.73 r = -0.43 score = 0.312

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. gauge_lon

z = +1.06 r = -0.25 score = 0.270

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. slope_mean

z = -0.64 r = +0.41 score = 0.263

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. runoff_ratio

z = +0.46 r = +0.56 score = 0.253

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. gauge_lat

z = +0.46 r = +0.51 score = 0.232

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03028000: West Branch Clarion River at Wilcox, PA

Dominant forcing: VP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.45 r = -0.50 score = 0.727

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.00 r = -0.47 score = 0.469

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.79 r = +0.50 score = 0.395

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. slope_fdc

z = +1.18 r = +0.24 score = 0.282

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. frac_forest

z = +0.91 r = +0.29 score = 0.265

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. low_prec_dur

z = -0.92 r = -0.27 score = 0.247

short dry spells limit basin drying and keep storage close to full.
7. pet_mean

z = -0.94 r = -0.25 score = 0.240

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. aridity

z = -0.66 r = -0.31 score = 0.206

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. slope_mean

z = -0.59 r = +0.27 score = 0.161

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. gauge_lat

z = +0.45 r = +0.35 score = 0.158

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03049000: Buffalo Creek near Freeport, PA

Dominant forcing: VP Peak TE: 0.0023 Peak lag: 19 day(s)
Summary: VP-dominant at lag 19 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high slope_fdc.

Top 10 driver attributes

1. low_prec_freq

z = -1.08 r = -0.50 score = 0.541

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.67 r = -0.47 score = 0.314

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. slope_fdc

z = +1.18 r = +0.24 score = 0.283

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. stream_elas

z = -0.77 r = -0.36 score = 0.279

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. low_prec_dur

z = -0.87 r = -0.27 score = 0.233

short dry spells limit basin drying and keep storage close to full.
6. pet_mean

z = -0.74 r = -0.25 score = 0.187

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. slope_mean

z = -0.64 r = +0.27 score = 0.174

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. aridity

z = -0.54 r = -0.31 score = 0.170

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. silt_frac

z = +1.14 r = +0.13 score = 0.154

high silt fraction holds water and slows infiltration.
10. gauge_lon

z = +0.99 r = -0.13 score = 0.131

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03049800: Little Pine Creek near Etna, PA

Dominant forcing: SRAD Peak TE: 0.0011 Peak lag: 30 day(s)
Summary: SRAD-dominant at lag 30 day(s); top drivers consistent with high gauge_lon, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. gauge_lon z = +0.98 r = -0.43 score = 0.422

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. gvf_diff z = +0.98 r = -0.33 score = 0.320

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur z = -1.08 r = +0.29 score = 0.310

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. low_prec_freq z = -0.92 r = -0.31 score = 0.287

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. frac_forest z = -1.03 r = +0.23 score = 0.243

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. glim_1st_class_frac z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. slope_mean z = -0.53 r = +0.37 score = 0.194

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. glim_2nd_class_frac z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
9. lai_diff z = +1.32 r = -0.14 score = 0.184

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
10. high_prec_freq z = -0.50 r = -0.33 score = 0.164

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03050000: TYGART VALLEY RIVER NEAR DAILEY, WV

Dominant forcing: PRCP Peak TE: 0.0060 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, high gvf_diff, high gauge_lon.

Top 10 driver attributes

1. low_prec_freq z = -1.71 r = -0.22 score = 0.382

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gvf_diff z = +1.39 r = -0.24 score = 0.337

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. gauge_lon z = +0.98 r = -0.27 score = 0.266

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. high_prec_freq z = -1.37 r = -0.19 score = 0.258

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. slope_fdc z = +1.22 r = +0.20 score = 0.244

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. high_prec_dur z = -1.08 r = +0.23 score = 0.244

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. aridity z = -0.68 r = -0.28 score = 0.187

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. soil_depth_statsgo z = -1.19 r = +0.13 score = 0.159

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
9. frac_forest z = +0.90 r = +0.17 score = 0.156

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. p_mean z = +0.44 r = +0.34 score = 0.150

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03066000: BLACKWATER R AT DAVIS, WV

Dominant forcing: PRCP Peak TE: 0.0020 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, low high_prec_dur.

Top 10 driver attributes

1. low_prec_freq z = -1.70 r = -0.22 score = 0.380

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq z = -1.61 r = -0.19 score = 0.304

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. high_prec_dur z = -1.30 r = +0.23 score = 0.295

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gauge_lon z = +1.01 r = -0.27 score = 0.273

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. gvf_diff z = +1.05 r = -0.24 score = 0.255

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. slope_fdc z = +1.18 r = +0.20 score = 0.237

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. p_mean z = +0.64 r = +0.34 score = 0.220

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. aridity z = -0.76 r = -0.28 score = 0.210

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. q95 z = +0.69 r = +0.31 score = 0.209

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. runoff_ratio z = +0.83 r = +0.23 score = 0.192

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03069500: CHEAT RIVER NEAR PARSONS, WV

Dominant forcing: VP Peak TE: 0.0028 Peak lag: 25 day(s)
Summary: VP-dominant at lag 25 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.84 r = -0.50 score = 0.918

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.71 r = -0.47 score = 0.805

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.92 r = +0.50 score = 0.457

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. q95

z = +0.65 r = +0.49 score = 0.318

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. q_mean

z = +0.62 r = +0.43 score = 0.265

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. slope_fdc

z = +1.10 r = +0.24 score = 0.264

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. low_prec_dur

z = -0.97 r = -0.27 score = 0.260

short dry spells limit basin drying and keep storage close to full.
8. frac_forest

z = +0.81 r = +0.29 score = 0.235

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. aridity

z = -0.74 r = -0.31 score = 0.232

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. pet_mean

z = -0.61 r = -0.25 score = 0.156

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03070500: BIG SANDY CREEK AT ROCKVILLE, WV

Dominant forcing: Tair Peak TE: 0.0056 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq z = -1.46 r = -0.50 score = 0.737

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq z = -1.25 r = -0.49 score = 0.615

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio z = +0.69 r = +0.56 score = 0.386

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. slope_fdc z = +1.51 r = +0.24 score = 0.364

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. gauge_lon z = +0.99 r = -0.25 score = 0.253

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. q95 z = +0.43 r = +0.51 score = 0.221

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. pet_mean z = -0.64 r = -0.33 score = 0.210

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. aridity z = -0.65 r = -0.30 score = 0.196

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. low_prec_dur z = -0.93 r = -0.20 score = 0.189

short dry spells limit basin drying and keep storage close to full.
10. glim_1st_class_frac z = +1.05 r = -0.16 score = 0.170

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03076600: BEAR CREEK AT FRIENDSVILLE, MD

Dominant forcing: Tair Peak TE: 0.0049 Peak lag: 22 day(s)
Summary: Tair-dominant at lag 22 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high slope_fdc.

Top 10 driver attributes

1. low_prec_freq

z = -1.36 r = -0.50 score = 0.687

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.06 r = -0.49 score = 0.523

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. slope_fdc

z = +1.11 r = +0.24 score = 0.267

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. gauge_lon

z = +1.01 r = -0.25 score = 0.257

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. runoff_ratio

z = +0.43 r = +0.56 score = 0.238

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. slope_mean

z = -0.52 r = +0.41 score = 0.214

gentle terrain slows surface runoff and produces a smoother streamflow signal.
7. pet_mean

z = -0.63 r = -0.33 score = 0.208

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. aridity

z = -0.63 r = -0.30 score = 0.191

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. stream_elas

z = -0.44 r = -0.43 score = 0.190

low streamflow elasticity means streamflow is buffered against rainfall variability.
10. low_prec_dur

z = -0.93 r = -0.20 score = 0.188

short dry spells limit basin drying and keep storage close to full.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03078000: CASSELMAN RIVER AT GRANTSVILLE, MD

Dominant forcing: Tair Peak TE: 0.0055 Peak lag: 25 day(s)
Summary: Tair-dominant at lag 25 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.14 r = -0.50 score = 0.577

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.83 r = -0.49 score = 0.409

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.66 r = +0.56 score = 0.369

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. slope_fdc

z = +1.41 r = +0.24 score = 0.341

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. gauge_lon

z = +1.03 r = -0.25 score = 0.262

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. pet_mean

z = -0.63 r = -0.33 score = 0.209

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. geol_permeability

z = -1.76 r = -0.11 score = 0.199

low geological permeability slows subsurface flow and reduces groundwater contribution.
8. frac_forest

z = +0.54 r = +0.35 score = 0.187

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. low_prec_dur

z = -0.91 r = -0.20 score = 0.185

short dry spells limit basin drying and keep storage close to full.
10. aridity

z = -0.59 r = -0.30 score = 0.178

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03140000: Mill Creek near Coshocton OH

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high slope_fdc, high gauge_lon, high gvf_diff.

Top 10 driver attributes

1. slope_fdc

z = +1.20

r = +0.20

score = 0.241

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. gauge_lon

z = +0.86

r = -0.27

score = 0.233

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. gvf_diff

z = +0.82

r = -0.24

score = 0.199

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. stream_elas

z = +1.30

r = -0.12

score = 0.161

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. sand_frac

z = -1.18

r = +0.12

score = 0.139

low sand fraction reduces infiltration and may favor surface runoff.
8. high_prec_dur

z = -0.56

r = +0.23

score = 0.128

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. elev_mean

z = -0.60

r = -0.20

score = 0.117

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. hfd_mean

z = -0.47

r = -0.24

score = 0.113

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03144000: Wakatomika Creek near Frazeyburg OH

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high gauge_lon, high slope_fdc.

Top 10 driver attributes

1. gvf_diff

z = +1.03 r = -0.24 score = 0.251

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. gauge_lon

z = +0.84 r = -0.27 score = 0.229

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. slope_fdc

z = +0.88 r = +0.20 score = 0.176

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. high_prec_dur

z = -0.67 r = +0.23 score = 0.151

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. elev_mean

z = -0.59 r = -0.20 score = 0.116

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. aridity

z = -0.39 r = -0.28 score = 0.107

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. sand_frac

z = -0.89 r = +0.12 score = 0.104

low sand fraction reduces infiltration and may favor surface runoff.
10. soil_depth_statsgo

z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03159540: Shade River near Chester OH

Dominant forcing: Tair Peak TE: 0.0034 Peak lag: 21 day(s)
Summary: Tair-dominant at lag 21 day(s); top drivers consistent with high clay_frac, low slope_mean, low geol_permeability.

Top 10 driver attributes

1. clay_frac

z = +1.65 r = -0.30 score = 0.486

high clay fraction reduces infiltration and increases surface runoff.
2. slope_mean

z = -0.77 r = +0.41 score = 0.312

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. geol_permeability

z = -2.21 r = -0.11 score = 0.250

low geological permeability slows subsurface flow and reduces groundwater contribution.
4. frac_snow

z = -0.49 r = +0.46 score = 0.224

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
5. q5

z = -0.60 r = +0.36 score = 0.218

low low-flow values reflect basins that nearly dry out during dry periods.
6. gauge_lon

z = +0.86 r = -0.25 score = 0.218

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. low_prec_freq

z = -0.41 r = -0.50 score = 0.209

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. baseflow_index

z = -0.91 r = +0.22 score = 0.204

a low baseflow index means streamflow responds quickly to recent rainfall.
9. elev_mean

z = -0.67 r = +0.30 score = 0.199

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. silt_frac

z = +1.47 r = +0.13 score = 0.190

high silt fraction holds water and slows infiltration.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03161000: SOUTH FORK NEW RIVER NEAR JEFFERSON, NC

Dominant forcing: SRAD Peak TE: 0.0020 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high gauge_lon, high baseflow_index.

Top 10 driver attributes

1. q5

z = +1.72

r = +0.31

score = 0.537

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lon

z = +0.89

r = -0.43

score = 0.383

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. baseflow_index

z = +1.45

r = +0.22

score = 0.317

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. high_prec_dur

z = -0.82

r = +0.29

score = 0.236

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. glim_1st_class_frac

z = +1.06

r = -0.21

score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. frac_forest

z = +0.96

r = +0.23

score = 0.224

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gvf_diff

z = +0.62

r = -0.33

score = 0.202

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. low_prec_freq

z = -0.63

r = -0.31

score = 0.195

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. glim_2nd_class_frac

z = -1.08

r = +0.18

score = 0.190

a small secondary geological class means the subsurface is more uniform.
10. runoff_ratio

z = +0.41

r = +0.43

score = 0.174

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03164000: NEW RIVER NEAR GALAX, VA

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high area_gages2, high area_geospa_fabric.

Top 10 driver attributes

1. gauge_lon z = +0.91 r = -0.27 score = 0.248

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. area_gages2 z = +1.27 r = +0.18 score = 0.229

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
3. area_geospa_fabric z = +1.26 r = +0.17 score = 0.215

a large geospatial drainage area integrates many sub-basin responses.
4. high_prec_dur z = -0.93 r = +0.23 score = 0.210

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. q5 z = +1.02 r = +0.18 score = 0.183

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. aridity z = -0.56 r = -0.28 score = 0.154

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. gvf_diff z = +0.60 r = -0.24 score = 0.145

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. frac_forest z = +0.79 r = +0.17 score = 0.137

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. low_prec_freq z = -0.55 r = -0.22 score = 0.123

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. p_mean z = +0.34 r = +0.34 score = 0.118

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03165000: CHESTNUT CREEK AT GALAX, VA

Dominant forcing: SRAD Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high gauge_lon, high baseflow_index.

Top 10 driver attributes

1. q5

z = +1.32

r = +0.31

score = 0.412

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lon

z = +0.92

r = -0.43

score = 0.396

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. baseflow_index

z = +1.28

r = +0.22

score = 0.281

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. glim_2nd_class_frac

z = +1.06

r = +0.18

score = 0.187

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. high_prec_dur

z = -0.64

r = +0.29

score = 0.184

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. gauge_lat

z = -0.50

r = +0.29

score = 0.146

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. soil_depth_pelletier

z = -0.61

r = -0.24

score = 0.145

shallow soils limit storage and produce a quicker streamflow response.
8. frac_forest

z = +0.55

r = +0.23

score = 0.129

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. runoff_ratio

z = +0.30

r = +0.43

score = 0.129

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. frac_snow

z = -0.50

r = +0.26

score = 0.128

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03170000: LITTLE RIVER AT GRAYSONTOWN, VA

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, low high_prec_dur, low q95.

Top 10 driver attributes

1. gauge_lon

z = +0.94 r = -0.27 score = 0.255

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. high_prec_dur

z = -0.81 r = +0.23 score = 0.184

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. q95

z = -0.52 r = +0.31 score = 0.160

low peak-flow values reflect a damped streamflow response with limited peaks.
4. gvf_diff

z = +0.53 r = -0.24 score = 0.130

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. frac_forest

z = +0.66 r = +0.17 score = 0.113

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. stream_elas

z = -0.88 r = -0.12 score = 0.108

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
8. aridity

z = -0.34 r = -0.28 score = 0.094

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. gvf_max

z = +0.74 r = +0.12 score = 0.085

high maximum green vegetation fraction reflects dense canopy cover.
10. slope_fdc

z = +0.40 r = +0.20 score = 0.081

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03173000: WALKER CREEK AT BANE, VA

Dominant forcing: PRCP Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +0.93 r = -0.27 score = 0.253

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.19 r = +0.20 score = 0.239

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gvf_diff

z = +0.82 r = -0.24 score = 0.199

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. high_prec_dur

z = -0.84 r = +0.23 score = 0.191

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. stream_elas

z = +1.38 r = -0.12 score = 0.171

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. low_prec_freq

z = -0.56 r = -0.22 score = 0.126

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. q95

z = -0.35 r = +0.31 score = 0.107

low peak-flow values reflect a damped streamflow response with limited peaks.
8. soil_depth_pelletier

z = -0.61 r = +0.16 score = 0.097

shallow soils limit storage and produce a quicker streamflow response.
9. gvf_max

z = +0.81 r = +0.12 score = 0.093

high maximum green vegetation fraction reflects dense canopy cover.
10. q_mean

z = -0.31 r = +0.28 score = 0.086

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03180500: GREENBRIER RIVER AT DURBIN, WV

Dominant forcing: VP Peak TE: 0.0023 Peak lag: 17 day(s)
Summary: VP-dominant at lag 17 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high slope_fdc.

Top 10 driver attributes

1. low_prec_freq

z = -1.44 r = -0.50 score = 0.720

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.35 r = -0.47 score = 0.635

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. slope_fdc

z = +1.40 r = +0.24 score = 0.335

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. runoff_ratio

z = +0.63 r = +0.50 score = 0.313

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. frac_forest

z = +0.93 r = +0.29 score = 0.271

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. low_prec_dur

z = -0.91 r = -0.27 score = 0.244

short dry spells limit basin drying and keep storage close to full.
7. aridity

z = -0.65 r = -0.31 score = 0.204

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. q95

z = +0.39 r = +0.49 score = 0.193

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. silt_frac

z = +1.42 r = +0.13 score = 0.191

high silt fraction holds water and slows infiltration.
10. q_mean

z = +0.34 r = +0.43 score = 0.146

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03182500: GREENBRIER RIVER AT BUCKEYE, WV

Dominant forcing: Tair Peak TE: 0.0031 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low_low_prec_freq, low_high_prec_freq, high_frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -1.19 r = -0.50 score = 0.601

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.71 r = -0.49 score = 0.350

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.95 r = +0.35 score = 0.327

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. slope_fdc

z = +1.33 r = +0.24 score = 0.320

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. stream_elas

z = -0.61 r = -0.43 score = 0.260

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. gauge_lon

z = +0.97 r = -0.25 score = 0.246

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. runoff_ratio

z = +0.36 r = +0.56 score = 0.199

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. low_prec_dur

z = -0.85 r = -0.20 score = 0.174

short dry spells limit basin drying and keep storage close to full.
9. aridity

z = -0.56 r = -0.30 score = 0.171

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. pet_mean

z = -0.48 r = -0.33 score = 0.160

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03186500: WILLIAMS RIVER AT DYER, WV

Dominant forcing: Tair Peak TE: 0.0048 Peak lag: 22 day(s)
Summary: Tair-dominant at lag 22 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = -1.36 r = -0.50 score = 0.687

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.26 r = -0.49 score = 0.620

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. runoff_ratio

z = +0.96 r = +0.56 score = 0.534

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. stream_elas

z = -0.90 r = -0.43 score = 0.384

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. q95

z = +0.73 r = +0.51 score = 0.375

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. frac_forest

z = +0.95 r = +0.35 score = 0.328

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. slope_fdc

z = +1.29 r = +0.24 score = 0.311

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. q_mean

z = +0.67 r = +0.45 score = 0.301

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
9. gauge_lon

z = +0.94 r = -0.25 score = 0.240

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. aridity

z = -0.73 r = -0.30 score = 0.221

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03187500: CRANBERRY RIVER NEAR RICHWOOD, WV

Dominant forcing: VP Peak TE: 0.0038 Peak lag: 29 day(s)
Summary: VP-dominant at lag 29 day(s); top drivers consistent with low low_prec_freq, high runoff_ratio, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq

z = -1.16 r = -0.50 score = 0.581

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. runoff_ratio

z = +1.09 r = +0.50 score = 0.543

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. high_prec_freq

z = -1.04 r = -0.47 score = 0.490

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. q95

z = +0.78 r = +0.49 score = 0.383

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. q_mean

z = +0.75 r = +0.43 score = 0.321

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. slope_fdc

z = +1.31 r = +0.24 score = 0.313

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. stream_elas

z = -0.86 r = -0.36 score = 0.311

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. frac_forest

z = +0.95 r = +0.29 score = 0.275

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. low_prec_dur

z = -0.88 r = -0.27 score = 0.235

short dry spells limit basin drying and keep storage close to full.
10. aridity

z = -0.73 r = -0.31 score = 0.228

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03213700: TUG FORK AT WILLIAMSON, WV

Dominant forcing: PRCP Peak TE: 0.0076 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high slope_fdc, low high_prec_dur.

Top 10 driver attributes

1. gvf_diff z = +1.53 r = -0.24 score = 0.372

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. slope_fdc z = +1.21 r = +0.20 score = 0.242

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. high_prec_dur z = -1.03 r = +0.23 score = 0.233

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gauge_lon z = +0.83 r = -0.27 score = 0.226

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. low_prec_freq z = -0.92 r = -0.22 score = 0.206

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. area_gages2 z = +0.96 r = +0.18 score = 0.173

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
7. area_geospa_fabric z = +0.95 r = +0.17 score = 0.162

a large geospatial drainage area integrates many sub-basin responses.
8. frac_forest z = +0.91 r = +0.17 score = 0.157

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. glim_2nd_class_frac z = -1.01 r = +0.15 score = 0.155

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac z = +0.98 r = -0.15 score = 0.143

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03237280: Upper Twin Creek at McGaw OH

Dominant forcing: VP Peak TE: 0.0029 Peak lag: 27 day(s)
Summary: VP-dominant at lag 27 day(s); top drivers consistent with high slope_fdc, high stream_elas, high silt_frac.

Top 10 driver attributes

1. slope_fdc

z = +1.58 r = +0.24 score = 0.378

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. stream_elas

z = +0.92 r = -0.36 score = 0.332

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. silt_frac

z = +2.12 r = +0.13 score = 0.286

high silt fraction holds water and slows infiltration.
4. frac_forest

z = +0.93 r = +0.29 score = 0.270

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. low_prec_freq

z = -0.45 r = -0.50 score = 0.225

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. q5

z = -0.64 r = +0.34 score = 0.214

low low-flow values reflect basins that nearly dry out during dry periods.
7. high_prec_freq

z = -0.41 r = -0.47 score = 0.195

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. low_prec_dur

z = -0.71 r = -0.27 score = 0.188

short dry spells limit basin drying and keep storage close to full.
9. aridity

z = -0.54 r = -0.31 score = 0.169

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. baseflow_index

z = -0.83 r = +0.17 score = 0.144

a low baseflow index means streamflow responds quickly to recent rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03237500: Ohio Brush Creek near West Union OH

Dominant forcing: Tair Peak TE: 0.0013 Peak lag: 29 day(s)
Summary: Tair-dominant at lag 29 day(s); top drivers consistent with high clay_frac, low slope_mean, low baseflow_index.

Top 10 driver attributes

1. clay_frac

z = +1.17 r = -0.30 score = 0.346

high clay fraction reduces infiltration and increases surface runoff.
2. slope_mean

z = -0.68 r = +0.41 score = 0.276

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. baseflow_index

z = -1.17 r = +0.22 score = 0.262

a low baseflow index means streamflow responds quickly to recent rainfall.
4. frac_snow

z = -0.49 r = +0.46 score = 0.224

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
5. q5

z = -0.61 r = +0.36 score = 0.224

low low-flow values reflect basins that nearly dry out during dry periods.
6. low_prec_freq

z = -0.43 r = -0.50 score = 0.218

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lon

z = +0.76 r = -0.25 score = 0.194

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. elev_mean

z = -0.62 r = +0.30 score = 0.184

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. soil_conductivity

z = -0.75 r = +0.20 score = 0.153

low soil conductivity restricts infiltration, often increasing surface runoff.
10. geol_permeability

z = -1.33 r = -0.11 score = 0.150

low geological permeability slows subsurface flow and reduces groundwater contribution.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03238500: White Oak Creek near Georgetown OH

Dominant forcing: Tair Peak TE: 0.0045 Peak lag: 6 day(s)
Summary: Tair-dominant at lag 6 day(s); top drivers consistent with high soil_depth_pelletier, low frac_forest, low baseflow_index.

Top 10 driver attributes

1. soil_depth_pelletier

z = +1.91 r = -0.26 score = 0.489

deep soils provide more storage and slower release of water to streamflow.
2. frac_forest

z = -1.25 r = +0.35 score = 0.431

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. baseflow_index

z = -1.76 r = +0.22 score = 0.394

a low baseflow index means streamflow responds quickly to recent rainfall.
4. slope_mean

z = -0.82 r = +0.41 score = 0.334

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. clay_frac

z = +1.07 r = -0.30 score = 0.316

high clay fraction reduces infiltration and increases surface runoff.
6. stream_elas

z = -0.58 r = -0.43 score = 0.247

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. frac_snow

z = -0.51 r = +0.46 score = 0.233

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
8. q5

z = -0.63 r = +0.36 score = 0.231

low low-flow values reflect basins that nearly dry out during dry periods.
9. low_prec_freq

z = -0.43 r = -0.50 score = 0.216

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. silt_frac

z = +1.54 r = +0.13 score = 0.200

high silt fraction holds water and slows infiltration.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03241500: Massies Creek at Wilberforce OH

Dominant forcing: PRCP Peak TE: 0.0041 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high carbonate_rocks_frac, high soil_depth_pelletier.

Top 10 driver attributes

1. gvf_diff z = +1.51 r = -0.24 score = 0.366

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. carbonate_rocks_frac z = +2.89 r = -0.12 score = 0.346

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. soil_depth_pelletier z = +2.03 r = +0.16 score = 0.325

deep soils provide more storage and slower release of water to streamflow.
4. frac_forest z = -1.65 r = +0.17 score = 0.284

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gauge_lon z = +0.73 r = -0.27 score = 0.199

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. high_prec_dur z = -0.84 r = +0.23 score = 0.190

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. slope_fdc z = +0.77 r = +0.20 score = 0.154

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. stream_elas z = +0.98 r = -0.12 score = 0.121

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. root_depth_99 z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.
10. elev_mean z = -0.56 r = -0.20 score = 0.109

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03280700: CUTSHIN CREEK AT WOOTON, KY

Dominant forcing: Tair Peak TE: 0.0037 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with high frac_forest, low low_prec_freq, low frac_snow.

Top 10 driver attributes

1. frac_forest

z = +0.94 r = +0.35 score = 0.326

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
2. low_prec_freq

z = -0.63 r = -0.50 score = 0.318

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. frac_snow

z = -0.59 r = +0.46 score = 0.270

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
4. high_prec_freq

z = -0.46 r = -0.49 score = 0.225

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. slope_fdc

z = +0.89 r = +0.24 score = 0.216

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. gauge_lat

z = -0.40 r = +0.51 score = 0.202

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. gauge_lon

z = +0.77 r = -0.25 score = 0.196

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. aridity

z = -0.60 r = -0.30 score = 0.183

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. q5

z = -0.48 r = +0.36 score = 0.176

low low-flow values reflect basins that nearly dry out during dry periods.
10. glim_1st_class_frac

z = +1.06 r = -0.16 score = 0.171

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03281100: GOOSE CREEK AT MANCHESTER, KY

Dominant forcing: Tair Peak TE: 0.0044 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with low frac_snow, high frac_forest, low geol_permeability.

Top 10 driver attributes

1. frac_snow

z = -0.67

r = +0.46

score = 0.308

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
2. frac_forest

z = +0.82

r = +0.35

score = 0.285

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
3. geol_permeability

z = -2.01

r = -0.11

score = 0.227

low geological permeability slows subsurface flow and reduces groundwater contribution.
4. clay_frac

z = +0.75

r = -0.30

score = 0.223

high clay fraction reduces infiltration and increases surface runoff.
5. slope_mean

z = -0.50

r = +0.41

score = 0.205

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. low_prec_freq

z = -0.40

r = -0.50

score = 0.203

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lat

z = -0.40

r = +0.51

score = 0.203

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. gauge_lon

z = +0.74

r = -0.25

score = 0.189

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. silt_frac

z = +1.41

r = +0.13

score = 0.182

high silt fraction holds water and slows infiltration.
10. aridity

z = -0.57

r = -0.30

score = 0.173

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03281500: SOUTH FORK KENTUCKY RIVER AT BOONEVILLE, KY

Dominant forcing: PRCP Peak TE: 0.0116 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high gauge_lon, low high_prec_dur.

Top 10 driver attributes

1. gvf_diff z = +1.49 r = -0.24 score = 0.363

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. gauge_lon z = +0.75 r = -0.27 score = 0.203

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. high_prec_dur z = -0.86 r = +0.23 score = 0.195

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. slope_fdc z = +0.92 r = +0.20 score = 0.185

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. hfd_mean z = -0.71 r = -0.24 score = 0.168

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. aridity z = -0.57 r = -0.28 score = 0.159

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. glim_2nd_class_frac z = -1.01 r = +0.15 score = 0.155

a small secondary geological class means the subsurface is more uniform.
8. frac_forest z = +0.87 r = +0.17 score = 0.151

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. glim_1st_class_frac z = +1.01 r = -0.15 score = 0.147

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. soil_depth_statsgo z = -1.04 r = +0.13 score = 0.139

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03285000: DIX RIVER NEAR DANVILLE, KY

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, low glim_1st_class_frac, high gauge_lon.

Top 10 driver attributes

1. hfd_mean

z = -0.86 r = -0.24 score = 0.204

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. glim_1st_class_frac

z = -1.32 r = -0.15 score = 0.192

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
3. gauge_lon

z = +0.69 r = -0.27 score = 0.186

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. carbonate_rocks_frac

z = +1.53 r = -0.12 score = 0.184

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
5. sand_frac

z = -1.41 r = +0.12 score = 0.166

low sand fraction reduces infiltration and may favor surface runoff.
6. high_prec_dur

z = -0.73 r = +0.23 score = 0.165

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. glim_2nd_class_frac

z = +0.95 r = +0.15 score = 0.146

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. gvf_diff

z = +0.58 r = -0.24 score = 0.141

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. aridity

z = -0.47 r = -0.28 score = 0.130

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. slope_fdc

z = +0.64 r = +0.20 score = 0.129

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03291780: INDIAN-KENTUCK CREEK NR CANAAN, IND.

Dominant forcing: PRCP Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high gvf_diff, high gauge_lon.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.12 score = 0.405

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gvf_diff

z = +1.06 r = -0.24 score = 0.257

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. gauge_lon

z = +0.65 r = -0.27 score = 0.176

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. aridity

z = -0.56 r = -0.28 score = 0.154

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. sand_frac

z = -1.30 r = +0.12 score = 0.152

low sand fraction reduces infiltration and may favor surface runoff.
8. elev_mean

z = -0.65 r = -0.20 score = 0.127

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. soil_depth_pelletier

z = +0.79 r = +0.16 score = 0.126

deep soils provide more storage and slower release of water to streamflow.
10. hfd_mean

z = -0.49 r = -0.24 score = 0.115

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03300400: BEECH FORK AT MAUD, KY

Dominant forcing: PRCP Peak TE: 0.0087 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low sand_frac, low soil_depth_statsgo, low hfd_mean.

Top 10 driver attributes

1. sand_frac

z = -1.81 r = +0.12 score = 0.212

low sand fraction reduces infiltration and may favor surface runoff.
2. soil_depth_statsgo

z = -1.35 r = +0.13 score = 0.180

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
3. hfd_mean

z = -0.75 r = -0.24 score = 0.178

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. gauge_lon

z = +0.65 r = -0.27 score = 0.176

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. high_prec_dur

z = -0.74 r = +0.23 score = 0.168

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. gvf_diff

z = +0.60 r = -0.24 score = 0.146

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. elev_mean

z = -0.64 r = -0.20 score = 0.126

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. soil_conductivity

z = -0.85 r = +0.15 score = 0.125

low soil conductivity restricts infiltration, often increasing surface runoff.
9. aridity

z = -0.45 r = -0.28 score = 0.124

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. q5

z = -0.63 r = +0.18 score = 0.114

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03338780: NORTH FORK VERMILION RIVER NEAR BISMARCK, IL

Dominant forcing: PRCP Peak TE: 0.0100 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff

z = +2.13 r = -0.24 score = 0.518

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier

z = +2.29 r = +0.16 score = 0.366

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest

z = -1.61 r = +0.17 score = 0.278

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. stream_elas

z = +1.58 r = -0.12 score = 0.195

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. p_seasonality

z = +0.70 r = -0.24 score = 0.172

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gauge_lon

z = +0.50 r = -0.27 score = 0.136

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. elev_mean

z = -0.69 r = -0.20 score = 0.136

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. high_prec_dur

z = -0.59 r = +0.23 score = 0.133

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. sand_frac

z = -1.10 r = +0.12 score = 0.129

low sand fraction reduces infiltration and may favor surface runoff.
10. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03340800: BIG RACCOON CREEK NEAR FINCASTLE IND

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low high_prec_dur.

Top 10 driver attributes

1. gvf_diff

z = +1.49 r = -0.24 score = 0.362

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier

z = +2.15 r = +0.16 score = 0.343

deep soils provide more storage and slower release of water to streamflow.
3. high_prec_dur

z = -0.87 r = +0.23 score = 0.198

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. frac_forest

z = -0.98 r = +0.17 score = 0.169

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gauge_lon

z = +0.54 r = -0.27 score = 0.148

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. glim_2nd_class_frac

z = -0.96 r = +0.15 score = 0.148

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac

z = +0.96 r = -0.15 score = 0.140

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. elev_mean

z = -0.63 r = -0.20 score = 0.122

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. aridity

z = -0.41 r = -0.28 score = 0.112

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03346000: NORTH FORK EMBARRAS RIVER NEAR OBLONG, IL

Dominant forcing: PRCP Peak TE: 0.0107 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gvf_diff, high glim_2nd_class_frac.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.15 r = +0.16 score = 0.344

deep soils provide more storage and slower release of water to streamflow.
2. gvf_diff

z = +1.11 r = -0.24 score = 0.269

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. glim_2nd_class_frac

z = +1.69 r = +0.15 score = 0.260

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. frac_forest

z = -1.46 r = +0.17 score = 0.252

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. elev_mean

z = -0.75 r = -0.20 score = 0.146

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. glim_1st_class_frac

z = -0.97 r = -0.15 score = 0.142

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. sand_frac

z = -1.18 r = +0.12 score = 0.139

low sand fraction reduces infiltration and may favor surface runoff.
8. gauge_lon

z = +0.48 r = -0.27 score = 0.131

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. carbonate_rocks_frac

z = +1.07 r = -0.12 score = 0.129

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
10. slope_fdc

z = -0.60 r = +0.20 score = 0.121

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03357350: PLUM CREEK NEAR BAINBRIDGE, IND.

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. soil_depth_pelletier

z = +1.86 r = +0.16 score = 0.297

deep soils provide more storage and slower release of water to streamflow.
2. gvf_diff

z = +1.17 r = -0.24 score = 0.285

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur

z = -0.95 r = +0.23 score = 0.216

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gauge_lon

z = +0.56 r = -0.27 score = 0.152

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. elev_mean

z = -0.64 r = -0.20 score = 0.124

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. aridity

z = -0.44 r = -0.28 score = 0.121

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. q5

z = -0.64 r = +0.18 score = 0.115

low low-flow values reflect basins that nearly dry out during dry periods.
8. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.
9. hfd_mean

z = -0.46 r = -0.24 score = 0.109

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. p_seasonality

z = +0.44 r = -0.24 score = 0.108

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03364500: CLIFTY CREEK AT HARTSVILLE, IN

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high soil_depth_pelletier, high gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.12 score = 0.405

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. soil_depth_pelletier

z = +2.31 r = +0.16 score = 0.370

deep soils provide more storage and slower release of water to streamflow.
3. gvf_diff

z = +1.23 r = -0.24 score = 0.299

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. high_prec_dur

z = -0.87 r = +0.23 score = 0.197

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. gauge_lon

z = +0.62 r = -0.27 score = 0.169

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. frac_forest

z = -0.97 r = +0.17 score = 0.168

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. aridity

z = -0.48 r = -0.28 score = 0.134

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. elev_mean

z = -0.62 r = -0.20 score = 0.120

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03366500: MUSCATATUCK RIVER NEAR DEPUTY, IN

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high gvf_diff, high soil_depth_pelletier.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.83 r = -0.12 score = 0.339

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gvf_diff

z = +1.08 r = -0.24 score = 0.263

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. soil_depth_pelletier

z = +1.59 r = +0.16 score = 0.253

deep soils provide more storage and slower release of water to streamflow.
4. gauge_lon

z = +0.62 r = -0.27 score = 0.169

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. aridity

z = -0.55 r = -0.28 score = 0.152

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. elev_mean

z = -0.66 r = -0.20 score = 0.129

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. sand_frac

z = -1.01 r = +0.12 score = 0.118

low sand fraction reduces infiltration and may favor surface runoff.
8. q5

z = -0.61 r = +0.18 score = 0.110

low low-flow values reflect basins that nearly dry out during dry periods.
9. hfd_mean

z = -0.45 r = -0.24 score = 0.106

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. soil_depth_statsgo

z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03368000: BRUSH CREEK NEAR NEBRASKA, IN

Dominant forcing: Tair Peak TE: 0.0031 Peak lag: 27 day(s)
Summary: Tair-dominant at lag 27 day(s); top drivers consistent with low stream_elas, high soil_depth_pelletier, low baseflow_index.

Top 10 driver attributes

1. stream_elas

z = -1.68 r = -0.43 score = 0.719

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. soil_depth_pelletier

z = +1.74 r = -0.26 score = 0.444

deep soils provide more storage and slower release of water to streamflow.
3. baseflow_index

z = -1.76 r = +0.22 score = 0.394

a low baseflow index means streamflow responds quickly to recent rainfall.
4. slope_mean

z = -0.89 r = +0.41 score = 0.363

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. frac_snow

z = -0.54 r = +0.46 score = 0.251

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
6. q5

z = -0.64 r = +0.36 score = 0.233

low low-flow values reflect basins that nearly dry out during dry periods.
7. clay_frac

z = +0.75 r = -0.30 score = 0.220

high clay fraction reduces infiltration and increases surface runoff.
8. geol_permeability

z = +1.76 r = -0.11 score = 0.200

high geological permeability favors fast subsurface flow and stronger groundwater contribution.
9. elev_mean

z = -0.63 r = +0.30 score = 0.187

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. slope_fdc

z = -0.74 r = +0.24 score = 0.178

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03384450: LUSK CREEK NEAR EDDYVILLE, IL

Dominant forcing: Tair Peak TE: 0.0031 Peak lag: 16 day(s)
Summary: Tair-dominant at lag 16 day(s); top drivers consistent with high stream_elas, low baseflow_index, low frac_snow.

Top 10 driver attributes

1. stream_elas

z = +1.37 r = -0.43 score = 0.585

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. baseflow_index

z = -1.37 r = +0.22 score = 0.308

a low baseflow index means streamflow responds quickly to recent rainfall.
3. frac_snow

z = -0.63 r = +0.46 score = 0.290

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
4. high_prec_freq

z = +0.57 r = -0.49 score = 0.278

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. slope_mean

z = -0.63 r = +0.41 score = 0.258

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. frac_forest

z = +0.73 r = +0.35 score = 0.254

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. silt_frac

z = +1.96 r = +0.13 score = 0.253

high silt fraction holds water and slows infiltration.
8. q5

z = -0.64 r = +0.36 score = 0.232

low low-flow values reflect basins that nearly dry out during dry periods.
9. elev_mean

z = -0.75 r = +0.30 score = 0.224

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. gauge_lat

z = -0.34 r = +0.51 score = 0.172

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03439000: FRENCH BROAD RIVER AT ROSMAN, NC

Dominant forcing: SRAD Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high q_mean, high gauge_lon.

Top 10 driver attributes

1. q5

z = +2.61

r = +0.31

score = 0.816

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. q_mean

z = +1.08

r = +0.35

score = 0.382

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. gauge_lon

z = +0.80

r = -0.43

score = 0.346

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. runoff_ratio

z = +0.78

r = +0.43

score = 0.333

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. p_mean

z = +1.63

r = +0.20

score = 0.328

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. baseflow_index

z = +1.41

r = +0.22

score = 0.309

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
7. glim_2nd_class_frac

z = +1.37

r = +0.18

score = 0.240

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. gauge_lat

z = -0.79

r = +0.29

score = 0.231

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. frac_forest

z = +0.95

r = +0.23

score = 0.224

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. high_prec_dur

z = -0.76

r = +0.29

score = 0.220

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03450000: BEETREE CREEK NEAR SWANNANOA, NC

Dominant forcing: PRCP Peak TE: 0.0020 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gauge_lon, high slope_fdc, high gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = +0.83 r = -0.27 score = 0.224

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
2. slope_fdc

z = +1.11 r = +0.20 score = 0.224

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. gvf_diff

z = +0.89 r = -0.24 score = 0.217

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. p_mean

z = +0.59 r = +0.34 score = 0.204

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. aridity

z = -0.64 r = -0.28 score = 0.177

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
6. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. frac_forest

z = +0.89 r = +0.17 score = 0.154

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. high_prec_freq

z = -0.74 r = -0.19 score = 0.140

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. soil_depth_statsgo

z = +0.77 r = +0.13 score = 0.103

deep soils (STATSGO) increase basin storage and delay streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03455500: W F PIGEON R ABOVE LAKE LOGAN NR HAZELWOOD, NC

Dominant forcing: SRAD Peak TE: 0.0036 Peak lag: 1 day(s)

Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q5, high q_mean, high p_mean.

Top 10 driver attributes

1. q5

z = +1.60

r = +0.31

score = 0.500

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. q_mean

z = +1.17

r = +0.35

score = 0.414

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean

z = +1.80

r = +0.20

score = 0.362

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. slope_mean

z = +0.97

r = +0.37

score = 0.355

steep terrain promotes fast surface runoff and a quick streamflow response.
5. gauge_lon

z = +0.79

r = -0.43

score = 0.343

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. runoff_ratio

z = +0.78

r = +0.43

score = 0.334

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. q95

z = +0.73

r = +0.43

score = 0.314

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. gvff_diff

z = +0.72

r = -0.33

score = 0.236

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. glim_1st_class_frac

z = +1.06

r = -0.21

score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. high_prec_freq

z = -0.69

r = -0.33

score = 0.225

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03456500: EAST FORK PIGEON RIVER NEAR CANTON, NC

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high gauge_lon, low aridity.

Top 10 driver attributes

1. p_mean

z = +1.08 r = +0.34 score = 0.374

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. gauge_lon

z = +0.80 r = -0.27 score = 0.216

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. aridity

z = -0.78 r = -0.28 score = 0.215

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. q_mean

z = +0.72 r = +0.28 score = 0.198

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
5. gvf_diff

z = +0.81 r = -0.24 score = 0.196

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. q5

z = +1.00 r = +0.18 score = 0.180

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. runoff_ratio

z = +0.67 r = +0.23 score = 0.154

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. frac_forest

z = +0.88 r = +0.17 score = 0.151

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03460000: CATALOOCHEE CREEK NEAR CATALOOCHEE, NC

Dominant forcing: Tair Peak TE: 0.0039 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with low high_prec_freq, high slope_mean, high q5.

Top 10 driver attributes

1. high_prec_freq

z = -1.27 r = -0.49 score = 0.626

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. slope_mean

z = +1.49 r = +0.41 score = 0.606

steep terrain promotes fast surface runoff and a quick streamflow response.
3. q5

z = +1.29 r = +0.36 score = 0.471

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. low_prec_freq

z = -0.80 r = -0.50 score = 0.405

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. gauge_lat

z = -0.69 r = +0.51 score = 0.347

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. frac_forest

z = +0.95 r = +0.35 score = 0.328

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. baseflow_index

z = +1.20 r = +0.22 score = 0.270

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
8. runoff_ratio

z = +0.43 r = +0.56 score = 0.240

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. aridity

z = -0.75 r = -0.30 score = 0.228

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. frac_snow

z = -0.49 r = +0.46 score = 0.228

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03463300: SOUTH TOE RIVER NEAR CELO, NC

Dominant forcing: PRCP Peak TE: 0.0016 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q_mean, high runoff_ratio.

Top 10 driver attributes

1. p_mean

z = +0.93

r = +0.34

score = 0.319

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q_mean

z = +1.01

r = +0.28

score = 0.281

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. runoff_ratio

z = +1.21

r = +0.23

score = 0.280

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. gauge_lon

z = +0.84

r = -0.27

score = 0.228

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. q5

z = +1.06

r = +0.18

score = 0.192

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. q95

z = +0.59

r = +0.31

score = 0.181

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. aridity

z = -0.63

r = -0.28

score = 0.174

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. frac_forest

z = +0.97

r = +0.17

score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03471500: S F HOLSTON RIVER AT RIVERSIDE NR CHILHOWIE, VA

Dominant forcing: PRCP Peak TE: 0.0028 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, high gauge_lon, high gvf_diff.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +1.59 r = +0.15 score = 0.245

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. gauge_lon

z = +0.87 r = -0.27 score = 0.237

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. gvf_diff

z = +0.96 r = -0.24 score = 0.233

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. high_prec_dur

z = -0.98 r = +0.23 score = 0.223

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. slope_fdc

z = +0.80 r = +0.20 score = 0.161

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. aridity

z = -0.48 r = -0.28 score = 0.132

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. glim_1st_class_frac

z = -0.84 r = -0.15 score = 0.122

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. carbonate_rocks_frac

z = +1.02 r = -0.12 score = 0.122

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
9. low_prec_freq

z = -0.47 r = -0.22 score = 0.104

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. gvf_max

z = +0.85 r = +0.12 score = 0.098

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03473000: S F HOLSTON RIVER NEAR DAMASCUS, VA

Dominant forcing: PRCP Peak TE: 0.0058 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low glim_1st_class_frac, low high_prec_dur, high gauge_lon.

Top 10 driver attributes

1. glim_1st_class_frac z = -1.69 r = -0.15 score = 0.245

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
2. high_prec_dur z = -1.04 r = +0.23 score = 0.236

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gauge_lon z = +0.86 r = -0.27 score = 0.234

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. gvf_diff z = +0.83 r = -0.24 score = 0.203

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. glim_2nd_class_frac z = +1.09 r = +0.15 score = 0.167

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. slope_fdc z = +0.81 r = +0.20 score = 0.163

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. low_prec_freq z = -0.65 r = -0.22 score = 0.145

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. aridity z = -0.51 r = -0.28 score = 0.142

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. high_prec_freq z = -0.67 r = -0.19 score = 0.126

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. frac_forest z = +0.60 r = +0.17 score = 0.103

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03479000: WATAUGA RIVER NEAR SUGAR GROVE, NC

Dominant forcing: VP Peak TE: 0.0017 Peak lag: 30 day(s)
Summary: VP-dominant at lag 30 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = -0.80 r = -0.50 score = 0.399

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.59 r = -0.47 score = 0.277

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_forest

z = +0.93 r = +0.29 score = 0.271

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
4. slope_mean

z = +0.76 r = +0.27 score = 0.207

steep terrain promotes fast surface runoff and a quick streamflow response.
5. gauge_lat

z = -0.58 r = +0.35 score = 0.203

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. q5

z = +0.60 r = +0.34 score = 0.200

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. aridity

z = -0.63 r = -0.31 score = 0.197

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. low_prec_dur

z = -0.72 r = -0.27 score = 0.193

short dry spells limit basin drying and keep storage close to full.
9. runoff_ratio

z = +0.30 r = +0.50 score = 0.149

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. frac_snow

z = -0.43 r = +0.31 score = 0.134

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03488000: N F HOLSTON RIVER NEAR SALTVILLE, VA

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high gauge_lon, low high_prec_dur.

Top 10 driver attributes

1. gvf_diff z = +1.05 r = -0.24 score = 0.254

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. gauge_lon z = +0.87 r = -0.27 score = 0.235

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. high_prec_dur z = -0.96 r = +0.23 score = 0.218

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. slope_fdc z = +1.06 r = +0.20 score = 0.214

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. low_prec_freq z = -0.85 r = -0.22 score = 0.189

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. soil_depth_statsgo z = -0.97 r = +0.13 score = 0.130

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. aridity z = -0.42 r = -0.28 score = 0.116

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. hfd_mean z = -0.48 r = -0.24 score = 0.114

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. high_prec_freq z = -0.58 r = -0.19 score = 0.109

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. soil_depth_pelletier z = -0.60 r = +0.16 score = 0.096

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03498500: LITTLE RIVER NEAR MARYVILLE, TN

Dominant forcing: PRCP Peak TE: 0.0045 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low low_prec_freq, low soil_depth_statsgo.

Top 10 driver attributes

1. p_mean

z = +0.76 r = +0.34 score = 0.262

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. low_prec_freq

z = -1.07 r = -0.22 score = 0.239

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. soil_depth_statsgo

z = -1.64 r = +0.13 score = 0.220

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
4. aridity

z = -0.72 r = -0.28 score = 0.200

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
5. gauge_lon

z = +0.73 r = -0.27 score = 0.199

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. high_prec_freq

z = -1.00 r = -0.19 score = 0.188

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. gvf_diff

z = +0.60 r = -0.24 score = 0.147

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. glim_1st_class_frac

z = -0.97 r = -0.15 score = 0.141

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. hfd_mean

z = -0.57 r = -0.24 score = 0.135

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. root_depth_99

z = +1.31 r = +0.10 score = 0.132

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03500000: LITTLE TENNESSEE RIVER NEAR PRENTISS, NC

Dominant forcing: PRCP Peak TE: 0.0067 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_{mean} , high q_5 , low aridity.

Top 10 driver attributes

1. **p_mean**

$z = +1.37$

$r = +0.34$

score = 0.473

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **q5**

$z = +1.57$

$r = +0.18$

score = 0.284

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. **aridity**

$z = -0.83$

$r = -0.28$

score = 0.230

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
4. **gauge_lon**

$z = +0.77$

$r = -0.27$

score = 0.208

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. **q_mean**

$z = +0.65$

$r = +0.28$

score = 0.181

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. **gvf_diff**

$z = +0.73$

$r = -0.24$

score = 0.177

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. **frac_forest**

$z = +0.85$

$r = +0.17$

score = 0.147

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. **low_prec_freq**

$z = -0.62$

$r = -0.22$

score = 0.138

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. **frac_snow**

$z = -0.73$

$r = -0.15$

score = 0.107

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. **gvf_max**

$z = +0.89$

$r = +0.12$

score = 0.103

high maximum green vegetation fraction reflects dense canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03500240: CARTOOGECHAYE CREEK NEAR FRANKLIN, NC

Dominant forcing: Tair Peak TE: 0.0029 Peak lag: 15 day(s)
Summary: Tair-dominant at lag 15 day(s); top drivers consistent with high q5, low gauge_lat, low frac_snow.

Top 10 driver attributes

1. q5

z = +1.29

r = +0.36

score = 0.471

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lat

z = -0.78

r = +0.51

score = 0.397

southern latitude is associated with warmer climates and rainfall-driven streamflow.
3. frac_snow

z = -0.73

r = +0.46

score = 0.336

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
4. slope_mean

z = +0.76

r = +0.41

score = 0.311

steep terrain promotes fast surface runoff and a quick streamflow response.
5. frac_forest

z = +0.86

r = +0.35

score = 0.297

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. baseflow_index

z = +1.32

r = +0.22

score = 0.295

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
7. low_prec_freq

z = -0.49

r = -0.50

score = 0.248

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. p_mean

z = +1.26

r = +0.19

score = 0.244

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
9. q_mean

z = +0.53

r = +0.45

score = 0.238

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
10. aridity

z = -0.73

r = -0.30

score = 0.221

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03504000: NANTAHALA RIVER NEAR RAINBOW SPRINGS, NC

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q5, high q_mean.

Top 10 driver attributes

1. p_mean

z = +1.66

r = +0.34

score = 0.573

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q5

z = +2.53

r = +0.18

score = 0.457

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. q_mean

z = +1.42

r = +0.28

score = 0.393

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio

z = +1.16

r = +0.23

score = 0.267

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gvf_diff

z = +1.09

r = -0.24

score = 0.265

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. aridity

z = -0.93

r = -0.28

score = 0.258

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. q95

z = +0.67

r = +0.31

score = 0.206

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. gauge_lon

z = +0.75

r = -0.27

score = 0.204

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03574500: PAINT ROCK RIVER NEAR WOODVILLE AL

Dominant forcing: PRCP Peak TE: 0.0117 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, high gvf_diff, high p_mean.

Top 10 driver attributes

1. hfd_mean

z = -1.27

r = -0.24

score = 0.301

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. gvf_diff

z = +1.12

r = -0.24

score = 0.273

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. p_mean

z = +0.73

r = +0.34

score = 0.252

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. glim_2nd_class_frac

z = +1.25

r = +0.15

score = 0.192

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. carbonate_rocks_frac

z = +1.58

r = -0.12

score = 0.189

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. glim_1st_class_frac

z = -1.27

r = -0.15

score = 0.184

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. q95

z = +0.60

r = +0.31

score = 0.182

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. slope_fdc

z = +0.88

r = +0.20

score = 0.176

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. aridity

z = -0.63

r = -0.28

score = 0.175

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gauge_lon

z = +0.58

r = -0.27

score = 0.159

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03592718: LITTLE YELLOW CREEK EAST NR BURNSVILLE, MS

Dominant forcing: PRCP Peak TE: 0.0016 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low hfd_mean, high pet_mean.

Top 10 driver attributes

1. p_mean

z = +0.72

r = +0.34

score = 0.249

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. hfd_mean

z = -0.96

r = -0.24

score = 0.229

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. pet_mean

z = +1.75

r = -0.12

score = 0.214

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. frac_forest

z = +0.85

r = +0.17

score = 0.147

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. elev_mean

z = -0.75

r = -0.20

score = 0.146

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. gauge_lon

z = +0.46

r = -0.27

score = 0.126

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. high_prec_dur

z = -0.52

r = +0.23

score = 0.117

short heavy-rainfall events produce sharper but briefer runoff peaks.
10. frac_snow

z = -0.80

r = -0.15

score = 0.117

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 03604000: BUFFALO RIVER NEAR FLAT WOODS, TN

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, low aridity, high gvf_diff.

Top 10 driver attributes

1. p_mean

z = +0.77

r = +0.34

score = 0.267

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. aridity

z = -0.61

r = -0.28

score = 0.168

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
3. gvf_diff

z = +0.68

r = -0.24

score = 0.166

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. hfd_mean

z = -0.67

r = -0.24

score = 0.159

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. sand_frac

z = -1.31

r = +0.12

score = 0.153

low sand fraction reduces infiltration and may favor surface runoff.
6. q5

z = +0.80

r = +0.18

score = 0.144

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. gauge_lon

z = +0.49

r = -0.27

score = 0.133

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. elev_mean

z = -0.62

r = -0.20

score = 0.122

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. high_prec_dur

z = -0.50

r = +0.23

score = 0.112

short heavy-rainfall events produce sharper but briefer runoff peaks.
10. frac_snow

z = -0.75

r = -0.15

score = 0.109

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04015330: KNIFE RIVER NEAR TWO HARBORS, MN

Dominant forcing: SRAD Peak TE: 0.0008 Peak lag: 6 day(s)
Summary: SRAD-dominant at lag 6 day(s); top drivers consistent with high p_seasonality, high gauge_lat, high gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = +1.20 r = -0.46 score = 0.547

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gauge_lat

z = +1.48 r = +0.29 score = 0.433

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. gvf_diff

z = +1.11 r = -0.33 score = 0.365

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. clay_frac

z = +1.38 r = -0.22 score = 0.305

high clay fraction reduces infiltration and increases surface runoff.
5. lai_diff

z = +1.63 r = -0.14 score = 0.227

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
6. glim_2nd_class_frac

z = +1.28 r = +0.18 score = 0.225

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. frac_forest

z = +0.94 r = +0.23 score = 0.222

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. slope_mean

z = -0.57 r = +0.37 score = 0.211

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. baseflow_index

z = -0.95 r = +0.22 score = 0.208

a low baseflow index means streamflow responds quickly to recent rainfall.
10. q5

z = -0.51 r = +0.31 score = 0.160

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04024430: NEMADJI RIVER NEAR SOUTH SUPERIOR, WI

Dominant forcing: PRCP Peak TE: 0.0092 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high gvf_diff, high p_seasonality.

Top 10 driver attributes

1. organic_frac

z = +2.10 r = +0.17 score = 0.357

high organic soil fraction increases water retention and supports baseflow.
2. gvf_diff

z = +1.44 r = -0.24 score = 0.349

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. p_seasonality

z = +1.37 r = -0.24 score = 0.335

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. p_mean

z = -0.73 r = +0.34 score = 0.250

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. soil_depth_pelletier

z = +1.48 r = +0.16 score = 0.237

deep soils provide more storage and slower release of water to streamflow.
6. gauge_lat

z = +1.42 r = +0.13 score = 0.179

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. glim_2nd_class_frac

z = +1.12 r = +0.15 score = 0.172

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. pet_mean

z = -1.40 r = -0.12 score = 0.170

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. frac_forest

z = +0.96 r = +0.17 score = 0.165

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. hfd_mean

z = +0.65 r = -0.24 score = 0.154

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04027000: BAD RIVER NEAR ODANAH, WI

Dominant forcing: SRAD Peak TE: 0.0034 Peak lag: 15 day(s)
Summary: SRAD-dominant at lag 15 day(s); top drivers consistent with high gvf_diff, low glim_1st_class_frac, high p_seasonality.

Top 10 driver attributes

1. gvf_diff z = +1.51 r = -0.33 score = 0.493

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. glim_1st_class_frac z = -2.16 r = -0.21 score = 0.462

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
3. p_seasonality z = +0.95 r = -0.46 score = 0.434

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. gauge_lat z = +1.39 r = +0.29 score = 0.408

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. slope_mean z = -0.78 r = +0.37 score = 0.287

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. lai_diff z = +1.66 r = -0.14 score = 0.232

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
7. high_prec_dur z = -0.79 r = +0.29 score = 0.227

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. frac_forest z = +0.96 r = +0.23 score = 0.225

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. soil_depth_pelletier z = +0.82 r = -0.24 score = 0.197

deep soils provide more storage and slower release of water to streamflow.
10. q95 z = -0.41 r = +0.43 score = 0.178

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04040500: STURGEON RIVER NEAR SIDNAW, MI

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, low low_prec_freq, low p_mean.

Top 10 driver attributes

1. organic_frac

z = +2.19 r = +0.17 score = 0.374

high organic soil fraction increases water retention and supports baseflow.
2. low_prec_freq

z = -1.05 r = -0.22 score = 0.235

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. p_mean

z = -0.58 r = +0.34 score = 0.200

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. gvf_diff

z = +0.78 r = -0.24 score = 0.191

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. gauge_lat

z = +1.41 r = +0.13 score = 0.178

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. root_depth_99

z = +1.75 r = +0.10 score = 0.177

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. high_prec_freq

z = -0.91 r = -0.19 score = 0.171

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. frac_forest

z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. p_seasonality

z = +0.66 r = -0.24 score = 0.161

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. soil_depth_pelletier

z = +0.97 r = +0.16 score = 0.155

deep soils provide more storage and slower release of water to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04043050: TRAP ROCK RIVER NEAR LAKE LINDEN, MI

Dominant forcing: SRAD Peak TE: 0.0082 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low stream_elas, high gauge_lat, high gvf_diff.

Top 10 driver attributes

1. stream_elas

z = -1.83 r = -0.26 score = 0.474

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. gauge_lat

z = +1.53 r = +0.29 score = 0.449

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. gvf_diff

z = +1.36 r = -0.33 score = 0.446

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. soil_conductivity

z = +1.64 r = +0.25 score = 0.417

high soil conductivity allows fast infiltration and subsurface flow.
5. runoff_ratio

z = +0.92 r = +0.43 score = 0.395

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. sand_frac

z = +2.18 r = +0.18 score = 0.384

high sand fraction promotes fast infiltration and subsurface drainage.
7. clay_frac

z = -1.34 r = -0.22 score = 0.295

low clay fraction allows easier infiltration and reduces surface runoff.
8. soil_porosity

z = -1.64 r = -0.17 score = 0.281

low soil porosity reduces water storage and produces a flashier response.
9. glim_2nd_class_frac

z = +1.50 r = +0.18 score = 0.263

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. slope_mean

z = -0.62 r = +0.37 score = 0.228

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04045500: TAHQUAMENON RIVER NEAR PARADISE, MI

Dominant forcing: Tair Peak TE: 0.0055 Peak lag: 2 day(s)
Summary: Tair-dominant at lag 2 day(s); top drivers consistent with high gauge_lat, high soil_conductivity, low low_prec_freq.

Top 10 driver attributes

1. gauge_lat z = +1.41 r = +0.51 score = 0.712

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
2. soil_conductivity z = +2.95 r = +0.20 score = 0.601

high soil conductivity allows fast infiltration and subsurface flow.
3. low_prec_freq z = -1.16 r = -0.50 score = 0.583

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. high_prec_freq z = -1.10 r = -0.49 score = 0.539

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. stream_elas z = -1.23 r = -0.43 score = 0.526

low streamflow elasticity means streamflow is buffered against rainfall variability.
6. clay_frac z = -1.76 r = -0.30 score = 0.521

low clay fraction allows easier infiltration and reduces surface runoff.
7. soil_depth_pelletier z = +1.69 r = -0.26 score = 0.431

deep soils provide more storage and slower release of water to streamflow.
8. slope_mean z = -0.84 r = +0.41 score = 0.344

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. frac_forest z = +0.92 r = +0.35 score = 0.318

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. frac_snow z = +0.60 r = +0.46 score = 0.278

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04056500: MANISTIQUE RIVER NEAR MANISTIQUE, MI

Dominant forcing: Tair Peak TE: 0.0036 Peak lag: 8 day(s)
Summary: Tair-dominant at lag 8 day(s); top drivers consistent with high soil_conductivity, high gauge_lat, low clay_frac.

Top 10 driver attributes

1. soil_conductivity

z = +5.03 r = +0.20 score = 1.026

high soil conductivity allows fast infiltration and subsurface flow.
2. gauge_lat

z = +1.30 r = +0.51 score = 0.659

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. clay_frac

z = -1.87 r = -0.30 score = 0.552

low clay fraction allows easier infiltration and reduces surface runoff.
4. soil_depth_pelletier

z = +1.96 r = -0.26 score = 0.501

deep soils provide more storage and slower release of water to streamflow.
5. pet_mean

z = -1.36 r = -0.33 score = 0.449

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. stream_elas

z = -1.04 r = -0.43 score = 0.444

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. low_prec_freq

z = -0.86 r = -0.50 score = 0.434

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. high_prec_freq

z = -0.85 r = -0.49 score = 0.420

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. slope_mean

z = -0.87 r = +0.41 score = 0.356

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. frac_forest

z = +0.94 r = +0.35 score = 0.324

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04057510: STURGEON RIVER NEAR NAHMA JUNCTION, MI

Dominant forcing: PRCP Peak TE: 0.0079 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high soil_conductivity, high carbonate_rocks_frac.

Top 10 driver attributes

1. organic_frac

z = +8.51 r = +0.17 score = 1.451

high organic soil fraction increases water retention and supports baseflow.
2. soil_conductivity

z = +6.03 r = +0.15 score = 0.888

high soil conductivity allows fast infiltration and subsurface flow.
3. carbonate_rocks_frac

z = +2.45 r = -0.12 score = 0.294

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. soil_depth_pelletier

z = +1.71 r = +0.16 score = 0.273

deep soils provide more storage and slower release of water to streamflow.
5. p_mean

z = -0.64 r = +0.34 score = 0.220

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
7. sand_frac

z = +1.50 r = +0.12 score = 0.175

high sand fraction promotes fast infiltration and subsurface drainage.
8. pet_mean

z = -1.35 r = -0.12 score = 0.165

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. gauge_lat

z = +1.29 r = +0.13 score = 0.162

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. frac_forest

z = +0.92 r = +0.17 score = 0.158

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04057800: MIDDLE BRANCH ESCANABA RIVER AT HUMBOLDT, MI

Dominant forcing: PRCP Peak TE: 0.0035 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, high root_depth_99, low p_mean.

Top 10 driver attributes

1. low_prec_freq

z = -1.03 r = -0.22 score = 0.231

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
3. p_mean

z = -0.54 r = +0.34 score = 0.187

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. gauge_lat

z = +1.39 r = +0.13 score = 0.176

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. frac_forest

z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. pet_mean

z = -1.33 r = -0.12 score = 0.162

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. gvf_diff

z = +0.65 r = -0.24 score = 0.159

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. p_seasonality

z = +0.57 r = -0.24 score = 0.138

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. gauge_lon

z = +0.49 r = -0.27 score = 0.132

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. hfd_mean

z = +0.55 r = -0.24 score = 0.132

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04059500: FORD RIVER NEAR HYDE, MI

Dominant forcing: PRCP Peak TE: 0.0050 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, low p_mean, high soil_depth_pelletier.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +1.78 r = +0.15 score = 0.273

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. p_mean

z = -0.76 r = +0.34 score = 0.262

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. soil_depth_pelletier

z = +1.52 r = +0.16 score = 0.243

deep soils provide more storage and slower release of water to streamflow.
4. glim_1st_class_frac

z = -1.60 r = -0.15 score = 0.233

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gvf_diff

z = +0.85 r = -0.24 score = 0.208

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. p_seasonality

z = +0.80 r = -0.24 score = 0.195

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. root_depth_99

z = +1.65 r = +0.10 score = 0.167

very deep maximum rooting allows vegetation to tap deep storage in droughts.
8. frac_forest

z = +0.95 r = +0.17 score = 0.165

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. gauge_lat

z = +1.25 r = +0.13 score = 0.158

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. organic_frac

z = +0.91 r = +0.17 score = 0.154

high organic soil fraction increases water retention and supports baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04063700: POPPLE RIVER NEAR FENCE, WI

Dominant forcing: PRCP Peak TE: 0.0054 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high p_seasonality, high gvf_diff.

Top 10 driver attributes

1. organic_frac

z = +3.69 r = +0.17 score = 0.630

high organic soil fraction increases water retention and supports baseflow.
2. p_seasonality

z = +1.06 r = -0.24 score = 0.260

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. gvf_diff

z = +0.96 r = -0.24 score = 0.234

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. p_mean

z = -0.68 r = +0.34 score = 0.234

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. soil_depth_pelletier

z = +1.38 r = +0.16 score = 0.220

deep soils provide more storage and slower release of water to streamflow.
6. soil_conductivity

z = +1.41 r = +0.15 score = 0.208

high soil conductivity allows fast infiltration and subsurface flow.
7. glim_2nd_class_frac

z = +1.26 r = +0.15 score = 0.193

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
9. hfd_mean

z = +0.77 r = -0.24 score = 0.183

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. q95

z = -0.59 r = +0.31 score = 0.181

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04074950: WOLF RIVER AT LANGLADE, WI

Dominant forcing: PRCP Peak TE: 0.0091 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high p_seasonality, low glim_1st_class_frac.

Top 10 driver attributes

1. gvf_diff z = +1.15 r = -0.24 score = 0.280

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. p_seasonality z = +1.09 r = -0.24 score = 0.266

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. glim_1st_class_frac z = -1.58 r = -0.15 score = 0.229

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
4. p_mean z = -0.63 r = +0.34 score = 0.216

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. sand_frac z = +1.81 r = +0.12 score = 0.211

high sand fraction promotes fast infiltration and subsurface drainage.
6. stream_elas z = -1.65 r = -0.12 score = 0.203

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. q95 z = -0.66 r = +0.31 score = 0.200

low peak-flow values reflect a damped streamflow response with limited peaks.
8. soil_conductivity z = +1.33 r = +0.15 score = 0.197

high soil conductivity allows fast infiltration and subsurface flow.
9. organic_frac z = +1.12 r = +0.17 score = 0.191

high organic soil fraction increases water retention and supports baseflow.
10. glim_2nd_class_frac z = +1.14 r = +0.15 score = 0.175

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04105700: AUGUSTA CREEK NEAR AUGUSTA, MI

Dominant forcing: PRCP Peak TE: 0.0175 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low stream_elas, high q5, high gvf_diff.

Top 10 driver attributes

1. stream_elas

z = -2.38 r = -0.12 score = 0.294

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. q5

z = +1.40 r = +0.18 score = 0.252

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. gvf_diff

z = +0.91 r = -0.24 score = 0.220

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. sand_frac

z = +1.76 r = +0.12 score = 0.206

high sand fraction promotes fast infiltration and subsurface drainage.
5. soil_depth_pelletier

z = +1.24 r = +0.16 score = 0.197

deep soils provide more storage and slower release of water to streamflow.
6. q95

z = -0.64 r = +0.31 score = 0.197

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gauge_lon

z = +0.64 r = -0.27 score = 0.175

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. high_prec_dur

z = -0.66 r = +0.23 score = 0.149

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. low_prec_freq

z = -0.62 r = -0.22 score = 0.138

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. slope_fdc

z = -0.63 r = +0.20 score = 0.127

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04115265: FISH CREEK NEAR CRYSTAL, MI

Dominant forcing: PRCP Peak TE: 0.0103 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high sand_frac, high soil_conductivity, low frac_forest.

Top 10 driver attributes

1. sand_frac

z = +2.28

r = +0.12

score = 0.266

high sand fraction promotes fast infiltration and subsurface drainage.
2. soil_conductivity

z = +1.65

r = +0.15

score = 0.243

high soil conductivity allows fast infiltration and subsurface flow.
3. frac_forest

z = -1.26

r = +0.17

score = 0.217

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. q95

z = -0.68

r = +0.31

score = 0.207

low peak-flow values reflect a damped streamflow response with limited peaks.
5. soil_depth_pelletier

z = +1.28

r = +0.16

score = 0.205

deep soils provide more storage and slower release of water to streamflow.
6. stream_elas

z = -1.64

r = -0.12

score = 0.203

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. gvf_diff

z = +0.76

r = -0.24

score = 0.186

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. p_mean

z = -0.54

r = +0.34

score = 0.186

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. gauge_lon

z = +0.67

r = -0.27

score = 0.181

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04122200: WHITE RIVER NEAR WHITEHALL, MI

Dominant forcing: PRCP Peak TE: 0.0092 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high sand_frac, high q5.

Top 10 driver attributes

1. soil_conductivity

z = +2.30 r = +0.15 score = 0.339

high soil conductivity allows fast infiltration and subsurface flow.
2. sand_frac

z = +2.67 r = +0.12 score = 0.312

high sand fraction promotes fast infiltration and subsurface drainage.
3. q5

z = +1.47 r = +0.18 score = 0.265

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. stream_elas

z = -2.08 r = -0.12 score = 0.257

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. gvf_diff

z = +1.03 r = -0.24 score = 0.250

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. soil_depth_pelletier

z = +1.38 r = +0.16 score = 0.220

deep soils provide more storage and slower release of water to streamflow.
7. q95

z = -0.61 r = +0.31 score = 0.187

low peak-flow values reflect a damped streamflow response with limited peaks.
8. gauge_lon

z = +0.59 r = -0.27 score = 0.160

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. p_mean

z = -0.44 r = +0.34 score = 0.151

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. slope_fdc

z = -0.67 r = +0.20 score = 0.134

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04122500: PERE MARQUETTE RIVER AT SCOTTVILLE, MI

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high soil_conductivity, high sand_frac, high q5.

Top 10 driver attributes

1. soil_conductivity

z = +2.99 r = +0.15 score = 0.441

high soil conductivity allows fast infiltration and subsurface flow.
2. sand_frac

z = +2.79 r = +0.12 score = 0.327

high sand fraction promotes fast infiltration and subsurface drainage.
3. q5

z = +1.58 r = +0.18 score = 0.285

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. stream_elas

z = -2.17 r = -0.12 score = 0.268

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. gvf_diff

z = +1.08 r = -0.24 score = 0.262

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. organic_frac

z = +1.22 r = +0.17 score = 0.209

high organic soil fraction increases water retention and supports baseflow.
7. soil_depth_pelletier

z = +1.30 r = +0.16 score = 0.208

deep soils provide more storage and slower release of water to streamflow.
8. q95

z = -0.61 r = +0.31 score = 0.187

low peak-flow values reflect a damped streamflow response with limited peaks.
9. gauge_lon

z = +0.59 r = -0.27 score = 0.159

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. p_mean

z = -0.46 r = +0.34 score = 0.157

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04124000: MANISTEE RIVER NEAR SHERMAN, MI

Dominant forcing: PRCP Peak TE: 0.0121 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_conductivity, high q5, high sand_frac.

Top 10 driver attributes

1. soil_conductivity

z = +3.62 r = +0.15 score = 0.533

high soil conductivity allows fast infiltration and subsurface flow.
2. q5

z = +2.42 r = +0.18 score = 0.436

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. sand_frac

z = +2.82 r = +0.12 score = 0.330

high sand fraction promotes fast infiltration and subsurface drainage.
4. organic_frac

z = +1.75 r = +0.17 score = 0.298

high organic soil fraction increases water retention and supports baseflow.
5. slope_fdc

z = -1.47 r = +0.20 score = 0.295

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. glim_2nd_class_frac

z = +1.56 r = +0.15 score = 0.239

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. stream_elas

z = -1.82 r = -0.12 score = 0.224

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. gvf_diff

z = +0.92 r = -0.24 score = 0.223

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
9. low_prec_freq

z = -0.97 r = -0.22 score = 0.216

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. q95

z = -0.66 r = +0.31 score = 0.200

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04127918: PINE RIVER NEAR RUDYARD, MI

Dominant forcing: PRCP Peak TE: 0.0098 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high soil_depth_pelletier, high carbonate_rocks_frac.

Top 10 driver attributes

1. organic_frac

z = +3.94 r = +0.17 score = 0.672

high organic soil fraction increases water retention and supports baseflow.
2. soil_depth_pelletier

z = +2.11 r = +0.16 score = 0.337

deep soils provide more storage and slower release of water to streamflow.
3. carbonate_rocks_frac

z = +2.41 r = -0.12 score = 0.288

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. low_prec_freq

z = -1.28 r = -0.22 score = 0.287

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. high_prec_freq

z = -1.25 r = -0.19 score = 0.235

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. high_prec_dur

z = -0.90 r = +0.23 score = 0.204

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. gauge_lon

z = +0.69 r = -0.27 score = 0.187

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
8. gauge_lat

z = +1.33 r = +0.13 score = 0.168

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. pet_mean

z = -1.37 r = -0.12 score = 0.167

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. p_mean

z = -0.45 r = +0.34 score = 0.155

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04127997: STURGEON RIVER AT WOLVERINE, MI

Dominant forcing: PRCP Peak TE: 0.0115 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high soil_conductivity, low slope_fdc.

Top 10 driver attributes

1. q5

z = +2.18

r = +0.18

score = 0.393

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. soil_conductivity

z = +2.42

r = +0.15

score = 0.357

high soil conductivity allows fast infiltration and subsurface flow.
3. slope_fdc

z = -1.55

r = +0.20

score = 0.311

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. gvf_diff

z = +1.22

r = -0.24

score = 0.297

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. low_prec_freq

z = -1.04

r = -0.22

score = 0.233

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. q95

z = -0.69

r = +0.31

score = 0.210

low peak-flow values reflect a damped streamflow response with limited peaks.
7. sand_frac

z = +1.74

r = +0.12

score = 0.204

high sand fraction promotes fast infiltration and subsurface drainage.
8. gauge_lon

z = +0.69

r = -0.27

score = 0.187

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. p_mean

z = -0.52

r = +0.34

score = 0.179

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. high_prec_freq

z = -0.92

r = -0.19

score = 0.173

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04161580: STONY CREEK NEAR ROMEO, MI

Dominant forcing: PRCP Peak TE: 0.0069 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low stream_elas, high gvf_diff, high gauge_lon.

Top 10 driver attributes

1. stream_elas

z = -2.33 r = -0.12 score = 0.288

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. gvf_diff

z = +0.90 r = -0.24 score = 0.219

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. gauge_lon

z = +0.78 r = -0.27 score = 0.213

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
4. q95

z = -0.65 r = +0.31 score = 0.200

low peak-flow values reflect a damped streamflow response with limited peaks.
5. p_mean

z = -0.55 r = +0.34 score = 0.190

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. soil_depth_pelletier

z = +1.14 r = +0.16 score = 0.182

deep soils provide more storage and slower release of water to streamflow.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. slope_fdc

z = +0.79 r = +0.20 score = 0.159

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. q_mean

z = -0.55 r = +0.28 score = 0.153

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04185000: Tiffin River at Stryker OH

Dominant forcing: PRCP Peak TE: 0.0125 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high glim_2nd_class_frac, high gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.02 r = +0.16 score = 0.323

deep soils provide more storage and slower release of water to streamflow.
2. glim_2nd_class_frac

z = +2.10 r = +0.15 score = 0.321

a large secondary geological class adds heterogeneity to the subsurface flow response.
3. gvf_diff

z = +1.10 r = -0.24 score = 0.268

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. frac_forest

z = -1.17 r = +0.17 score = 0.201

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gauge_lon

z = +0.70 r = -0.27 score = 0.190

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. glim_1st_class_frac

z = -1.20 r = -0.15 score = 0.174

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. p_mean

z = -0.47 r = +0.34 score = 0.164

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. high_prec_dur

z = -0.63 r = +0.23 score = 0.142

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. hfd_mean

z = -0.57 r = -0.24 score = 0.135

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. slope_fdc

z = +0.64 r = +0.20 score = 0.129

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04196800: Tymochtee Creek at Crawford OH

Dominant forcing: PRCP Peak TE: 0.0158 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.12 score = 0.405

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. soil_depth_pelletier

z = +2.29 r = +0.16 score = 0.366

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest

z = -1.70 r = +0.17 score = 0.294

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. gvf_diff

z = +1.16 r = -0.24 score = 0.282

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. gauge_lon

z = +0.77 r = -0.27 score = 0.208

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. stream_elas

z = +1.35 r = -0.12 score = 0.167

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
9. p_mean

z = -0.41 r = +0.34 score = 0.142

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. sand_frac

z = -1.21 r = +0.12 score = 0.142

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04197100: Honey Creek at Melmore OH

Dominant forcing: PRCP Peak TE: 0.0100 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, low frac_forest, high gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.17 r = +0.16 score = 0.346

deep soils provide more storage and slower release of water to streamflow.
2. frac_forest

z = -1.70 r = +0.17 score = 0.293

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. gvf_diff

z = +1.11 r = -0.24 score = 0.271

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. gauge_lon

z = +0.78 r = -0.27 score = 0.212

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. hfd_mean

z = -0.67 r = -0.24 score = 0.160

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. high_prec_dur

z = -0.66 r = +0.23 score = 0.149

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. sand_frac

z = -1.06 r = +0.12 score = 0.124

low sand fraction reduces infiltration and may favor surface runoff.
8. p_mean

z = -0.35 r = +0.34 score = 0.122

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. elev_mean

z = -0.59 r = -0.20 score = 0.115

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q5

z = -0.61 r = +0.18 score = 0.110

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04197170: Rock Creek at Tiffin OH

Dominant forcing: Tair Peak TE: 0.0056 Peak lag: 24 day(s)
Summary: Tair-dominant at lag 24 day(s); top drivers consistent with low frac_forest, high soil_depth_pelletier, low slope_mean.

Top 10 driver attributes

1. frac_forest

z = -1.70 r = +0.35 score = 0.588

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. soil_depth_pelletier

z = +2.10 r = -0.26 score = 0.537

deep soils provide more storage and slower release of water to streamflow.
3. slope_mean

z = -0.91 r = +0.41 score = 0.371

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. baseflow_index

z = -1.65 r = +0.22 score = 0.369

a low baseflow index means streamflow responds quickly to recent rainfall.
5. clay_frac

z = +1.04 r = -0.30 score = 0.309

high clay fraction reduces infiltration and increases surface runoff.
6. slope_fdc

z = -1.04 r = +0.24 score = 0.251

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. pet_mean

z = -0.68 r = -0.33 score = 0.224

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. gauge_lon

z = +0.78 r = -0.25 score = 0.198

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. q5

z = -0.53 r = +0.36 score = 0.194

low low-flow values reflect basins that nearly dry out during dry periods.
10. q_mean

z = -0.41 r = +0.45 score = 0.185

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04213000: Conneaut Creek at Conneaut OH

Dominant forcing: PRCP Peak TE: 0.0110 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, low low_prec_freq, low hfd_mean.

Top 10 driver attributes

1. gvf_diff z = +1.32 r = -0.24 score = 0.320

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. low_prec_freq z = -1.28 r = -0.22 score = 0.286

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. hfd_mean z = -1.11 r = -0.24 score = 0.264

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. gauge_lon z = +0.94 r = -0.27 score = 0.254

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. high_prec_dur z = -0.86 r = +0.23 score = 0.196

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. high_prec_freq z = -0.86 r = -0.19 score = 0.163

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. aridity z = -0.57 r = -0.28 score = 0.156

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. slope_fdc z = +0.77 r = +0.20 score = 0.155

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04213075: Brandy Run near Girard, PA

Dominant forcing: VP Peak TE: 0.0023 Peak lag: 29 day(s)
Summary: VP-dominant at lag 29 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high silt_frac.

Top 10 driver attributes

1. low_prec_freq

z = -1.26 r = -0.50 score = 0.631

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.94 r = -0.47 score = 0.443

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. silt_frac

z = +2.52 r = +0.13 score = 0.340

high silt fraction holds water and slows infiltration.
4. frac_forest

z = +0.88 r = +0.29 score = 0.257

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. low_prec_dur

z = -0.83 r = -0.27 score = 0.222

short dry spells limit basin drying and keep storage close to full.
6. pet_mean

z = -0.86 r = -0.25 score = 0.219

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. slope_mean

z = -0.69 r = +0.27 score = 0.186

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. gauge_lat

z = +0.53 r = +0.35 score = 0.186

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. soil_depth_pelletier

z = +1.14 r = -0.16 score = 0.182

deep soils provide more storage and slower release of water to streamflow.
10. aridity

z = -0.57 r = -0.31 score = 0.177

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04216418: TONAWANDA CREEK AT ATTICA NY

Dominant forcing: Tair Peak TE: 0.0061 Peak lag: 28 day(s)
Summary: Tair-dominant at lag 28 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high gauge_lat.

Top 10 driver attributes

1. low_prec_freq

z = -1.48 r = -0.50 score = 0.744

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.25 r = -0.49 score = 0.615

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. gauge_lat

z = +0.70 r = +0.51 score = 0.351

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. stream_elas

z = -0.79 r = -0.43 score = 0.338

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. pet_mean

z = -1.00 r = -0.33 score = 0.331

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. gauge_lon

z = +1.08 r = -0.25 score = 0.275

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. frac_forest

z = +0.75 r = +0.35 score = 0.258

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. geol_permeability

z = -1.96 r = -0.11 score = 0.223

low geological permeability slows subsurface flow and reduces groundwater contribution.
9. slope_mean

z = -0.51 r = +0.41 score = 0.207

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. slope_fdc

z = +0.83 r = +0.24 score = 0.201

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04221000: GENESEE RIVER AT WELLSVILLE NY

Dominant forcing: Tair Peak TE: 0.0031 Peak lag: 15 day(s)
Summary: Tair-dominant at lag 15 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, low pet_mean.

Top 10 driver attributes

1. low_prec_freq

z = -1.20 r = -0.50 score = 0.606

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -0.90 r = -0.49 score = 0.442

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. pet_mean

z = -0.93 r = -0.33 score = 0.308

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. frac_forest

z = +0.88 r = +0.35 score = 0.306

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. gauge_lon

z = +1.10 r = -0.25 score = 0.280

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
6. gauge_lat

z = +0.55 r = +0.51 score = 0.279

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. slope_fdc

z = +1.12 r = +0.24 score = 0.272

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. stream_elas

z = -0.50 r = -0.43 score = 0.214

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. geol_permeability

z = -1.84 r = -0.11 score = 0.208

low geological permeability slows subsurface flow and reduces groundwater contribution.
10. p_seasonality

z = +0.49 r = -0.38 score = 0.190

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04224775: CANASERAGA CREEK ABOVE DANSVILLE NY

Dominant forcing: Tair Peak TE: 0.0029 Peak lag: 19 day(s)
Summary: Tair-dominant at lag 19 day(s); top drivers consistent with low stream_elas, low low_prec_freq, low high_prec_freq.

Top 10 driver attributes

1. stream_elas

z = -2.32 r = -0.43 score = 0.994

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. low_prec_freq

z = -1.05 r = -0.50 score = 0.530

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. high_prec_freq

z = -0.82 r = -0.49 score = 0.404

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. pet_mean

z = -0.98 r = -0.33 score = 0.324

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
5. gauge_lat

z = +0.63 r = +0.51 score = 0.319

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. gauge_lon

z = +1.12 r = -0.25 score = 0.284

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. geol_permeability

z = -1.99 r = -0.11 score = 0.226

low geological permeability slows subsurface flow and reduces groundwater contribution.
8. silt_frac

z = +1.70 r = +0.13 score = 0.220

high silt fraction holds water and slows infiltration.
9. slope_fdc

z = +0.91 r = +0.24 score = 0.220

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. frac_forest

z = +0.61 r = +0.35 score = 0.212

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04233000: CAYUGA INLET NEAR ITHACA NY

Dominant forcing: Tair Peak TE: 0.0021 Peak lag: 12 day(s)
Summary: Tair-dominant at lag 12 day(s); top drivers consistent with low low_prec_freq, low stream_elas, low pet_mean.

Top 10 driver attributes

1. low_prec_freq

z = -0.82 r = -0.50 score = 0.411

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. stream_elas

z = -0.89 r = -0.43 score = 0.382

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. pet_mean

z = -0.98 r = -0.33 score = 0.324

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
4. frac_forest

z = +0.92 r = +0.35 score = 0.319

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
5. gauge_lat

z = +0.60 r = +0.51 score = 0.306

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. gauge_lon

z = +1.19 r = -0.25 score = 0.302

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
7. geol_permeability

z = -2.20 r = -0.11 score = 0.249

low geological permeability slows subsurface flow and reduces groundwater contribution.
8. high_prec_freq

z = -0.47 r = -0.49 score = 0.231

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. silt_frac

z = +1.66 r = +0.13 score = 0.214

high silt fraction holds water and slows infiltration.
10. p_seasonality

z = +0.53 r = -0.38 score = 0.203

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04256000: INDEPENDENCE RIVER AT DONNATTSBURG NY

Dominant forcing: PRCP Peak TE: 0.0070 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, high gauge_lon, low high_prec_dur.

Top 10 driver attributes

1. low_prec_freq z = -1.68 r = -0.22 score = 0.375

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gauge_lon z = +1.26 r = -0.27 score = 0.343

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
3. high_prec_dur z = -1.30 r = +0.23 score = 0.294

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. gvf_diff z = +1.01 r = -0.24 score = 0.245

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. high_prec_freq z = -1.28 r = -0.19 score = 0.241

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. runoff_ratio z = +1.01 r = +0.23 score = 0.233

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. aridity z = -0.74 r = -0.28 score = 0.204

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. sand_frac z = +1.44 r = +0.12 score = 0.169

high sand fraction promotes fast infiltration and subsurface drainage.
9. frac_forest z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. q5 z = +0.81 r = +0.18 score = 0.146

high low-flow values reflect strong baseflow and steady year-round streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 04296000: BLACK RIVER AT COVENTRY, VT

Dominant forcing: Tair Peak TE: 0.0071 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low high_prec_freq, high gauge_lat.

Top 10 driver attributes

1. low_prec_freq

z = -1.96 r = -0.50 score = 0.987

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. high_prec_freq

z = -1.60 r = -0.49 score = 0.788

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. gauge_lat

z = +1.08 r = +0.51 score = 0.546

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. gauge_lon

z = +1.45 r = -0.25 score = 0.369

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
5. pet_mean

z = -1.07 r = -0.33 score = 0.356

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
6. frac_forest

z = +0.97 r = +0.35 score = 0.335

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. runoff_ratio

z = +0.52 r = +0.56 score = 0.289

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. clay_frac

z = -0.84 r = -0.30 score = 0.248

low clay fraction allows easier infiltration and reduces surface runoff.
9. aridity

z = -0.73 r = -0.30 score = 0.222

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. frac_snow

z = +0.47 r = +0.46 score = 0.217

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05056000: SHEYENNE RIVER NR WARWICK, ND

Dominant forcing: Tair Peak TE: 0.0021 Peak lag: 6 day(s)
Summary: Tair-dominant at lag 6 day(s); top drivers consistent with high gauge_lat, low runoff_ratio, high p_seasonality.

Top 10 driver attributes

1. gauge_lat z = +1.64 r = +0.51 score = 0.831

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
2. runoff_ratio z = -1.47 r = +0.56 score = 0.815

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. p_seasonality z = +1.75 r = -0.38 score = 0.670

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. low_prec_freq z = +1.19 r = -0.50 score = 0.600

many dry days indicate a drier climate where streamflow depends on stored water.
5. frac_forest z = -1.70 r = +0.35 score = 0.590

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q95 z = -0.98 r = +0.51 score = 0.500

low peak-flow values reflect a damped streamflow response with limited peaks.
7. soil_depth_pelletier z = +1.89 r = -0.26 score = 0.482

deep soils provide more storage and slower release of water to streamflow.
8. pet_mean z = -1.35 r = -0.33 score = 0.446

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. q_mean z = -0.93 r = +0.45 score = 0.416

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. high_prec_freq z = +0.74 r = -0.49 score = 0.365

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05057000: SHEYENNE RIVER NR COOPERSTOWN, ND

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high gauge_lat.

Top 10 driver attributes

1. p_seasonality

z = +1.76 r = -0.46 score = 0.803

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.43 r = +0.43 score = 0.611

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gauge_lat

z = +1.57 r = +0.29 score = 0.461

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. soil_depth_pelletier

z = +1.89 r = -0.24 score = 0.453

deep soils provide more storage and slower release of water to streamflow.
5. q95

z = -0.96 r = +0.43 score = 0.416

low peak-flow values reflect a damped streamflow response with limited peaks.
6. frac_forest

z = -1.70 r = +0.23 score = 0.399

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. gvf_diff

z = +1.12 r = -0.33 score = 0.366

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. slope_mean

z = -0.89 r = +0.37 score = 0.328

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. q_mean

z = -0.92 r = +0.35 score = 0.326

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. low_prec_freq

z = +0.97 r = -0.31 score = 0.303

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05057200: BALDHILL CREEK NR DAZEY, ND

Dominant forcing: SRAD Peak TE: 0.0023 Peak lag: 13 day(s)
Summary: SRAD-dominant at lag 13 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high soil_depth_pelletier.

Top 10 driver attributes

1. p_seasonality

z = +1.82 r = -0.46 score = 0.830

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.47 r = +0.43 score = 0.630

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. soil_depth_pelletier

z = +2.34 r = -0.24 score = 0.561

deep soils provide more storage and slower release of water to streamflow.
4. gvf_diff

z = +1.49 r = -0.33 score = 0.489

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. gauge_lat

z = +1.53 r = +0.29 score = 0.449

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. q95

z = -0.98 r = +0.43 score = 0.421

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.67 r = +0.23 score = 0.393

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. stream_elas

z = +1.51 r = -0.26 score = 0.391

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. low_prec_freq

z = +1.20 r = -0.31 score = 0.375

many dry days indicate a drier climate where streamflow depends on stored water.
10. slope_mean

z = -0.93 r = +0.37 score = 0.342

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05062500: WILD RICE RIVER AT TWIN VALLEY, MN

Dominant forcing: SRAD Peak TE: 0.0023 Peak lag: 7 day(s)
Summary: SRAD-dominant at lag 7 day(s); top drivers consistent with high p_seasonality, high gvf_diff, high gauge_lat.

Top 10 driver attributes

1. p_seasonality

z = +1.62 r = -0.46 score = 0.739

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gvf_diff

z = +1.58 r = -0.33 score = 0.517

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. gauge_lat

z = +1.54 r = +0.29 score = 0.451

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. soil_depth_pelletier

z = +1.79 r = -0.24 score = 0.428

deep soils provide more storage and slower release of water to streamflow.
5. runoff_ratio

z = -0.90 r = +0.43 score = 0.383

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -0.78 r = +0.43 score = 0.336

low peak-flow values reflect a damped streamflow response with limited peaks.
7. slope_mean

z = -0.86 r = +0.37 score = 0.315

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. q_mean

z = -0.75 r = +0.35 score = 0.265

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. glim_1st_class_frac

z = -0.93 r = -0.21 score = 0.200

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. p_mean

z = -0.97 r = +0.20 score = 0.195

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05087500: MIDDLE RIVER AT ARGYLE, MN

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high p_seasonality, low p_mean, high soil_depth_pelletier.

Top 10 driver attributes

1. p_seasonality

z = +1.67 r = -0.24 score = 0.408

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. p_mean

z = -1.15 r = +0.34 score = 0.396

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. soil_depth_pelletier

z = +2.41 r = +0.16 score = 0.385

deep soils provide more storage and slower release of water to streamflow.
4. gvf_diff

z = +1.38 r = -0.24 score = 0.336

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. hfd_mean

z = +1.36 r = -0.24 score = 0.324

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
6. glim_1st_class_frac

z = -2.05 r = -0.15 score = 0.298

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. frac_forest

z = -1.33 r = +0.17 score = 0.229

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. gauge_lat

z = +1.75 r = +0.13 score = 0.220

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. q95

z = -0.69 r = +0.31 score = 0.212

low peak-flow values reflect a damped streamflow response with limited peaks.
10. q_mean

z = -0.75 r = +0.28 score = 0.208

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05120500: WINTERING RIVER NR KARLSRUHE, ND

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high p_seasonality, high stream_elas, low runoff_ratio.

Top 10 driver attributes

1. p_seasonality

z = +1.60 r = -0.46 score = 0.730

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. stream_elas

z = +2.72 r = -0.26 score = 0.704

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. runoff_ratio

z = -1.53 r = +0.43 score = 0.654

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. gauge_lat

z = +1.71 r = +0.29 score = 0.500

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. soil_depth_pelletier

z = +1.93 r = -0.24 score = 0.462

deep soils provide more storage and slower release of water to streamflow.
6. low_prec_freq

z = +1.41 r = -0.31 score = 0.440

many dry days indicate a drier climate where streamflow depends on stored water.
7. q95

z = -0.98 r = +0.43 score = 0.425

low peak-flow values reflect a damped streamflow response with limited peaks.
8. frac_forest

z = -1.71 r = +0.23 score = 0.402

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. q_mean

z = -0.94 r = +0.35 score = 0.333

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. gvf_diff

z = +0.99 r = -0.33 score = 0.324

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05123400: WILLOW CREEK NR WILLOW CITY, ND

Dominant forcing: SRAD Peak TE: 0.0027 Peak lag: 25 day(s)
Summary: SRAD-dominant at lag 25 day(s); top drivers consistent with high high_q_dur, high stream_elas, high zero_q_freq.

Top 10 driver attributes

1. high_q_dur

z = +8.51

r = +0.16

score = 1.333

long high-flow events suggest sustained wet conditions and slow recession.
2. stream_elas

z = +4.66

r = -0.26

score = 1.205

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. zero_q_freq

z = +3.86

r = -0.23

score = 0.874

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
4. p_seasonality

z = +1.74

r = -0.46

score = 0.795

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. runoff_ratio

z = -1.49

r = +0.43

score = 0.637

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. gauge_lat

z = +1.79

r = +0.29

score = 0.526

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. soil_depth_pelletier

z = +1.92

r = -0.24

score = 0.461

deep soils provide more storage and slower release of water to streamflow.
8. q95

z = -0.96

r = +0.43

score = 0.414

low peak-flow values reflect a damped streamflow response with limited peaks.
9. gvf_diff

z = +1.22

r = -0.33

score = 0.401

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
10. frac_forest

z = -1.59

r = +0.23

score = 0.374

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05129115: VERMILION RIVER NR CRANE LAKE, MN

Dominant forcing: PRCP Peak TE: 0.0047 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low p_mean, high organic_frac.

Top 10 driver attributes

1. p_seasonality

z = +1.52 r = -0.24 score = 0.370

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. p_mean

z = -0.91 r = +0.34 score = 0.314

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. organic_frac

z = +1.78 r = +0.17 score = 0.303

high organic soil fraction increases water retention and supports baseflow.
4. hfd_mean

z = +1.03 r = -0.24 score = 0.245

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. gauge_lat

z = +1.73 r = +0.13 score = 0.219

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. pet_mean

z = -1.60 r = -0.12 score = 0.195

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. q95

z = -0.63 r = +0.31 score = 0.193

low peak-flow values reflect a damped streamflow response with limited peaks.
8. area_gages2

z = +0.92 r = +0.18 score = 0.166

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
9. slope_fdc

z = +0.81 r = +0.20 score = 0.163

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. area_geospa_fabric

z = +0.92 r = +0.17 score = 0.156

a large geospatial drainage area integrates many sub-basin responses.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05131500: LITTLE FORK RIVER AT LITTLEFORK, MN

Dominant forcing: PRCP Peak TE: 0.0089 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high organic_frac, high area_gages2, high p_seasonality.

Top 10 driver attributes

1. organic_frac

z = +5.08 r = +0.17 score = 0.866

high organic soil fraction increases water retention and supports baseflow.
2. area_gages2

z = +2.11 r = +0.18 score = 0.381

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
3. p_seasonality

z = +1.52 r = -0.24 score = 0.371

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. area_geospa_fabric

z = +2.07 r = +0.17 score = 0.352

a large geospatial drainage area integrates many sub-basin responses.
5. p_mean

z = -0.89 r = +0.34 score = 0.308

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. soil_depth_pelletier

z = +1.71 r = +0.16 score = 0.273

deep soils provide more storage and slower release of water to streamflow.
7. glim_1st_class_frac

z = -1.85 r = -0.15 score = 0.269

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. hfd_mean

z = +1.01 r = -0.24 score = 0.241

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. gauge_lat

z = +1.76 r = +0.13 score = 0.222

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. pet_mean

z = -1.56 r = -0.12 score = 0.190

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05291000: WHETSTONE RIVER NEAR BIG STONE CITY, SD

Dominant forcing: PRCP Peak TE: 0.0059 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, high gvf_diff.

Top 10 driver attributes

1. p_mean

z = -1.08 r = +0.34 score = 0.372

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.50 r = -0.24 score = 0.367

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. gvf_diff

z = +1.28 r = -0.24 score = 0.311

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. frac_forest

z = -1.70 r = +0.17 score = 0.293

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. soil_depth_pelletier

z = +1.71 r = +0.16 score = 0.273

deep soils provide more storage and slower release of water to streamflow.
6. q95

z = -0.85 r = +0.31 score = 0.260

low peak-flow values reflect a damped streamflow response with limited peaks.
7. runoff_ratio

z = -1.12 r = +0.23 score = 0.259

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q_mean

z = -0.83 r = +0.28 score = 0.229

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. low_prec_freq

z = +0.95 r = -0.22 score = 0.213

many dry days indicate a drier climate where streamflow depends on stored water.
10. stream_elas

z = +1.70 r = -0.12 score = 0.210

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05362000: JUMP RIVER AT SHELDON, WI

Dominant forcing: PRCP Peak TE: 0.0121 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high p_seasonality, high soil_depth_pelletier.

Top 10 driver attributes

1. gvf_diff z = +1.63 r = -0.24 score = 0.396

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. p_seasonality z = +1.26 r = -0.24 score = 0.308

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. soil_depth_pelletier z = +1.83 r = +0.16 score = 0.292

deep soils provide more storage and slower release of water to streamflow.
4. p_mean z = -0.64 r = +0.34 score = 0.220

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. frac_forest z = +0.94 r = +0.17 score = 0.163

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. pet_mean z = -1.27 r = -0.12 score = 0.155

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
7. gauge_lat z = +1.16 r = +0.13 score = 0.147

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. glim_1st_class_frac z = -0.99 r = -0.15 score = 0.144

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. hfd_mean z = +0.57 r = -0.24 score = 0.136

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. glim_2nd_class_frac z = +0.87 r = +0.15 score = 0.133

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05393500: SPIRIT RIVER AT SPIRIT FALLS, WI

Dominant forcing: PRCP Peak TE: 0.0079 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high p_seasonality, high soil_depth_pelletier.

Top 10 driver attributes

1. gvf_diff z = +1.50 r = -0.24 score = 0.365

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. p_seasonality z = +1.19 r = -0.24 score = 0.289

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. soil_depth_pelletier z = +1.56 r = +0.16 score = 0.248

deep soils provide more storage and slower release of water to streamflow.
4. p_mean z = -0.62 r = +0.34 score = 0.212

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. frac_forest z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. pet_mean z = -1.26 r = -0.12 score = 0.154

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
9. hfd_mean z = +0.63 r = -0.24 score = 0.151

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. gauge_lat z = +1.19 r = +0.13 score = 0.150

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05399500: BIG EAU PLEINE RIVER AT STRATFORD, WI

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, high soil_depth_pelletier, high gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = +1.30 r = -0.24 score = 0.316

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. soil_depth_pelletier

z = +1.91 r = +0.16 score = 0.306

deep soils provide more storage and slower release of water to streamflow.
3. gvf_diff

z = +1.21 r = -0.24 score = 0.293

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. glim_1st_class_frac

z = -1.76 r = -0.15 score = 0.256

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. p_mean

z = -0.69 r = +0.34 score = 0.238

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. glim_2nd_class_frac

z = +1.47 r = +0.15 score = 0.226

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. slope_fdc

z = -1.11 r = +0.20 score = 0.223

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. high_prec_dur

z = -0.82 r = +0.23 score = 0.185

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. pet_mean

z = -1.19 r = -0.12 score = 0.145

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. q_mean

z = -0.52 r = +0.28 score = 0.145

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05408000: KICKAPOO RIVER AT LA FARGE, WI

Dominant forcing: SRAD Peak TE: 0.0004 Peak lag: 24 day(s)
Summary: SRAD-dominant at lag 24 day(s); top drivers consistent with high p_seasonality, high gvf_diff, low stream_elas.

Top 10 driver attributes

1. p_seasonality

z = +1.26 r = -0.46 score = 0.578

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gvf_diff

z = +1.59 r = -0.33 score = 0.521

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. stream_elas

z = -1.85 r = -0.26 score = 0.480

low streamflow elasticity means streamflow is buffered against rainfall variability.
4. baseflow_index

z = +1.70 r = +0.22 score = 0.373

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
5. glim_2nd_class_frac

z = +2.01 r = +0.18 score = 0.353

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. q95

z = -0.76 r = +0.43 score = 0.328

low peak-flow values reflect a damped streamflow response with limited peaks.
7. glim_1st_class_frac

z = -1.14 r = -0.21 score = 0.244

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. gauge_lat

z = +0.83 r = +0.29 score = 0.244

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. slope_mean

z = -0.59 r = +0.37 score = 0.216

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. runoff_ratio

z = -0.50 r = +0.43 score = 0.213

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05412500: Turkey River at Garber, IA

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high p_seasonality, high area_gages2.

Top 10 driver attributes

1. gvf_diff

z = +1.76 r = -0.24 score = 0.427

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. p_seasonality

z = +1.34 r = -0.24 score = 0.328

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. area_gages2

z = +1.80 r = +0.18 score = 0.325

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
4. area_geospa_fabric

z = +1.88 r = +0.17 score = 0.321

a large geospatial drainage area integrates many sub-basin responses.
5. carbonate_rocks_frac

z = +2.23 r = -0.12 score = 0.268

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. hfd_mean

z = +0.98 r = -0.24 score = 0.232

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
7. high_prec_dur

z = -0.72 r = +0.23 score = 0.163

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. glim_2nd_class_frac

z = +1.00 r = +0.15 score = 0.153

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. q95

z = -0.48 r = +0.31 score = 0.145

low peak-flow values reflect a damped streamflow response with limited peaks.
10. p_mean

z = -0.40 r = +0.34 score = 0.139

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05413500: GRANT RIVER AT BURTON, WI

Dominant forcing: Tair Peak TE: 0.0017 Peak lag: 10 day(s)
Summary: Tair-dominant at lag 10 day(s); top drivers consistent with low stream_elas, high p_seasonality, high baseflow_index.

Top 10 driver attributes

1. stream_elas

z = -1.29 r = -0.43 score = 0.552

low streamflow elasticity means streamflow is buffered against rainfall variability.
2. p_seasonality

z = +1.23 r = -0.38 score = 0.471

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. baseflow_index

z = +1.73 r = +0.22 score = 0.389

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. q95

z = -0.71 r = +0.51 score = 0.363

low peak-flow values reflect a damped streamflow response with limited peaks.
5. gauge_lat

z = +0.67 r = +0.51 score = 0.337

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. runoff_ratio

z = -0.55 r = +0.56 score = 0.304

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. pet_mean

z = -0.91 r = -0.33 score = 0.300

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
8. slope_mean

z = -0.68 r = +0.41 score = 0.276

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. q_mean

z = -0.52 r = +0.45 score = 0.234

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. silt_frac

z = +1.61 r = +0.13 score = 0.208

high silt fraction holds water and slows infiltration.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05414000: PLATTE RIVER NEAR ROCKVILLE, WI

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 18 day(s)
Summary: SRAD-dominant at lag 18 day(s); top drivers consistent with high p_seasonality, high gvf_diff, high baseflow_index.

Top 10 driver attributes

1. p_seasonality

z = +1.21 r = -0.46 score = 0.551

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gvf_diff

z = +1.11 r = -0.33 score = 0.363

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. baseflow_index

z = +1.57 r = +0.22 score = 0.344

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. q95

z = -0.67 r = +0.43 score = 0.290

low peak-flow values reflect a damped streamflow response with limited peaks.
5. slope_mean

z = -0.73 r = +0.37 score = 0.269

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. high_prec_dur

z = -0.93 r = +0.29 score = 0.268

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. sand_frac

z = -1.42 r = +0.18 score = 0.250

low sand fraction reduces infiltration and may favor surface runoff.
8. carbonate_rocks_frac

z = +2.19 r = -0.11 score = 0.247

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
9. soil_porosity

z = +1.28 r = -0.17 score = 0.220

high soil porosity increases water storage capacity and dampens streamflow response.
10. runoff_ratio

z = -0.50 r = +0.43 score = 0.215

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05444000: ELKHORN CREEK NEAR PENROSE, IL

Dominant forcing: Tair Peak TE: 0.0037 Peak lag: 30 day(s)
Summary: Tair-dominant at lag 30 day(s); top drivers consistent with low frac_forest, high p_seasonality, low slope_mean.

Top 10 driver attributes

1. frac_forest

z = -1.40 r = +0.35 score = 0.484

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. p_seasonality

z = +0.93 r = -0.38 score = 0.356

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. slope_mean

z = -0.85 r = +0.41 score = 0.345

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. q95

z = -0.62 r = +0.51 score = 0.316

low peak-flow values reflect a damped streamflow response with limited peaks.
5. gauge_lat

z = +0.51 r = +0.51 score = 0.258

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. silt_frac

z = +1.99 r = +0.13 score = 0.258

high silt fraction holds water and slows infiltration.
7. baseflow_index

z = +1.10 r = +0.22 score = 0.246

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
8. soil_depth_pelletier

z = +0.95 r = -0.26 score = 0.244

deep soils provide more storage and slower release of water to streamflow.
9. pet_mean

z = -0.73 r = -0.33 score = 0.241

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. runoff_ratio

z = -0.40 r = +0.56 score = 0.221

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05454000: Rapid Creek near Iowa City, IA

Dominant forcing: VP Peak TE: 0.0027 Peak lag: 4 day(s)
Summary: VP-dominant at lag 4 day(s); top drivers consistent with low frac_forest, high stream_elas, high p_seasonality.

Top 10 driver attributes

1. frac_forest

z = -1.47 r = +0.29 score = 0.428

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. stream_elas

z = +0.91 r = -0.36 score = 0.328

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. p_seasonality

z = +1.16 r = -0.27 score = 0.311

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. soil_depth_pelletier

z = +1.73 r = -0.16 score = 0.276

deep soils provide more storage and slower release of water to streamflow.
5. q95

z = -0.51 r = +0.49 score = 0.251

low peak-flow values reflect a damped streamflow response with limited peaks.
6. silt_frac

z = +1.78 r = +0.13 score = 0.240

high silt fraction holds water and slows infiltration.
7. slope_mean

z = -0.88 r = +0.27 score = 0.239

gentle terrain slows surface runoff and produces a smoother streamflow signal.
8. runoff_ratio

z = -0.46 r = +0.50 score = 0.230

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. high_prec_freq

z = +0.49 r = -0.47 score = 0.230

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q5

z = -0.62 r = +0.34 score = 0.208

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05458000: Little Cedar River near Ionia, IA

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high carbonate_rocks_frac, high p_seasonality.

Top 10 driver attributes

1. gvf_diff z = +2.22 r = -0.24 score = 0.539

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. carbonate_rocks_frac z = +3.32 r = -0.12 score = 0.398

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. p_seasonality z = +1.46 r = -0.24 score = 0.356

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. soil_depth_pelletier z = +2.12 r = +0.16 score = 0.339

deep soils provide more storage and slower release of water to streamflow.
5. hfd_mean z = +1.25 r = -0.24 score = 0.297

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
6. frac_forest z = -1.04 r = +0.17 score = 0.179

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. p_mean z = -0.49 r = +0.34 score = 0.168

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. q95 z = -0.52 r = +0.31 score = 0.157

low peak-flow values reflect a damped streamflow response with limited peaks.
9. glim_2nd_class_frac z = -0.98 r = +0.15 score = 0.150

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac z = +0.99 r = -0.15 score = 0.144

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05466500: EDWARDS RIVER NEAR NEW BOSTON, IL

Dominant forcing: PRCP Peak TE: 0.0116 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff z = +1.80 r = -0.24 score = 0.437

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +1.68 r = +0.16 score = 0.268

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest z = -1.46 r = +0.17 score = 0.252

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. p_seasonality z = +0.86 r = -0.24 score = 0.209

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. glim_2nd_class_frac z = +1.19 r = +0.15 score = 0.182

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. sand_frac z = -1.36 r = +0.12 score = 0.159

low sand fraction reduces infiltration and may favor surface runoff.
7. slope_fdc z = +0.74 r = +0.20 score = 0.149

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. hfd_mean z = +0.57 r = -0.24 score = 0.136

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. elev_mean z = -0.68 r = -0.20 score = 0.134

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. p_mean z = -0.37 r = +0.34 score = 0.129

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05487980: White Breast Creek near Dallas, IA

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high hfd_mean, high p_seasonality, high glim_2nd_class_frac.

Top 10 driver attributes

1. hfd_mean

z = +1.42

r = -0.24

score = 0.338

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
2. p_seasonality

z = +1.24

r = -0.24

score = 0.303

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. glim_2nd_class_frac

z = +1.74

r = +0.15

score = 0.267

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. glim_1st_class_frac

z = -1.58

r = -0.15

score = 0.230

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gvf_diff

z = +0.92

r = -0.24

score = 0.223

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. frac_forest

z = -1.02

r = +0.17

score = 0.175

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. soil_depth_pelletier

z = +1.05

r = +0.16

score = 0.168

deep soils provide more storage and slower release of water to streamflow.
8. high_prec_dur

z = -0.72

r = +0.23

score = 0.162

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. slope_fdc

z = -0.75

r = +0.20

score = 0.151

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. q_mean

z = -0.51

r = +0.28

score = 0.141

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05488200: English Creek near Knoxville, IA

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, high gvf_diff, high hfd_mean.

Top 10 driver attributes

1. p_seasonality

z = +1.23 r = -0.24 score = 0.300

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gvf_diff

z = +1.15 r = -0.24 score = 0.278

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. hfd_mean

z = +1.05 r = -0.24 score = 0.249

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. soil_depth_pelletier

z = +1.25 r = +0.16 score = 0.200

deep soils provide more storage and slower release of water to streamflow.
5. sand_frac

z = -1.45 r = +0.12 score = 0.170

low sand fraction reduces infiltration and may favor surface runoff.
6. high_prec_dur

z = -0.74 r = +0.23 score = 0.167

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. p_mean

z = -0.42 r = +0.34 score = 0.145

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. glim_2nd_class_frac

z = -0.94 r = +0.15 score = 0.144

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +0.96 r = -0.15 score = 0.139

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. q95

z = -0.45 r = +0.31 score = 0.138

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05489000: Cedar Creek near Bussey, IA

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, high hfd_mean, high gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = +1.22 r = -0.24 score = 0.298

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. hfd_mean

z = +1.14 r = -0.24 score = 0.271

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. gvf_diff

z = +1.04 r = -0.24 score = 0.253

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. high_prec_dur

z = -0.72 r = +0.23 score = 0.164

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. soil_depth_pelletier

z = +0.98 r = +0.16 score = 0.157

deep soils provide more storage and slower release of water to streamflow.
6. q95

z = -0.48 r = +0.31 score = 0.147

low peak-flow values reflect a damped streamflow response with limited peaks.
7. q_mean

z = -0.52 r = +0.28 score = 0.145

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. frac_forest

z = -0.83 r = +0.17 score = 0.143

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. p_mean

z = -0.41 r = +0.34 score = 0.141

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. glim_2nd_class_frac

z = -0.91 r = +0.15 score = 0.140

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05495000: Fox River at Wayland, MO

Dominant forcing: PRCP Peak TE: 0.0109 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, high soil_depth_pelletier, high gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = +1.00 r = -0.24 score = 0.245

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. soil_depth_pelletier

z = +1.53 r = +0.16 score = 0.244

deep soils provide more storage and slower release of water to streamflow.
3. gvf_diff

z = +0.98 r = -0.24 score = 0.238

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. hfd_mean

z = +0.92 r = -0.24 score = 0.219

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. high_prec_dur

z = -0.86 r = +0.23 score = 0.194

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. frac_forest

z = -0.95 r = +0.17 score = 0.164

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. slope_fdc

z = -0.79 r = +0.20 score = 0.159

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. stream_elas

z = +1.14 r = -0.12 score = 0.141

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. q_mean

z = -0.49 r = +0.28 score = 0.135

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. elev_mean

z = -0.67 r = -0.20 score = 0.130

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05495500: BEAR CREEK NEAR MARCELLINE, IL

Dominant forcing: PRCP Peak TE: 0.0046 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high gvf_diff, high carbonate_rocks_frac.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.05 r = +0.16 score = 0.328

deep soils provide more storage and slower release of water to streamflow.
2. gvf_diff

z = +1.34 r = -0.24 score = 0.325

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. carbonate_rocks_frac

z = +2.57 r = -0.12 score = 0.308

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. frac_forest

z = -1.32 r = +0.17 score = 0.228

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. slope_fdc

z = -1.00 r = +0.20 score = 0.201

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. stream_elas

z = +1.58 r = -0.12 score = 0.196

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. p_seasonality

z = +0.80 r = -0.24 score = 0.195

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. sand_frac

z = -1.41 r = +0.12 score = 0.165

low sand fraction reduces infiltration and may favor surface runoff.
9. elev_mean

z = -0.71 r = -0.20 score = 0.138

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q_mean

z = -0.49 r = +0.28 score = 0.135

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05501000: North River at Palmyra, MO

Dominant forcing: PRCP Peak TE: 0.0054 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.01 r = -0.12 score = 0.361

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. soil_depth_pelletier

z = +1.75 r = +0.16 score = 0.280

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest

z = -1.41 r = +0.17 score = 0.242

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. gvf_diff

z = +0.87 r = -0.24 score = 0.212

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. p_seasonality

z = +0.79 r = -0.24 score = 0.192

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. high_prec_dur

z = -0.72 r = +0.23 score = 0.163

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. slope_fdc

z = -0.81 r = +0.20 score = 0.163

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. sand_frac

z = -1.24 r = +0.12 score = 0.145

low sand fraction reduces infiltration and may favor surface runoff.
9. elev_mean

z = -0.69 r = -0.20 score = 0.135

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. hfd_mean

z = +0.57 r = -0.24 score = 0.135

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05503800: Crooked Creek near Paris, MO

Dominant forcing: PRCP Peak TE: 0.0045 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, low slope_fdc, low frac_forest.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.28 r = +0.16 score = 0.364

deep soils provide more storage and slower release of water to streamflow.
2. slope_fdc

z = -1.61 r = +0.20 score = 0.323

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. frac_forest

z = -1.51 r = +0.17 score = 0.261

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. p_seasonality

z = +0.79 r = -0.24 score = 0.193

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. gvf_diff

z = +0.78 r = -0.24 score = 0.189

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. sand_frac

z = -1.25 r = +0.12 score = 0.146

low sand fraction reduces infiltration and may favor surface runoff.
7. hfd_mean

z = +0.61 r = -0.24 score = 0.145

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
8. high_prec_freq

z = +0.73 r = -0.19 score = 0.137

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. elev_mean

z = -0.67 r = -0.20 score = 0.131

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q_mean

z = -0.47 r = +0.28 score = 0.131

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05507600: Lick Creek at Perry, MO

Dominant forcing: PRCP Peak TE: 0.0021 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, low slope_fdc, high carbonate_rocks_frac.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.32 r = +0.16 score = 0.370

deep soils provide more storage and slower release of water to streamflow.
2. slope_fdc

z = -1.76 r = +0.20 score = 0.353

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. carbonate_rocks_frac

z = +2.72 r = -0.12 score = 0.326

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. frac_forest

z = -1.61 r = +0.17 score = 0.278

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gvf_diff

z = +0.79 r = -0.24 score = 0.193

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. sand_frac

z = -1.53 r = +0.12 score = 0.179

low sand fraction reduces infiltration and may favor surface runoff.
7. pet_mean

z = +1.41 r = -0.12 score = 0.172

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
8. p_seasonality

z = +0.70 r = -0.24 score = 0.170

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. stream_elas

z = +1.27 r = -0.12 score = 0.157

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. q95

z = -0.50 r = +0.31 score = 0.151

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05508805: Spencer Cr. bl Plum Cr. nr Frankford, MO

Dominant forcing: PRCP Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low frac_forest, high carbonate_rocks_frac, high soil_depth_pelletier.

Top 10 driver attributes

1. frac_forest

z = -1.49 r = +0.17 score = 0.256

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. carbonate_rocks_frac

z = +2.10 r = -0.12 score = 0.252

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. soil_depth_pelletier

z = +1.36 r = +0.16 score = 0.218

deep soils provide more storage and slower release of water to streamflow.
4. gvf_diff

z = +0.84 r = -0.24 score = 0.204

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. glim_2nd_class_frac

z = +1.23 r = +0.15 score = 0.189

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. slope_fdc

z = -0.90 r = +0.20 score = 0.180

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. p_seasonality

z = +0.69 r = -0.24 score = 0.168

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. sand_frac

z = -1.43 r = +0.12 score = 0.168

low sand fraction reduces infiltration and may favor surface runoff.
9. q95

z = -0.49 r = +0.31 score = 0.149

low peak-flow values reflect a damped streamflow response with limited peaks.
10. runoff_ratio

z = -0.63 r = +0.23 score = 0.146

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05514500: CUIVRE RIVER NEAR TROY, MO

Dominant forcing: PRCP Peak TE: 0.0110 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high soil_depth_pelletier, low slope_fdc.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.42 r = -0.12 score = 0.290

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. soil_depth_pelletier

z = +1.26 r = +0.16 score = 0.202

deep soils provide more storage and slower release of water to streamflow.
3. slope_fdc

z = -0.96 r = +0.20 score = 0.193

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. frac_forest

z = -1.09 r = +0.17 score = 0.188

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. area_gages2

z = +0.95 r = +0.18 score = 0.171

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
6. area_geospa_fabric

z = +0.94 r = +0.17 score = 0.160

a large geospatial drainage area integrates many sub-basin responses.
7. gvf_diff

z = +0.64 r = -0.24 score = 0.156

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. stream_elas

z = +1.26 r = -0.12 score = 0.156

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. p_seasonality

z = +0.62 r = -0.24 score = 0.151

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. sand_frac

z = -1.26 r = +0.12 score = 0.148

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05525500: SUGAR CREEK AT MILFORD, IL

Dominant forcing: PRCP Peak TE: 0.0140 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff z = +2.11 r = -0.24 score = 0.513

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +2.37 r = +0.16 score = 0.378

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest z = -1.64 r = +0.17 score = 0.283

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. glim_1st_class_frac z = -1.50 r = -0.15 score = 0.218

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. high_prec_dur z = -0.81 r = +0.23 score = 0.184

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. carbonate_rocks_frac z = +1.40 r = -0.12 score = 0.167

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
7. p_seasonality z = +0.69 r = -0.24 score = 0.167

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. sand_frac z = -1.23 r = +0.12 score = 0.143

low sand fraction reduces infiltration and may favor surface runoff.
9. gauge_lon z = +0.50 r = -0.27 score = 0.135

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
10. elev_mean z = -0.69 r = -0.20 score = 0.135

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05556500: BIG BUREAU CREEK AT PRINCETON, IL

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff

z = +2.16 r = -0.24 score = 0.524

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier

z = +1.99 r = +0.16 score = 0.318

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest

z = -1.39 r = +0.17 score = 0.239

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. p_seasonality

z = +0.88 r = -0.24 score = 0.215

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. slope_fdc

z = +1.04 r = +0.20 score = 0.209

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. carbonate_rocks_frac

z = +1.57 r = -0.12 score = 0.189

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
7. glim_1st_class_frac

z = -1.27 r = -0.15 score = 0.184

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. sand_frac

z = -1.28 r = +0.12 score = 0.150

low sand fraction reduces infiltration and may favor surface runoff.
9. stream_elas

z = +1.17 r = -0.12 score = 0.144

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. glim_2nd_class_frac

z = +0.89 r = +0.15 score = 0.137

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05584500: LA MOINE RIVER AT COLMAR, IL

Dominant forcing: PRCP Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff z = +1.70 r = -0.24 score = 0.414

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +2.17 r = +0.16 score = 0.347

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest z = -1.31 r = +0.17 score = 0.227

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. p_seasonality z = +0.81 r = -0.24 score = 0.198

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. carbonate_rocks_frac z = +1.61 r = -0.12 score = 0.193

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. glim_1st_class_frac z = -1.22 r = -0.15 score = 0.177

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. sand_frac z = -1.44 r = +0.12 score = 0.169

low sand fraction reduces infiltration and may favor surface runoff.
8. stream_elas z = +1.13 r = -0.12 score = 0.140

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. elev_mean z = -0.71 r = -0.20 score = 0.140

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q_mean z = -0.49 r = +0.28 score = 0.136

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05585000: LA MOINE RIVER AT RIPLEY, IL

Dominant forcing: PRCP Peak TE: 0.0064 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, high area_gages2.

Top 10 driver attributes

1. gvf_diff z = +1.56 r = -0.24 score = 0.379

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +2.00 r = +0.16 score = 0.320

deep soils provide more storage and slower release of water to streamflow.
3. area_gages2 z = +1.51 r = +0.18 score = 0.272

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
4. area_geospa_fabric z = +1.50 r = +0.17 score = 0.255

a large geospatial drainage area integrates many sub-basin responses.
5. glim_1st_class_frac z = -1.69 r = -0.15 score = 0.245

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. frac_forest z = -1.17 r = +0.17 score = 0.202

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. p_seasonality z = +0.78 r = -0.24 score = 0.190

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. sand_frac z = -1.43 r = +0.12 score = 0.167

low sand fraction reduces infiltration and may favor surface runoff.
9. stream_elas z = +1.30 r = -0.12 score = 0.160

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. glim_2nd_class_frac z = +1.04 r = +0.15 score = 0.160

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05591550: WHITLEY CREEK NEAR ALLENVILLE, IL

Dominant forcing: PRCP Peak TE: 0.0071 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff z = +2.01 r = -0.24 score = 0.488

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +2.36 r = +0.16 score = 0.377

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest z = -1.68 r = +0.17 score = 0.289

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. sand_frac z = -1.20 r = +0.12 score = 0.140

low sand fraction reduces infiltration and may favor surface runoff.
7. elev_mean z = -0.70 r = -0.20 score = 0.136

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. gauge_lon z = +0.45 r = -0.27 score = 0.122

eastern longitude (less negative) is generally associated with humid conditions and stronger rainfall response.
9. q5 z = -0.64 r = +0.18 score = 0.115

low low-flow values reflect basins that nearly dry out during dry periods.
10. p_seasonality z = +0.45 r = -0.24 score = 0.111

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05592050: ROBINSON CREEK NEAR SHELBYVILLE, IL

Dominant forcing: PRCP Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. gvf_diff z = +1.57 r = -0.24 score = 0.382

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. soil_depth_pelletier z = +2.20 r = +0.16 score = 0.351

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest z = -1.53 r = +0.17 score = 0.263

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. stream_elas z = +1.40 r = -0.12 score = 0.172

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. glim_2nd_class_frac z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
6. glim_1st_class_frac z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. sand_frac z = -1.31 r = +0.12 score = 0.153

low sand fraction reduces infiltration and may favor surface runoff.
8. high_prec_dur z = -0.66 r = +0.23 score = 0.150

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. elev_mean z = -0.72 r = -0.20 score = 0.141

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q_mean z = -0.46 r = +0.28 score = 0.127

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05592575: HICKORY CREEK NEAR BROWNSTOWN, IL

Dominant forcing: Tair Peak TE: 0.0031 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with high soil_depth_pelletier, low frac_forest, low baseflow_index.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.12 r = -0.26 score = 0.541

deep soils provide more storage and slower release of water to streamflow.
2. frac_forest

z = -1.55 r = +0.35 score = 0.536

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. baseflow_index

z = -1.93 r = +0.22 score = 0.432

a low baseflow index means streamflow responds quickly to recent rainfall.
4. high_prec_freq

z = +0.85 r = -0.49 score = 0.419

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. slope_mean

z = -0.95 r = +0.41 score = 0.386

gentle terrain slows surface runoff and produces a smoother streamflow signal.
6. zero_q_freq

z = +1.26 r = -0.29 score = 0.368

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
7. slope_fdc

z = -1.26 r = +0.24 score = 0.303

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. frac_snow

z = -0.56 r = +0.46 score = 0.258

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. silt_frac

z = +1.94 r = +0.13 score = 0.251

high silt fraction holds water and slows infiltration.
10. q5

z = -0.64 r = +0.36 score = 0.233

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05593575: LITTLE CROOKED CREEK NEAR NEW MINDEN, IL

Dominant forcing: PRCP Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, low slope_fdc, high carbonate_rocks_frac.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.22 r = +0.16 score = 0.355

deep soils provide more storage and slower release of water to streamflow.
2. slope_fdc

z = -1.63 r = +0.20 score = 0.327

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. carbonate_rocks_frac

z = +2.48 r = -0.12 score = 0.297

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. frac_forest

z = -1.52 r = +0.17 score = 0.262

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. high_prec_dur

z = -0.96 r = +0.23 score = 0.218

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. sand_frac

z = -1.35 r = +0.12 score = 0.158

low sand fraction reduces infiltration and may favor surface runoff.
7. elev_mean

z = -0.77 r = -0.20 score = 0.151

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. high_prec_freq

z = +0.80 r = -0.19 score = 0.150

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. gvf_diff

z = +0.51 r = -0.24 score = 0.123

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
10. q5

z = -0.62 r = +0.18 score = 0.113

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05593900: EAST FORK SHOAL CREEK NEAR COFFEEN, IL

Dominant forcing: PRCP Peak TE: 0.0037 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high soil_depth_pelletier, high stream_elas.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.37 r = -0.12 score = 0.404

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. soil_depth_pelletier

z = +2.19 r = +0.16 score = 0.350

deep soils provide more storage and slower release of water to streamflow.
3. stream_elas

z = +2.35 r = -0.12 score = 0.290

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. frac_forest

z = -1.52 r = +0.17 score = 0.263

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gvf_diff

z = +1.06 r = -0.24 score = 0.257

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. slope_fdc

z = -1.08 r = +0.20 score = 0.217

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. sand_frac

z = -1.56 r = +0.12 score = 0.183

low sand fraction reduces infiltration and may favor surface runoff.
8. glim_2nd_class_frac

z = -1.06 r = +0.15 score = 0.163

a small secondary geological class means the subsurface is more uniform.
9. high_prec_dur

z = -0.68 r = +0.23 score = 0.154

short heavy-rainfall events produce sharper but briefer runoff peaks.
10. glim_1st_class_frac

z = +1.05 r = -0.15 score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 05595730: RAYSE CREEK NEAR WALTONVILLE, IL

Dominant forcing: PRCP Peak TE: 0.0035 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high carbonate_rocks_frac, low slope_fdc.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.15 r = +0.16 score = 0.343

deep soils provide more storage and slower release of water to streamflow.
2. carbonate_rocks_frac

z = +2.42 r = -0.12 score = 0.290

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. slope_fdc

z = -1.38 r = +0.20 score = 0.278

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. frac_forest

z = -1.48 r = +0.17 score = 0.255

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. high_prec_dur

z = -0.83 r = +0.23 score = 0.188

short heavy-rainfall events produce sharper but briefer runoff peaks.
6. gvf_diff

z = +0.75 r = -0.24 score = 0.181

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. elev_mean

z = -0.77 r = -0.20 score = 0.151

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. sand_frac

z = -1.27 r = +0.12 score = 0.149

low sand fraction reduces infiltration and may favor surface runoff.
9. high_prec_freq

z = +0.68 r = -0.19 score = 0.127

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q5

z = -0.64 r = +0.18 score = 0.115

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06037500: Madison River near West Yellowstone MT

Dominant forcing: SRAD Peak TE: 0.0091 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high q5, high frac_snow, high baseflow_index.

Top 10 driver attributes

1. q5

z = +2.20

r = +0.31

score = 0.687

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. frac_snow

z = +2.32

r = +0.26

score = 0.591

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. baseflow_index

z = +2.40

r = +0.22

score = 0.525

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. elev_mean

z = +2.10

r = +0.24

score = 0.507

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. high_prec_freq

z = -1.50

r = -0.33

score = 0.490

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. gauge_lon

z = -0.94

r = -0.43

score = 0.407

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. stream_elas

z = -1.31

r = -0.26

score = 0.340

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. gvf_diff

z = -0.99

r = -0.33

score = 0.325

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. p_seasonality

z = -0.69

r = -0.46

score = 0.316

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. gauge_lat

z = +1.04

r = +0.29

score = 0.305

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06043500: Gallatin River near Gallatin Gateway MT

Dominant forcing: SRAD Peak TE: 0.0096 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, high slope_mean, low glim_1st_class_frac.

Top 10 driver attributes

1. high_prec_freq

z = -1.95 r = -0.33 score = 0.637

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. slope_mean

z = +1.49 r = +0.37 score = 0.549

steep terrain promotes fast surface runoff and a quick streamflow response.
3. glim_1st_class_frac

z = -2.42 r = -0.21 score = 0.518

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
4. frac_snow

z = +1.98 r = +0.26 score = 0.506

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. elev_mean

z = +2.05 r = +0.24 score = 0.496

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. gauge_lon

z = -0.95 r = -0.43 score = 0.413

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. low_prec_freq

z = -1.16 r = -0.31 score = 0.363

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. gauge_lat

z = +1.20 r = +0.29 score = 0.352

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. stream_elas

z = -1.10 r = -0.26 score = 0.284

low streamflow elasticity means streamflow is buffered against rainfall variability.
10. baseflow_index

z = +1.15 r = +0.22 score = 0.252

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06154410: Little Peoples Creek near Hays MT

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, high hfd_mean.

Top 10 driver attributes

1. p_mean

z = -1.39 r = +0.34 score = 0.480

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.52 r = -0.24 score = 0.372

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. hfd_mean

z = +1.54 r = -0.24 score = 0.365

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. high_prec_dur

z = +1.60 r = +0.23 score = 0.363

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. low_prec_freq

z = +1.49 r = -0.22 score = 0.334

many dry days indicate a drier climate where streamflow depends on stored water.
6. q95

z = -0.88 r = +0.31 score = 0.267

low peak-flow values reflect a damped streamflow response with limited peaks.
7. glim_1st_class_frac

z = -1.76 r = -0.15 score = 0.257

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. aridity

z = +0.88 r = -0.28 score = 0.244

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. high_prec_freq

z = +1.28 r = -0.19 score = 0.241

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q_mean

z = -0.80 r = +0.28 score = 0.221

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06188000: LAMAR RIVER NR TOWER FALLS RANGER STATION, YNP

Dominant forcing: SRAD Peak TE: 0.0137 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low high_prec_freq, high frac_snow, high elev_mean.

Top 10 driver attributes

1. high_prec_freq

z = -2.39 r = -0.33 score = 0.781

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. frac_snow

z = +2.18 r = +0.26 score = 0.557

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. elev_mean

z = +2.30 r = +0.24 score = 0.556

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. slope_mean

z = +1.24 r = +0.37 score = 0.457

steep terrain promotes fast surface runoff and a quick streamflow response.
5. low_prec_freq

z = -1.30 r = -0.31 score = 0.405

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. runoff_ratio

z = +0.91 r = +0.43 score = 0.389

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gauge_lon

z = -0.90 r = -0.43 score = 0.389

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. gauge_lat

z = +1.09 r = +0.29 score = 0.320

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. high_q_dur

z = +1.83 r = +0.16 score = 0.287

long high-flow events suggest sustained wet conditions and slow recession.
10. glim_1st_class_frac

z = -1.16 r = -0.21 score = 0.248

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06191500: Yellowstone River at Corwin Springs MT

Dominant forcing: SRAD Peak TE: 0.0101 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low high_prec_freq, high frac_snow, high elev_mean.

Top 10 driver attributes

1. high_prec_freq

z = -2.46 r = -0.33 score = 0.802

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. frac_snow

z = +2.22 r = +0.26 score = 0.567

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. elev_mean

z = +2.28 r = +0.24 score = 0.549

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. low_prec_freq

z = -1.45 r = -0.31 score = 0.453

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. glim_1st_class_frac

z = -1.98 r = -0.21 score = 0.425

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. gauge_lon

z = -0.93 r = -0.43 score = 0.400

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. slope_mean

z = +1.02 r = +0.37 score = 0.373

steep terrain promotes fast surface runoff and a quick streamflow response.
8. gauge_lat

z = +1.13 r = +0.29 score = 0.330

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. runoff_ratio

z = +0.61 r = +0.43 score = 0.260

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. baseflow_index

z = +1.12 r = +0.22 score = 0.246

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06221400: DINWOODY CREEK ABOVE LAKES, NEAR BURRIS, WYO.

Dominant forcing: Tair Peak TE: 0.0212 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high runoff_ratio, high frac_snow, high elev_mean.

Top 10 driver attributes

1. runoff_ratio

z = +2.39 r = +0.56 score = 1.325

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. frac_snow

z = +2.64 r = +0.46 score = 1.219

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. elev_mean

z = +3.28 r = +0.30 score = 0.975

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. stream_elas

z = -1.69 r = -0.43 score = 0.725

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. slope_mean

z = +1.72 r = +0.41 score = 0.700

steep terrain promotes fast surface runoff and a quick streamflow response.
6. other_frac

z = +5.32 r = +0.11 score = 0.562

a large 'other' land fraction reflects mixed or atypical surface cover.
7. slope_fdc

z = +2.16 r = +0.24 score = 0.523

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. aridity

z = +1.43 r = -0.30 score = 0.434

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. gauge_lat

z = +0.79 r = +0.51 score = 0.398

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. gvf_max

z = -2.82 r = +0.14 score = 0.387

low maximum green vegetation fraction reflects sparse canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06224000: BULL LAKE CREEK ABOVE BULL LAKE, WY

Dominant forcing: SRAD Peak TE: 0.0119 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high runoff_ratio, high elev_mean, high other_frac.

Top 10 driver attributes

1. runoff_ratio

z = +2.24 r = +0.43 score = 0.960

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. elev_mean

z = +3.00 r = +0.24 score = 0.724

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. other_frac

z = +5.09 r = +0.14 score = 0.708

a large 'other' land fraction reflects mixed or atypical surface cover.
4. frac_snow

z = +2.54 r = +0.26 score = 0.648

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. slope_mean

z = +1.25 r = +0.37 score = 0.460

steep terrain promotes fast surface runoff and a quick streamflow response.
6. gvf_diff

z = -1.32 r = -0.33 score = 0.432

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lon

z = -0.83 r = -0.43 score = 0.357

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. stream_elas

z = -1.21 r = -0.26 score = 0.314

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. glim_1st_class_frac

z = -1.26 r = -0.21 score = 0.271

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. glim_2nd_class_frac

z = +1.49 r = +0.18 score = 0.261

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06278300: SHELL CREEK ABOVE SHELL CREEK RESERVOIR, WY

Dominant forcing: SRAD Peak TE: 0.0117 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, high runoff_ratio.

Top 10 driver attributes

1. frac_snow

z = +2.73

r = +0.26

score = 0.697

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.79

r = +0.24

score = 0.673

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. runoff_ratio

z = +1.13

r = +0.43

score = 0.485

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. other_frac

z = +3.39

r = +0.14

score = 0.471

a large 'other' land fraction reflects mixed or atypical surface cover.
5. slope_mean

z = +0.86

r = +0.37

score = 0.316

steep terrain promotes fast surface runoff and a quick streamflow response.
6. clay_frac

z = -1.43

r = -0.22

score = 0.316

low clay fraction allows easier infiltration and reduces surface runoff.
7. gauge_lon

z = -0.72

r = -0.43

score = 0.310

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. gvf_diff

z = -0.92

r = -0.33

score = 0.301

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. gauge_lat

z = +1.01

r = +0.29

score = 0.296

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. high_prec_freq

z = -0.90

r = -0.33

score = 0.293

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06280300: SOUTH FORK SHOSHONE RIVER NEAR VALLEY, WY

Dominant forcing: SRAD Peak TE: 0.0149 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean z = +2.02 r = +0.37 score = 0.741

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean z = +2.86 r = +0.24 score = 0.689

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow z = +2.41 r = +0.26 score = 0.614

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. runoff_ratio z = +0.93 r = +0.43 score = 0.398

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gvf_diff z = -1.15 r = -0.33 score = 0.375

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. gauge_lon z = -0.85 r = -0.43 score = 0.367

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_freq z = -1.10 r = -0.33 score = 0.357

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. gauge_lat z = +0.95 r = +0.29 score = 0.279

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. other_frac z = +1.63 r = +0.14 score = 0.227

a large 'other' land fraction reflects mixed or atypical surface cover.
10. root_depth_50 z = -1.80 r = -0.13 score = 0.226

shallow median rooting limits vegetation water access to surface soils.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06289000: Little Bighorn River at State Line nr Wyola MT

Dominant forcing: SRAD Peak TE: 0.0056 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, high gauge_lat.

Top 10 driver attributes

1. elev_mean

z = +2.22

r = +0.24

score = 0.537

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +2.00

r = +0.26

score = 0.510

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. gauge_lat

z = +1.11

r = +0.29

score = 0.324

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. gauge_lon

z = -0.73

r = -0.43

score = 0.315

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. slope_mean

z = +0.85

r = +0.37

score = 0.314

steep terrain promotes fast surface runoff and a quick streamflow response.
6. baseflow_index

z = +1.31

r = +0.22

score = 0.286

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
7. carbonate_rocks_frac

z = +2.45

r = -0.11

score = 0.276

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
8. q95

z = -0.52

r = +0.43

score = 0.223

low peak-flow values reflect a damped streamflow response with limited peaks.
9. p_mean

z = -1.09

r = +0.20

score = 0.219

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. other_frac

z = +1.35

r = +0.14

score = 0.188

a large 'other' land fraction reflects mixed or atypical surface cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06291500: Lodge Grass Cr ab Willow C Div nr Wyola MT

Dominant forcing: SRAD Peak TE: 0.0052 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, low stream_elas, high slope_mean.

Top 10 driver attributes

1. elev_mean

z = +1.88

r = +0.24

score = 0.454

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. stream_elas

z = -1.67

r = -0.26

score = 0.433

low streamflow elasticity means streamflow is buffered against rainfall variability.
3. slope_mean

z = +1.16

r = +0.37

score = 0.426

steep terrain promotes fast surface runoff and a quick streamflow response.
4. glim_1st_class_frac

z = -1.87

r = -0.21

score = 0.401

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gauge_lat

z = +1.13

r = +0.29

score = 0.331

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. high_prec_dur

z = +1.13

r = +0.29

score = 0.325

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. gauge_lon

z = -0.73

r = -0.43

score = 0.315

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. frac_snow

z = +1.12

r = +0.26

score = 0.285

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
9. q95

z = -0.61

r = +0.43

score = 0.265

low peak-flow values reflect a damped streamflow response with limited peaks.
10. aridity

z = +1.51

r = -0.16

score = 0.249

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06311000: NORTH FORK POWDER RIVER NEAR HAZELTON, WY

Dominant forcing: SRAD Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, low gauge_lon.

Top 10 driver attributes

1. elev_mean z = +2.23 r = +0.24 score = 0.539

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow z = +1.95 r = +0.26 score = 0.498

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. gauge_lon z = -0.70 r = -0.43 score = 0.301

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gauge_lat z = +0.92 r = +0.29 score = 0.269

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. p_seasonality z = +0.59 r = -0.46 score = 0.269

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. p_mean z = -1.28 r = +0.20 score = 0.256

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. root_depth_50 z = -1.90 r = -0.13 score = 0.238

shallow median rooting limits vegetation water access to surface soils.
8. gvff_diff z = -0.71 r = -0.33 score = 0.234

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. low_prec_freq z = +0.70 r = -0.31 score = 0.218

many dry days indicate a drier climate where streamflow depends on stored water.
10. q_mean z = -0.57 r = +0.35 score = 0.202

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06332515: BEAR DEN CREEK NR MANDAREE, ND

Dominant forcing: SRAD Peak TE: 0.0028 Peak lag: 8 day(s)
Summary: SRAD-dominant at lag 8 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, low baseflow_index.

Top 10 driver attributes

1. p_seasonality

z = +1.63 r = -0.46 score = 0.745

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.51 r = +0.43 score = 0.648

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. baseflow_index

z = -2.23 r = +0.22 score = 0.488

a low baseflow index means streamflow responds quickly to recent rainfall.
4. gauge_lat

z = +1.64 r = +0.29 score = 0.481

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. q95

z = -1.01 r = +0.43 score = 0.435

low peak-flow values reflect a damped streamflow response with limited peaks.
6. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. low_prec_freq

z = +1.10 r = -0.31 score = 0.343

many dry days indicate a drier climate where streamflow depends on stored water.
8. q_mean

z = -0.94 r = +0.35 score = 0.333

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. p_mean

z = -1.45 r = +0.20 score = 0.291

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. other_frac

z = +1.99 r = +0.14 score = 0.278

a large 'other' land fraction reflects mixed or atypical surface cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06339100: KNIFE RIVER AT MANNING, ND

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, low slope_fdc.

Top 10 driver attributes

1. p_mean

z = -1.49 r = +0.34 score = 0.513

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.67 r = -0.24 score = 0.407

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. slope_fdc

z = -1.79 r = +0.20 score = 0.360

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. aridity

z = +1.26 r = -0.28 score = 0.348

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. runoff_ratio

z = -1.47 r = +0.23 score = 0.339

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -1.00 r = +0.31 score = 0.305

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. low_prec_freq

z = +1.22 r = -0.22 score = 0.272

many dry days indicate a drier climate where streamflow depends on stored water.
9. q_mean

z = -0.93 r = +0.28 score = 0.259

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. gauge_lat

z = +1.53 r = +0.13 score = 0.194

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06339500: KNIFE RIVER NR GOLDEN VALLEY, ND

Dominant forcing: SRAD Peak TE: 0.0016 Peak lag: 14 day(s)
Summary: SRAD-dominant at lag 14 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high gauge_lat.

Top 10 driver attributes

1. p_seasonality

z = +1.66 r = -0.46 score = 0.759

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.51 r = +0.43 score = 0.645

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gauge_lat

z = +1.52 r = +0.29 score = 0.445

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. q95

z = -1.00 r = +0.43 score = 0.431

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. low_prec_freq

z = +1.25 r = -0.31 score = 0.391

many dry days indicate a drier climate where streamflow depends on stored water.
7. q_mean

z = -0.94 r = +0.35 score = 0.333

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. baseflow_index

z = -1.49 r = +0.22 score = 0.326

a low baseflow index means streamflow responds quickly to recent rainfall.
9. p_mean

z = -1.50 r = +0.20 score = 0.302

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. slope_mean

z = -0.76 r = +0.37 score = 0.278

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06344600: GREEN RIVER NR NEW HRADEC, ND

Dominant forcing: PRCP Peak TE: 0.0021 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, low slope_fdc.

Top 10 driver attributes

1. p_mean

z = -1.50

r = +0.34

score = 0.518

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.66

r = -0.24

score = 0.405

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. slope_fdc

z = -1.87

r = +0.20

score = 0.376

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. aridity

z = +1.33

r = -0.28

score = 0.366

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. runoff_ratio

z = -1.47

r = +0.23

score = 0.338

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -1.00

r = +0.31

score = 0.305

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.72

r = +0.17

score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. low_prec_freq

z = +1.23

r = -0.22

score = 0.275

many dry days indicate a drier climate where streamflow depends on stored water.
9. q_mean

z = -0.93

r = +0.28

score = 0.259

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. high_prec_dur

z = +1.13

r = +0.23

score = 0.255

long heavy-rainfall events allow more soil saturation and stronger runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06350000: CANNONBALL RIVER AT REGENT, ND

Dominant forcing: Tair Peak TE: 0.0024 Peak lag: 17 day(s)
Summary: Tair-dominant at lag 17 day(s); top drivers consistent with low runoff_ratio, high gauge_lat, high low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = -1.48 r = +0.56 score = 0.823

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. gauge_lat

z = +1.38 r = +0.51 score = 0.697

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. low_prec_freq

z = +1.36 r = -0.50 score = 0.685

many dry days indicate a drier climate where streamflow depends on stored water.
4. frac_forest

z = -1.72 r = +0.35 score = 0.594

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. p_seasonality

z = +1.54 r = -0.38 score = 0.591

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. q95

z = -1.00 r = +0.51 score = 0.513

low peak-flow values reflect a damped streamflow response with limited peaks.
7. aridity

z = +1.56 r = -0.30 score = 0.473

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. slope_fdc

z = -1.76 r = +0.24 score = 0.426

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. q_mean

z = -0.94 r = +0.45 score = 0.420

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. pet_mean

z = -1.10 r = -0.33 score = 0.365

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06352000: CEDAR CREEK NR HAYNES, ND

Dominant forcing: SRAD Peak TE: 0.0023 Peak lag: 12 day(s)
Summary: SRAD-dominant at lag 12 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, low q95.

Top 10 driver attributes

1. p_seasonality

z = +1.50 r = -0.46 score = 0.687

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.46 r = +0.43 score = 0.623

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. q95

z = -1.00 r = +0.43 score = 0.430

low peak-flow values reflect a damped streamflow response with limited peaks.
4. low_prec_freq

z = +1.34 r = -0.31 score = 0.417

many dry days indicate a drier climate where streamflow depends on stored water.
5. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. gauge_lat

z = +1.33 r = +0.29 score = 0.389

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. baseflow_index

z = -1.57 r = +0.22 score = 0.344

a low baseflow index means streamflow responds quickly to recent rainfall.
8. q_mean

z = -0.93 r = +0.35 score = 0.330

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. p_mean

z = -1.53 r = +0.20 score = 0.308

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. slope_mean

z = -0.83 r = +0.37 score = 0.304

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06353000: CEDAR CREEK NR RALEIGH, ND

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with low runoff_ratio, high aridity, high gauge_lat.

Top 10 driver attributes

1. runoff_ratio

z = -1.50 r = +0.56 score = 0.833

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. aridity

z = +2.64 r = -0.30 score = 0.799

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. gauge_lat

z = +1.31 r = +0.51 score = 0.665

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. low_prec_freq

z = +1.31 r = -0.50 score = 0.659

many dry days indicate a drier climate where streamflow depends on stored water.
5. p_seasonality

z = +1.56 r = -0.38 score = 0.596

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. frac_forest

z = -1.72 r = +0.35 score = 0.594

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. q95

z = -1.00 r = +0.51 score = 0.510

low peak-flow values reflect a damped streamflow response with limited peaks.
8. stream_elas

z = +1.14 r = -0.43 score = 0.488

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. q_mean

z = -0.94 r = +0.45 score = 0.421

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. slope_fdc

z = -1.45 r = +0.24 score = 0.351

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06354000: CANNONBALL RIVER AT BREIEN, ND

Dominant forcing: Tair Peak TE: 0.0025 Peak lag: 23 day(s)
Summary: Tair-dominant at lag 23 day(s); top drivers consistent with low runoff_ratio, high gauge_lat, high aridity.

Top 10 driver attributes

1. runoff_ratio

z = -1.47 r = +0.56 score = 0.818

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. gauge_lat

z = +1.37 r = +0.51 score = 0.692

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. aridity

z = +2.05 r = -0.30 score = 0.622

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. low_prec_freq

z = +1.20 r = -0.50 score = 0.605

many dry days indicate a drier climate where streamflow depends on stored water.
5. p_seasonality

z = +1.57 r = -0.38 score = 0.602

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. frac_forest

z = -1.72 r = +0.35 score = 0.594

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. q95

z = -0.99 r = +0.51 score = 0.507

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.94 r = +0.45 score = 0.418

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. slope_mean

z = -0.77 r = +0.41 score = 0.315

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. slope_fdc

z = -1.30 r = +0.24 score = 0.315

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06360500: MOREAU R NEAR WHITEHORSE SD

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, low p_mean.

Top 10 driver attributes

1. area_gages2

z = +6.97 r = +0.18 score = 1.259

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +6.90 r = +0.17 score = 1.175

a large geospatial drainage area integrates many sub-basin responses.
3. p_mean

z = -1.47 r = +0.34 score = 0.506

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. aridity

z = +1.40 r = -0.28 score = 0.387

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. stream_elas

z = +2.92 r = -0.12 score = 0.360

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. p_seasonality

z = +1.45 r = -0.24 score = 0.354

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. runoff_ratio

z = -1.44 r = +0.23 score = 0.333

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. slope_fdc

z = -1.56 r = +0.20 score = 0.313

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. q95

z = -0.97 r = +0.31 score = 0.297

low peak-flow values reflect a damped streamflow response with limited peaks.
10. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06404000: BATTLE CR NEAR KEYSTONE SD

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, high stream_elas.

Top 10 driver attributes

1. p_mean

z = -1.22

r = +0.34

score = 0.419

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.49

r = -0.24

score = 0.363

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. stream_elas

z = +2.88

r = -0.12

score = 0.355

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. hfd_mean

z = +1.36

r = -0.24

score = 0.323

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. gvf_diff

z = -1.22

r = -0.24

score = 0.296

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. runoff_ratio

z = -1.21

r = +0.23

score = 0.280

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. q95

z = -0.91

r = +0.31

score = 0.278

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.86

r = +0.28

score = 0.239

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. aridity

z = +0.82

r = -0.28

score = 0.227

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
10. low_prec_freq

z = +0.86

r = -0.22

score = 0.193

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06406000: BATTLE CR AT HERMOSA SD

Dominant forcing: PRCP Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, high slope_fdc.

Top 10 driver attributes

1. p_mean z = -1.30 r = +0.34 score = 0.447

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality z = +1.50 r = -0.24 score = 0.365

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. slope_fdc z = +1.67 r = +0.20 score = 0.335

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. runoff_ratio z = -1.31 r = +0.23 score = 0.303

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. q95 z = -0.96 r = +0.31 score = 0.292

low peak-flow values reflect a damped streamflow response with limited peaks.
6. aridity z = +0.98 r = -0.28 score = 0.271

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
7. q_mean z = -0.89 r = +0.28 score = 0.247

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. glim_1st_class_frac z = -1.65 r = -0.15 score = 0.240

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. gvf_diff z = -0.97 r = -0.24 score = 0.235

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. hfd_mean z = +0.96 r = -0.24 score = 0.228

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06408700: RHOADS FORK NEAR ROCHFORD SD

Dominant forcing: SRAD Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high baseflow_index, high p_seasonality, high elev_mean.

Top 10 driver attributes

1. baseflow_index

z = +2.99 r = +0.22 score = 0.655

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
2. p_seasonality

z = +1.01 r = -0.46 score = 0.462

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. elev_mean

z = +1.60 r = +0.24 score = 0.386

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. q95

z = -0.80 r = +0.43 score = 0.344

low peak-flow values reflect a damped streamflow response with limited peaks.
5. carbonate_rocks_frac

z = +2.76 r = -0.11 score = 0.311

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. gauge_lat

z = +0.94 r = +0.29 score = 0.275

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. q5

z = +0.80 r = +0.31 score = 0.251

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. gvf_diff

z = -0.76 r = -0.33 score = 0.249

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. stream_elas

z = -0.96 r = -0.26 score = 0.247

low streamflow elasticity means streamflow is buffered against rainfall variability.
10. frac_snow

z = +0.96 r = +0.26 score = 0.246

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06409000: CASTLE CR ABOVE DEERFIELD RES NEAR HILL CITY SD

Dominant forcing: SRAD Peak TE: 0.0052 Peak lag: 21 day(s)
Summary: SRAD-dominant at lag 21 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high baseflow_index.

Top 10 driver attributes

1. p_seasonality

z = +1.14 r = -0.46 score = 0.522

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.21 r = +0.43 score = 0.518

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. baseflow_index

z = +2.31 r = +0.22 score = 0.506

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
4. q95

z = -0.94 r = +0.43 score = 0.408

low peak-flow values reflect a damped streamflow response with limited peaks.
5. elev_mean

z = +1.53 r = +0.24 score = 0.369

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. q_mean

z = -0.85 r = +0.35 score = 0.301

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. carbonate_rocks_frac

z = +2.48 r = -0.11 score = 0.279

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
8. gauge_lat

z = +0.92 r = +0.29 score = 0.268

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. p_mean

z = -1.08 r = +0.20 score = 0.217

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. gauge_lon

z = -0.50 r = -0.43 score = 0.214

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06431500: SPEARFISH CR AT SPEARFISH SD

Dominant forcing: PRCP Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high elev_mean, low q95.

Top 10 driver attributes

1. p_mean

z = -0.92 r = +0.34 score = 0.316

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. elev_mean

z = +1.44 r = -0.20 score = 0.281

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. q95

z = -0.89 r = +0.31 score = 0.272

low peak-flow values reflect a damped streamflow response with limited peaks.
4. carbonate_rocks_frac

z = +1.93 r = -0.12 score = 0.232

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
5. p_seasonality

z = +0.82 r = -0.24 score = 0.201

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. q_mean

z = -0.72 r = +0.28 score = 0.200

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. runoff_ratio

z = -0.84 r = +0.23 score = 0.194

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. aridity

z = +0.69 r = -0.28 score = 0.189

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. hfd_mean

z = +0.60 r = -0.24 score = 0.142

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. glim_2nd_class_frac

z = +0.92 r = +0.15 score = 0.141

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06440200: SOUTH FORK BAD R NEAR COTTONWOOD,SD

Dominant forcing: PRCP Peak TE: 0.0059 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, low slope_fdc, high zero_q_freq.

Top 10 driver attributes

1. p_mean

z = -1.38

r = +0.34

score = 0.474

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. slope_fdc

z = -2.26

r = +0.20

score = 0.454

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. zero_q_freq

z = +3.15

r = -0.13

score = 0.421

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
4. aridity

z = +1.20

r = -0.28

score = 0.332

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. runoff_ratio

z = -1.42

r = +0.23

score = 0.329

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. p_seasonality

z = +1.33

r = -0.24

score = 0.326

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. q95

z = -0.98

r = +0.31

score = 0.299

low peak-flow values reflect a damped streamflow response with limited peaks.
8. frac_forest

z = -1.72

r = +0.17

score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. hfd_mean

z = +1.15

r = -0.24

score = 0.273

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. low_prec_freq

z = +1.15

r = -0.22

score = 0.258

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06441500: BAD R NEAR FORT PIERRE SD

Dominant forcing: PRCP Peak TE: 0.0054 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, low p_mean.

Top 10 driver attributes

1. area_gages2

z = +4.32 r = +0.18 score = 0.781

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +4.29 r = +0.17 score = 0.731

a large geospatial drainage area integrates many sub-basin responses.
3. p_mean

z = -1.38 r = +0.34 score = 0.476

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. slope_fdc

z = -1.79 r = +0.20 score = 0.360

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
5. runoff_ratio

z = -1.46 r = +0.23 score = 0.338

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. glim_2nd_class_frac

z = +2.20 r = +0.15 score = 0.337

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. p_seasonality

z = +1.37 r = -0.24 score = 0.335

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. aridity

z = +1.18 r = -0.28 score = 0.326

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. frac_forest

z = -1.71 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. q95

z = -0.97 r = +0.31 score = 0.295

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06447000: WHITE R NEAR KADOKA SD

Dominant forcing: PRCP Peak TE: 0.0146 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, low p_mean.

Top 10 driver attributes

1. area_gages2

z = +7.08 r = +0.18 score = 1.279

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +7.05 r = +0.17 score = 1.201

a large geospatial drainage area integrates many sub-basin responses.
3. p_mean

z = -1.38 r = +0.34 score = 0.475

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. aridity

z = +1.66 r = -0.28 score = 0.457

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. runoff_ratio

z = -1.50 r = +0.23 score = 0.347

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. p_seasonality

z = +1.37 r = -0.24 score = 0.333

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. q95

z = -0.98 r = +0.31 score = 0.300

low peak-flow values reflect a damped streamflow response with limited peaks.
8. hfd_mean

z = +1.19 r = -0.24 score = 0.282

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. frac_forest

z = -1.62 r = +0.17 score = 0.279

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. q_mean

z = -0.94 r = +0.28 score = 0.259

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06447500: LITTLE WHITE R NEAR MARTIN SD

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 16 day(s)
Summary: SRAD-dominant at lag 16 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, low q95.

Top 10 driver attributes

1. p_seasonality

z = +1.37 r = -0.46 score = 0.624

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.44 r = +0.43 score = 0.618

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. q95

z = -0.99 r = +0.43 score = 0.425

low peak-flow values reflect a damped streamflow response with limited peaks.
4. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. soil_depth_pelletier

z = +1.59 r = -0.24 score = 0.380

deep soils provide more storage and slower release of water to streamflow.
6. low_prec_freq

z = +1.08 r = -0.31 score = 0.336

many dry days indicate a drier climate where streamflow depends on stored water.
7. glim_2nd_class_frac

z = +1.86 r = +0.18 score = 0.327

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. q_mean

z = -0.92 r = +0.35 score = 0.326

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. high_prec_freq

z = +0.95 r = -0.33 score = 0.310

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. slope_mean

z = -0.84 r = +0.37 score = 0.310

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06450500: LITTLE WHITE R BELOW WHITE RIVER SD

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high area_gages2, high p_seasonality.

Top 10 driver attributes

1. p_mean

z = -1.27

r = +0.34

score = 0.436

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. area_gages2

z = +1.95

r = +0.18

score = 0.352

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
3. p_seasonality

z = +1.39

r = -0.24

score = 0.339

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. area_geospa_fabric

z = +1.91

r = +0.17

score = 0.325

a large geospatial drainage area integrates many sub-basin responses.
5. runoff_ratio

z = -1.37

r = +0.23

score = 0.317

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -0.98

r = +0.31

score = 0.298

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.72

r = +0.17

score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. glim_2nd_class_frac

z = +1.90

r = +0.15

score = 0.291

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. q_mean

z = -0.90

r = +0.28

score = 0.250

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. aridity

z = +0.90

r = -0.28

score = 0.248

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06452000: WHITE R NEAR OACOMA SD

Dominant forcing: PRCP Peak TE: 0.0066 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, low p_mean.

Top 10 driver attributes

1. area_gages2

z = +14.69 r = +0.18 score = 2.653

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +14.63 r = +0.17 score = 2.491

a large geospatial drainage area integrates many sub-basin responses.
3. p_mean

z = -1.33 r = +0.34 score = 0.458

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. runoff_ratio

z = -1.46 r = +0.23 score = 0.336

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. p_seasonality

z = +1.37 r = -0.24 score = 0.335

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. glim_2nd_class_frac

z = +2.06 r = +0.15 score = 0.317

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. aridity

z = +1.12 r = -0.28 score = 0.308

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. q95

z = -0.97 r = +0.31 score = 0.295

low peak-flow values reflect a damped streamflow response with limited peaks.
9. frac_forest

z = -1.65 r = +0.17 score = 0.285

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. q_mean

z = -0.93 r = +0.28 score = 0.256

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06453600: PONCA CREEK AT VERDEL, NEBR.

Dominant forcing: PRCP Peak TE: 0.0041 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, low runoff_ratio.

Top 10 driver attributes

1. p_mean

z = -1.09

r = +0.34

score = 0.376

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.41

r = -0.24

score = 0.343

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. runoff_ratio

z = -1.37

r = +0.23

score = 0.315

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. q95

z = -0.94

r = +0.31

score = 0.286

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.66

r = +0.17

score = 0.286

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. stream_elas

z = +2.01

r = -0.12

score = 0.248

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. q_mean

z = -0.89

r = +0.28

score = 0.247

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. low_prec_freq

z = +1.00

r = -0.22

score = 0.224

many dry days indicate a drier climate where streamflow depends on stored water.
9. hfd_mean

z = +0.92

r = -0.24

score = 0.219

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. high_prec_freq

z = +0.95

r = -0.19

score = 0.179

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06464500: KEYA PAHA R AT WEWELA SD

Dominant forcing: PRCP Peak TE: 0.0070 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high p_seasonality, low runoff_ratio.

Top 10 driver attributes

1. p_mean

z = -1.21

r = +0.34

score = 0.418

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. p_seasonality

z = +1.47

r = -0.24

score = 0.358

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. runoff_ratio

z = -1.43

r = +0.23

score = 0.329

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. q95

z = -0.97

r = +0.31

score = 0.297

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.69

r = +0.17

score = 0.291

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q_mean

z = -0.91

r = +0.28

score = 0.253

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. low_prec_freq

z = +1.03

r = -0.22

score = 0.230

many dry days indicate a drier climate where streamflow depends on stored water.
8. area_gages2

z = +1.25

r = +0.18

score = 0.225

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
9. aridity

z = +0.81

r = -0.28

score = 0.223

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
10. area_geospa_fabric

z = +1.24

r = +0.17

score = 0.211

a large geospatial drainage area integrates many sub-basin responses.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06468170: JAMES RIVER NR GRACE CITY, ND

Dominant forcing: Tair Peak TE: 0.0018 Peak lag: 16 day(s)
Summary: Tair-dominant at lag 16 day(s); top drivers consistent with high gauge_lat, low runoff_ratio, high p_seasonality.

Top 10 driver attributes

1. gauge_lat z = +1.60 r = +0.51 score = 0.807

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
2. runoff_ratio z = -1.41 r = +0.56 score = 0.782

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. p_seasonality z = +1.82 r = -0.38 score = 0.699

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. low_prec_freq z = +1.33 r = -0.50 score = 0.671

many dry days indicate a drier climate where streamflow depends on stored water.
5. frac_forest z = -1.71 r = +0.35 score = 0.591

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. soil_depth_pelletier z = +2.17 r = -0.26 score = 0.555

deep soils provide more storage and slower release of water to streamflow.
7. q95 z = -0.95 r = +0.51 score = 0.486

low peak-flow values reflect a damped streamflow response with limited peaks.
8. high_prec_freq z = +0.91 r = -0.49 score = 0.446

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. pet_mean z = -1.30 r = -0.33 score = 0.431

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. q_mean z = -0.92 r = +0.45 score = 0.411

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06468250: JAMES RIVER ABOVE ARROWWOOD LAKE NR KENSAL, ND

Dominant forcing: SRAD Peak TE: 0.0022 Peak lag: 25 day(s)
Summary: SRAD-dominant at lag 25 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high soil_depth_pelletier.

Top 10 driver attributes

1. p_seasonality

z = +1.83 r = -0.46 score = 0.834

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.41 r = +0.43 score = 0.604

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. soil_depth_pelletier

z = +2.17 r = -0.24 score = 0.521

deep soils provide more storage and slower release of water to streamflow.
4. gauge_lat

z = +1.57 r = +0.29 score = 0.459

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. q95

z = -0.94 r = +0.43 score = 0.408

low peak-flow values reflect a damped streamflow response with limited peaks.
6. gvf_diff

z = +1.23 r = -0.33 score = 0.404

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. frac_forest

z = -1.71 r = +0.23 score = 0.401

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. low_prec_freq

z = +1.27 r = -0.31 score = 0.397

many dry days indicate a drier climate where streamflow depends on stored water.
9. high_q_dur

z = +2.33 r = +0.16 score = 0.364

long high-flow events suggest sustained wet conditions and slow recession.
10. slope_mean

z = -0.88 r = +0.37 score = 0.325

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06470800: BEAR CREEK NR OAKES, ND

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 22 day(s)
Summary: SRAD-dominant at lag 22 day(s); top drivers consistent with high stream_elas, high p_seasonality, low runoff_ratio.

Top 10 driver attributes

1. stream_elas

z = +3.57 r = -0.26 score = 0.923

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. p_seasonality

z = +1.64 r = -0.46 score = 0.750

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. runoff_ratio

z = -1.51 r = +0.43 score = 0.645

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. soil_depth_pelletier

z = +2.40 r = -0.24 score = 0.576

deep soils provide more storage and slower release of water to streamflow.
5. gvf_diff

z = +1.59 r = -0.33 score = 0.520

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. q95

z = -0.97 r = +0.43 score = 0.421

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.70 r = +0.23 score = 0.399

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. gauge_lat

z = +1.34 r = +0.29 score = 0.393

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. low_prec_freq

z = +1.20 r = -0.31 score = 0.373

many dry days indicate a drier climate where streamflow depends on stored water.
10. slope_mean

z = -0.91 r = +0.37 score = 0.333

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06477500: FIRESTEEL CR NEAR MOUNT VERNON SD

Dominant forcing: PRCP Peak TE: 0.0065 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low p_mean, low runoff_ratio.

Top 10 driver attributes

1. slope_fdc

z = -2.04 r = +0.20 score = 0.411

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. p_mean

z = -1.12 r = +0.34 score = 0.385

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. runoff_ratio

z = -1.49 r = +0.23 score = 0.344

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. p_seasonality

z = +1.40 r = -0.24 score = 0.343

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. soil_depth_pelletier

z = +2.10 r = +0.16 score = 0.335

deep soils provide more storage and slower release of water to streamflow.
6. hfd_mean

z = +1.28 r = -0.24 score = 0.304

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
7. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. q95

z = -0.96 r = +0.31 score = 0.294

low peak-flow values reflect a damped streamflow response with limited peaks.
9. stream_elas

z = +2.18 r = -0.12 score = 0.269

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. low_prec_freq

z = +1.17 r = -0.22 score = 0.261

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06479215: BIG SIOUX RIVER NR FLORENCE SD

Dominant forcing: SRAD Peak TE: 0.0023 Peak lag: 19 day(s)
Summary: SRAD-dominant at lag 19 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high stream_elas.

Top 10 driver attributes

1. p_seasonality

z = +1.52 r = -0.46 score = 0.695

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.15 r = +0.43 score = 0.493

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. stream_elas

z = +1.80 r = -0.26 score = 0.467

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. soil_depth_pelletier

z = +1.91 r = -0.24 score = 0.459

deep soils provide more storage and slower release of water to streamflow.
5. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q95

z = -0.84 r = +0.43 score = 0.364

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gvf_diff

z = +1.05 r = -0.33 score = 0.343

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. high_prec_freq

z = +1.05 r = -0.33 score = 0.342

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. gauge_lat

z = +1.14 r = +0.29 score = 0.334

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. slope_mean

z = -0.84 r = +0.37 score = 0.310

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06479438: BIG SIOUX R NEAR WATERTOWN SD

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 17 day(s)
Summary: SRAD-dominant at lag 17 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high soil_depth_pelletier.

Top 10 driver attributes

1. p_seasonality

z = +1.54 r = -0.46 score = 0.703

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.31 r = +0.43 score = 0.562

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. soil_depth_pelletier

z = +2.00 r = -0.24 score = 0.479

deep soils provide more storage and slower release of water to streamflow.
4. frac_forest

z = -1.71 r = +0.23 score = 0.402

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. q95

z = -0.91 r = +0.43 score = 0.392

low peak-flow values reflect a damped streamflow response with limited peaks.
6. high_prec_freq

z = +1.14 r = -0.33 score = 0.371

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. gvf_diff

z = +1.07 r = -0.33 score = 0.349

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. gauge_lat

z = +1.11 r = +0.29 score = 0.324

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. slope_mean

z = -0.87 r = +0.37 score = 0.321

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. q_mean

z = -0.88 r = +0.35 score = 0.310

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06601000: OMAHA CR AT HOMER, NEBR

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 9 day(s)
Summary: SRAD-dominant at lag 9 day(s); top drivers consistent with high p_seasonality, high gvf_diff, low runoff_ratio.

Top 10 driver attributes

1. p_seasonality

z = +1.41 r = -0.46 score = 0.644

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. gvf_diff

z = +1.68 r = -0.33 score = 0.551

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. runoff_ratio

z = -0.94 r = +0.43 score = 0.401

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. frac_forest

z = -1.62 r = +0.23 score = 0.380

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. q95

z = -0.86 r = +0.43 score = 0.373

low peak-flow values reflect a damped streamflow response with limited peaks.
6. low_prec_freq

z = +1.04 r = -0.31 score = 0.323

many dry days indicate a drier climate where streamflow depends on stored water.
7. glim_2nd_class_frac

z = +1.84 r = +0.18 score = 0.322

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. high_prec_freq

z = +0.98 r = -0.33 score = 0.321

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. slope_mean

z = -0.79 r = +0.37 score = 0.290

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. baseflow_index

z = +1.19 r = +0.22 score = 0.261

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06614800: MICHIGAN RIVER NEAR CAMERON PASS, CO.

Dominant forcing: SRAD Peak TE: 0.0059 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high slope_mean, high frac_snow.

Top 10 driver attributes

1. elev_mean

z = +3.23

r = +0.24

score = 0.779

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. slope_mean

z = +2.11

r = +0.37

score = 0.777

steep terrain promotes fast surface runoff and a quick streamflow response.
3. frac_snow

z = +2.87

r = +0.26

score = 0.732

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. high_prec_freq

z = -1.93

r = -0.33

score = 0.630

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. runoff_ratio

z = +1.00

r = +0.43

score = 0.429

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. glim_1st_class_frac

z = -1.84

r = -0.21

score = 0.393

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. q95

z = +0.82

r = +0.43

score = 0.354

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. clay_frac

z = -1.42

r = -0.22

score = 0.313

low clay fraction allows easier infiltration and reduces surface runoff.
9. other_frac

z = +2.19

r = +0.14

score = 0.305

a large 'other' land fraction reflects mixed or atypical surface cover.
10. high_q_dur

z = +1.82

r = +0.16

score = 0.286

long high-flow events suggest sustained wet conditions and slow recession.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06622700: NORTH BRUSH CREEK NEAR SARATOGA, WY

Dominant forcing: SRAD Peak TE: 0.0081 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low high_prec_freq.

Top 10 driver attributes

1. frac_snow

z = +2.66 r = +0.26 score = 0.678

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.64 r = +0.24 score = 0.638

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = -1.46 r = -0.33 score = 0.476

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. glim_1st_class_frac

z = -1.66 r = -0.21 score = 0.356

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. p_seasonality

z = -0.64 r = -0.46 score = 0.291

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. gauge_lon

z = -0.66 r = -0.43 score = 0.286

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_q_dur

z = +1.68 r = +0.16 score = 0.263

long high-flow events suggest sustained wet conditions and slow recession.
8. runoff_ratio

z = +0.58 r = +0.43 score = 0.249

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. gvf_diff

z = -0.70 r = -0.33 score = 0.228

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. slope_mean

z = +0.53 r = +0.37 score = 0.196

steep terrain promotes fast surface runoff and a quick streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06623800: ENCAMPMENT RIVER AB HOG PARK CR, NR ENCAMPMENT, WY

Dominant forcing: SRAD Peak TE: 0.0064 Peak lag: 1 day(s)

Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low p_seasonality.

Top 10 driver attributes

1. frac_snow

z = +2.84

r = +0.26

score = 0.724

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.81

r = +0.24

score = 0.679

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. p_seasonality

z = -1.01

r = -0.46

score = 0.462

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. high_prec_freq

z = -1.34

r = -0.33

score = 0.436

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. slope_mean

z = +0.95

r = +0.37

score = 0.348

steep terrain promotes fast surface runoff and a quick streamflow response.
6. high_q_dur

z = +2.19

r = +0.16

score = 0.344

long high-flow events suggest sustained wet conditions and slow recession.
7. gauge_lon

z = -0.68

r = -0.43

score = 0.294

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. glim_1st_class_frac

z = -1.37

r = -0.21

score = 0.293

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. q95

z = +0.54

r = +0.43

score = 0.233

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. low_prec_freq

z = -0.73

r = -0.31

score = 0.227

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06632400: ROCK CREEK AB KING CANYON CANAL, NR ARLINGTON, WY

Dominant forcing: SRAD Peak TE: 0.0069 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low high_prec_freq.

Top 10 driver attributes

1. frac_snow

z = +2.79 r = +0.26 score = 0.712

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.85 r = +0.24 score = 0.688

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = -1.15 r = -0.33 score = 0.375

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. high_q_dur

z = +2.30 r = +0.16 score = 0.360

long high-flow events suggest sustained wet conditions and slow recession.
5. gauge_lon

z = -0.64 r = -0.43 score = 0.278

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. p_seasonality

z = -0.59 r = -0.46 score = 0.271

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. gvf_diff

z = -0.73 r = -0.33 score = 0.238

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. glim_1st_class_frac

z = -1.07 r = -0.21 score = 0.228

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. runoff_ratio

z = +0.44 r = +0.43 score = 0.189

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. root_depth_50

z = -1.50 r = -0.13 score = 0.188

shallow median rooting limits vegetation water access to surface soils.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06746095: JOE WRIGHT CREEK ABOVE JOE WRIGHT RESERVOIR, CO.

Dominant forcing: SRAD Peak TE: 0.0063 Peak lag: 2 day(s)

Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high runoff_ratio, high q95, high elev_mean.

Top 10 driver attributes

1. runoff_ratio

z = +3.19 r = +0.43 score = 1.363

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. q95

z = +2.32 r = +0.43 score = 1.000

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. elev_mean

z = +3.32 r = +0.24 score = 0.801

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. frac_snow

z = +2.71 r = +0.26 score = 0.692

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. high_prec_freq

z = -1.81 r = -0.33 score = 0.591

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. slope_mean

z = +1.60 r = +0.37 score = 0.587

steep terrain promotes fast surface runoff and a quick streamflow response.
7. high_q_dur

z = +2.50 r = +0.16 score = 0.391

long high-flow events suggest sustained wet conditions and slow recession.
8. q_mean

z = +1.05 r = +0.35 score = 0.371

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
9. clay_frac

z = -1.24 r = -0.22 score = 0.274

low clay fraction allows easier infiltration and reduces surface runoff.
10. gauge_lon

z = -0.62 r = -0.43 score = 0.269

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06784000: SOUTH LOUP R AT ST. MICHAEL, NEBR.

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_geospa_fabric, high area_gages2, low p_mean.

Top 10 driver attributes

1. area_geospa_fabric

z = +3.26 r = +0.17 score = 0.556

a large geospatial drainage area integrates many sub-basin responses.
2. area_gages2

z = +3.07 r = +0.18 score = 0.555

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
3. p_mean

z = -1.07 r = +0.34 score = 0.369

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. p_seasonality

z = +1.49 r = -0.24 score = 0.365

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. runoff_ratio

z = -1.44 r = +0.23 score = 0.331

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -0.99 r = +0.31 score = 0.301

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.68 r = +0.17 score = 0.289

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. q_mean

z = -0.91 r = +0.28 score = 0.252

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. low_prec_freq

z = +0.92 r = -0.22 score = 0.206

many dry days indicate a drier climate where streamflow depends on stored water.
10. soil_depth_pelletier

z = +1.17 r = +0.16 score = 0.187

deep soils provide more storage and slower release of water to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06803510: LITTLE SALT CREEK NEAR LINCOLN, NEBR.

Dominant forcing: VP Peak TE: 0.0013 Peak lag: 2 day(s)
Summary: VP-dominant at lag 2 day(s); top drivers consistent with low runoff_ratio, low frac_forest, low q95.

Top 10 driver attributes

1. runoff_ratio

z = -1.10 r = +0.50 score = 0.550

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. frac_forest

z = -1.72 r = +0.29 score = 0.499

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. q95

z = -0.92 r = +0.49 score = 0.453

low peak-flow values reflect a damped streamflow response with limited peaks.
4. high_prec_freq

z = +0.87 r = -0.47 score = 0.411

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. low_prec_freq

z = +0.79 r = -0.50 score = 0.397

many dry days indicate a drier climate where streamflow depends on stored water.
6. p_seasonality

z = +1.38 r = -0.27 score = 0.370

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. clay_frac

z = +1.55 r = -0.22 score = 0.341

high clay fraction reduces infiltration and increases surface runoff.
8. q_mean

z = -0.79 r = +0.43 score = 0.339

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. slope_fdc

z = -1.02 r = +0.24 score = 0.245

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. slope_mean

z = -0.81 r = +0.27 score = 0.219

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06803530: ROCK CREEK NEAR CERESCO, NEBR.

Dominant forcing: VP Peak TE: 0.0015 Peak lag: 5 day(s)

Summary: VP-dominant at lag 5 day(s); top drivers consistent with low runoff_ratio, low frac_forest, low q95.

Top 10 driver attributes

1. runoff_ratio

z = -1.12 r = +0.50 score = 0.557

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. frac_forest

z = -1.72 r = +0.29 score = 0.499

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. q95

z = -0.90 r = +0.49 score = 0.442

low peak-flow values reflect a damped streamflow response with limited peaks.
4. low_prec_freq

z = +0.78 r = -0.50 score = 0.392

many dry days indicate a drier climate where streamflow depends on stored water.
5. p_seasonality

z = +1.37 r = -0.27 score = 0.366

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. high_prec_freq

z = +0.74 r = -0.47 score = 0.349

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. q_mean

z = -0.79 r = +0.43 score = 0.340

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. clay_frac

z = +1.41 r = -0.22 score = 0.309

high clay fraction reduces infiltration and increases surface runoff.
9. slope_fdc

z = -1.22 r = +0.24 score = 0.292

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. slope_mean

z = -0.80 r = +0.27 score = 0.218

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06814000: TURKEY C NR SENECA, KS

Dominant forcing: VP Peak TE: 0.0037 Peak lag: 4 day(s)
Summary: VP-dominant at lag 4 day(s); top drivers consistent with high stream_elas, low frac_forest, low runoff_ratio.

Top 10 driver attributes

1. stream_elas

z = +1.94 r = -0.36 score = 0.703

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. frac_forest

z = -1.70 r = +0.29 score = 0.495

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. runoff_ratio

z = -0.91 r = +0.50 score = 0.453

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. clay_frac

z = +1.95 r = -0.22 score = 0.430

high clay fraction reduces infiltration and increases surface runoff.
5. low_prec_freq

z = +0.81 r = -0.50 score = 0.407

many dry days indicate a drier climate where streamflow depends on stored water.
6. high_prec_freq

z = +0.79 r = -0.47 score = 0.369

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. q95

z = -0.74 r = +0.49 score = 0.367

low peak-flow values reflect a damped streamflow response with limited peaks.
8. p_seasonality

z = +1.29 r = -0.27 score = 0.346

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. q_mean

z = -0.70 r = +0.43 score = 0.298

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. baseflow_index

z = -1.30 r = +0.17 score = 0.226

a low baseflow index means streamflow responds quickly to recent rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06847900: PRAIRIE DOG C AB KEITH SEBELIUS LAKE, KS

Dominant forcing: SRAD Peak TE: 0.0011 Peak lag: 28 day(s)
Summary: SRAD-dominant at lag 28 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, high soil_depth_pelletier.

Top 10 driver attributes

1. p_seasonality

z = +1.60 r = -0.46 score = 0.733

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.63 r = +0.43 score = 0.696

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. soil_depth_pelletier

z = +1.86 r = -0.24 score = 0.447

deep soils provide more storage and slower release of water to streamflow.
4. q95

z = -1.02 r = +0.43 score = 0.440

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.71 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. low_prec_freq

z = +1.21 r = -0.31 score = 0.379

many dry days indicate a drier climate where streamflow depends on stored water.
7. q_mean

z = -0.96 r = +0.35 score = 0.339

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. slope_mean

z = -0.84 r = +0.37 score = 0.309

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. aridity

z = +1.69 r = -0.16 score = 0.279

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
10. high_prec_freq

z = +0.82 r = -0.33 score = 0.267

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06853800: WHITE ROCK C NR BURR OAK, KS

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low runoff_ratio, low p_mean.

Top 10 driver attributes

1. p_seasonality

z = +1.38 r = -0.24 score = 0.338

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. runoff_ratio

z = -1.43 r = +0.23 score = 0.330

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. p_mean

z = -0.93 r = +0.34 score = 0.320

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. q95

z = -0.97 r = +0.31 score = 0.296

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q_mean

z = -0.90 r = +0.28 score = 0.249

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. soil_depth_pelletier

z = +1.55 r = +0.16 score = 0.248

deep soils provide more storage and slower release of water to streamflow.
8. hfd_mean

z = +1.03 r = -0.24 score = 0.245

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. low_prec_freq

z = +1.07 r = -0.22 score = 0.238

many dry days indicate a drier climate where streamflow depends on stored water.
10. high_prec_dur

z = -0.83 r = +0.23 score = 0.187

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06876700: SALT C NR ADA, KS

Dominant forcing: PRCP Peak TE: 0.0041 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low glim_1st_class_frac, high hfd_mean.

Top 10 driver attributes

1. p_seasonality

z = +1.43 r = -0.24 score = 0.349

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. glim_1st_class_frac

z = -2.33 r = -0.15 score = 0.338

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
3. hfd_mean

z = +1.34 r = -0.24 score = 0.319

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. runoff_ratio

z = -1.33 r = +0.23 score = 0.306

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. p_mean

z = -0.85 r = +0.34 score = 0.294

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. stream_elas

z = +2.36 r = -0.12 score = 0.291

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. q95

z = -0.91 r = +0.31 score = 0.277

low peak-flow values reflect a damped streamflow response with limited peaks.
9. low_prec_freq

z = +1.12 r = -0.22 score = 0.252

many dry days indicate a drier climate where streamflow depends on stored water.
10. q_mean

z = -0.86 r = +0.28 score = 0.239

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06878000: CHAPMAN C NR CHAPMAN, KS

Dominant forcing: PRCP Peak TE: 0.0101 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high hfd_mean, high p_seasonality, low frac_forest.

Top 10 driver attributes

1. hfd_mean

z = +1.37 r = -0.24 score = 0.326

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
2. p_seasonality

z = +1.31 r = -0.24 score = 0.320

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
3. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. glim_2nd_class_frac

z = +1.73 r = +0.15 score = 0.266

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. stream_elas

z = +2.15 r = -0.12 score = 0.265

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. q95

z = -0.85 r = +0.31 score = 0.259

low peak-flow values reflect a damped streamflow response with limited peaks.
7. runoff_ratio

z = -1.11 r = +0.23 score = 0.257

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. p_mean

z = -0.73 r = +0.34 score = 0.251

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. glim_1st_class_frac

z = -1.53 r = -0.15 score = 0.223

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. q_mean

z = -0.78 r = +0.28 score = 0.217

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06879650: KINGS C NR MANHATTAN, KS

Dominant forcing: SRAD Peak TE: 0.0018 Peak lag: 28 day(s)
Summary: SRAD-dominant at lag 28 day(s); top drivers consistent with high_q_dur, high zero_q_freq, high p_seasonality.

Top 10 driver attributes

1. high_q_dur

z = +8.35

r = +0.16

score = 1.308

long high-flow events suggest sustained wet conditions and slow recession.
2. zero_q_freq

z = +4.31

r = -0.23

score = 0.974

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. p_seasonality

z = +1.34

r = -0.46

score = 0.613

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. clay_frac

z = +2.45

r = -0.22

score = 0.539

high clay fraction reduces infiltration and increases surface runoff.
5. gvf_diff

z = +1.31

r = -0.33

score = 0.429

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. frac_forest

z = -1.63

r = +0.23

score = 0.384

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. runoff_ratio

z = -0.88

r = +0.43

score = 0.377

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. glim_2nd_class_frac

z = +1.79

r = +0.18

score = 0.314

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. q95

z = -0.68

r = +0.43

score = 0.295

low peak-flow values reflect a damped streamflow response with limited peaks.
10. sand_frac

z = -1.66

r = +0.18

score = 0.292

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06885500: BLACK VERMILLION R NR FRANKFORT, KS

Dominant forcing: PRCP Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low frac_forest, low slope_fdc.

Top 10 driver attributes

1. p_seasonality

z = +1.34 r = -0.24 score = 0.327

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. frac_forest

z = -1.70 r = +0.17 score = 0.292

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. slope_fdc

z = -1.35 r = +0.20 score = 0.271

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. hfd_mean

z = +1.08 r = -0.24 score = 0.258

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. stream_elas

z = +1.87 r = -0.12 score = 0.231

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. soil_depth_pelletier

z = +1.44 r = +0.16 score = 0.231

deep soils provide more storage and slower release of water to streamflow.
7. q95

z = -0.70 r = +0.31 score = 0.214

low peak-flow values reflect a damped streamflow response with limited peaks.
8. runoff_ratio

z = -0.89 r = +0.23 score = 0.205

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. p_mean

z = -0.58 r = +0.34 score = 0.198

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. low_prec_freq

z = +0.87 r = -0.22 score = 0.195

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06888500: MILL C NR PAXICO, KS

Dominant forcing: VP Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with high clay_frac, low frac_forest, high low_prec_freq.

Top 10 driver attributes

1. clay_frac

z = +2.31 r = -0.22 score = 0.508

high clay fraction reduces infiltration and increases surface runoff.
2. frac_forest

z = -1.68 r = +0.29 score = 0.488

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. low_prec_freq

z = +0.75 r = -0.50 score = 0.374

many dry days indicate a drier climate where streamflow depends on stored water.
4. high_prec_freq

z = +0.76 r = -0.47 score = 0.359

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. p_seasonality

z = +1.27 r = -0.27 score = 0.341

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. runoff_ratio

z = -0.68 r = +0.50 score = 0.340

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. q95

z = -0.66 r = +0.49 score = 0.327

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.58 r = +0.43 score = 0.248

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. stream_elas

z = +0.61 r = -0.36 score = 0.221

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. slope_mean

z = -0.76 r = +0.27 score = 0.205

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06889200: SOLDIER C NR DELIA, KS

Dominant forcing: PRCP Peak TE: 0.0017 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low frac_forest, low q95.

Top 10 driver attributes

1. p_seasonality

z = +1.28 r = -0.24 score = 0.312

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. frac_forest

z = -1.67 r = +0.17 score = 0.288

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. q95

z = -0.67 r = +0.31 score = 0.204

low peak-flow values reflect a damped streamflow response with limited peaks.
4. gvf_diff

z = +0.83 r = -0.24 score = 0.202

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. glim_1st_class_frac

z = -1.32 r = -0.15 score = 0.192

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. hfd_mean

z = +0.77 r = -0.24 score = 0.183

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
7. soil_depth_pelletier

z = +1.12 r = +0.16 score = 0.179

deep soils provide more storage and slower release of water to streamflow.
8. low_prec_freq

z = +0.79 r = -0.22 score = 0.177

many dry days indicate a drier climate where streamflow depends on stored water.
9. slope_fdc

z = -0.83 r = +0.20 score = 0.167

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. q_mean

z = -0.60 r = +0.28 score = 0.166

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06889500: SOLDIER C NR TOPEKA, KS

Dominant forcing: PRCP Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low frac_forest, high gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = +1.26 r = -0.24 score = 0.307

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. frac_forest

z = -1.68 r = +0.17 score = 0.289

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. gvf_diff

z = +0.86 r = -0.24 score = 0.210

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. hfd_mean

z = +0.88 r = -0.24 score = 0.208

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. q95

z = -0.67 r = +0.31 score = 0.203

low peak-flow values reflect a damped streamflow response with limited peaks.
6. soil_depth_pelletier

z = +1.18 r = +0.16 score = 0.188

deep soils provide more storage and slower release of water to streamflow.
7. slope_fdc

z = -0.86 r = +0.20 score = 0.173

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. high_prec_freq

z = +0.90 r = -0.19 score = 0.168

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. runoff_ratio

z = -0.73 r = +0.23 score = 0.168

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. q_mean

z = -0.60 r = +0.28 score = 0.166

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06892000: STRANGER C NR TONGANOXIE, KS

Dominant forcing: PRCP Peak TE: 0.0075 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_seasonality, low frac_forest, high hfd_mean.

Top 10 driver attributes

1. p_seasonality

z = +1.19 r = -0.24 score = 0.290

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. frac_forest

z = -1.54 r = +0.17 score = 0.265

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. hfd_mean

z = +0.96 r = -0.24 score = 0.227

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. soil_depth_pelletier

z = +1.23 r = +0.16 score = 0.196

deep soils provide more storage and slower release of water to streamflow.
5. slope_fdc

z = -0.93 r = +0.20 score = 0.188

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. high_prec_freq

z = +0.87 r = -0.19 score = 0.164

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. runoff_ratio

z = -0.68 r = +0.23 score = 0.158

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q_mean

z = -0.56 r = +0.28 score = 0.155

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. gvf_diff

z = +0.60 r = -0.24 score = 0.146

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
10. sand_frac

z = -1.21 r = +0.12 score = 0.141

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06903400: Chariton River near Chariton, IA

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high hfd_mean, high soil_depth_pelletier, high p_seasonality.

Top 10 driver attributes

1. hfd_mean

z = +1.35 r = -0.24 score = 0.320

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
2. soil_depth_pelletier

z = +1.97 r = +0.16 score = 0.315

deep soils provide more storage and slower release of water to streamflow.
3. p_seasonality

z = +1.22 r = -0.24 score = 0.298

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. slope_fdc

z = -1.32 r = +0.20 score = 0.266

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
5. gvf_diff

z = +0.98 r = -0.24 score = 0.237

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. frac_forest

z = -1.12 r = +0.17 score = 0.193

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. stream_elas

z = +1.44 r = -0.12 score = 0.177

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. high_prec_dur

z = -0.75 r = +0.23 score = 0.170

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. q_mean

z = -0.50 r = +0.28 score = 0.140

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. glim_1st_class_frac

z = +0.89 r = -0.15 score = 0.130

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06906800: Lamine River near Otterville, MO

Dominant forcing: PRCP Peak TE: 0.0083 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, high carbonate_rocks_frac, low slope_fdc.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +1.72 r = +0.15 score = 0.264

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. carbonate_rocks_frac

z = +1.83 r = -0.12 score = 0.220

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. slope_fdc

z = -1.04 r = +0.20 score = 0.209

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. p_seasonality

z = +0.77 r = -0.24 score = 0.188

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. sand_frac

z = -1.49 r = +0.12 score = 0.175

low sand fraction reduces infiltration and may favor surface runoff.
6. gvf_diff

z = +0.56 r = -0.24 score = 0.136

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. glim_1st_class_frac

z = -0.93 r = -0.15 score = 0.136

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. frac_forest

z = -0.77 r = +0.17 score = 0.133

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. high_prec_freq

z = +0.69 r = -0.19 score = 0.129

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q95

z = -0.40 r = +0.31 score = 0.122

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06910800: MARAIS DES CYGNES R NR READING, KS

Dominant forcing: Tair Peak TE: 0.0014 Peak lag: 7 day(s)
Summary: Tair-dominant at lag 7 day(s); top drivers consistent with high clay_frac, low frac_forest, high low_prec_freq.

Top 10 driver attributes

1. clay_frac

z = +2.34 r = -0.30 score = 0.691

high clay fraction reduces infiltration and increases surface runoff.
2. frac_forest

z = -1.70 r = +0.35 score = 0.587

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. low_prec_freq

z = +0.92 r = -0.50 score = 0.462

many dry days indicate a drier climate where streamflow depends on stored water.
4. p_seasonality

z = +1.16 r = -0.38 score = 0.446

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. high_prec_freq

z = +0.87 r = -0.49 score = 0.430

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. runoff_ratio

z = -0.73 r = +0.56 score = 0.407

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. stream_elas

z = +0.89 r = -0.43 score = 0.383

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. baseflow_index

z = -1.69 r = +0.22 score = 0.378

a low baseflow index means streamflow responds quickly to recent rainfall.
9. slope_mean

z = -0.84 r = +0.41 score = 0.344

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. q95

z = -0.62 r = +0.51 score = 0.317

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06911900: DRAGOON C NR BURLINGAME, KS

Dominant forcing: VP Peak TE: 0.0015 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with high clay_frac, low frac_forest, high stream_elas.

Top 10 driver attributes

1. clay_frac

z = +2.25 r = -0.22 score = 0.494

high clay fraction reduces infiltration and increases surface runoff.
2. frac_forest

z = -1.67 r = +0.29 score = 0.485

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. stream_elas

z = +1.31 r = -0.36 score = 0.472

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. low_prec_freq

z = +0.92 r = -0.50 score = 0.458

many dry days indicate a drier climate where streamflow depends on stored water.
5. high_prec_freq

z = +0.95 r = -0.47 score = 0.447

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. runoff_ratio

z = -0.73 r = +0.50 score = 0.364

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. q95

z = -0.66 r = +0.49 score = 0.326

low peak-flow values reflect a damped streamflow response with limited peaks.
8. p_seasonality

z = +1.16 r = -0.27 score = 0.311

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. baseflow_index

z = -1.77 r = +0.17 score = 0.308

a low baseflow index means streamflow responds quickly to recent rainfall.
10. q_mean

z = -0.59 r = +0.43 score = 0.254

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06917000: L OSAGE R AT FULTON, KS

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low frac_forest, high carbonate_rocks_frac, high p_seasonality.

Top 10 driver attributes

1. frac_forest

z = -1.56 r = +0.17 score = 0.268

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
2. carbonate_rocks_frac

z = +2.00 r = -0.12 score = 0.239

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. p_seasonality

z = +0.95 r = -0.24 score = 0.233

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. hfd_mean

z = +0.81 r = -0.24 score = 0.192

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. sand_frac

z = -1.53 r = +0.12 score = 0.179

low sand fraction reduces infiltration and may favor surface runoff.
6. high_prec_freq

z = +0.92 r = -0.19 score = 0.173

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. low_prec_freq

z = +0.65 r = -0.22 score = 0.145

many dry days indicate a drier climate where streamflow depends on stored water.
8. slope_fdc

z = -0.66 r = +0.20 score = 0.133

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. elev_mean

z = -0.59 r = -0.20 score = 0.115

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. soil_conductivity

z = -0.78 r = +0.15 score = 0.115

low soil conductivity restricts infiltration, often increasing surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06918460: Turnback Creek above Greenfield, MO

Dominant forcing: PRCP Peak TE: 0.0028 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, low frac_forest, high slope_fdc.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.72 r = -0.12 score = 0.326

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. frac_forest

z = -1.41 r = +0.17 score = 0.243

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. slope_fdc

z = +0.82 r = +0.20 score = 0.166

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
4. sand_frac

z = -1.34 r = +0.12 score = 0.157

low sand fraction reduces infiltration and may favor surface runoff.
5. stream_elas

z = +1.21 r = -0.12 score = 0.149

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. p_seasonality

z = +0.55 r = -0.24 score = 0.134

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. gvff_diff

z = +0.52 r = -0.24 score = 0.127

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. high_prec_freq

z = +0.66 r = -0.19 score = 0.125

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. q95

z = -0.39 r = +0.31 score = 0.119

low peak-flow values reflect a damped streamflow response with limited peaks.
10. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06919500: Cedar Creek near Pleasant View, MO

Dominant forcing: PRCP Peak TE: 0.0118 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, low high_prec_dur, low frac_forest.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +1.98 r = +0.15 score = 0.303

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. high_prec_dur

z = -0.87 r = +0.23 score = 0.198

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. frac_forest

z = -1.06 r = +0.17 score = 0.183

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. p_seasonality

z = +0.71 r = -0.24 score = 0.174

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. glim_1st_class_frac

z = -1.11 r = -0.15 score = 0.162

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. gvf_diff

z = +0.65 r = -0.24 score = 0.158

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
7. carbonate_rocks_frac

z = +1.23 r = -0.12 score = 0.147

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
8. runoff_ratio

z = -0.64 r = +0.23 score = 0.147

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. sand_frac

z = -1.20 r = +0.12 score = 0.140

low sand fraction reduces infiltration and may favor surface runoff.
10. q_mean

z = -0.46 r = +0.28 score = 0.126

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06921070: Pomme de Terre River near Polk, MO

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, low frac_forest, low sand_frac.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.34 r = -0.12 score = 0.401

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. frac_forest

z = -0.97 r = +0.17 score = 0.167

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. sand_frac

z = -1.37 r = +0.12 score = 0.160

low sand fraction reduces infiltration and may favor surface runoff.
4. glim_2nd_class_frac

z = -1.02 r = +0.15 score = 0.156

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac

z = +1.02 r = -0.15 score = 0.148

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. p_seasonality

z = +0.54 r = -0.24 score = 0.131

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. gvf_diff

z = +0.54 r = -0.24 score = 0.131

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
8. high_prec_freq

z = +0.65 r = -0.19 score = 0.123

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. root_depth_99

z = -1.09 r = +0.10 score = 0.110

shallow maximum rooting limits vegetation reliance on deep storage.
10. soil_conductivity

z = -0.74 r = +0.15 score = 0.110

low soil conductivity restricts infiltration, often increasing surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06921200: Lindley Creek near Polk, MO

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, low frac_forest, high gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.12 score = 0.405

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. frac_forest

z = -1.04 r = +0.17 score = 0.179

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. gvf_diff

z = +0.73 r = -0.24 score = 0.177

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. stream_elas

z = +1.43 r = -0.12 score = 0.177

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
6. sand_frac

z = -1.41 r = +0.12 score = 0.165

low sand fraction reduces infiltration and may favor surface runoff.
7. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. q95

z = -0.50 r = +0.31 score = 0.152

low peak-flow values reflect a damped streamflow response with limited peaks.
9. high_prec_freq

z = +0.77 r = -0.19 score = 0.146

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. p_seasonality

z = +0.59 r = -0.24 score = 0.144

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 06934000: Gasconade River near Rich Fountain, MO

Dominant forcing: PRCP Peak TE: 0.0089 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, high carbonate_rocks_frac.

Top 10 driver attributes

1. area_gages2

z = +4.39 r = +0.18 score = 0.793

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +4.36 r = +0.17 score = 0.743

a large geospatial drainage area integrates many sub-basin responses.
3. carbonate_rocks_frac

z = +3.09 r = -0.12 score = 0.371

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. gvf_diff

z = +0.87 r = -0.24 score = 0.211

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. stream_elas

z = +1.32 r = -0.12 score = 0.163

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. sand_frac

z = -1.34 r = +0.12 score = 0.157

low sand fraction reduces infiltration and may favor surface runoff.
7. high_prec_dur

z = -0.62 r = +0.23 score = 0.140

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. q95

z = -0.41 r = +0.31 score = 0.124

low peak-flow values reflect a damped streamflow response with limited peaks.
9. soil_conductivity

z = -0.75 r = +0.15 score = 0.110

low soil conductivity restricts infiltration, often increasing surface runoff.
10. elev_mean

z = -0.54 r = -0.20 score = 0.106

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07014500: Meramec River near Sullivan, MO

Dominant forcing: PRCP Peak TE: 0.0159 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high glim_2nd_class_frac, high area_geospa_fabric.

Top 10 driver attributes

1. area_gages2

z = +1.79 r = +0.18 score = 0.324

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. glim_2nd_class_frac

z = +2.03 r = +0.15 score = 0.311

a large secondary geological class adds heterogeneity to the subsurface flow response.
3. area_geospa_fabric

z = +1.78 r = +0.17 score = 0.303

a large geospatial drainage area integrates many sub-basin responses.
4. gvf_diff

z = +1.15 r = -0.24 score = 0.279

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. carbonate_rocks_frac

z = +1.59 r = -0.12 score = 0.190

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. glim_1st_class_frac

z = -1.25 r = -0.15 score = 0.182

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. q95

z = -0.53 r = +0.31 score = 0.163

low peak-flow values reflect a damped streamflow response with limited peaks.
8. stream_elas

z = +1.26 r = -0.12 score = 0.156

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. sand_frac

z = -1.25 r = +0.12 score = 0.146

low sand fraction reduces infiltration and may favor surface runoff.
10. high_prec_dur

z = -0.61 r = +0.23 score = 0.138

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07056000: Buffalo River near St. Joe, AR

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high slope_fdc, high area_gages2.

Top 10 driver attributes

1. gvf_diff z = +1.14 r = -0.24 score = 0.277

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. slope_fdc z = +0.85 r = +0.20 score = 0.172

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. area_gages2 z = +0.80 r = +0.18 score = 0.144

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
4. glim_2nd_class_frac z = +0.88 r = +0.15 score = 0.135

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. area_geospa_fabric z = +0.78 r = +0.17 score = 0.133

a large geospatial drainage area integrates many sub-basin responses.
6. aridity z = -0.46 r = -0.28 score = 0.126

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. high_prec_dur z = -0.52 r = +0.23 score = 0.118

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. high_prec_freq z = +0.62 r = -0.19 score = 0.117

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. frac_forest z = +0.66 r = +0.17 score = 0.113

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. sand_frac z = -0.89 r = +0.12 score = 0.105

low sand fraction reduces infiltration and may favor surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07057500: North Fork River near Tecumseh, MO

Dominant forcing: PRCP Peak TE: 0.0037 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high q5, high high_prec_freq.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.37 r = -0.12 score = 0.404

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. q5

z = +1.10 r = +0.18 score = 0.198

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. high_prec_freq

z = +1.02 r = -0.19 score = 0.191

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. gvf_diff

z = +0.74 r = -0.24 score = 0.179

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. glim_2nd_class_frac

z = -1.07 r = +0.15 score = 0.164

a small secondary geological class means the subsurface is more uniform.
6. glim_1st_class_frac

z = +1.05 r = -0.15 score = 0.153

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. sand_frac

z = -1.13 r = +0.12 score = 0.132

low sand fraction reduces infiltration and may favor surface runoff.
8. low_prec_freq

z = +0.55 r = -0.22 score = 0.122

many dry days indicate a drier climate where streamflow depends on stored water.
9. q95

z = -0.39 r = +0.31 score = 0.119

low peak-flow values reflect a damped streamflow response with limited peaks.
10. elev_mean

z = -0.55 r = -0.20 score = 0.108

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07060710: North Sylamore Creek near Fifty Six, AR

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high gvf_diff, low runoff_ratio.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.83 r = -0.12 score = 0.339

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gvf_diff

z = +1.09 r = -0.24 score = 0.265

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. runoff_ratio

z = -0.77 r = +0.23 score = 0.178

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. high_prec_dur

z = -0.77 r = +0.23 score = 0.175

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. q95

z = -0.51 r = +0.31 score = 0.155

low peak-flow values reflect a damped streamflow response with limited peaks.
6. q_mean

z = -0.47 r = +0.28 score = 0.130

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. high_prec_freq

z = +0.69 r = -0.19 score = 0.129

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. gvf_max

z = +1.05 r = +0.12 score = 0.122

high maximum green vegetation fraction reflects dense canopy cover.
9. elev_mean

z = -0.62 r = -0.20 score = 0.121

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. slope_fdc

z = -0.55 r = +0.20 score = 0.110

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07066000: Jacks Fork at Eminence, MO

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high gvf_diff, high high_prec_freq, high carbonate_rocks_frac.

Top 10 driver attributes

1. gvf_diff

z = +1.05

r = -0.24

score = 0.255

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
2. high_prec_freq

z = +1.31

r = -0.19

score = 0.247

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. carbonate_rocks_frac

z = +1.98

r = -0.12

score = 0.238

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. glim_2nd_class_frac

z = +1.45

r = +0.15

score = 0.223

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. sand_frac

z = -1.36

r = +0.12

score = 0.159

low sand fraction reduces infiltration and may favor surface runoff.
6. low_prec_freq

z = +0.71

r = -0.22

score = 0.159

many dry days indicate a drier climate where streamflow depends on stored water.
7. soil_conductivity

z = -0.76

r = +0.15

score = 0.111

low soil conductivity restricts infiltration, often increasing surface runoff.
8. q95

z = -0.36

r = +0.31

score = 0.109

low peak-flow values reflect a damped streamflow response with limited peaks.
9. glim_1st_class_frac

z = -0.74

r = -0.15

score = 0.108

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. elev_mean

z = -0.53

r = -0.20

score = 0.103

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07067000: Current River at Van Buren, MO

Dominant forcing: PRCP Peak TE: 0.0085 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, high glim_2nd_class_frac.

Top 10 driver attributes

1. area_gages2

z = +2.09 r = +0.18 score = 0.377

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +2.05 r = +0.17 score = 0.349

a large geospatial drainage area integrates many sub-basin responses.
3. glim_2nd_class_frac

z = +2.13 r = +0.15 score = 0.327

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. gvf_diff

z = +1.10 r = -0.24 score = 0.268

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. carbonate_rocks_frac

z = +1.57 r = -0.12 score = 0.188

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. glim_1st_class_frac

z = -1.28 r = -0.15 score = 0.186

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. high_prec_freq

z = +0.92 r = -0.19 score = 0.173

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. sand_frac

z = -1.32 r = +0.12 score = 0.155

low sand fraction reduces infiltration and may favor surface runoff.
9. q5

z = +0.80 r = +0.18 score = 0.144

high low-flow values reflect strong baseflow and steady year-round streamflow.
10. low_prec_freq

z = +0.57 r = -0.22 score = 0.128

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07068000: Current River at Doniphan, MO

Dominant forcing: PRCP Peak TE: 0.0131 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, high glim_2nd_class_frac.

Top 10 driver attributes

1. area_gages2

z = +2.66 r = +0.18 score = 0.480

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +2.63 r = +0.17 score = 0.449

a large geospatial drainage area integrates many sub-basin responses.
3. glim_2nd_class_frac

z = +2.23 r = +0.15 score = 0.343

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. gvf_diff

z = +1.12 r = -0.24 score = 0.273

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
5. q5

z = +1.38 r = +0.18 score = 0.248

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. glim_1st_class_frac

z = -1.34 r = -0.15 score = 0.195

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. carbonate_rocks_frac

z = +1.37 r = -0.12 score = 0.164

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
8. sand_frac

z = -1.32 r = +0.12 score = 0.154

low sand fraction reduces infiltration and may favor surface runoff.
9. q95

z = -0.39 r = +0.31 score = 0.120

low peak-flow values reflect a damped streamflow response with limited peaks.
10. high_prec_dur

z = -0.53 r = +0.23 score = 0.119

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07071500: Eleven Point River near Bardley, MO

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, high gvf_diff, low q95.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +1.96 r = +0.15 score = 0.301

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. gvf_diff

z = +0.87 r = -0.24 score = 0.211

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. q95

z = -0.60 r = +0.31 score = 0.182

low peak-flow values reflect a damped streamflow response with limited peaks.
4. high_prec_freq

z = +0.93 r = -0.19 score = 0.175

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. glim_1st_class_frac

z = -1.10 r = -0.15 score = 0.160

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. carbonate_rocks_frac

z = +1.22 r = -0.12 score = 0.146

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
7. sand_frac

z = -1.23 r = +0.12 score = 0.144

low sand fraction reduces infiltration and may favor surface runoff.
8. area_gages2

z = +0.72 r = +0.18 score = 0.131

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
9. area_geospa_fabric

z = +0.71 r = +0.17 score = 0.122

a large geospatial drainage area integrates many sub-basin responses.
10. elev_mean

z = -0.61 r = -0.20 score = 0.118

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07083000: HALFMOON CREEK NEAR MALTA, CO.

Dominant forcing: SRAD Peak TE: 0.0094 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean

z = +2.39 r = +0.37 score = 0.879

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +3.43 r = +0.24 score = 0.828

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow

z = +2.62 r = +0.26 score = 0.669

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. gvf_diff

z = -1.14 r = -0.33 score = 0.372

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. high_prec_freq

z = -1.06 r = -0.33 score = 0.347

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. runoff_ratio

z = +0.79 r = +0.43 score = 0.338

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. glim_1st_class_frac

z = -1.39 r = -0.21 score = 0.298

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. gauge_lon

z = -0.65 r = -0.43 score = 0.283

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. root_depth_50

z = -1.90 r = -0.13 score = 0.238

shallow median rooting limits vegetation water access to surface soils.
10. clay_frac

z = -1.00 r = -0.22 score = 0.220

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07142300: RATTLESNAKE C NR MACKSVILLE, KS

Dominant forcing: PRCP Peak TE: 0.0017 Peak lag: 3 day(s)
Summary: PRCP-dominant at lag 3 day(s); top drivers consistent with high soil_depth_pelletier, low runoff_ratio, low p_mean.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.38 r = +0.16 score = 0.380

deep soils provide more storage and slower release of water to streamflow.
2. runoff_ratio

z = -1.61 r = +0.23 score = 0.373

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. p_mean

z = -1.03 r = +0.34 score = 0.354

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. p_seasonality

z = +1.30 r = -0.24 score = 0.317

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
5. low_prec_freq

z = +1.41 r = -0.22 score = 0.315

many dry days indicate a drier climate where streamflow depends on stored water.
6. q95

z = -1.01 r = +0.31 score = 0.309

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. slope_fdc

z = +1.42 r = +0.20 score = 0.286

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. q_mean

z = -0.96 r = +0.28 score = 0.264

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. aridity

z = +0.83 r = -0.28 score = 0.229

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07145700: SLATE C AT WELLINGTON, KS

Dominant forcing: PRCP Peak TE: 0.0082 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, high hfd_mean, high soil_depth_pelletier.

Top 10 driver attributes

1. slope_fdc

z = -1.65 r = +0.20 score = 0.332

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. hfd_mean

z = +1.24 r = -0.24 score = 0.295

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. soil_depth_pelletier

z = +1.79 r = +0.16 score = 0.286

deep soils provide more storage and slower release of water to streamflow.
4. frac_forest

z = -1.62 r = +0.17 score = 0.280

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. p_seasonality

z = +1.08 r = -0.24 score = 0.265

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. low_prec_freq

z = +1.06 r = -0.22 score = 0.238

many dry days indicate a drier climate where streamflow depends on stored water.
7. high_prec_freq

z = +1.16 r = -0.19 score = 0.218

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. p_mean

z = -0.53 r = +0.34 score = 0.182

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. q95

z = -0.55 r = +0.31 score = 0.168

low peak-flow values reflect a damped streamflow response with limited peaks.
10. q_mean

z = -0.60 r = +0.28 score = 0.166

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07148400: Salt Fork Arkansas River nr Alva, OK

Dominant forcing: PRCP Peak TE: 0.0082 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, low runoff_ratio, high low_prec_freq.

Top 10 driver attributes

1. p_mean z = -0.96 r = +0.34 score = 0.329

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. runoff_ratio z = -1.39 r = +0.23 score = 0.321

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. low_prec_freq z = +1.36 r = -0.22 score = 0.303

many dry days indicate a drier climate where streamflow depends on stored water.
4. frac_forest z = -1.71 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. q95 z = -0.94 r = +0.31 score = 0.288

low peak-flow values reflect a damped streamflow response with limited peaks.
6. p_seasonality z = +1.15 r = -0.24 score = 0.281

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. aridity z = +1.01 r = -0.28 score = 0.278

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. q_mean z = -0.89 r = +0.28 score = 0.247

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. high_prec_freq z = +1.13 r = -0.19 score = 0.212

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. glim_2nd_class_frac z = +1.37 r = +0.15 score = 0.210

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07149000: MEDICINE LODGE R NR KIOWA, KS

Dominant forcing: PRCP Peak TE: 0.0094 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, low p_mean, high low_prec_freq.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +2.32 r = +0.15 score = 0.356

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. p_mean

z = -0.91 r = +0.34 score = 0.313

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. low_prec_freq

z = +1.32 r = -0.22 score = 0.296

many dry days indicate a drier climate where streamflow depends on stored water.
4. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. runoff_ratio

z = -1.26 r = +0.23 score = 0.291

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. p_seasonality

z = +1.18 r = -0.24 score = 0.288

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
7. q95

z = -0.92 r = +0.31 score = 0.280

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.85 r = +0.28 score = 0.235

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. aridity

z = +0.76 r = -0.28 score = 0.209

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
10. high_prec_freq

z = +1.10 r = -0.19 score = 0.208

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07151500: CHIKASKIA R NR CORBIN, KS

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, low frac_forest, high p_seasonality.

Top 10 driver attributes

1. soil_depth_pelletier

z = +1.84 r = +0.16 score = 0.294

deep soils provide more storage and slower release of water to streamflow.
2. frac_forest

z = -1.69 r = +0.17 score = 0.291

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. p_seasonality

z = +1.08 r = -0.24 score = 0.263

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. q95

z = -0.79 r = +0.31 score = 0.242

low peak-flow values reflect a damped streamflow response with limited peaks.
5. p_mean

z = -0.67 r = +0.34 score = 0.230

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. low_prec_freq

z = +1.02 r = -0.22 score = 0.228

many dry days indicate a drier climate where streamflow depends on stored water.
7. runoff_ratio

z = -0.97 r = +0.23 score = 0.223

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q_mean

z = -0.73 r = +0.28 score = 0.201

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. high_prec_freq

z = +1.06 r = -0.19 score = 0.200

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. hfd_mean

z = +0.80 r = -0.24 score = 0.191

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07167500: OTTER C AT CLIMAX, KS

Dominant forcing: SRAD Peak TE: 0.0016 Peak lag: 24 day(s)
Summary: SRAD-dominant at lag 24 day(s); top drivers consistent with high p_seasonality, high clay_frac, low frac_forest.

Top 10 driver attributes

1. p_seasonality

z = +1.12 r = -0.46 score = 0.509

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
2. clay_frac

z = +2.30 r = -0.22 score = 0.508

high clay fraction reduces infiltration and increases surface runoff.
3. frac_forest

z = -1.67 r = +0.23 score = 0.393

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. baseflow_index

z = -1.72 r = +0.22 score = 0.377

a low baseflow index means streamflow responds quickly to recent rainfall.
5. gvf_diff

z = +0.94 r = -0.33 score = 0.308

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
6. sand_frac

z = -1.73 r = +0.18 score = 0.304

low sand fraction reduces infiltration and may favor surface runoff.
7. low_prec_freq

z = +0.92 r = -0.31 score = 0.287

many dry days indicate a drier climate where streamflow depends on stored water.
8. high_prec_freq

z = +0.87 r = -0.33 score = 0.285

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. slope_mean

z = -0.75 r = +0.37 score = 0.275

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. soil_porosity

z = +1.42 r = -0.17 score = 0.244

high soil porosity increases water storage capacity and dampens streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07180500: CEDAR C NR CEDAR POINT, KS

Dominant forcing: VP Peak TE: 0.0020 Peak lag: 7 day(s)
Summary: VP-dominant at lag 7 day(s); top drivers consistent with high low_prec_freq, high high_prec_freq, low frac_forest.

Top 10 driver attributes

1. low_prec_freq

z = +1.26 r = -0.50 score = 0.630

many dry days indicate a drier climate where streamflow depends on stored water.
2. high_prec_freq

z = +1.16 r = -0.47 score = 0.545

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. frac_forest

z = -1.71 r = +0.29 score = 0.498

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. clay_frac

z = +2.24 r = -0.22 score = 0.493

high clay fraction reduces infiltration and increases surface runoff.
5. p_seasonality

z = +1.18 r = -0.27 score = 0.317

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. q95

z = -0.63 r = +0.49 score = 0.312

low peak-flow values reflect a damped streamflow response with limited peaks.
7. runoff_ratio

z = -0.58 r = +0.50 score = 0.290

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. slope_mean

z = -0.88 r = +0.27 score = 0.239

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. q_mean

z = -0.54 r = +0.43 score = 0.233

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. q5

z = -0.58 r = +0.34 score = 0.197

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07184000: LIGHTNING C NR MCCUNE, KS

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low frac_forest, high p_seasonality.

Top 10 driver attributes

1. slope_fdc

z = -1.61 r = +0.20 score = 0.323

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. frac_forest

z = -1.70 r = +0.17 score = 0.292

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. p_seasonality

z = +0.85 r = -0.24 score = 0.207

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. stream_elas

z = +1.52 r = -0.12 score = 0.187

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. glim_2nd_class_frac

z = +1.15 r = +0.15 score = 0.177

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. hfd_mean

z = +0.70 r = -0.24 score = 0.167

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
7. glim_1st_class_frac

z = -1.06 r = -0.15 score = 0.154

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. sand_frac

z = -1.21 r = +0.12 score = 0.142

low sand fraction reduces infiltration and may favor surface runoff.
9. elev_mean

z = -0.61 r = -0.20 score = 0.119

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. q5

z = -0.64 r = +0.18 score = 0.115

low low-flow values reflect basins that nearly dry out during dry periods.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07195800: Flint Creek at Springtown, AR

Dominant forcing: SRAD Peak TE: 0.0018 Peak lag: 25 day(s)
Summary: SRAD-dominant at lag 25 day(s); top drivers consistent with high carbonate_rocks_frac, high high_prec_freq, high clay_frac.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.11 score = 0.380

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. high_prec_freq

z = +1.12 r = -0.33 score = 0.364

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. clay_frac

z = +1.27 r = -0.22 score = 0.280

high clay fraction reduces infiltration and increases surface runoff.
4. slope_mean

z = -0.76 r = +0.37 score = 0.279

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. frac_forest

z = -1.12 r = +0.23 score = 0.263

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. sand_frac

z = -1.34 r = +0.18 score = 0.236

low sand fraction reduces infiltration and may favor surface runoff.
7. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. low_prec_freq

z = +0.67 r = -0.31 score = 0.208

many dry days indicate a drier climate where streamflow depends on stored water.
9. soil_porosity

z = +1.21 r = -0.17 score = 0.208

high soil porosity increases water storage capacity and dampens streamflow response.
10. q95

z = -0.44 r = +0.43 score = 0.190

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07196900: Baron Fork at Dutch Mills, AR

Dominant forcing: SRAD Peak TE: 0.0025 Peak lag: 26 day(s)
Summary: SRAD-dominant at lag 26 day(s); top drivers consistent with high high_prec_freq, low high_prec_dur, high gvf_diff.

Top 10 driver attributes

1. high_prec_freq

z = +0.88 r = -0.33 score = 0.288

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. high_prec_dur

z = -0.80 r = +0.29 score = 0.231

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. gvf_diff

z = +0.65 r = -0.33 score = 0.212

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
4. gauge_lat

z = -0.65 r = +0.29 score = 0.189

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. low_prec_freq

z = +0.60 r = -0.31 score = 0.188

many dry days indicate a drier climate where streamflow depends on stored water.
6. q5

z = -0.57 r = +0.31 score = 0.179

low low-flow values reflect basins that nearly dry out during dry periods.
7. other_frac

z = +1.21 r = +0.14 score = 0.169

a large 'other' land fraction reflects mixed or atypical surface cover.
8. frac_snow

z = -0.65 r = +0.26 score = 0.165

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. baseflow_index

z = -0.74 r = +0.22 score = 0.162

a low baseflow index means streamflow responds quickly to recent rainfall.
10. soil_depth_pelletier

z = -0.61 r = -0.24 score = 0.146

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07197000: Baron Fork at Eldon, OK

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high gvf_diff, low high_prec_dur.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +2.21 r = -0.12 score = 0.264

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gvf_diff

z = +0.82 r = -0.24 score = 0.199

strong seasonal vegetation cycle is associated with deciduous forests and a clear ET seasonality.
3. high_prec_dur

z = -0.75 r = +0.23 score = 0.169

short heavy-rainfall events produce sharper but briefer runoff peaks.
4. high_prec_freq

z = +0.84 r = -0.19 score = 0.158

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. sand_frac

z = -1.16 r = +0.12 score = 0.136

low sand fraction reduces infiltration and may favor surface runoff.
6. aridity

z = -0.43 r = -0.28 score = 0.118

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
7. low_prec_freq

z = +0.52 r = -0.22 score = 0.116

many dry days indicate a drier climate where streamflow depends on stored water.
8. elev_mean

z = -0.52 r = -0.20 score = 0.102

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. soil_conductivity

z = -0.68 r = +0.15 score = 0.100

low soil conductivity restricts infiltration, often increasing surface runoff.
10. frac_snow

z = -0.67 r = -0.15 score = 0.098

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07208500: RAYADO CREEK NEAR CIMARRON, NM

Dominant forcing: SRAD Peak TE: 0.0038 Peak lag: 13 day(s)
Summary: SRAD-dominant at lag 13 day(s); top drivers consistent with high elev_mean, high high_prec_freq, high p_seasonality.

Top 10 driver attributes

1. elev_mean

z = +2.44

r = +0.24

score = 0.588

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. high_prec_freq

z = +1.73

r = -0.33

score = 0.565

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. p_seasonality

z = +1.12

r = -0.46

score = 0.512

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
4. runoff_ratio

z = -1.11

r = +0.43

score = 0.473

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. q95

z = -0.86

r = +0.43

score = 0.372

low peak-flow values reflect a damped streamflow response with limited peaks.
6. glim_1st_class_frac

z = -1.53

r = -0.21

score = 0.327

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. low_prec_freq

z = +1.05

r = -0.31

score = 0.327

many dry days indicate a drier climate where streamflow depends on stored water.
8. gvfdiff

z = -0.91

r = -0.33

score = 0.296

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. q_mean

z = -0.83

r = +0.35

score = 0.293

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. glim_2nd_class_frac

z = +1.67

r = +0.18

score = 0.293

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07226500: UTE CREEK NEAR LOGAN, NM

Dominant forcing: PRCP Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, high hfd_mean, high aridity.

Top 10 driver attributes

1. zero_q_freq

z = +6.63 r = -0.13 score = 0.886

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. hfd_mean

z = +2.81 r = -0.24 score = 0.667

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
3. aridity

z = +2.38 r = -0.28 score = 0.657

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. p_mean

z = -1.55 r = +0.34 score = 0.533

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. slope_fdc

z = -2.44 r = +0.20 score = 0.490

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. area_gages2

z = +2.61 r = +0.18 score = 0.472

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
7. area_geospa_fabric

z = +2.60 r = +0.17 score = 0.442

a large geospatial drainage area integrates many sub-basin responses.
8. p_seasonality

z = +1.77 r = -0.24 score = 0.431

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. stream_elas

z = -3.16 r = -0.12 score = 0.391

low streamflow elasticity means streamflow is buffered against rainfall variability.
10. runoff_ratio

z = -1.64 r = +0.23 score = 0.379

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07261000: Cadron Creek near Guy, AR

Dominant forcing: SRAD Peak TE: 0.0052 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low slope_mean, low frac_forest, high high_prec_freq.

Top 10 driver attributes

1. slope_mean

z = -0.68 r = +0.37 score = 0.248

gentle terrain slows surface runoff and produces a smoother streamflow signal.
2. frac_forest

z = -1.04 r = +0.23 score = 0.243

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. high_prec_freq

z = +0.72 r = -0.33 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. high_prec_dur

z = -0.79 r = +0.29 score = 0.229

short heavy-rainfall events produce sharper but briefer runoff peaks.
5. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. gauge_lat

z = -0.76 r = +0.29 score = 0.222

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. q5

z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.
8. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
9. frac_snow

z = -0.69 r = +0.26 score = 0.176

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. elev_mean

z = -0.71 r = +0.24 score = 0.172

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07263295: Maumelle River at Williams Junction, AR

Dominant forcing: SRAD Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high high_prec_freq, high root_depth_50, high clay_frac.

Top 10 driver attributes

1. high_prec_freq

z = +0.92 r = -0.33 score = 0.299

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. root_depth_50

z = +2.30 r = -0.13 score = 0.288

deep median rooting allows vegetation to access stored soil water during dry periods.
3. clay_frac

z = +1.25 r = -0.22 score = 0.276

high clay fraction reduces infiltration and increases surface runoff.
4. baseflow_index

z = -1.20 r = +0.22 score = 0.263

a low baseflow index means streamflow responds quickly to recent rainfall.
5. gauge_lat

z = -0.84 r = +0.29 score = 0.246

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. zero_q_freq

z = +0.91 r = -0.23 score = 0.205

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
8. q5

z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.
9. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
10. frac_snow

z = -0.72 r = +0.26 score = 0.184

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07290650: BAYOU PIERRE NR WILLOWS, MS

Dominant forcing: PRCP Peak TE: 0.0067 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high soil_depth_pelletier, low gauge_lat.

Top 10 driver attributes

1. p_mean

z = +0.64 r = +0.34 score = 0.219

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. soil_depth_pelletier

z = +1.15 r = +0.16 score = 0.184

deep soils provide more storage and slower release of water to streamflow.
3. gauge_lat

z = -1.39 r = +0.13 score = 0.175

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. elev_mean

z = -0.83 r = -0.20 score = 0.162

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. hfd_mean

z = -0.63 r = -0.24 score = 0.150

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. pet_mean

z = +1.15 r = -0.12 score = 0.140

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
9. frac_snow

z = -0.85 r = -0.15 score = 0.125

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. slope_fdc

z = -0.61 r = +0.20 score = 0.122

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07291000: HOMOCHITTO RIVER AT EDDICETON, MS

Dominant forcing: SRAD Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, low slope_mean, low gvf_diff.

Top 10 driver attributes

1. gauge_lat

z = -1.49 r = +0.29 score = 0.436

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. slope_mean

z = -0.82 r = +0.37 score = 0.301

gentle terrain slows surface runoff and produces a smoother streamflow signal.
3. gvf_diff

z = -0.84 r = -0.33 score = 0.274

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_pelletier

z = +1.13 r = -0.24 score = 0.270

deep soils provide more storage and slower release of water to streamflow.
5. root_depth_50

z = +2.00 r = -0.13 score = 0.251

deep median rooting allows vegetation to access stored soil water during dry periods.
6. high_prec_freq

z = +0.75 r = -0.33 score = 0.245

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. frac_snow

z = -0.86 r = +0.26 score = 0.219

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. frac_forest

z = +0.91 r = +0.23 score = 0.213

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. elev_mean

z = -0.80 r = +0.24 score = 0.193

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07292500: HOMOCHITTO RIVER AT ROSETTA, MS

Dominant forcing: SRAD Peak TE: 0.0046 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, high stream_elas, low slope_mean.

Top 10 driver attributes

1. gauge_lat

z = -1.52 r = +0.29 score = 0.446

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. stream_elas

z = +1.38 r = -0.26 score = 0.358

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. slope_mean

z = -0.83 r = +0.37 score = 0.305

gentle terrain slows surface runoff and produces a smoother streamflow signal.
4. gvf_diff

z = -0.93 r = -0.33 score = 0.303

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. root_depth_50

z = +2.05 r = -0.13 score = 0.257

deep median rooting allows vegetation to access stored soil water during dry periods.
6. soil_depth_pelletier

z = +1.01 r = -0.24 score = 0.242

deep soils provide more storage and slower release of water to streamflow.
7. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. frac_snow

z = -0.86 r = +0.26 score = 0.220

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. high_prec_freq

z = +0.63 r = -0.33 score = 0.206

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. elev_mean

z = -0.82 r = +0.24 score = 0.198

low elevation is associated with warmer temperatures and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07295000: BUFFALO RIVER NR WOODVILLE, MS

Dominant forcing: Tair Peak TE: 0.0038 Peak lag: 26 day(s)
Summary: Tair-dominant at lag 26 day(s); top drivers consistent with low gauge_lat, high high_prec_freq, high pet_mean.

Top 10 driver attributes

1. gauge_lat z = -1.54 r = +0.51 score = 0.778

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. high_prec_freq z = +0.95 r = -0.49 score = 0.467

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. pet_mean z = +1.38 r = -0.33 score = 0.457

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. frac_snow z = -0.86 r = +0.46 score = 0.399

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
5. low_prec_freq z = +0.70 r = -0.50 score = 0.355

many dry days indicate a drier climate where streamflow depends on stored water.
6. slope_mean z = -0.84 r = +0.41 score = 0.340

gentle terrain slows surface runoff and produces a smoother streamflow signal.
7. frac_forest z = +0.81 r = +0.35 score = 0.279

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. soil_depth_pelletier z = +1.00 r = -0.26 score = 0.255

deep soils provide more storage and slower release of water to streamflow.
9. elev_mean z = -0.84 r = +0.30 score = 0.251

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. baseflow_index z = -0.99 r = +0.22 score = 0.222

a low baseflow index means streamflow responds quickly to recent rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07299670: Groesbeck Ck at SH 6 nr Quanah, TX

Dominant forcing: PRCP Peak TE: 0.0047 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, high low_prec_freq.

Top 10 driver attributes

1. aridity

z = +2.08

r = -0.28

score = 0.574

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.05

r = +0.34

score = 0.363

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. low_prec_freq

z = +1.55

r = -0.22

score = 0.348

many dry days indicate a drier climate where streamflow depends on stored water.
4. runoff_ratio

z = -1.41

r = +0.23

score = 0.326

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. q95

z = -0.99

r = +0.31

score = 0.303

low peak-flow values reflect a damped streamflow response with limited peaks.
6. pet_mean

z = +2.44

r = -0.12

score = 0.298

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. frac_forest

z = -1.71

r = +0.17

score = 0.295

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. gvf_diff

z = -1.04

r = -0.24

score = 0.253

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. q_mean

z = -0.90

r = +0.28

score = 0.250

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. gvf_max

z = -2.07

r = +0.12

score = 0.239

low maximum green vegetation fraction reflects sparse canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07301410: Sweetwater Ck nr Kelton, TX

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_mean, high low_prec_freq, low runoff_ratio.

Top 10 driver attributes

1. p_mean

z = -1.11

r = +0.34

score = 0.383

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
2. low_prec_freq

z = +1.65

r = -0.22

score = 0.370

many dry days indicate a drier climate where streamflow depends on stored water.
3. runoff_ratio

z = -1.56

r = +0.23

score = 0.359

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. q95

z = -1.00

r = +0.31

score = 0.306

low peak-flow values reflect a damped streamflow response with limited peaks.
5. aridity

z = +1.08

r = -0.28

score = 0.298

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
6. frac_forest

z = -1.72

r = +0.17

score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. p_seasonality

z = +1.11

r = -0.24

score = 0.272

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. q_mean

z = -0.94

r = +0.28

score = 0.261

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. sand_frac

z = +1.79

r = +0.12

score = 0.209

high sand fraction promotes fast infiltration and subsurface drainage.
10. gvf_max

z = -1.77

r = +0.12

score = 0.205

low maximum green vegetation fraction reflects sparse canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07301500: North Fork Red River near Carter, OK

Dominant forcing: PRCP Peak TE: 0.0027 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high area_gages2, high area_geospa_fabric, low p_mean.

Top 10 driver attributes

1. area_gages2

z = +3.58 r = +0.18 score = 0.646

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
2. area_geospa_fabric

z = +3.55 r = +0.17 score = 0.605

a large geospatial drainage area integrates many sub-basin responses.
3. p_mean

z = -1.08 r = +0.34 score = 0.374

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. runoff_ratio

z = -1.54 r = +0.23 score = 0.356

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. aridity

z = +1.20 r = -0.28 score = 0.331

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
6. q95

z = -1.00 r = +0.31 score = 0.305

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_forest

z = -1.71 r = +0.17 score = 0.295

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. p_seasonality

z = +1.13 r = -0.24 score = 0.275

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. low_prec_freq

z = +1.22 r = -0.22 score = 0.274

many dry days indicate a drier climate where streamflow depends on stored water.
10. q_mean

z = -0.94 r = +0.28 score = 0.260

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07315200: E Fk Little Wichita Rv nr Henrietta, TX

Dominant forcing: PRCP Peak TE: 0.0031 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, high zero_q_freq, low runoff_ratio.

Top 10 driver attributes

1. slope_fdc

z = -2.35 r = +0.20 score = 0.471

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. zero_q_freq

z = +3.34 r = -0.13 score = 0.447

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. runoff_ratio

z = -1.38 r = +0.23 score = 0.318

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. stream_elas

z = +2.48 r = -0.12 score = 0.306

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
5. q95

z = -0.97 r = +0.31 score = 0.297

low peak-flow values reflect a damped streamflow response with limited peaks.
6. aridity

z = +1.06 r = -0.28 score = 0.292

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
7. low_prec_freq

z = +1.30 r = -0.22 score = 0.290

many dry days indicate a drier climate where streamflow depends on stored water.
8. frac_forest

z = -1.47 r = +0.17 score = 0.253

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. p_mean

z = -0.73 r = +0.34 score = 0.251

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. high_prec_freq

z = +1.29 r = -0.19 score = 0.243

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07315700: Mud Creek near Courtney, OK

Dominant forcing: PRCP Peak TE: 0.0062 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low frac_forest, high low_prec_freq.

Top 10 driver attributes

1. slope_fdc

z = -1.56 r = +0.20 score = 0.314

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. frac_forest

z = -1.60 r = +0.17 score = 0.276

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. low_prec_freq

z = +1.15 r = -0.22 score = 0.257

many dry days indicate a drier climate where streamflow depends on stored water.
4. runoff_ratio

z = -1.05 r = +0.23 score = 0.241

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. q95

z = -0.70 r = +0.31 score = 0.213

low peak-flow values reflect a damped streamflow response with limited peaks.
6. q_mean

z = -0.74 r = +0.28 score = 0.204

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
7. high_prec_freq

z = +1.02 r = -0.19 score = 0.191

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. p_mean

z = -0.54 r = +0.34 score = 0.185

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. hfd_mean

z = +0.69 r = -0.24 score = 0.163

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. stream_elas

z = +1.17 r = -0.12 score = 0.145

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07335700: Kiamichi River near Big Cedar, OK

Dominant forcing: SRAD Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high high_prec_freq, high clay_frac, low gauge_lat.

Top 10 driver attributes

1. high_prec_freq

z = +1.04 r = -0.33 score = 0.339

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. clay_frac

z = +1.28 r = -0.22 score = 0.282

high clay fraction reduces infiltration and increases surface runoff.
3. gauge_lat

z = -0.88 r = +0.29 score = 0.259

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. baseflow_index

z = -1.05 r = +0.22 score = 0.229

a low baseflow index means streamflow responds quickly to recent rainfall.
5. low_prec_freq

z = +0.71 r = -0.31 score = 0.222

many dry days indicate a drier climate where streamflow depends on stored water.
6. frac_forest

z = +0.89 r = +0.23 score = 0.210

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. lai_diff

z = +1.50 r = -0.14 score = 0.209

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
8. runoff_ratio

z = +0.48 r = +0.43 score = 0.207

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. q5

z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.
10. q95

z = +0.44 r = +0.43 score = 0.192

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07340300: Cossatot River near Vandervoort, AR

Dominant forcing: SRAD Peak TE: 0.0056 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, high glim_1st_class_frac, high high_prec_freq.

Top 10 driver attributes

1. gauge_lat

z = -0.93 r = +0.29 score = 0.274

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
3. high_prec_freq

z = +0.66 r = -0.33 score = 0.217

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. baseflow_index

z = -0.96 r = +0.22 score = 0.209

a low baseflow index means streamflow responds quickly to recent rainfall.
5. clay_frac

z = +0.95 r = -0.22 score = 0.208

high clay fraction reduces infiltration and increases surface runoff.
6. high_prec_dur

z = -0.72 r = +0.29 score = 0.207

short heavy-rainfall events produce sharper but briefer runoff peaks.
7. lai_diff

z = +1.45 r = -0.14 score = 0.203

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
8. frac_forest

z = +0.86 r = +0.23 score = 0.201

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. slope_mean

z = -0.54 r = +0.37 score = 0.199

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07346045: Black Cypress Bayou at Jefferson, TX

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 3 day(s)
Summary: PRCP-dominant at lag 3 day(s); top drivers consistent with high slope_fdc, high high_prec_freq, high sand_frac.

Top 10 driver attributes

1. slope_fdc

z = +1.77

r = +0.20

score = 0.356

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. high_prec_freq

z = +1.07

r = -0.19

score = 0.202

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. sand_frac

z = +1.42

r = +0.12

score = 0.166

high sand fraction promotes fast infiltration and subsurface drainage.
4. hfd_mean

z = -0.69

r = -0.24

score = 0.165

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. elev_mean

z = -0.83

r = -0.20

score = 0.163

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. gauge_lat

z = -1.24

r = +0.13

score = 0.157

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. low_prec_freq

z = +0.64

r = -0.22

score = 0.142

many dry days indicate a drier climate where streamflow depends on stored water.
8. stream_elas

z = +1.10

r = -0.12

score = 0.135

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
9. runoff_ratio

z = -0.58

r = +0.23

score = 0.134

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. frac_snow

z = -0.79

r = -0.15

score = 0.116

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07359610: Caddo River near Caddo Gap, AR

Dominant forcing: SRAD Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, low high_prec_dur, high clay_frac.

Top 10 driver attributes

1. gauge_lat

z = -0.93 r = +0.29 score = 0.274

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. high_prec_dur

z = -0.82 r = +0.29 score = 0.236

short heavy-rainfall events produce sharper but briefer runoff peaks.
3. clay_frac

z = +1.00 r = -0.22 score = 0.220

high clay fraction reduces infiltration and increases surface runoff.
4. high_prec_freq

z = +0.62 r = -0.33 score = 0.202

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. frac_forest

z = +0.82 r = +0.23 score = 0.193

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
6. frac_snow

z = -0.73 r = +0.26 score = 0.185

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
7. lai_diff

z = +1.14 r = -0.14 score = 0.159

large leaf area seasonality reflects strong vegetation phenology, which drives a seasonal ET cycle.
8. slope_mean

z = -0.42 r = +0.37 score = 0.154

gentle terrain slows surface runoff and produces a smoother streamflow signal.
9. p_mean

z = +0.73 r = +0.20 score = 0.146

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. soil_depth_pelletier

z = -0.60 r = -0.24 score = 0.145

shallow soils limit storage and produce a quicker streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07362100: SMACKOVER CREEK NEAR SMACKOVER, ARK.

Dominant forcing: PRCP Peak TE: 0.0069 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high root_depth_99, high stream_elas, low elev_mean.

Top 10 driver attributes

1. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
2. stream_elas

z = +1.50 r = -0.12 score = 0.185

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. elev_mean

z = -0.86 r = -0.20 score = 0.168

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
4. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
5. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
6. frac_forest

z = +0.89 r = +0.17 score = 0.153

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
7. gauge_lat

z = -1.13 r = +0.13 score = 0.142

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. p_mean

z = +0.39 r = +0.34 score = 0.134

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
9. hfd_mean

z = -0.51 r = -0.24 score = 0.121

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. high_prec_freq

z = +0.63 r = -0.19 score = 0.119

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07362587: Alum Fork Saline River near Reform, AR

Dominant forcing: SRAD Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low baseflow_index, high clay_frac, high root_depth_50.

Top 10 driver attributes

1. baseflow_index

z = -1.48 r = +0.22 score = 0.324

a low baseflow index means streamflow responds quickly to recent rainfall.
2. clay_frac

z = +1.43 r = -0.22 score = 0.316

high clay fraction reduces infiltration and increases surface runoff.
3. root_depth_50

z = +2.30 r = -0.13 score = 0.288

deep median rooting allows vegetation to access stored soil water during dry periods.
4. gauge_lat

z = -0.85 r = +0.29 score = 0.250

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. high_prec_freq

z = +0.72 r = -0.33 score = 0.235

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. frac_forest

z = +0.96 r = +0.23 score = 0.225

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. q5

z = -0.64 r = +0.31 score = 0.199

low low-flow values reflect basins that nearly dry out during dry periods.
9. glim_2nd_class_frac

z = -1.08 r = +0.18 score = 0.190

a small secondary geological class means the subsurface is more uniform.
10. high_prec_dur

z = -0.63 r = +0.29 score = 0.183

short heavy-rainfall events produce sharper but briefer runoff peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07373000: Big Creek at Pollock, LA

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high p_mean, low hfd_mean.

Top 10 driver attributes

1. soil_depth_pelletier

z = +1.66 r = +0.16 score = 0.266

deep soils provide more storage and slower release of water to streamflow.
2. p_mean

z = +0.70 r = +0.34 score = 0.241

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. hfd_mean

z = -1.00 r = -0.24 score = 0.238

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
4. gvf_diff

z = -0.88 r = -0.24 score = 0.213

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. root_depth_99

z = +1.89 r = +0.10 score = 0.191

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. gauge_lat

z = -1.48 r = +0.13 score = 0.187

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. elev_mean

z = -0.89 r = -0.20 score = 0.173

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.165

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. pet_mean

z = +1.13 r = -0.12 score = 0.138

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07375000: Tchefuncta River near Folsom, LA

Dominant forcing: PRCP Peak TE: 0.0059 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high p_mean, low slope_fdc.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.15 r = +0.16 score = 0.343

deep soils provide more storage and slower release of water to streamflow.
2. p_mean

z = +0.96 r = +0.34 score = 0.331

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. slope_fdc

z = -1.11 r = +0.20 score = 0.223

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. gauge_lat

z = -1.66 r = +0.13 score = 0.209

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. pet_mean

z = +1.67 r = -0.12 score = 0.204

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
6. gvf_diff

z = -0.83 r = -0.24 score = 0.203

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. elev_mean

z = -0.90 r = -0.20 score = 0.175

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. hfd_mean

z = -0.58 r = -0.24 score = 0.137

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 07376000: Tickfaw River at Holden, LA

Dominant forcing: PRCP Peak TE: 0.0105 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high p_mean, high soil_depth_pelletier, high pet_mean.

Top 10 driver attributes

1. p_mean

z = +0.94

r = +0.34

score = 0.325

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. soil_depth_pelletier

z = +2.00

r = +0.16

score = 0.319

deep soils provide more storage and slower release of water to streamflow.
3. pet_mean

z = +2.07

r = -0.12

score = 0.252

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. gvf_diff

z = -0.99

r = -0.24

score = 0.241

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. slope_fdc

z = -1.06

r = +0.20

score = 0.214

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. gauge_lat

z = -1.68

r = +0.13

score = 0.212

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. elev_mean

z = -0.88

r = -0.20

score = 0.173

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. hfd_mean

z = -0.67

r = -0.24

score = 0.160

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08013000: CALCASIEU RIVER NR GLENMORA, LA

Dominant forcing: PRCP Peak TE: 0.0075 Peak lag: 3 day(s)
Summary: PRCP-dominant at lag 3 day(s); top drivers consistent with high soil_depth_pelletier, low hfd_mean, low gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.05 r = +0.16 score = 0.328

deep soils provide more storage and slower release of water to streamflow.
2. hfd_mean

z = -1.24 r = -0.24 score = 0.296

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. gvf_diff

z = -1.09 r = -0.24 score = 0.265

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. p_mean

z = +0.75 r = +0.34 score = 0.259

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gauge_lat

z = -1.58 r = +0.13 score = 0.200

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. elev_mean

z = -0.87 r = -0.20 score = 0.170

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. aridity

z = -0.51 r = -0.28 score = 0.140

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. high_prec_dur

z = -0.61 r = +0.23 score = 0.138

short heavy-rainfall events produce sharper but briefer runoff peaks.
9. root_depth_99

z = +1.36 r = +0.10 score = 0.137

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. frac_forest

z = +0.74 r = +0.17 score = 0.127

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08014500: Whiskey Chitto Creek Near Oberlin, LA

Dominant forcing: PRCP Peak TE: 0.0077 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high soil_depth_pelletier, high p_mean, low gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.08 r = +0.16 score = 0.332

deep soils provide more storage and slower release of water to streamflow.
2. p_mean

z = +0.86 r = +0.34 score = 0.297

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. gvf_diff

z = -1.20 r = -0.24 score = 0.291

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. hfd_mean

z = -1.13 r = -0.24 score = 0.269

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. gauge_lat

z = -1.64 r = +0.13 score = 0.207

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. elev_mean

z = -0.89 r = -0.20 score = 0.174

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
7. high_prec_dur

z = -0.69 r = +0.23 score = 0.157

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. aridity

z = -0.53 r = -0.28 score = 0.146

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. root_depth_99

z = +1.39 r = +0.10 score = 0.140

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. frac_snow

z = -0.86 r = -0.15 score = 0.126

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08023080: Bayou Grand Cane near Stanley, LA

Dominant forcing: PRCP Peak TE: 0.0154 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low hfd_mean, high p_mean.

Top 10 driver attributes

1. slope_fdc

z = -1.51

r = +0.20

score = 0.304

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. hfd_mean

z = -1.02

r = -0.24

score = 0.243

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
3. p_mean

z = +0.54

r = +0.34

score = 0.187

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. gauge_lat

z = -1.40

r = +0.13

score = 0.176

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. elev_mean

z = -0.86

r = -0.20

score = 0.167

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
7. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
8. gvf_diff

z = -0.58

r = -0.24

score = 0.140

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. stream_elas

z = +1.09

r = -0.12

score = 0.135

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. zero_q_freq

z = +0.95

r = -0.13

score = 0.127

frequent zero-flow days indicate an ephemeral or highly water-limited basin.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08025500: Bayou Toro near Toro, LA

Dominant forcing: PRCP Peak TE: 0.0062 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, low gvf_diff, low gauge_lat.

Top 10 driver attributes

1. hfd_mean

z = -1.16

r = -0.24

score = 0.276

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. gvf_diff

z = -1.12

r = -0.24

score = 0.273

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. gauge_lat

z = -1.52

r = +0.13

score = 0.192

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. p_mean

z = +0.52

r = +0.34

score = 0.180

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. elev_mean

z = -0.84

r = -0.20

score = 0.165

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
6. high_prec_freq

z = +0.74

r = -0.19

score = 0.140

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. high_prec_dur

z = -0.61

r = +0.23

score = 0.138

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. frac_forest

z = +0.78

r = +0.17

score = 0.135

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. root_depth_99

z = +1.28

r = +0.10

score = 0.130

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. frac_snow

z = -0.86

r = -0.15

score = 0.126

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08029500: Big Cow Ck nr Newton, TX

Dominant forcing: SRAD Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, high soil_depth_pelletier, low gvf_diff.

Top 10 driver attributes

1. gauge_lat

z = -1.62 r = +0.29 score = 0.474

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. soil_depth_pelletier

z = +1.47 r = -0.24 score = 0.353

deep soils provide more storage and slower release of water to streamflow.
3. gvf_diff

z = -1.05 r = -0.33 score = 0.344

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. slope_mean

z = -0.87 r = +0.37 score = 0.321

gentle terrain slows surface runoff and produces a smoother streamflow signal.
5. sand_frac

z = +1.35 r = +0.18 score = 0.237

high sand fraction promotes fast infiltration and subsurface drainage.
6. glim_1st_class_frac

z = +1.06 r = -0.21 score = 0.227

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
7. high_prec_dur

z = -0.79 r = +0.29 score = 0.227

short heavy-rainfall events produce sharper but briefer runoff peaks.
8. frac_snow

z = -0.86 r = +0.26 score = 0.220

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
9. elev_mean

z = -0.87 r = +0.24 score = 0.209

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
10. runoff_ratio

z = -0.49 r = +0.43 score = 0.209

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08050800: Timber Ck nr Collinsville, TX

Dominant forcing: VP Peak TE: 0.0016 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, high stream_elas, high high_prec_freq.

Top 10 driver attributes

1. zero_q_freq

z = +3.76 r = -0.30 score = 1.122

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. stream_elas

z = +2.86 r = -0.36 score = 1.033

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. high_prec_freq

z = +1.20 r = -0.47 score = 0.566

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. low_prec_freq

z = +1.13 r = -0.50 score = 0.565

many dry days indicate a drier climate where streamflow depends on stored water.
5. runoff_ratio

z = -0.90 r = +0.50 score = 0.447

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. slope_fdc

z = -1.76 r = +0.24 score = 0.420

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. frac_forest

z = -1.37 r = +0.29 score = 0.399

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. baseflow_index

z = -2.28 r = +0.17 score = 0.396

a low baseflow index means streamflow responds quickly to recent rainfall.
9. gauge_lat

z = -1.09 r = +0.35 score = 0.385

southern latitude is associated with warmer climates and rainfall-driven streamflow.
10. q95

z = -0.78 r = +0.49 score = 0.384

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08066200: Long King Ck at Livingston, TX

Dominant forcing: PRCP Peak TE: 0.0016 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, low slope_fdc, high pet_mean.

Top 10 driver attributes

1. hfd_mean

z = -1.41

r = -0.24

score = 0.336

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. slope_fdc

z = -1.38

r = +0.20

score = 0.278

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. pet_mean

z = +1.98

r = -0.12

score = 0.241

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. gvf_diff

z = -0.98

r = -0.24

score = 0.237

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. soil_depth_pelletier

z = +1.30

r = +0.16

score = 0.208

deep soils provide more storage and slower release of water to streamflow.
6. gauge_lat

z = -1.64

r = +0.13

score = 0.207

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. stream_elas

z = +1.36

r = -0.12

score = 0.168

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. elev_mean

z = -0.86

r = -0.20

score = 0.167

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. high_prec_freq

z = +0.76

r = -0.19

score = 0.144

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. runoff_ratio

z = -0.62

r = +0.23

score = 0.143

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08066300: Menard Ck nr Rye, TX

Dominant forcing: PRCP Peak TE: 0.0048 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high soil_depth_pelletier, high pet_mean, low gvf_diff.

Top 10 driver attributes

1. soil_depth_pelletier

z = +2.06 r = +0.16 score = 0.329

deep soils provide more storage and slower release of water to streamflow.
2. pet_mean

z = +2.53 r = -0.12 score = 0.308

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
3. gvf_diff

z = -1.25 r = -0.24 score = 0.304

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. hfd_mean

z = -1.09 r = -0.24 score = 0.259

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. gauge_lat

z = -1.68 r = +0.13 score = 0.212

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. p_mean

z = +0.51 r = +0.34 score = 0.175

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
7. elev_mean

z = -0.88 r = -0.20 score = 0.172

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
8. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. runoff_ratio

z = -0.66 r = +0.23 score = 0.152

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08070000: E Fk San Jacinto Rv nr Cleveland, TX

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, high soil_depth_pelletier, low gvf_diff.

Top 10 driver attributes

1. hfd_mean

z = -1.40

r = -0.24

score = 0.333

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. soil_depth_pelletier

z = +1.57

r = +0.16

score = 0.251

deep soils provide more storage and slower release of water to streamflow.
3. gvf_diff

z = -0.95

r = -0.24

score = 0.232

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lat

z = -1.71

r = +0.13

score = 0.216

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. pet_mean

z = +1.61

r = -0.12

score = 0.196

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
6. slope_fdc

z = -0.94

r = +0.20

score = 0.189

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. stream_elas

z = +1.51

r = -0.12

score = 0.186

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. elev_mean

z = -0.86

r = -0.20

score = 0.168

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. runoff_ratio

z = -0.72

r = +0.23

score = 0.166

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08070200: E Fk San Jacinto Rv nr New Caney, TX

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low hfd_mean, high soil_depth_pelletier, high pet_mean.

Top 10 driver attributes

1. hfd_mean

z = -1.29

r = -0.24

score = 0.306

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
2. soil_depth_pelletier

z = +1.70

r = +0.16

score = 0.272

deep soils provide more storage and slower release of water to streamflow.
3. pet_mean

z = +2.23

r = -0.12

score = 0.272

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
4. gvf_diff

z = -0.91

r = -0.24

score = 0.222

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. gauge_lat

z = -1.75

r = +0.13

score = 0.221

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. stream_elas

z = +1.51

r = -0.12

score = 0.186

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. runoff_ratio

z = -0.74

r = +0.23

score = 0.171

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. elev_mean

z = -0.88

r = -0.20

score = 0.171

low elevation is associated with warmer temperatures and rainfall-driven streamflow.
9. glim_2nd_class_frac

z = -1.08

r = +0.15

score = 0.166

a small secondary geological class means the subsurface is more uniform.
10. glim_1st_class_frac

z = +1.06

r = -0.15

score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08079600: DMF Brazos Rv at Justiceburg, TX

Dominant forcing: PRCP Peak TE: 0.0013 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high zero_q_freq, low slope_fdc.

Top 10 driver attributes

1. aridity

z = +2.52

r = -0.28

score = 0.696

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. zero_q_freq

z = +4.59

r = -0.13

score = 0.614

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. slope_fdc

z = -2.42

r = +0.20

score = 0.486

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. p_mean

z = -1.34

r = +0.34

score = 0.460

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. hfd_mean

z = +1.75

r = -0.24

score = 0.415

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
6. runoff_ratio

z = -1.59

r = +0.23

score = 0.367

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. low_prec_freq

z = +1.54

r = -0.22

score = 0.343

many dry days indicate a drier climate where streamflow depends on stored water.
8. q95

z = -1.01

r = +0.31

score = 0.309

low peak-flow values reflect a damped streamflow response with limited peaks.
9. soil_depth_pelletier

z = +1.86

r = +0.16

score = 0.297

deep soils provide more storage and slower release of water to streamflow.
10. frac_forest

z = -1.72

r = +0.17

score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08082700: Millers Ck nr Munday, TX

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, low slope_fdc, high aridity.

Top 10 driver attributes

1. zero_q_freq

z = +5.98 r = -0.13 score = 0.799

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. slope_fdc

z = -2.44 r = +0.20 score = 0.490

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. aridity

z = +1.53 r = -0.28 score = 0.423

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. hfd_mean

z = +1.76 r = -0.24 score = 0.418

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
5. low_prec_freq

z = +1.66 r = -0.22 score = 0.372

many dry days indicate a drier climate where streamflow depends on stored water.
6. p_mean

z = -1.01 r = +0.34 score = 0.349

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. runoff_ratio

z = -1.49 r = +0.23 score = 0.345

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. gvf_diff

z = -1.35 r = -0.24 score = 0.329

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. q95

z = -1.01 r = +0.31 score = 0.307

low peak-flow values reflect a damped streamflow response with limited peaks.
10. frac_forest

z = -1.72 r = +0.17 score = 0.296

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08086212: Hubbard Ck bl Albany, TX

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low runoff_ratio, low p_mean.

Top 10 driver attributes

1. slope_fdc

z = -2.14 r = +0.20 score = 0.430

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. runoff_ratio

z = -1.46 r = +0.23 score = 0.337

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. p_mean

z = -0.96 r = +0.34 score = 0.330

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. low_prec_freq

z = +1.44 r = -0.22 score = 0.322

many dry days indicate a drier climate where streamflow depends on stored water.
5. q95

z = -0.99 r = +0.31 score = 0.302

low peak-flow values reflect a damped streamflow response with limited peaks.
6. frac_forest

z = -1.71 r = +0.17 score = 0.294

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. aridity

z = +1.06 r = -0.28 score = 0.293

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. zero_q_freq

z = +2.07 r = -0.13 score = 0.276

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
9. hfd_mean

z = +1.09 r = -0.24 score = 0.260

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. q_mean

z = -0.91 r = +0.28 score = 0.252

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08086290: Big Sandy Ck abv Breckenridge, TX

Dominant forcing: PRCP Peak TE: 0.0013 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low runoff_ratio, high zero_q_freq.

Top 10 driver attributes

1. slope_fdc

z = -2.38 r = +0.20 score = 0.479

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. runoff_ratio

z = -1.44 r = +0.23 score = 0.333

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. zero_q_freq

z = +2.49 r = -0.13 score = 0.332

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
4. low_prec_freq

z = +1.47 r = -0.22 score = 0.328

many dry days indicate a drier climate where streamflow depends on stored water.
5. hfd_mean

z = +1.35 r = -0.24 score = 0.321

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
6. aridity

z = +1.16 r = -0.28 score = 0.321

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
7. p_mean

z = -0.88 r = +0.34 score = 0.303

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. q95

z = -0.99 r = +0.31 score = 0.302

low peak-flow values reflect a damped streamflow response with limited peaks.
9. frac_forest

z = -1.70 r = +0.17 score = 0.293

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. q_mean

z = -0.90 r = +0.28 score = 0.250

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08101000: Cowhouse Ck at Pidcoke, TX

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high stream_elas, low runoff_ratio, low gauge_lat.

Top 10 driver attributes

1. stream_elas

z = +2.05 r = -0.26 score = 0.530

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. runoff_ratio

z = -1.17 r = +0.43 score = 0.500

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gauge_lat

z = -1.53 r = +0.29 score = 0.448

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. frac_forest

z = -1.70 r = +0.23 score = 0.399

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. high_prec_freq

z = +1.12 r = -0.33 score = 0.364

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. q95

z = -0.84 r = +0.43 score = 0.362

low peak-flow values reflect a damped streamflow response with limited peaks.
7. low_prec_freq

z = +1.10 r = -0.31 score = 0.342

many dry days indicate a drier climate where streamflow depends on stored water.
8. q_mean

z = -0.79 r = +0.35 score = 0.279

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. gvfdiff

z = -0.85 r = -0.33 score = 0.277

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. slope_mean

z = -0.74 r = +0.37 score = 0.272

gentle terrain slows surface runoff and produces a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08103900: S Fk Rocky Ck nr Briggs, TX

Dominant forcing: VP Peak TE: 0.0022 Peak lag: 11 day(s)
Summary: VP-dominant at lag 11 day(s); top drivers consistent with high zero_q_freq, high low_prec_freq, low gauge_lat.

Top 10 driver attributes

1. zero_q_freq

z = +3.07 r = -0.30 score = 0.917

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. low_prec_freq

z = +1.25 r = -0.50 score = 0.624

many dry days indicate a drier climate where streamflow depends on stored water.
3. gauge_lat

z = -1.60 r = +0.35 score = 0.563

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. runoff_ratio

z = -1.08 r = +0.50 score = 0.539

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. high_prec_freq

z = +1.14 r = -0.47 score = 0.535

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. frac_forest

z = -1.66 r = +0.29 score = 0.483

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. stream_elas

z = +1.24 r = -0.36 score = 0.450

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. q95

z = -0.75 r = +0.49 score = 0.371

low peak-flow values reflect a damped streamflow response with limited peaks.
9. q_mean

z = -0.75 r = +0.43 score = 0.323

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. frac_snow

z = -0.83 r = +0.31 score = 0.259

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08104900: S Fk San Gabriel Rv at Georgetown, TX

Dominant forcing: SRAD Peak TE: 0.0014 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, high stream_elas, low runoff_ratio.

Top 10 driver attributes

1. gauge_lat z = -1.66 r = +0.29 score = 0.485

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. stream_elas z = +1.75 r = -0.26 score = 0.452

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. runoff_ratio z = -1.00 r = +0.43 score = 0.428

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. high_prec_freq z = +1.29 r = -0.33 score = 0.421

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. frac_forest z = -1.65 r = +0.23 score = 0.387

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. low_prec_freq z = +1.20 r = -0.31 score = 0.373

many dry days indicate a drier climate where streamflow depends on stored water.
7. q95 z = -0.77 r = +0.43 score = 0.332

low peak-flow values reflect a damped streamflow response with limited peaks.
8. gvf_diff z = -0.99 r = -0.33 score = 0.324

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. slope_mean z = -0.72 r = +0.37 score = 0.263

gentle terrain slows surface runoff and produces a smoother streamflow signal.
10. q_mean z = -0.72 r = +0.35 score = 0.253

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08109700: Middle Yegua Ck nr Dime Box, TX

Dominant forcing: PRCP Peak TE: 0.0071 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high glim_2nd_class_frac, low slope_fdc, low frac_forest.

Top 10 driver attributes

1. glim_2nd_class_frac

z = +2.09 r = +0.15 score = 0.320

a large secondary geological class adds heterogeneity to the subsurface flow response.
2. slope_fdc

z = -1.43 r = +0.20 score = 0.287

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. frac_forest

z = -1.64 r = +0.17 score = 0.283

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. runoff_ratio

z = -1.18 r = +0.23 score = 0.272

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. low_prec_freq

z = +1.11 r = -0.22 score = 0.248

many dry days indicate a drier climate where streamflow depends on stored water.
6. q95

z = -0.80 r = +0.31 score = 0.245

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gauge_lat

z = -1.71 r = +0.13 score = 0.216

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. q_mean

z = -0.78 r = +0.28 score = 0.215

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. hfd_mean

z = -0.90 r = -0.24 score = 0.213

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. high_prec_freq

z = +1.09 r = -0.19 score = 0.206

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08150800: Beaver Ck nr Mason, TX

Dominant forcing: PRCP Peak TE: 0.0013 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low runoff_ratio, high carbonate_rocks_frac, high low_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = -1.47 r = +0.23 score = 0.338

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. carbonate_rocks_frac

z = +2.66 r = -0.12 score = 0.319

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. low_prec_freq

z = +1.39 r = -0.22 score = 0.310

many dry days indicate a drier climate where streamflow depends on stored water.
4. q95

z = -0.98 r = +0.31 score = 0.298

low peak-flow values reflect a damped streamflow response with limited peaks.
5. p_mean

z = -0.82 r = +0.34 score = 0.281

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. frac_forest

z = -1.61 r = +0.17 score = 0.278

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. q_mean

z = -0.91 r = +0.28 score = 0.251

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. aridity

z = +0.87 r = -0.28 score = 0.240

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. gvf_diff

z = -0.97 r = -0.24 score = 0.236

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. slope_fdc

z = -1.07 r = +0.20 score = 0.214

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08155200: Barton Ck at SH 71 nr Oak Hill, TX

Dominant forcing: PRCP Peak TE: 0.0029 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, low gvf_diff, low frac_forest.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.38 r = -0.12 score = 0.405

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. gvf_diff

z = -1.28 r = -0.24 score = 0.310

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. frac_forest

z = -1.48 r = +0.17 score = 0.255

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. low_prec_freq

z = +1.11 r = -0.22 score = 0.247

many dry days indicate a drier climate where streamflow depends on stored water.
5. gauge_lat

z = -1.72 r = +0.13 score = 0.217

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. runoff_ratio

z = -0.89 r = +0.23 score = 0.206

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. q95

z = -0.65 r = +0.31 score = 0.197

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.68 r = +0.28 score = 0.187

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. p_mean

z = -0.53 r = +0.34 score = 0.184

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. high_prec_freq

z = +0.95 r = -0.19 score = 0.179

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08158700: Onion Ck nr Driftwood, TX

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high carbonate_rocks_frac, low frac_forest, low gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.27 r = -0.12 score = 0.391

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. frac_forest

z = -1.61 r = +0.17 score = 0.277

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. gvf_diff

z = -1.10 r = -0.24 score = 0.268

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. low_prec_freq

z = +1.13 r = -0.22 score = 0.253

many dry days indicate a drier climate where streamflow depends on stored water.
5. gauge_lat

z = -1.76 r = +0.13 score = 0.222

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. runoff_ratio

z = -0.95 r = +0.23 score = 0.219

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. q95

z = -0.67 r = +0.31 score = 0.205

low peak-flow values reflect a damped streamflow response with limited peaks.
8. q_mean

z = -0.69 r = +0.28 score = 0.192

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. high_prec_freq

z = +0.95 r = -0.19 score = 0.179

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. p_mean

z = -0.49 r = +0.34 score = 0.170

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08158810: Bear Ck bl FM 1826 nr Driftwood, TX

Dominant forcing: SRAD Peak TE: 0.0014 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gauge_lat, high carbonate_rocks_frac, high stream_elas.

Top 10 driver attributes

1. gauge_lat z = -1.75 r = +0.29 score = 0.511

southern latitude is associated with warmer climates and rainfall-driven streamflow.
2. carbonate_rocks_frac z = +3.38 r = -0.11 score = 0.380

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
3. stream_elas z = +1.45 r = -0.26 score = 0.374

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
4. gvf_diff z = -1.13 r = -0.33 score = 0.369

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. frac_forest z = -1.57 r = +0.23 score = 0.369

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. zero_q_freq z = +1.60 r = -0.23 score = 0.363

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
7. low_prec_freq z = +1.13 r = -0.31 score = 0.353

many dry days indicate a drier climate where streamflow depends on stored water.
8. runoff_ratio z = -0.81 r = +0.43 score = 0.348

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. high_prec_freq z = +0.91 r = -0.33 score = 0.296

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. high_q_dur z = +1.81 r = +0.16 score = 0.283

long high-flow events suggest sustained wet conditions and slow recession.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08164000: Lavaca Rv nr Edna, TX

Dominant forcing: PRCP Peak TE: 0.0033 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, low slope_fdc, high soil_depth_pelletier.

Top 10 driver attributes

1. pet_mean

z = +3.53

r = -0.12

score = 0.430

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. slope_fdc

z = -1.50

r = +0.20

score = 0.302

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. soil_depth_pelletier

z = +1.75

r = +0.16

score = 0.280

deep soils provide more storage and slower release of water to streamflow.
4. frac_forest

z = -1.52

r = +0.17

score = 0.262

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. aridity

z = +0.91

r = -0.28

score = 0.252

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
6. gauge_lat

z = -1.98

r = +0.13

score = 0.249

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. stream_elas

z = +1.72

r = -0.12

score = 0.213

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. high_prec_dur

z = +0.80

r = +0.23

score = 0.182

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. gvf_diff

z = -0.73

r = -0.24

score = 0.178

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. runoff_ratio

z = -0.77

r = +0.23

score = 0.178

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08164300: Navidad Rv nr Hallettsville, TX

Dominant forcing: PRCP Peak TE: 0.0054 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, low slope_fdc, low frac_forest.

Top 10 driver attributes

1. pet_mean

z = +3.48

r = -0.12

score = 0.424

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. slope_fdc

z = -1.55

r = +0.20

score = 0.312

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. frac_forest

z = -1.68

r = +0.17

score = 0.290

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. aridity

z = +0.97

r = -0.28

score = 0.269

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. q95

z = -0.80

r = +0.31

score = 0.244

low peak-flow values reflect a damped streamflow response with limited peaks.
6. gauge_lat

z = -1.88

r = +0.13

score = 0.237

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. runoff_ratio

z = -0.98

r = +0.23

score = 0.225

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. low_prec_freq

z = +0.94

r = -0.22

score = 0.211

many dry days indicate a drier climate where streamflow depends on stored water.
9. soil_depth_pelletier

z = +1.28

r = +0.16

score = 0.204

deep soils provide more storage and slower release of water to streamflow.
10. gvf_diff

z = -0.77

r = -0.24

score = 0.188

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08164600: Garcitas Ck nr Inez, TX

Dominant forcing: PRCP Peak TE: 0.0085 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, high soil_depth_pelletier, low frac_forest.

Top 10 driver attributes

1. slope_fdc

z = -2.00

r = +0.20

score = 0.401

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. soil_depth_pelletier

z = +2.37

r = +0.16

score = 0.378

deep soils provide more storage and slower release of water to streamflow.
3. frac_forest

z = -1.56

r = +0.17

score = 0.269

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
4. gauge_lat

z = -1.99

r = +0.13

score = 0.251

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. stream_elas

z = +2.00

r = -0.12

score = 0.247

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
6. high_prec_dur

z = +0.99

r = +0.23

score = 0.224

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. pet_mean

z = +1.78

r = -0.12

score = 0.217

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
8. low_prec_freq

z = +0.94

r = -0.22

score = 0.210

many dry days indicate a drier climate where streamflow depends on stored water.
9. q95

z = -0.64

r = +0.31

score = 0.196

low peak-flow values reflect a damped streamflow response with limited peaks.
10. runoff_ratio

z = -0.80

r = +0.23

score = 0.184

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08165300: N Fk Guadalupe Rv nr Hunt, TX

Dominant forcing: VP Peak TE: 0.0013 Peak lag: 15 day(s)
Summary: VP-dominant at lag 15 day(s); top drivers consistent with high low_prec_freq, low runoff_ratio, low gauge_lat.

Top 10 driver attributes

1. low_prec_freq

z = +1.52 r = -0.50 score = 0.762

many dry days indicate a drier climate where streamflow depends on stored water.
2. runoff_ratio

z = -1.27 r = +0.50 score = 0.634

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gauge_lat

z = -1.76 r = +0.35 score = 0.621

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. high_prec_freq

z = +1.02 r = -0.47 score = 0.478

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. q95

z = -0.96 r = +0.49 score = 0.471

low peak-flow values reflect a damped streamflow response with limited peaks.
6. frac_forest

z = -1.32 r = +0.29 score = 0.384

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. q_mean

z = -0.84 r = +0.43 score = 0.359

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. low_prec_dur

z = +1.04 r = -0.27 score = 0.279

long dry spells lead to drawdown of basin storage and slower streamflow response.
9. frac_snow

z = -0.85 r = +0.31 score = 0.264

very little precipitation falls as snow, so streamflow is dominated by rainfall-driven processes.
10. pet_mean

z = +1.01 r = -0.25 score = 0.256

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08171300: Blanco Rv nr Kyle, TX

Dominant forcing: PRCP Peak TE: 0.0022 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high pet_mean, high aridity, high carbonate_rocks_frac.

Top 10 driver attributes

1. pet_mean

z = +3.40

r = -0.12

score = 0.415

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. aridity

z = +1.25

r = -0.28

score = 0.345

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. carbonate_rocks_frac

z = +2.75

r = -0.12

score = 0.330

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. frac_forest

z = -1.56

r = +0.17

score = 0.269

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
5. gvf_diff

z = -1.04

r = -0.24

score = 0.252

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. low_prec_freq

z = +1.10

r = -0.22

score = 0.245

many dry days indicate a drier climate where streamflow depends on stored water.
7. gauge_lat

z = -1.78

r = +0.13

score = 0.225

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. q95

z = -0.73

r = +0.31

score = 0.221

low peak-flow values reflect a damped streamflow response with limited peaks.
9. runoff_ratio

z = -0.95

r = +0.23

score = 0.218

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. q_mean

z = -0.69

r = +0.28

score = 0.192

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08175000: Sandies Ck nr Westhoff, TX

Dominant forcing: PRCP Peak TE: 0.0046 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low runoff_ratio, high low_prec_freq.

Top 10 driver attributes

1. slope_fdc

z = -1.99 r = +0.20 score = 0.400

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. runoff_ratio

z = -1.21 r = +0.23 score = 0.279

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. low_prec_freq

z = +1.22 r = -0.22 score = 0.272

many dry days indicate a drier climate where streamflow depends on stored water.
4. glim_2nd_class_frac

z = +1.73 r = +0.15 score = 0.266

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. frac_forest

z = -1.53 r = +0.17 score = 0.263

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q95

z = -0.82 r = +0.31 score = 0.251

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gauge_lat

z = -1.93 r = +0.13 score = 0.243

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. q_mean

z = -0.80 r = +0.28 score = 0.222

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. p_mean

z = -0.60 r = +0.34 score = 0.206

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. high_prec_dur

z = +0.89 r = +0.23 score = 0.203

long heavy-rainfall events allow more soil saturation and stronger runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08176900: Coletto Ck at Arnold Rd nr Schroeder, TX

Dominant forcing: PRCP Peak TE: 0.0048 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, low runoff_ratio, high high_prec_dur.

Top 10 driver attributes

1. slope_fdc

z = -1.59

r = +0.20

score = 0.320

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. runoff_ratio

z = -1.25

r = +0.23

score = 0.289

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. high_prec_dur

z = +1.26

r = +0.23

score = 0.286

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. q95

z = -0.91

r = +0.31

score = 0.278

low peak-flow values reflect a damped streamflow response with limited peaks.
5. low_prec_freq

z = +1.22

r = -0.22

score = 0.273

many dry days indicate a drier climate where streamflow depends on stored water.
6. soil_depth_pelletier

z = +1.64

r = +0.16

score = 0.262

deep soils provide more storage and slower release of water to streamflow.
7. gauge_lat

z = -1.99

r = +0.13

score = 0.252

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. pet_mean

z = +1.97

r = -0.12

score = 0.240

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
9. q_mean

z = -0.81

r = +0.28

score = 0.225

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. aridity

z = +0.81

r = -0.28

score = 0.225

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08178880: Medina Rv at Bandera, TX

Dominant forcing: PRCP Peak TE: 0.0013 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high low_prec_freq, low gauge_lat.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.32 r = -0.12 score = 0.398

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. low_prec_freq

z = +1.32 r = -0.22 score = 0.296

many dry days indicate a drier climate where streamflow depends on stored water.
3. gauge_lat

z = -1.83 r = +0.13 score = 0.231

southern latitude is associated with warmer climates and rainfall-driven streamflow.
4. q95

z = -0.74 r = +0.31 score = 0.225

low peak-flow values reflect a damped streamflow response with limited peaks.
5. gvf_diff

z = -0.90 r = -0.24 score = 0.219

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. soil_depth_statsgo

z = -1.56 r = +0.13 score = 0.209

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. sand_frac

z = -1.69 r = +0.12 score = 0.198

low sand fraction reduces infiltration and may favor surface runoff.
8. p_mean

z = -0.55 r = +0.34 score = 0.191

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. runoff_ratio

z = -0.82 r = +0.23 score = 0.190

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. q_mean

z = -0.65 r = +0.28 score = 0.181

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08189500: Mission Rv at Refugio, TX

Dominant forcing: PRCP Peak TE: 0.0075 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high pet_mean, high aridity, low slope_fdc.

Top 10 driver attributes

1. pet_mean

z = +3.51

r = -0.12

score = 0.428

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
2. aridity

z = +1.52

r = -0.28

score = 0.421

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. slope_fdc

z = -1.83

r = +0.20

score = 0.368

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
4. soil_depth_pelletier

z = +1.87

r = +0.16

score = 0.298

deep soils provide more storage and slower release of water to streamflow.
5. runoff_ratio

z = -1.27

r = +0.23

score = 0.294

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. q95

z = -0.92

r = +0.31

score = 0.281

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gauge_lat

z = -2.10

r = +0.13

score = 0.265

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. frac_forest

z = -1.44

r = +0.17

score = 0.247

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
9. low_prec_freq

z = +1.09

r = -0.22

score = 0.244

many dry days indicate a drier climate where streamflow depends on stored water.
10. q_mean

z = -0.83

r = +0.28

score = 0.230

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08190000: Nueces Rv at Laguna, TX

Dominant forcing: PRCP Peak TE: 0.0016 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high carbonate_rocks_frac, high low_prec_freq, low p_mean.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.09 r = -0.12 score = 0.370

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. low_prec_freq

z = +1.61 r = -0.22 score = 0.360

many dry days indicate a drier climate where streamflow depends on stored water.
3. p_mean

z = -0.95 r = +0.34 score = 0.327

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. aridity

z = +1.16 r = -0.28 score = 0.321

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. gvf_diff

z = -1.32 r = -0.24 score = 0.320

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. q95

z = -0.88 r = +0.31 score = 0.268

low peak-flow values reflect a damped streamflow response with limited peaks.
7. runoff_ratio

z = -1.07 r = +0.23 score = 0.247

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. gauge_lat

z = -1.89 r = +0.13 score = 0.238

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. soil_depth_statsgo

z = -1.68 r = +0.13 score = 0.225

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. q_mean

z = -0.80 r = +0.28 score = 0.221

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08190500: W Nueces Rv nr Brackettville, TX

Dominant forcing: PRCP Peak TE: 0.0012 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low slope_fdc, high carbonate_rocks_frac.

Top 10 driver attributes

1. aridity

z = +2.02

r = -0.28

score = 0.559

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. slope_fdc

z = -2.21

r = +0.20

score = 0.445

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. carbonate_rocks_frac

z = +3.29

r = -0.12

score = 0.395

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
4. low_prec_freq

z = +1.73

r = -0.22

score = 0.386

many dry days indicate a drier climate where streamflow depends on stored water.
5. runoff_ratio

z = -1.54

r = +0.23

score = 0.356

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. p_mean

z = -1.01

r = +0.34

score = 0.349

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. gvf_diff

z = -1.38

r = -0.24

score = 0.336

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. pet_mean

z = +2.56

r = -0.12

score = 0.312

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
9. q95

z = -1.00

r = +0.31

score = 0.307

low peak-flow values reflect a damped streamflow response with limited peaks.
10. q_mean

z = -0.94

r = +0.28

score = 0.259

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08194200: San Casimiro Ck nr Freer, TX

Dominant forcing: PRCP Peak TE: 0.0044 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high hfd_mean, high zero_q_freq, high aridity.

Top 10 driver attributes

1. hfd_mean

z = +3.14

r = -0.24

score = 0.746

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
2. zero_q_freq

z = +5.26

r = -0.13

score = 0.703

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. aridity

z = +2.49

r = -0.28

score = 0.688

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. slope_fdc

z = -2.44

r = +0.20

score = 0.490

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
5. p_mean

z = -1.32

r = +0.34

score = 0.454

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. low_prec_freq

z = +1.93

r = -0.22

score = 0.432

many dry days indicate a drier climate where streamflow depends on stored water.
7. runoff_ratio

z = -1.44

r = +0.23

score = 0.332

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q95

z = -1.00

r = +0.31

score = 0.305

low peak-flow values reflect a damped streamflow response with limited peaks.
9. frac_forest

z = -1.71

r = +0.17

score = 0.295

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. gauge_lat

z = -2.17

r = +0.13

score = 0.273

southern latitude is associated with warmer climates and rainfall-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08195000: Frio Rv at Concan, TX

Dominant forcing: PRCP Peak TE: 0.0022 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high low_prec_freq, low gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.09 r = -0.12 score = 0.371

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. low_prec_freq

z = +1.56 r = -0.22 score = 0.349

many dry days indicate a drier climate where streamflow depends on stored water.
3. gvf_diff

z = -1.13 r = -0.24 score = 0.275

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. soil_depth_statsgo

z = -1.99 r = +0.13 score = 0.266

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
5. p_mean

z = -0.74 r = +0.34 score = 0.256

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. q95

z = -0.81 r = +0.31 score = 0.248

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gauge_lat

z = -1.87 r = +0.13 score = 0.236

southern latitude is associated with warmer climates and rainfall-driven streamflow.
8. runoff_ratio

z = -0.96 r = +0.23 score = 0.221

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. aridity

z = +0.77 r = -0.28 score = 0.213

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
10. q_mean

z = -0.73 r = +0.28 score = 0.203

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08196000: Dry Frio Rv nr Reagan Wells, TX

Dominant forcing: PRCP Peak TE: 0.0011 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high low_prec_freq, low gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.30 r = -0.12 score = 0.395

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. low_prec_freq

z = +1.73 r = -0.22 score = 0.387

many dry days indicate a drier climate where streamflow depends on stored water.
3. gvf_diff

z = -1.24 r = -0.24 score = 0.301

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. p_mean

z = -0.83 r = +0.34 score = 0.286

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. aridity

z = +0.95 r = -0.28 score = 0.262

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
6. q95

z = -0.85 r = +0.31 score = 0.260

low peak-flow values reflect a damped streamflow response with limited peaks.
7. soil_depth_statsgo

z = -1.93 r = +0.13 score = 0.259

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. runoff_ratio

z = -1.10 r = +0.23 score = 0.254

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. gauge_lat

z = -1.87 r = +0.13 score = 0.236

southern latitude is associated with warmer climates and rainfall-driven streamflow.
10. q_mean

z = -0.79 r = +0.28 score = 0.219

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08198500: Sabinal Rv at Sabinal, TX

Dominant forcing: PRCP Peak TE: 0.0014 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low slope_fdc, high low_prec_freq, low runoff_ratio.

Top 10 driver attributes

1. slope_fdc

z = -2.08

r = +0.20

score = 0.419

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
2. low_prec_freq

z = +1.48

r = -0.22

score = 0.332

many dry days indicate a drier climate where streamflow depends on stored water.
3. runoff_ratio

z = -1.34

r = +0.23

score = 0.310

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. q95

z = -0.92

r = +0.31

score = 0.280

low peak-flow values reflect a damped streamflow response with limited peaks.
5. carbonate_rocks_frac

z = +2.31

r = -0.12

score = 0.277

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
6. gvf_diff

z = -1.05

r = -0.24

score = 0.256

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. p_mean

z = -0.71

r = +0.34

score = 0.245

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. gauge_lat

z = -1.91

r = +0.13

score = 0.241

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. q_mean

z = -0.86

r = +0.28

score = 0.238

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. aridity

z = +0.75

r = -0.28

score = 0.206

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08200000: Hondo Ck nr Tarpley, TX

Dominant forcing: PRCP Peak TE: 0.0006 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high carbonate_rocks_frac, high low_prec_freq, low gvf_diff.

Top 10 driver attributes

1. carbonate_rocks_frac

z = +3.12 r = -0.12 score = 0.374

a high carbonate rock fraction is associated with karst-style flow paths and rapid subsurface drainage.
2. low_prec_freq

z = +1.44 r = -0.22 score = 0.322

many dry days indicate a drier climate where streamflow depends on stored water.
3. gvf_diff

z = -1.04 r = -0.24 score = 0.254

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lat

z = -1.86 r = +0.13 score = 0.234

southern latitude is associated with warmer climates and rainfall-driven streamflow.
5. q95

z = -0.71 r = +0.31 score = 0.217

low peak-flow values reflect a damped streamflow response with limited peaks.
6. p_mean

z = -0.61 r = +0.34 score = 0.211

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. sand_frac

z = -1.59 r = +0.12 score = 0.186

low sand fraction reduces infiltration and may favor surface runoff.
8. soil_depth_statsgo

z = -1.33 r = +0.13 score = 0.178

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
9. q_mean

z = -0.64 r = +0.28 score = 0.177

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. runoff_ratio

z = -0.75 r = +0.23 score = 0.173

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08202700: Seco Ck at Rowe Ranch nr D'Hanis, TX

Dominant forcing: VP Peak TE: 0.0012 Peak lag: 18 day(s)
Summary: VP-dominant at lag 18 day(s); top drivers consistent with high zero_q_freq, high low_q_dur, low runoff_ratio.

Top 10 driver attributes

1. zero_q_freq

z = +8.14 r = -0.30 score = 2.429

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. low_q_dur

z = +8.66 r = -0.11 score = 0.929

long low-flow events indicate prolonged dry periods and limited storage recharge.
3. runoff_ratio

z = -1.55 r = +0.50 score = 0.773

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. low_prec_freq

z = +1.44 r = -0.50 score = 0.718

many dry days indicate a drier climate where streamflow depends on stored water.
5. gauge_lat

z = -1.90 r = +0.35 score = 0.668

southern latitude is associated with warmer climates and rainfall-driven streamflow.
6. slope_fdc

z = -2.44 r = +0.24 score = 0.584

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. baseflow_index

z = -2.98 r = +0.17 score = 0.518

a low baseflow index means streamflow responds quickly to recent rainfall.
8. q95

z = -1.02 r = +0.49 score = 0.505

low peak-flow values reflect a damped streamflow response with limited peaks.
9. q_mean

z = -0.93 r = +0.43 score = 0.399

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. high_prec_freq

z = +0.82 r = -0.47 score = 0.385

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08267500: RIO HONDO NEAR VALDEZ, NM

Dominant forcing: SRAD Peak TE: 0.0065 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high slope_mean, high elev_mean, high high_prec_freq.

Top 10 driver attributes

1. slope_mean

z = +2.18 r = +0.37 score = 0.802

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +2.86 r = +0.24 score = 0.690

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = +1.31 r = -0.33 score = 0.428

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. frac_snow

z = +1.45 r = +0.26 score = 0.369

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. glim_1st_class_frac

z = -1.66 r = -0.21 score = 0.355

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. glim_2nd_class_frac

z = +1.67 r = +0.18 score = 0.293

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. p_seasonality

z = +0.64 r = -0.46 score = 0.291

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. aridity

z = +1.73 r = -0.16 score = 0.284

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. baseflow_index

z = +1.24 r = +0.22 score = 0.271

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. low_prec_freq

z = +0.87 r = -0.31 score = 0.270

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08269000: RIO PUEBLO DE TAOS NEAR TAOS, NM

Dominant forcing: SRAD Peak TE: 0.0063 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high elev_mean, high slope_mean, high high_prec_freq.

Top 10 driver attributes

1. elev_mean

z = +2.59

r = +0.24

score = 0.624

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. slope_mean

z = +1.51

r = +0.37

score = 0.554

steep terrain promotes fast surface runoff and a quick streamflow response.
3. high_prec_freq

z = +1.59

r = -0.33

score = 0.518

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. aridity

z = +2.27

r = -0.16

score = 0.374

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. p_seasonality

z = +0.74

r = -0.46

score = 0.339

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. q95

z = -0.67

r = +0.43

score = 0.290

low peak-flow values reflect a damped streamflow response with limited peaks.
7. low_prec_freq

z = +0.92

r = -0.31

score = 0.287

many dry days indicate a drier climate where streamflow depends on stored water.
8. gvf_diff

z = -0.86

r = -0.33

score = 0.282

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. frac_snow

z = +1.10

r = +0.26

score = 0.281

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
10. gauge_lon

z = -0.60

r = -0.43

score = 0.259

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08271000: RIO LUCERO NEAR ARROYO SECO, NM

Dominant forcing: SRAD Peak TE: 0.0076 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high runoff_ratio.

Top 10 driver attributes

1. slope_mean

z = +2.02 r = +0.37 score = 0.741

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +2.50 r = +0.24 score = 0.603

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. runoff_ratio

z = +1.20 r = +0.43 score = 0.512

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. high_prec_freq

z = +1.56 r = -0.33 score = 0.507

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. frac_snow

z = +1.51 r = +0.26 score = 0.385

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. aridity

z = +2.15 r = -0.16 score = 0.353

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
7. p_seasonality

z = +0.64 r = -0.46 score = 0.292

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
8. low_prec_freq

z = +0.94 r = -0.31 score = 0.292

many dry days indicate a drier climate where streamflow depends on stored water.
9. baseflow_index

z = +1.24 r = +0.22 score = 0.271

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. gvf_diff

z = -0.81 r = -0.33 score = 0.265

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08324000: JEMEZ RIVER NEAR JEMEZ, NM

Dominant forcing: SRAD Peak TE: 0.0048 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low runoff_ratio, high elev_mean, high high_prec_freq.

Top 10 driver attributes

1. runoff_ratio

z = -1.26 r = +0.43 score = 0.541

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. elev_mean

z = +2.16 r = +0.24 score = 0.522

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = +1.49 r = -0.33 score = 0.486

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. glim_1st_class_frac

z = -1.94 r = -0.21 score = 0.416

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. q95

z = -0.90 r = +0.43 score = 0.390

low peak-flow values reflect a damped streamflow response with limited peaks.
6. gvf_diff

z = -1.09 r = -0.33 score = 0.357

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. high_prec_dur

z = +1.22 r = +0.29 score = 0.351

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. stream_elas

z = -1.22 r = -0.26 score = 0.315

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. q_mean

z = -0.88 r = +0.35 score = 0.310

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. gauge_lon

z = -0.68 r = -0.43 score = 0.292

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08377900: RIO MORA NEAR TERRERO, NM

Dominant forcing: Tair Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, high high_prec_freq.

Top 10 driver attributes

1. elev_mean

z = +2.87

r = +0.30

score = 0.854

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +1.42

r = +0.46

score = 0.655

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. high_prec_freq

z = +1.10

r = -0.49

score = 0.543

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. slope_mean

z = +1.25

r = +0.41

score = 0.511

steep terrain promotes fast surface runoff and a quick streamflow response.
5. low_prec_freq

z = +0.83

r = -0.50

score = 0.416

many dry days indicate a drier climate where streamflow depends on stored water.
6. gauge_lat

z = -0.67

r = +0.51

score = 0.337

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. frac_forest

z = +0.90

r = +0.35

score = 0.311

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. glim_1st_class_frac

z = -1.89

r = -0.16

score = 0.306

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. p_seasonality

z = +0.69

r = -0.38

score = 0.266

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
10. q_mean

z = -0.56

r = +0.45

score = 0.249

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08378500: PECOS RIVER NEAR PECOS, NM

Dominant forcing: SRAD Peak TE: 0.0043 Peak lag: 19 day(s)
Summary: SRAD-dominant at lag 19 day(s); top drivers consistent with high elev_mean, high slope_mean, high high_prec_freq.

Top 10 driver attributes

1. elev_mean

z = +2.81

r = +0.24

score = 0.677

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. slope_mean

z = +1.24

r = +0.37

score = 0.455

steep terrain promotes fast surface runoff and a quick streamflow response.
3. high_prec_freq

z = +1.10

r = -0.33

score = 0.360

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. frac_snow

z = +1.34

r = +0.26

score = 0.342

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. p_seasonality

z = +0.68

r = -0.46

score = 0.310

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
6. gauge_lon

z = -0.61

r = -0.43

score = 0.264

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. gvf_diff

z = -0.77

r = -0.33

score = 0.253

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. low_prec_freq

z = +0.80

r = -0.31

score = 0.249

many dry days indicate a drier climate where streamflow depends on stored water.
9. q95

z = -0.55

r = +0.43

score = 0.236

low peak-flow values reflect a damped streamflow response with limited peaks.
10. glim_1st_class_frac

z = -1.10

r = -0.21

score = 0.236

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 08380500: GALLINAS CREEK NEAR MONTEZUMA, NM

Dominant forcing: PRCP Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high elev_mean, low p_mean, high hfd_mean.

Top 10 driver attributes

1. elev_mean

z = +2.24

r = -0.20

score = 0.438

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. p_mean

z = -1.07

r = +0.34

score = 0.369

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. hfd_mean

z = +1.43

r = -0.24

score = 0.339

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. aridity

z = +1.12

r = -0.28

score = 0.310

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
5. gvf_diff

z = -1.19

r = -0.24

score = 0.288

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. glim_1st_class_frac

z = -1.86

r = -0.15

score = 0.271

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. high_prec_freq

z = +1.35

r = -0.19

score = 0.253

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. high_prec_dur

z = +1.12

r = +0.23

score = 0.253

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. q95

z = -0.81

r = +0.31

score = 0.248

low peak-flow values reflect a damped streamflow response with limited peaks.
10. p_seasonality

z = +1.00

r = -0.24

score = 0.245

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09034900: BOBTAIL CREEK NEAR JONES PASS, CO.

Dominant forcing: SRAD Peak TE: 0.0063 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, low high_prec_freq.

Top 10 driver attributes

1. elev_mean

z = +3.58

r = +0.24

score = 0.863

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +2.70

r = +0.26

score = 0.688

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. high_prec_freq

z = -1.72

r = -0.33

score = 0.562

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. runoff_ratio

z = +1.05

r = +0.43

score = 0.451

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. slope_mean

z = +1.20

r = +0.37

score = 0.441

steep terrain promotes fast surface runoff and a quick streamflow response.
6. clay_frac

z = -1.42

r = -0.22

score = 0.313

low clay fraction allows easier infiltration and reduces surface runoff.
7. q95

z = +0.72

r = +0.43

score = 0.310

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. high_q_dur

z = +1.95

r = +0.16

score = 0.305

long high-flow events suggest sustained wet conditions and slow recession.
9. other_frac

z = +2.19

r = +0.14

score = 0.305

a large 'other' land fraction reflects mixed or atypical surface cover.
10. gauge_lon

z = -0.62

r = -0.43

score = 0.270

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09035800: DARLING CREEK NEAR LEAL, CO.

Dominant forcing: SRAD Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean

z = +2.11 r = +0.37 score = 0.776

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +3.20 r = +0.24 score = 0.771

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow

z = +2.59 r = +0.26 score = 0.661

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. high_prec_freq

z = -1.58 r = -0.33 score = 0.515

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. gauge_lon

z = -0.63 r = -0.43 score = 0.273

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. clay_frac

z = -1.18 r = -0.22 score = 0.261

low clay fraction allows easier infiltration and reduces surface runoff.
7. gvff_diff

z = -0.67 r = -0.33 score = 0.218

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. high_q_dur

z = +1.39 r = +0.16 score = 0.218

long high-flow events suggest sustained wet conditions and slow recession.
9. p_seasonality

z = -0.41 r = -0.46 score = 0.188

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. soil_conductivity

z = +0.74 r = +0.25 score = 0.188

high soil conductivity allows fast infiltration and subsurface flow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09035900: SOUTH FORK OF WILLIAMS FORK NEAR LEAL, CO.

Dominant forcing: SRAD Peak TE: 0.0058 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, high slope_mean.

Top 10 driver attributes

1. elev_mean

z = +3.16

r = +0.24

score = 0.762

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +2.52

r = +0.26

score = 0.643

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. slope_mean

z = +1.65

r = +0.37

score = 0.606

steep terrain promotes fast surface runoff and a quick streamflow response.
4. high_prec_freq

z = -1.72

r = -0.33

score = 0.562

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. clay_frac

z = -1.31

r = -0.22

score = 0.289

low clay fraction allows easier infiltration and reduces surface runoff.
6. gauge_lon

z = -0.63

r = -0.43

score = 0.273

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. gvff_diff

z = -0.67

r = -0.33

score = 0.218

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. low_prec_freq

z = -0.67

r = -0.31

score = 0.208

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. other_frac

z = +1.36

r = +0.14

score = 0.190

a large 'other' land fraction reflects mixed or atypical surface cover.
10. stream_elas

z = -0.70

r = -0.26

score = 0.181

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09047700: KEYSTONE GULCH NEAR DILLON, CO.

Dominant forcing: SRAD Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, low high_prec_freq, high frac_snow.

Top 10 driver attributes

1. elev_mean

z = +3.28

r = +0.24

score = 0.790

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. high_prec_freq

z = -1.87

r = -0.33

score = 0.609

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. frac_snow

z = +2.21

r = +0.26

score = 0.564

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. slope_mean

z = +1.22

r = +0.37

score = 0.449

steep terrain promotes fast surface runoff and a quick streamflow response.
5. gvf_diff

z = -1.01

r = -0.33

score = 0.329

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. gauge_lon

z = -0.63

r = -0.43

score = 0.271

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. clay_frac

z = -1.21

r = -0.22

score = 0.266

low clay fraction allows easier infiltration and reduces surface runoff.
8. aridity

z = +1.41

r = -0.16

score = 0.232

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. baseflow_index

z = +1.03

r = +0.22

score = 0.226

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. glim_1st_class_frac

z = +1.04

r = -0.21

score = 0.222

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09065500: GORE CREEK AT UPPER STATION, NEAR MINTURN, CO.

Dominant forcing: SRAD Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, low high_prec_freq.

Top 10 driver attributes

1. elev_mean z = +3.30 r = +0.24 score = 0.795

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow z = +2.65 r = +0.26 score = 0.676

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. high_prec_freq z = -2.04 r = -0.33 score = 0.666

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. runoff_ratio z = +1.55 r = +0.43 score = 0.661

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. slope_mean z = +1.41 r = +0.37 score = 0.519

steep terrain promotes fast surface runoff and a quick streamflow response.
6. q95 z = +1.00 r = +0.43 score = 0.430

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. high_q_dur z = +2.07 r = +0.16 score = 0.325

long high-flow events suggest sustained wet conditions and slow recession.
8. gauge_lon z = -0.65 r = -0.43 score = 0.280

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. clay_frac z = -1.22 r = -0.22 score = 0.269

low clay fraction allows easier infiltration and reduces surface runoff.
10. gvf_diff z = -0.71 r = -0.33 score = 0.233

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09066000: BLACK GORE CREEK NEAR MINTURN, CO.

Dominant forcing: SRAD Peak TE: 0.0061 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high elev_mean, high frac_snow, low high_prec_freq.

Top 10 driver attributes

1. elev_mean

z = +3.37

r = +0.24

score = 0.812

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +2.48

r = +0.26

score = 0.634

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. high_prec_freq

z = -1.89

r = -0.33

score = 0.616

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. high_q_dur

z = +2.85

r = +0.16

score = 0.447

long high-flow events suggest sustained wet conditions and slow recession.
5. slope_mean

z = +1.07

r = +0.37

score = 0.393

steep terrain promotes fast surface runoff and a quick streamflow response.
6. gauge_lon

z = -0.65

r = -0.43

score = 0.279

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. runoff_ratio

z = +0.65

r = +0.43

score = 0.278

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. root_depth_50

z = -1.66

r = -0.13

score = 0.208

shallow median rooting limits vegetation water access to surface soils.
9. p_seasonality

z = -0.39

r = -0.46

score = 0.180

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. gvf_diff

z = -0.54

r = -0.33

score = 0.176

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09066200: BOOTH CREEK NEAR MINTURN, CO.

Dominant forcing: SRAD Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean

z = +2.10 r = +0.37 score = 0.773

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +2.94 r = +0.24 score = 0.710

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow

z = +2.63 r = +0.26 score = 0.671

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. runoff_ratio

z = +1.12 r = +0.43 score = 0.478

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. high_prec_freq

z = -1.39 r = -0.33 score = 0.454

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. q95

z = +0.88 r = +0.43 score = 0.379

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. high_q_dur

z = +2.02 r = +0.16 score = 0.317

long high-flow events suggest sustained wet conditions and slow recession.
8. gauge_lon

z = -0.65 r = -0.43 score = 0.281

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. clay_frac

z = -1.25 r = -0.22 score = 0.276

low clay fraction allows easier infiltration and reduces surface runoff.
10. stream_elas

z = -0.98 r = -0.26 score = 0.254

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09066300: MIDDLE CREEK NEAR MINTURN, CO.

Dominant forcing: SRAD Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean

z = +2.07 r = +0.37 score = 0.761

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean

z = +2.78 r = +0.24 score = 0.670

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow

z = +2.51 r = +0.26 score = 0.641

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. high_q_dur

z = +3.40 r = +0.16 score = 0.532

long high-flow events suggest sustained wet conditions and slow recession.
5. high_prec_freq

z = -1.17 r = -0.33 score = 0.383

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. gauge_lon

z = -0.65 r = -0.43 score = 0.282

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. clay_frac

z = -1.15 r = -0.22 score = 0.254

low clay fraction allows easier infiltration and reduces surface runoff.
8. p_seasonality

z = -0.47 r = -0.46 score = 0.213

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. soil_conductivity

z = +0.79 r = +0.25 score = 0.200

high soil conductivity allows fast infiltration and subsurface flow.
10. sand_frac

z = +1.02 r = +0.18 score = 0.179

high sand fraction promotes fast infiltration and subsurface drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09081600: CRYSTAL RIVER AB AVALANCHE C, NEAR REDSTONE, CO.

Dominant forcing: SRAD Peak TE: 0.0126 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high elev_mean, high frac_snow.

Top 10 driver attributes

1. slope_mean z = +2.22 r = +0.37 score = 0.814

steep terrain promotes fast surface runoff and a quick streamflow response.
2. elev_mean z = +2.84 r = +0.24 score = 0.685

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. frac_snow z = +2.40 r = +0.26 score = 0.612

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. high_prec_freq z = -1.29 r = -0.33 score = 0.422

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. runoff_ratio z = +0.97 r = +0.43 score = 0.414

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. gauge_lon z = -0.71 r = -0.43 score = 0.305

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. p_seasonality z = -0.65 r = -0.46 score = 0.295

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. low_prec_freq z = -0.63 r = -0.31 score = 0.197

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. q95 z = +0.45 r = +0.43 score = 0.195

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. root_depth_50 z = -1.37 r = -0.13 score = 0.172

shallow median rooting limits vegetation water access to surface soils.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09107000: TAYLOR RIVER AT TAYLOR PARK, CO.

Dominant forcing: SRAD Peak TE: 0.0098 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high elev_mean, high frac_snow, high slope_mean.

Top 10 driver attributes

1. elev_mean z = +3.23 r = +0.24 score = 0.780

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow z = +2.43 r = +0.26 score = 0.622

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. slope_mean z = +1.21 r = +0.37 score = 0.446

steep terrain promotes fast surface runoff and a quick streamflow response.
4. glim_1st_class_frac z = -1.70 r = -0.21 score = 0.364

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gvf_diff z = -1.08 r = -0.33 score = 0.354

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. gauge_lon z = -0.66 r = -0.43 score = 0.287

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_freq z = -0.88 r = -0.33 score = 0.286

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. glim_2nd_class_frac z = +1.42 r = +0.18 score = 0.250

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. baseflow_index z = +1.08 r = +0.22 score = 0.236

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. root_depth_50 z = -1.86 r = -0.13 score = 0.233

shallow median rooting limits vegetation water access to surface soils.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09210500: FONTENELLE C NR HERSCHLER RANCH, NR FONTENELLE, WY

Dominant forcing: SRAD Peak TE: 0.0076 Peak lag: 2 day(s)

Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high frac_snow, high elev_mean, low gauge_lon.

Top 10 driver attributes

1. frac_snow

z = +2.35

r = +0.26

score = 0.599

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.14

r = +0.24

score = 0.516

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. gauge_lon

z = -0.90

r = -0.43

score = 0.390

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. p_seasonality

z = -0.79

r = -0.46

score = 0.359

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. q95

z = -0.73

r = +0.43

score = 0.315

low peak-flow values reflect a damped streamflow response with limited peaks.
6. p_mean

z = -1.33

r = +0.20

score = 0.267

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. q_mean

z = -0.73

r = +0.35

score = 0.256

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. glim_1st_class_frac

z = -1.20

r = -0.21

score = 0.256

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. baseflow_index

z = +1.16

r = +0.22

score = 0.255

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. gvf_diff

z = -0.77

r = -0.33

score = 0.251

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09223000: HAMS FORK BELOW POLE CREEK, NEAR FRONTIER, WY

Dominant forcing: SRAD Peak TE: 0.0057 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high frac_snow, high elev_mean, low p_seasonality.

Top 10 driver attributes

1. frac_snow z = +2.56 r = +0.26 score = 0.654

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean z = +2.15 r = +0.24 score = 0.519

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. p_seasonality z = -0.90 r = -0.46 score = 0.412

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. gauge_lon z = -0.92 r = -0.43 score = 0.398

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. glim_2nd_class_frac z = +1.34 r = +0.18 score = 0.234

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. glim_1st_class_frac z = -1.02 r = -0.21 score = 0.219

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. p_mean z = -1.08 r = +0.20 score = 0.217

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. q_mean z = -0.60 r = +0.35 score = 0.211

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. q95 z = -0.46 r = +0.43 score = 0.199

low peak-flow values reflect a damped streamflow response with limited peaks.
10. high_q_dur z = +1.14 r = +0.16 score = 0.178

long high-flow events suggest sustained wet conditions and slow recession.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09306242: CORRAL GULCH NEAR RANGELY, CO.

Dominant forcing: SRAD Peak TE: 0.0013 Peak lag: 14 day(s)
Summary: SRAD-dominant at lag 14 day(s); top drivers consistent with low runoff_ratio, high high_prec_freq, high aridity.

Top 10 driver attributes

1. runoff_ratio

z = -1.55 r = +0.43 score = 0.662

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. high_prec_freq

z = +1.82 r = -0.33 score = 0.593

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. aridity

z = +3.00 r = -0.16 score = 0.494

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. elev_mean

z = +1.89 r = +0.24 score = 0.456

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. q95

z = -1.00 r = +0.43 score = 0.432

low peak-flow values reflect a damped streamflow response with limited peaks.
6. frac_forest

z = -1.72 r = +0.23 score = 0.403

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. gvf_diff

z = -1.09 r = -0.33 score = 0.356

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. low_prec_freq

z = +1.14 r = -0.31 score = 0.356

many dry days indicate a drier climate where streamflow depends on stored water.
9. gauge_lon

z = -0.78 r = -0.43 score = 0.338

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
10. q_mean

z = -0.95 r = +0.35 score = 0.335

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09312600: WHITE RIVER BL TABBYUNE C NEAR SOLDIER SUMMIT, UT

Dominant forcing: SRAD Peak TE: 0.0027 Peak lag: 9 day(s)
Summary: SRAD-dominant at lag 9 day(s); top drivers consistent with high stream_elas, high frac_snow, high elev_mean.

Top 10 driver attributes

1. stream_elas

z = +2.25 r = -0.26 score = 0.582

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. frac_snow

z = +2.10 r = +0.26 score = 0.536

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. elev_mean

z = +2.13 r = +0.24 score = 0.514

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. gauge_lon

z = -0.94 r = -0.43 score = 0.406

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. runoff_ratio

z = -0.92 r = +0.43 score = 0.394

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. frac_forest

z = -1.59 r = +0.23 score = 0.375

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. p_seasonality

z = -0.81 r = -0.46 score = 0.369

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. aridity

z = +2.13 r = -0.16 score = 0.351

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. q95

z = -0.76 r = +0.43 score = 0.327

low peak-flow values reflect a damped streamflow response with limited peaks.
10. q_mean

z = -0.79 r = +0.35 score = 0.280

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09352900: VALLECITO CREEK NEAR BAYFIELD, CO.

Dominant forcing: Tair Peak TE: 0.0105 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, high slope_mean.

Top 10 driver attributes

1. frac_snow

z = +2.24

r = +0.46

score = 1.035

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +3.21

r = +0.30

score = 0.954

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. slope_mean

z = +2.33

r = +0.41

score = 0.950

steep terrain promotes fast surface runoff and a quick streamflow response.
4. runoff_ratio

z = +1.47

r = +0.56

score = 0.816

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. q95

z = +0.48

r = +0.51

score = 0.246

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. gvf_max

z = -1.75

r = +0.14

score = 0.241

low maximum green vegetation fraction reflects sparse canopy cover.
7. high_prec_freq

z = -0.48

r = -0.49

score = 0.236

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. stream_elas

z = -0.51

r = -0.43

score = 0.219

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. pet_mean

z = +0.63

r = -0.33

score = 0.207

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
10. other_frac

z = +1.92

r = +0.11

score = 0.202

a large 'other' land fraction reflects mixed or atypical surface cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09378170: SOUTH CREEK ABOVE RESERVOIR NEAR MONTICELLO, UT

Dominant forcing: SRAD Peak TE: 0.0057 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high high_prec_freq, high high_prec_dur, low runoff_ratio.

Top 10 driver attributes

1. high_prec_freq

z = +1.85 r = -0.33 score = 0.604

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. high_prec_dur

z = +1.86 r = +0.29 score = 0.537

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. runoff_ratio

z = -1.21 r = +0.43 score = 0.519

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. elev_mean

z = +1.97 r = +0.24 score = 0.475

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. low_prec_freq

z = +1.39 r = -0.31 score = 0.433

many dry days indicate a drier climate where streamflow depends on stored water.
6. frac_snow

z = +1.61 r = +0.26 score = 0.410

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
7. stream_elas

z = +1.47 r = -0.26 score = 0.380

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. glim_1st_class_frac

z = -1.76 r = -0.21 score = 0.377

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. q95

z = -0.86 r = +0.43 score = 0.370

low peak-flow values reflect a damped streamflow response with limited peaks.
10. gauge_lon

z = -0.84 r = -0.43 score = 0.362

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09378630: RECAPTURE CREEK NEAR BLANDING, UT

Dominant forcing: SRAD Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high stream_elas, high zero_q_freq, high high_prec_freq.

Top 10 driver attributes

1. stream_elas

z = +3.16 r = -0.26 score = 0.817

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
2. zero_q_freq

z = +3.06 r = -0.23 score = 0.692

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. high_prec_freq

z = +1.89 r = -0.33 score = 0.615

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. high_prec_dur

z = +1.99 r = +0.29 score = 0.573

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. runoff_ratio

z = -1.09 r = +0.43 score = 0.466

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. high_q_dur

z = +2.85 r = +0.16 score = 0.446

long high-flow events suggest sustained wet conditions and slow recession.
7. low_prec_freq

z = +1.41 r = -0.31 score = 0.440

many dry days indicate a drier climate where streamflow depends on stored water.
8. elev_mean

z = +1.74 r = +0.24 score = 0.419

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. frac_snow

z = +1.60 r = +0.26 score = 0.409

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
10. gauge_lon

z = -0.84 r = -0.43 score = 0.365

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09386900: RIO NUTRIA NEAR RAMAH, NM

Dominant forcing: SRAD Peak TE: 0.0017 Peak lag: 8 day(s)
Summary: SRAD-dominant at lag 8 day(s); top drivers consistent with high high_prec_freq, high stream_elas, low runoff_ratio.

Top 10 driver attributes

1. high_prec_freq

z = +2.59 r = -0.33 score = 0.845

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. stream_elas

z = +2.45 r = -0.26 score = 0.633

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
3. runoff_ratio

z = -1.45 r = +0.43 score = 0.622

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. gvf_diff

z = -1.50 r = -0.33 score = 0.491

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. elev_mean

z = +2.01 r = +0.24 score = 0.486

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. low_prec_freq

z = +1.53 r = -0.31 score = 0.477

many dry days indicate a drier climate where streamflow depends on stored water.
7. q95

z = -0.98 r = +0.43 score = 0.422

low peak-flow values reflect a damped streamflow response with limited peaks.
8. aridity

z = +2.31 r = -0.16 score = 0.379

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. high_prec_dur

z = +1.28 r = +0.29 score = 0.370

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. baseflow_index

z = -1.68 r = +0.22 score = 0.368

a low baseflow index means streamflow responds quickly to recent rainfall.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09404450: EAST FORK VIRGIN RIVER NEAR GLENDALE, UT

Dominant forcing: SRAD Peak TE: 0.0019 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high aridity, low runoff_ratio, low gauge_lon.

Top 10 driver attributes

1. aridity

z = +2.93

r = -0.16

score = 0.482

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. runoff_ratio

z = -1.09

r = +0.43

score = 0.468

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gauge_lon

z = -1.04

r = -0.43

score = 0.448

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. high_prec_freq

z = +1.31

r = -0.33

score = 0.428

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. low_prec_freq

z = +1.36

r = -0.31

score = 0.424

many dry days indicate a drier climate where streamflow depends on stored water.
6. q95

z = -0.94

r = +0.43

score = 0.407

low peak-flow values reflect a damped streamflow response with limited peaks.
7. elev_mean

z = +1.67

r = +0.24

score = 0.402

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
8. baseflow_index

z = +1.78

r = +0.22

score = 0.389

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
9. gvf_diff

z = -1.16

r = -0.33

score = 0.380

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. high_prec_dur

z = +1.31

r = +0.29

score = 0.379

long heavy-rainfall events allow more soil saturation and stronger runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09430500: GILA RIVER NEAR GILA, NM

Dominant forcing: PRCP Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, high area_gages2.

Top 10 driver attributes

1. aridity

z = +2.63

r = -0.28

score = 0.725

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.39

r = +0.34

score = 0.480

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. area_gages2

z = +2.36

r = +0.18

score = 0.426

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.
4. area_geospa_fabric

z = +2.34

r = +0.17

score = 0.399

a large geospatial drainage area integrates many sub-basin responses.
5. high_prec_dur

z = +1.65

r = +0.23

score = 0.374

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. elev_mean

z = +1.87

r = -0.20

score = 0.365

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
7. gvf_diff

z = -1.39

r = -0.24

score = 0.337

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. runoff_ratio

z = -1.36

r = +0.23

score = 0.314

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. q95

z = -0.96

r = +0.31

score = 0.294

low peak-flow values reflect a damped streamflow response with limited peaks.
10. glim_1st_class_frac

z = -1.92

r = -0.15

score = 0.280

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09430600: MOGOLLON CREEK NEAR CLIFF, NM

Dominant forcing: PRCP Peak TE: 0.0026 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, high elev_mean.

Top 10 driver attributes

1. aridity

z = +2.64

r = -0.28

score = 0.729

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.18

r = +0.34

score = 0.408

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. elev_mean

z = +1.74

r = -0.20

score = 0.340

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. high_prec_freq

z = +1.73

r = -0.19

score = 0.326

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. pet_mean

z = +2.66

r = -0.12

score = 0.324

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
6. gvf_diff

z = -1.26

r = -0.24

score = 0.306

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. low_prec_freq

z = +1.30

r = -0.22

score = 0.292

many dry days indicate a drier climate where streamflow depends on stored water.
8. glim_1st_class_frac

z = -1.72

r = -0.15

score = 0.250

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. high_prec_dur

z = +1.08

r = +0.23

score = 0.245

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. glim_2nd_class_frac

z = +1.58

r = +0.15

score = 0.243

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09447800: BONITA CREEK NEAR MORENCI, AZ.

Dominant forcing: PRCP Peak TE: 0.0015 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, low gvf_diff.

Top 10 driver attributes

1. aridity

z = +3.22

r = -0.28

score = 0.889

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.55

r = +0.34

score = 0.534

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. gvf_diff

z = -1.62

r = -0.24

score = 0.393

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. low_prec_freq

z = +1.59

r = -0.22

score = 0.355

many dry days indicate a drier climate where streamflow depends on stored water.
5. runoff_ratio

z = -1.52

r = +0.23

score = 0.350

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. high_prec_dur

z = +1.51

r = +0.23

score = 0.343

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. glim_2nd_class_frac

z = +2.12

r = +0.15

score = 0.326

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. q95

z = -1.02

r = +0.31

score = 0.310

low peak-flow values reflect a damped streamflow response with limited peaks.
9. gvf_max

z = -2.63

r = +0.12

score = 0.304

low maximum green vegetation fraction reflects sparse canopy cover.
10. slope_fdc

z = -1.51

r = +0.20

score = 0.303

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09480000: SANTA CRUZ RIVER NEAR LOCHIEL, AZ.

Dominant forcing: PRCP Peak TE: 0.0032 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high high_prec_dur, low p_mean.

Top 10 driver attributes

1. aridity

z = +2.27 r = -0.28 score = 0.628

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. high_prec_dur

z = +2.07 r = +0.23 score = 0.469

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_mean

z = -1.30 r = +0.34 score = 0.447

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. high_prec_freq

z = +1.96 r = -0.19 score = 0.369

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. runoff_ratio

z = -1.56 r = +0.23 score = 0.361

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. low_prec_freq

z = +1.50 r = -0.22 score = 0.335

many dry days indicate a drier climate where streamflow depends on stored water.
7. q95

z = -1.01 r = +0.31 score = 0.307

low peak-flow values reflect a damped streamflow response with limited peaks.
8. p_seasonality

z = +1.22 r = -0.24 score = 0.298

strong precipitation seasonality means rainfall is concentrated in part of the year, which sharpens seasonal streamflow response.
9. slope_fdc

z = -1.38 r = +0.20 score = 0.278

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. q_mean

z = -0.95 r = +0.28 score = 0.262

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09484000: SABINO CREEK NEAR TUCSON, AZ.

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high low_prec_freq, low p_mean.

Top 10 driver attributes

1. aridity

z = +3.06

r = -0.28

score = 0.844

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. low_prec_freq

z = +1.86

r = -0.22

score = 0.416

many dry days indicate a drier climate where streamflow depends on stored water.
3. p_mean

z = -1.20

r = +0.34

score = 0.414

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. soil_depth_statsgo

z = -3.09

r = +0.13

score = 0.413

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
5. pet_mean

z = +3.26

r = -0.12

score = 0.398

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
6. gvf_diff

z = -1.46

r = -0.24

score = 0.354

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. zero_q_freq

z = +2.51

r = -0.13

score = 0.335

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
8. high_prec_freq

z = +1.52

r = -0.19

score = 0.287

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. high_prec_dur

z = +1.17

r = +0.23

score = 0.265

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. root_depth_99

z = +2.60

r = +0.10

score = 0.263

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09484600: PANTANO WASH NEAR VAIL, AZ.

Dominant forcing: PRCP Peak TE: 0.0071 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, high hfd_mean.

Top 10 driver attributes

1. aridity

z = +3.05

r = -0.28

score = 0.841

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.41

r = +0.34

score = 0.487

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. hfd_mean

z = +1.92

r = -0.24

score = 0.456

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
4. high_prec_dur

z = +1.67

r = +0.23

score = 0.379

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. runoff_ratio

z = -1.62

r = +0.23

score = 0.374

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. slope_fdc

z = -1.76

r = +0.20

score = 0.354

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. high_prec_freq

z = +1.73

r = -0.19

score = 0.326

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. low_prec_freq

z = +1.45

r = -0.22

score = 0.324

many dry days indicate a drier climate where streamflow depends on stored water.
9. q95

z = -1.02

r = +0.31

score = 0.310

low peak-flow values reflect a damped streamflow response with limited peaks.
10. gvf_diff

z = -1.23

r = -0.24

score = 0.299

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09492400: EAST FORK WHITE RIVER NEAR FORT APACHE, AZ.

Dominant forcing: SRAD Peak TE: 0.0025 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high high_prec_dur, high elev_mean, high high_prec_freq.

Top 10 driver attributes

1. high_prec_dur

z = +1.84 r = +0.29 score = 0.531

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. elev_mean

z = +2.18 r = +0.24 score = 0.525

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = +1.24 r = -0.33 score = 0.403

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
4. gauge_lon

z = -0.86 r = -0.43 score = 0.374

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. low_prec_freq

z = +1.12 r = -0.31 score = 0.349

many dry days indicate a drier climate where streamflow depends on stored water.
6. gauge_lat

z = -1.04 r = +0.29 score = 0.305

southern latitude is associated with warmer climates and rainfall-driven streamflow.
7. q95

z = -0.57 r = +0.43 score = 0.247

low peak-flow values reflect a damped streamflow response with limited peaks.
8. gvf_diff

z = -0.74 r = -0.33 score = 0.243

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. frac_forest

z = +0.94 r = +0.23 score = 0.222

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. aridity

z = +1.33 r = -0.16 score = 0.219

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09494000: WHITE RIVER NEAR FORT APACHE, AZ.

Dominant forcing: SRAD Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low runoff_ratio, high elev_mean.

Top 10 driver attributes

1. high_prec_dur

z = +1.59 r = +0.29 score = 0.460

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. runoff_ratio

z = -1.04 r = +0.43 score = 0.443

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. elev_mean

z = +1.77 r = +0.24 score = 0.428

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. glim_1st_class_frac

z = -1.84 r = -0.21 score = 0.394

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. gauge_lon

z = -0.89 r = -0.43 score = 0.383

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. q95

z = -0.82 r = +0.43 score = 0.354

low peak-flow values reflect a damped streamflow response with limited peaks.
7. gvf_diff

z = -1.03 r = -0.33 score = 0.338

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. gauge_lat

z = -1.06 r = +0.29 score = 0.310

southern latitude is associated with warmer climates and rainfall-driven streamflow.
9. q_mean

z = -0.81 r = +0.35 score = 0.285

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. low_prec_freq

z = +0.78 r = -0.31 score = 0.242

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09497800: CIBECUE CREEK NEAR CHYSOTILE, AZ.

Dominant forcing: PRCP Peak TE: 0.0046 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, low gvf_diff.

Top 10 driver attributes

1. aridity

z = +2.67

r = -0.28

score = 0.738

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.23

r = +0.34

score = 0.423

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. gvf_diff

z = -1.57

r = -0.24

score = 0.383

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. runoff_ratio

z = -1.28

r = +0.23

score = 0.295

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. high_prec_dur

z = +1.30

r = +0.23

score = 0.294

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. pet_mean

z = +2.41

r = -0.12

score = 0.294

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. slope_fdc

z = -1.44

r = +0.20

score = 0.289

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. q95

z = -0.94

r = +0.31

score = 0.286

low peak-flow values reflect a damped streamflow response with limited peaks.
9. low_prec_freq

z = +1.22

r = -0.22

score = 0.272

many dry days indicate a drier climate where streamflow depends on stored water.
10. gauge_lon

z = -0.91

r = -0.27

score = 0.247

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09497980: CHERRY CREEK NEAR GLOBE, AZ.

Dominant forcing: PRCP Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, low gvf_diff.

Top 10 driver attributes

1. aridity

z = +1.78 r = -0.28 score = 0.492

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.14 r = +0.34 score = 0.394

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. gvf_diff

z = -1.52 r = -0.24 score = 0.370

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. high_prec_dur

z = +1.61 r = +0.23 score = 0.364

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. slope_fdc

z = -1.80 r = +0.20 score = 0.362

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. low_prec_freq

z = +1.44 r = -0.22 score = 0.323

many dry days indicate a drier climate where streamflow depends on stored water.
7. runoff_ratio

z = -1.31 r = +0.23 score = 0.303

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
8. q95

z = -0.93 r = +0.31 score = 0.284

low peak-flow values reflect a damped streamflow response with limited peaks.
9. gauge_lon

z = -0.93 r = -0.27 score = 0.252

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
10. q_mean

z = -0.88 r = +0.28 score = 0.244

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09505200: WET BEAVER CREEK NEAR RIMROCK, AZ.

Dominant forcing: PRCP Peak TE: 0.0030 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low slope_fdc, low p_mean.

Top 10 driver attributes

1. aridity

z = +1.72

r = -0.28

score = 0.474

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. slope_fdc

z = -2.23

r = +0.20

score = 0.448

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
3. p_mean

z = -1.12

r = +0.34

score = 0.386

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. high_prec_dur

z = +1.63

r = +0.23

score = 0.369

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. gvf_diff

z = -1.40

r = -0.24

score = 0.341

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. low_prec_freq

z = +1.30

r = -0.22

score = 0.292

many dry days indicate a drier climate where streamflow depends on stored water.
7. elev_mean

z = +1.38

r = -0.20

score = 0.270

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
8. gauge_lon

z = -0.98

r = -0.27

score = 0.266

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. q95

z = -0.81

r = +0.31

score = 0.249

low peak-flow values reflect a damped streamflow response with limited peaks.
10. runoff_ratio

z = -1.04

r = +0.23

score = 0.239

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09505350: DRY BEAVER CREEK NEAR RIMROCK, AZ.

Dominant forcing: SRAD Peak TE: 0.0044 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, high high_prec_dur, low gvf_diff.

Top 10 driver attributes

- 1. zero_q_freq z = +5.78 r = -0.23 score = 1.308**
frequent zero-flow days indicate an ephemeral or highly water-limited basin.
- 2. high_prec_dur z = +1.77 r = +0.29 score = 0.511**
long heavy-rainfall events allow more soil saturation and stronger runoff response.
- 3. gvf_diff z = -1.51 r = -0.33 score = 0.493**
weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
- 4. clay_frac z = +2.17 r = -0.22 score = 0.479**
high clay fraction reduces infiltration and increases surface runoff.
- 5. runoff_ratio z = -1.03 r = +0.43 score = 0.439**
a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
- 6. gauge_lon z = -0.99 r = -0.43 score = 0.426**
western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
- 7. high_prec_freq z = +1.30 r = -0.33 score = 0.425**
frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
- 8. low_prec_freq z = +1.35 r = -0.31 score = 0.422**
many dry days indicate a drier climate where streamflow depends on stored water.
- 9. baseflow_index z = -1.56 r = +0.22 score = 0.342**
a low baseflow index means streamflow responds quickly to recent rainfall.
- 10. q95 z = -0.77 r = +0.43 score = 0.334**
low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09505800: WEST CLEAR CREEK NEAR CAMP VERDE, AZ.

Dominant forcing: SRAD Peak TE: 0.0026 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low runoff_ratio, low gvf_diff.

Top 10 driver attributes

1. high_prec_dur

z = +1.86 r = +0.29 score = 0.537

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. runoff_ratio

z = -1.14 r = +0.43 score = 0.488

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
3. gvf_diff

z = -1.37 r = -0.33 score = 0.448

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lon

z = -0.98 r = -0.43 score = 0.424

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. q95

z = -0.88 r = +0.43 score = 0.379

low peak-flow values reflect a damped streamflow response with limited peaks.
6. low_prec_freq

z = +1.21 r = -0.31 score = 0.377

many dry days indicate a drier climate where streamflow depends on stored water.
7. clay_frac

z = +1.70 r = -0.22 score = 0.375

high clay fraction reduces infiltration and increases surface runoff.
8. elev_mean

z = +1.41 r = +0.24 score = 0.339

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. high_prec_freq

z = +0.95 r = -0.33 score = 0.310

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. q_mean

z = -0.83 r = +0.35 score = 0.293

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09508300: WET BOTTOM CREEK NEAR CHILDS, AZ.

Dominant forcing: SRAD Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low gvf_diff, high high_prec_dur, high low_prec_freq.

Top 10 driver attributes

1. gvf_diff z = -1.72 r = -0.33 score = 0.563

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
2. high_prec_dur z = +1.90 r = +0.29 score = 0.547

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. low_prec_freq z = +1.62 r = -0.31 score = 0.506

many dry days indicate a drier climate where streamflow depends on stored water.
4. zero_q_freq z = +1.98 r = -0.23 score = 0.447

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
5. high_prec_freq z = +1.37 r = -0.33 score = 0.446

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
6. gauge_lon z = -0.98 r = -0.43 score = 0.424

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. baseflow_index z = -1.92 r = +0.22 score = 0.420

a low baseflow index means streamflow responds quickly to recent rainfall.
8. aridity z = +2.37 r = -0.16 score = 0.390

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. stream_elas z = +1.42 r = -0.26 score = 0.368

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. q95 z = -0.77 r = +0.43 score = 0.332

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09510200: SYCAMORE CREEK NEAR FORT MCDOWELL, AZ.

Dominant forcing: SRAD Peak TE: 0.0047 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high low_prec_freq, low gvf_diff, low runoff_ratio.

Top 10 driver attributes

1. low_prec_freq

z = +1.90 r = -0.31 score = 0.592

many dry days indicate a drier climate where streamflow depends on stored water.
2. gvf_diff

z = -1.72 r = -0.33 score = 0.564

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. runoff_ratio

z = -1.13 r = +0.43 score = 0.484

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. high_prec_dur

z = +1.62 r = +0.29 score = 0.468

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. aridity

z = +2.79 r = -0.16 score = 0.458

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
6. gauge_lon

z = -0.97 r = -0.43 score = 0.420

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. p_seasonality

z = -0.88 r = -0.46 score = 0.403

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q95

z = -0.92 r = +0.43 score = 0.395

low peak-flow values reflect a damped streamflow response with limited peaks.
9. high_prec_freq

z = +1.20 r = -0.33 score = 0.393

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. slope_mean

z = +0.99 r = +0.37 score = 0.364

steep terrain promotes fast surface runoff and a quick streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09512280: CAVE CREEK BLW COTTONWOOD CR NEAR CAVE CREEK, AZ.

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high zero_q_freq, low p_mean.

Top 10 driver attributes

1. aridity

z = +3.04

r = -0.28

score = 0.839

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. zero_q_freq

z = +4.33

r = -0.13

score = 0.579

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. p_mean

z = -1.47

r = +0.34

score = 0.507

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. low_prec_freq

z = +2.04

r = -0.22

score = 0.456

many dry days indicate a drier climate where streamflow depends on stored water.
5. root_depth_99

z = +4.14

r = +0.10

score = 0.419

very deep maximum rooting allows vegetation to tap deep storage in droughts.
6. slope_fdc

z = -2.03

r = +0.20

score = 0.407

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. gvf_diff

z = -1.63

r = -0.24

score = 0.396

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. high_prec_dur

z = +1.62

r = +0.23

score = 0.367

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. runoff_ratio

z = -1.32

r = +0.23

score = 0.305

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. q95

z = -0.99

r = +0.31

score = 0.303

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 09513780: NEW RIVER NEAR ROCK SPRINGS, AZ.

Dominant forcing: PRCP Peak TE: 0.0037 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high zero_q_freq, low p_mean.

Top 10 driver attributes

1. aridity

z = +3.13

r = -0.28

score = 0.865

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. zero_q_freq

z = +5.26

r = -0.13

score = 0.702

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. p_mean

z = -1.47

r = +0.34

score = 0.507

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. slope_fdc

z = -2.40

r = +0.20

score = 0.483

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
5. low_prec_freq

z = +2.02

r = -0.22

score = 0.451

many dry days indicate a drier climate where streamflow depends on stored water.
6. gvf_diff

z = -1.63

r = -0.24

score = 0.397

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. high_prec_dur

z = +1.56

r = +0.23

score = 0.354

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. root_depth_99

z = +3.49

r = +0.10

score = 0.353

very deep maximum rooting allows vegetation to tap deep storage in droughts.
9. q95

z = -0.96

r = +0.31

score = 0.292

low peak-flow values reflect a damped streamflow response with limited peaks.
10. gvf_max

z = -2.52

r = +0.12

score = 0.292

low maximum green vegetation fraction reflects sparse canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10023000: BIG CREEK NEAR RANDOLPH, UT

Dominant forcing: SRAD Peak TE: 0.0036 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high frac_snow, high baseflow_index, high elev_mean.

Top 10 driver attributes

1. frac_snow

z = +1.82 r = +0.26 score = 0.466

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. baseflow_index

z = +2.11 r = +0.22 score = 0.462

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. elev_mean

z = +1.91 r = +0.24 score = 0.460

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. runoff_ratio

z = -1.07 r = +0.43 score = 0.458

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. gauge_lon

z = -0.95 r = -0.43 score = 0.412

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. q95

z = -0.89 r = +0.43 score = 0.384

low peak-flow values reflect a damped streamflow response with limited peaks.
7. p_seasonality

z = -0.72 r = -0.46 score = 0.329

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q_mean

z = -0.83 r = +0.35 score = 0.293

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. frac_forest

z = -1.19 r = +0.23 score = 0.280

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
10. gvf_diff

z = -0.74 r = -0.33 score = 0.243

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10166430: WEST CANYON CREEK NEAR CEDAR FORT, UT

Dominant forcing: SRAD Peak TE: 0.0038 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with low runoff_ratio, low gvf_diff, low gauge_lon.

Top 10 driver attributes

1. runoff_ratio

z = -1.30 r = +0.43 score = 0.556

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. gvf_diff

z = -1.47 r = -0.33 score = 0.481

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. gauge_lon

z = -1.01 r = -0.43 score = 0.435

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. q95

z = -0.90 r = +0.43 score = 0.390

low peak-flow values reflect a damped streamflow response with limited peaks.
5. frac_forest

z = -1.63 r = +0.23 score = 0.383

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. stream_elas

z = +1.37 r = -0.26 score = 0.355

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
7. q_mean

z = -0.89 r = +0.35 score = 0.316

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
8. high_q_dur

z = +1.72 r = +0.16 score = 0.270

long high-flow events suggest sustained wet conditions and slow recession.
9. p_mean

z = -1.34 r = +0.20 score = 0.269

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. elev_mean

z = +1.04 r = +0.24 score = 0.251

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10172700: VERNON CREEK NEAR VERNON, UT

Dominant forcing: SRAD Peak TE: 0.0031 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with low runoff_ratio, high baseflow_index, low gvf_diff.

Top 10 driver attributes

1. runoff_ratio

z = -1.29 r = +0.43 score = 0.554

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
2. baseflow_index

z = +2.37 r = +0.22 score = 0.518

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. gvf_diff

z = -1.41 r = -0.33 score = 0.463

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lon

z = -1.02 r = -0.43 score = 0.442

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. q95

z = -0.98 r = +0.43 score = 0.421

low peak-flow values reflect a damped streamflow response with limited peaks.
6. high_prec_freq

z = +1.25 r = -0.33 score = 0.407

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. frac_forest

z = -1.71 r = +0.23 score = 0.402

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. p_seasonality

z = -0.81 r = -0.46 score = 0.368

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. low_prec_freq

z = +1.11 r = -0.31 score = 0.345

many dry days indicate a drier climate where streamflow depends on stored water.
10. elev_mean

z = +1.39 r = +0.24 score = 0.336

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10172800: SOUTH WILLOW CREEK NEAR GRANTSVILLE, UT

Dominant forcing: SRAD Peak TE: 0.0061 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high slope_mean, high runoff_ratio, low gauge_lon.

Top 10 driver attributes

1. slope_mean

z = +1.90 r = +0.37 score = 0.700

steep terrain promotes fast surface runoff and a quick streamflow response.
2. runoff_ratio

z = +1.36 r = +0.43 score = 0.581

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. gauge_lon

z = -1.04 r = -0.43 score = 0.447

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. glim_1st_class_frac

z = -2.09 r = -0.21 score = 0.447

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
5. elev_mean

z = +1.69 r = +0.24 score = 0.407

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. frac_forest

z = -1.62 r = +0.23 score = 0.380

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. frac_snow

z = +1.45 r = +0.26 score = 0.371

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. baseflow_index

z = +1.43 r = +0.22 score = 0.313

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
9. gvf_diff

z = -0.92 r = -0.33 score = 0.302

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. p_seasonality

z = -0.59 r = -0.46 score = 0.271

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10173450: MAMMOTH CREEK ABV WEST HATCH DITCH, NEAR HATCH, UT

Dominant forcing: SRAD Peak TE: 0.0054 Peak lag: 1 day(s)

Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low gauge_lon.

Top 10 driver attributes

1. frac_snow

z = +2.27

r = +0.26

score = 0.579

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.39

r = +0.24

score = 0.577

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. gauge_lon

z = -1.03

r = -0.43

score = 0.446

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gvf_diff

z = -1.31

r = -0.33

score = 0.428

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. glim_1st_class_frac

z = -1.71

r = -0.21

score = 0.366

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. high_prec_freq

z = +1.05

r = -0.33

score = 0.342

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. high_prec_dur

z = +1.19

r = +0.29

score = 0.342

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. p_seasonality

z = -0.70

r = -0.46

score = 0.320

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. low_prec_freq

z = +0.94

r = -0.31

score = 0.292

many dry days indicate a drier climate where streamflow depends on stored water.
10. q95

z = -0.66

r = +0.43

score = 0.285

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10205030: SALINA CREEK NEAR EMERY, UT

Dominant forcing: SRAD Peak TE: 0.0034 Peak lag: 26 day(s)
Summary: SRAD-dominant at lag 26 day(s); top drivers consistent with high elev_mean, high frac_snow, low runoff_ratio.

Top 10 driver attributes

1. elev_mean

z = +2.20

r = +0.24

score = 0.531

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
2. frac_snow

z = +1.99

r = +0.26

score = 0.509

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. runoff_ratio

z = -1.00

r = +0.43

score = 0.430

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
4. gauge_lon

z = -0.97

r = -0.43

score = 0.420

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. frac_forest

z = -1.63

r = +0.23

score = 0.383

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. aridity

z = +2.29

r = -0.16

score = 0.377

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
7. glim_2nd_class_frac

z = +2.12

r = +0.18

score = 0.373

a large secondary geological class adds heterogeneity to the subsurface flow response.
8. q95

z = -0.86

r = +0.43

score = 0.371

low peak-flow values reflect a damped streamflow response with limited peaks.
9. q_mean

z = -0.81

r = +0.35

score = 0.284

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. baseflow_index

z = +1.30

r = +0.22

score = 0.284

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10234500: BEAVER RIVER NEAR BEAVER, UT

Dominant forcing: SRAD Peak TE: 0.0050 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low gauge_lon.

Top 10 driver attributes

1. frac_snow

z = +2.24

r = +0.26

score = 0.571

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.21

r = +0.24

score = 0.534

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. gauge_lon

z = -1.03

r = -0.43

score = 0.447

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. high_prec_freq

z = +1.31

r = -0.33

score = 0.428

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. slope_mean

z = +1.04

r = +0.37

score = 0.382

steep terrain promotes fast surface runoff and a quick streamflow response.
6. gvf_diff

z = -1.11

r = -0.33

score = 0.362

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. q95

z = -0.71

r = +0.43

score = 0.307

low peak-flow values reflect a damped streamflow response with limited peaks.
8. low_prec_freq

z = +0.96

r = -0.31

score = 0.298

many dry days indicate a drier climate where streamflow depends on stored water.
9. baseflow_index

z = +1.30

r = +0.22

score = 0.284

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. aridity

z = +1.64

r = -0.16

score = 0.269

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10242000: COAL CREEK NEAR CEDAR CITY, UT

Dominant forcing: SRAD Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high aridity, low gvf_diff, low gauge_lon.

Top 10 driver attributes

1. aridity

z = +3.71 r = -0.16 score = 0.610

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. gvf_diff

z = -1.56 r = -0.33 score = 0.511

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
3. gauge_lon

z = -1.06 r = -0.43 score = 0.460

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. high_prec_freq

z = +1.37 r = -0.33 score = 0.446

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. glim_1st_class_frac

z = -1.94 r = -0.21 score = 0.417

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. high_prec_dur

z = +1.37 r = +0.29 score = 0.394

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. elev_mean

z = +1.62 r = +0.24 score = 0.392

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
8. low_prec_freq

z = +1.17 r = -0.31 score = 0.365

many dry days indicate a drier climate where streamflow depends on stored water.
9. p_seasonality

z = -0.70 r = -0.46 score = 0.321

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. q95

z = -0.71 r = +0.43 score = 0.305

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10244950: STEPTOE C NR ELY, NV

Dominant forcing: SRAD Peak TE: 0.0032 Peak lag: 21 day(s)
Summary: SRAD-dominant at lag 21 day(s); top drivers consistent with high high_prec_freq, high high_prec_dur, high frac_snow.

Top 10 driver attributes

1. high_prec_freq

z = +2.24 r = -0.33 score = 0.730

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
2. high_prec_dur

z = +2.13 r = +0.29 score = 0.614

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. frac_snow

z = +2.36 r = +0.26 score = 0.601

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. elev_mean

z = +2.09 r = +0.24 score = 0.505

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. gauge_lon

z = -1.17 r = -0.43 score = 0.504

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. baseflow_index

z = +2.11 r = +0.22 score = 0.462

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
7. gvf_diff

z = -1.40 r = -0.33 score = 0.458

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. low_prec_freq

z = +1.45 r = -0.31 score = 0.451

many dry days indicate a drier climate where streamflow depends on stored water.
9. slope_mean

z = +1.17 r = +0.37 score = 0.428

steep terrain promotes fast surface runoff and a quick streamflow response.
10. aridity

z = +2.52 r = -0.16 score = 0.415

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10249300: S TWIN R NR ROUND MOUNTAIN, NV

Dominant forcing: SRAD Peak TE: 0.0021 Peak lag: 29 day(s)
Summary: SRAD-dominant at lag 29 day(s); top drivers consistent with high aridity, high high_prec_freq, low gvf_diff.

Top 10 driver attributes

1. aridity

z = +6.75

r = -0.16

score = 1.110

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. high_prec_freq

z = +1.92

r = -0.33

score = 0.626

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
3. gvf_diff

z = -1.84

r = -0.33

score = 0.601

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. low_prec_freq

z = +1.87

r = -0.31

score = 0.584

many dry days indicate a drier climate where streamflow depends on stored water.
5. gauge_lon

z = -1.32

r = -0.43

score = 0.572

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. elev_mean

z = +1.90

r = +0.24

score = 0.460

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
7. slope_mean

z = +1.12

r = +0.37

score = 0.411

steep terrain promotes fast surface runoff and a quick streamflow response.
8. p_mean

z = -1.85

r = +0.20

score = 0.372

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. other_frac

z = +2.62

r = +0.14

score = 0.364

a large 'other' land fraction reflects mixed or atypical surface cover.
10. frac_forest

z = -1.55

r = +0.23

score = 0.364

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10258000: TAHQUITZ C NR PALM SPRINGS CA

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_seasonality, high low_prec_freq.

Top 10 driver attributes

1. aridity

z = +3.26 r = -0.28 score = 0.900

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_seasonality

z = -2.45 r = -0.24 score = 0.598

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. low_prec_freq

z = +2.47 r = -0.22 score = 0.553

many dry days indicate a drier climate where streamflow depends on stored water.
4. high_prec_dur

z = +2.35 r = +0.23 score = 0.534

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. p_mean

z = -1.53 r = +0.34 score = 0.527

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. gvf_diff

z = -1.59 r = -0.24 score = 0.387

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lon

z = -1.28 r = -0.27 score = 0.348

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. zero_q_freq

z = +2.16 r = -0.13 score = 0.289

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
9. soil_conductivity

z = +1.88 r = +0.15 score = 0.277

high soil conductivity allows fast infiltration and subsurface flow.
10. soil_depth_statsgo

z = -1.84 r = +0.13 score = 0.246

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10258500: PALM CYN C NR PALM SPRINGS CA

Dominant forcing: PRCP Peak TE: 0.0024 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, high zero_q_freq, low p_seasonality.

Top 10 driver attributes

1. aridity

z = +4.49

r = -0.28

score = 1.239

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. zero_q_freq

z = +5.52

r = -0.13

score = 0.738

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. p_seasonality

z = -2.47

r = -0.24

score = 0.603

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. p_mean

z = -1.73

r = +0.34

score = 0.595

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. low_prec_freq

z = +2.54

r = -0.22

score = 0.567

many dry days indicate a drier climate where streamflow depends on stored water.
6. slope_fdc

z = -2.44

r = +0.20

score = 0.490

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. gvff_diff

z = -1.91

r = -0.24

score = 0.463

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. high_prec_dur

z = +2.04

r = +0.23

score = 0.462

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. soil_depth_statsgo

z = -3.22

r = +0.13

score = 0.430

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. root_depth_99

z = +4.13

r = +0.10

score = 0.417

very deep maximum rooting allows vegetation to tap deep storage in droughts.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10259000: ANDREAS C NR PALM SPRINGS CA

Dominant forcing: PRCP Peak TE: 0.0038 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_seasonality, high low_prec_freq.

Top 10 driver attributes

1. aridity

z = +3.14

r = -0.28

score = 0.867

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_seasonality

z = -2.47

r = -0.24

score = 0.602

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. low_prec_freq

z = +2.53

r = -0.22

score = 0.566

many dry days indicate a drier climate where streamflow depends on stored water.
4. p_mean

z = -1.57

r = +0.34

score = 0.540

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. high_prec_dur

z = +2.14

r = +0.23

score = 0.484

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. gvf_diff

z = -1.77

r = -0.24

score = 0.430

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. soil_depth_statsgo

z = -3.08

r = +0.13

score = 0.413

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. gauge_lon

z = -1.28

r = -0.27

score = 0.348

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. q95

z = -0.86

r = +0.31

score = 0.263

low peak-flow values reflect a damped streamflow response with limited peaks.
10. gvf_max

z = -2.10

r = +0.12

score = 0.244

low maximum green vegetation fraction reflects sparse canopy cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10259200: DEEP C NR PALM DESERT CA

Dominant forcing: PRCP Peak TE: 0.0036 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, low p_seasonality.

Top 10 driver attributes

1. aridity

z = +6.02 r = -0.28 score = 1.662

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.84 r = +0.34 score = 0.635

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. p_seasonality

z = -2.49 r = -0.24 score = 0.608

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. low_prec_freq

z = +2.68 r = -0.22 score = 0.599

many dry days indicate a drier climate where streamflow depends on stored water.
5. zero_q_freq

z = +4.04 r = -0.13 score = 0.540

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
6. gvf_diff

z = -1.98 r = -0.24 score = 0.481

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. soil_depth_statsgo

z = -3.23 r = +0.13 score = 0.433

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. slope_fdc

z = -2.14 r = +0.20 score = 0.431

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. root_depth_99

z = +4.20 r = +0.10 score = 0.425

very deep maximum rooting allows vegetation to tap deep storage in droughts.
10. high_prec_dur

z = +1.74 r = +0.23 score = 0.394

long heavy-rainfall events allow more soil saturation and stronger runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10263500: BIG ROCK C NR VALYERMO CA

Dominant forcing: PRCP Peak TE: 0.0043 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high aridity, high low_prec_freq.

Top 10 driver attributes

1. p_seasonality

z = -2.52 r = -0.24 score = 0.616

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. aridity

z = +2.14 r = -0.28 score = 0.590

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. low_prec_freq

z = +2.24 r = -0.22 score = 0.502

many dry days indicate a drier climate where streamflow depends on stored water.
4. p_mean

z = -1.37 r = +0.34 score = 0.473

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. high_prec_dur

z = +2.08 r = +0.23 score = 0.471

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. gvf_diff

z = -1.88 r = -0.24 score = 0.458

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. gauge_lon

z = -1.36 r = -0.27 score = 0.369

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. root_depth_99

z = +3.08 r = +0.10 score = 0.312

very deep maximum rooting allows vegetation to tap deep storage in droughts.
9. gvf_max

z = -2.51 r = +0.12 score = 0.290

low maximum green vegetation fraction reflects sparse canopy cover.
10. glim_1st_class_frac

z = -1.62 r = -0.15 score = 0.235

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10310500: CLEAR CREEK NEAR CARSON CITY, NV

Dominant forcing: PRCP Peak TE: 0.0019 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high aridity, high high_prec_dur, low p_seasonality.

Top 10 driver attributes

1. aridity

z = +2.62 r = -0.28 score = 0.724

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. high_prec_dur

z = +2.26 r = +0.23 score = 0.511

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_seasonality

z = -2.04 r = -0.24 score = 0.498

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. p_mean

z = -1.27 r = +0.34 score = 0.436

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. gauge_lon

z = -1.48 r = -0.27 score = 0.402

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. soil_conductivity

z = +2.61 r = +0.15 score = 0.385

high soil conductivity allows fast infiltration and subsurface flow.
7. low_prec_freq

z = +1.51 r = -0.22 score = 0.337

many dry days indicate a drier climate where streamflow depends on stored water.
8. gvff_diff

z = -1.37 r = -0.24 score = 0.333

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. soil_depth_statsgo

z = -2.31 r = +0.13 score = 0.309

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. elev_mean

z = +1.45 r = -0.20 score = 0.284

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10316500: LAMOILLE C NR LAMOILLE, NV

Dominant forcing: SRAD Peak TE: 0.0082 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high runoff_ratio, high slope_mean.

Top 10 driver attributes

1. frac_snow

z = +2.82

r = +0.26

score = 0.720

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. runoff_ratio

z = +1.61

r = +0.43

score = 0.690

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. slope_mean

z = +1.82

r = +0.37

score = 0.670

steep terrain promotes fast surface runoff and a quick streamflow response.
4. elev_mean

z = +2.20

r = +0.24

score = 0.531

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. high_q_dur

z = +3.39

r = +0.16

score = 0.531

long high-flow events suggest sustained wet conditions and slow recession.
6. gauge_lon

z = -1.21

r = -0.43

score = 0.525

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_freq

z = +1.42

r = -0.33

score = 0.464

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. high_prec_dur

z = +1.55

r = +0.29

score = 0.446

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. p_seasonality

z = -0.96

r = -0.46

score = 0.437

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. frac_forest

z = -1.68

r = +0.23

score = 0.395

low forest cover reduces evapotranspiration and may produce stronger surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10329500: MARTIN C NR PARADISE VALLEY, NV

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_mean, low gauge_lon.

Top 10 driver attributes

1. aridity

z = +2.37 r = -0.28 score = 0.654

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_mean

z = -1.35 r = +0.34 score = 0.464

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
3. gauge_lon

z = -1.33 r = -0.27 score = 0.362

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. high_prec_freq

z = +1.64 r = -0.19 score = 0.309

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
5. gvf_max

z = -2.62 r = +0.12 score = 0.303

low maximum green vegetation fraction reflects sparse canopy cover.
6. p_seasonality

z = -1.24 r = -0.24 score = 0.303

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. frac_forest

z = -1.71 r = +0.17 score = 0.295

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
8. q95

z = -0.87 r = +0.31 score = 0.265

low peak-flow values reflect a damped streamflow response with limited peaks.
9. gvf_diff

z = -1.08 r = -0.24 score = 0.262

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. runoff_ratio

z = -1.09 r = +0.23 score = 0.251

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10336645: GENERAL C NR MEEKS BAY CA

Dominant forcing: PRCP Peak TE: 0.0075 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +2.71 r = +0.23 score = 0.614

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.07 r = -0.24 score = 0.505

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.50 r = -0.27 score = 0.408

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. elev_mean

z = +2.03 r = -0.20 score = 0.397

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
5. frac_snow

z = +2.40 r = -0.15 score = 0.351

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. gvf_diff

z = -1.23 r = -0.24 score = 0.298

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. q95

z = +0.95 r = +0.31 score = 0.289

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. soil_conductivity

z = +1.65 r = +0.15 score = 0.243

high soil conductivity allows fast infiltration and subsurface flow.
9. hfd_mean

z = +0.98 r = -0.24 score = 0.234

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. glim_2nd_class_frac

z = +1.51 r = +0.15 score = 0.232

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10336660: BLACKWOOD C NR TAHOE CITY CA

Dominant forcing: PRCP Peak TE: 0.0044 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, high q95.

Top 10 driver attributes

1. high_prec_dur

z = +2.95 r = +0.23 score = 0.669

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.09 r = -0.24 score = 0.509

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. q95

z = +1.57 r = +0.31 score = 0.479

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. gauge_lon

z = -1.50 r = -0.27 score = 0.408

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. runoff_ratio

z = +1.53 r = +0.23 score = 0.354

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. frac_snow

z = +2.42 r = -0.15 score = 0.354

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
7. elev_mean

z = +1.72 r = -0.20 score = 0.336

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
8. hfd_mean

z = +1.20 r = -0.24 score = 0.284

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. gvf_diff

z = -1.13 r = -0.24 score = 0.274

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. glim_1st_class_frac

z = -1.85 r = -0.15 score = 0.268

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10336740: Logan House Ck nr Glenbrook, NV

Dominant forcing: SRAD Peak TE: 0.0062 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low p_seasonality, high frac_snow, high high_prec_dur.

Top 10 driver attributes

1. **p_seasonality**

z = -2.04

r = -0.46

score = 0.930

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **frac_snow**

z = +2.76

r = +0.26

score = 0.704

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. **high_prec_dur**

z = +2.31

r = +0.29

score = 0.666

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. **gauge_lon**

z = -1.49

r = -0.43

score = 0.644

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. **runoff_ratio**

z = -1.10

r = +0.43

score = 0.470

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. **low_prec_freq**

z = +1.43

r = -0.31

score = 0.445

many dry days indicate a drier climate where streamflow depends on stored water.
7. **soil_conductivity**

z = +1.70

r = +0.25

score = 0.430

high soil conductivity allows fast infiltration and subsurface flow.
8. **clay_frac**

z = -1.91

r = -0.22

score = 0.421

low clay fraction allows easier infiltration and reduces surface runoff.
9. **elev_mean**

z = +1.73

r = +0.24

score = 0.417

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. **gvf_diff**

z = -1.27

r = -0.33

score = 0.414

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10343500: SAGEHEN C NR TRUCKEE CA

Dominant forcing: PRCP Peak TE: 0.0034 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +2.37 r = +0.23 score = 0.536

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.07 r = -0.24 score = 0.505

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.51 r = -0.27 score = 0.410

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. frac_snow

z = +2.64 r = -0.15 score = 0.387

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. elev_mean

z = +1.78 r = -0.20 score = 0.347

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. glim_2nd_class_frac

z = +1.81 r = +0.15 score = 0.278

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. gvff_diff

z = -1.07 r = -0.24 score = 0.261

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. low_prec_freq

z = +0.99 r = -0.22 score = 0.221

many dry days indicate a drier climate where streamflow depends on stored water.
9. hfd_mean

z = +0.78 r = -0.24 score = 0.184

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
10. p_mean

z = -0.49 r = +0.34 score = 0.168

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10348850: GALENA C AT GALENA C STATE PARK

Dominant forcing: PRCP Peak TE: 0.0010 Peak lag: 11 day(s)
Summary: PRCP-dominant at lag 11 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, high frac_snow.

Top 10 driver attributes

1. high_prec_dur

z = +2.51 r = +0.23 score = 0.568

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.12 r = -0.24 score = 0.517

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. frac_snow

z = +3.20 r = -0.15 score = 0.469

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. gauge_lon

z = -1.48 r = -0.27 score = 0.403

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. elev_mean

z = +1.63 r = -0.20 score = 0.319

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. gvf_diff

z = -1.26 r = -0.24 score = 0.306

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. low_prec_freq

z = +1.13 r = -0.22 score = 0.253

many dry days indicate a drier climate where streamflow depends on stored water.
8. hfd_mean

z = +0.98 r = -0.24 score = 0.234

a late half-flow date indicates streamflow is concentrated later in the water year, often due to snowmelt.
9. glim_1st_class_frac

z = -1.53 r = -0.15 score = 0.222

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. glim_2nd_class_frac

z = +1.33 r = +0.15 score = 0.203

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 10396000: DONNER UND BLITZEN RIVER NR FRENCHGLEN OR

Dominant forcing: VP Peak TE: 0.0069 Peak lag: 1 day(s)
Summary: VP-dominant at lag 1 day(s); top drivers consistent with high aridity, low frac_forest, high frac_snow.

Top 10 driver attributes

1. aridity

z = +1.82

r = -0.31

score = 0.570

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. frac_forest

z = -1.70

r = +0.29

score = 0.495

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
3. frac_snow

z = +1.42

r = +0.31

score = 0.442

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. clay_frac

z = +1.75

r = -0.22

score = 0.384

high clay fraction reduces infiltration and increases surface runoff.
5. gvf_max

z = -2.32

r = +0.14

score = 0.331

low maximum green vegetation fraction reflects sparse canopy cover.
6. low_prec_freq

z = +0.66

r = -0.50

score = 0.331

many dry days indicate a drier climate where streamflow depends on stored water.
7. high_prec_freq

z = +0.68

r = -0.47

score = 0.318

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. q95

z = -0.63

r = +0.49

score = 0.310

low peak-flow values reflect a damped streamflow response with limited peaks.
9. p_mean

z = -1.20

r = +0.24

score = 0.291

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. pet_mean

z = +1.09

r = -0.25

score = 0.278

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11098000: ARROYO SECO NR PASADENA CA

Dominant forcing: PRCP Peak TE: 0.0039 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, high low_prec_freq.

Top 10 driver attributes

1. p_seasonality

z = -2.60 r = -0.24 score = 0.635

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. high_prec_dur

z = +2.52 r = +0.23 score = 0.571

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. low_prec_freq

z = +2.13 r = -0.22 score = 0.477

many dry days indicate a drier climate where streamflow depends on stored water.
4. soil_depth_statsgo

z = -3.31 r = +0.13 score = 0.443

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
5. gvfdiff

z = -1.71 r = -0.24 score = 0.415

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. gauge_lon

z = -1.38 r = -0.27 score = 0.375

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. aridity

z = +1.25 r = -0.28 score = 0.344

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. p_mean

z = -0.92 r = +0.34 score = 0.316

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
9. hfd_mean

z = -1.05 r = -0.24 score = 0.249

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. slope_fdc

z = -1.13 r = +0.20 score = 0.227

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11124500: SANTA CRUZ C NR SANTA YNEZ CA

Dominant forcing: PRCP Peak TE: 0.0044 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, high low_prec_freq.

Top 10 driver attributes

1. high_prec_dur

z = +3.77 r = +0.23 score = 0.855

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.64 r = -0.24 score = 0.645

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. low_prec_freq

z = +2.09 r = -0.22 score = 0.468

many dry days indicate a drier climate where streamflow depends on stored water.
4. gauge_lon

z = -1.49 r = -0.27 score = 0.404

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. soil_depth_statsgo

z = -2.66 r = +0.13 score = 0.356

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
6. gvf_diff

z = -1.35 r = -0.24 score = 0.328

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. p_mean

z = -0.85 r = +0.34 score = 0.294

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. zero_q_freq

z = +1.99 r = -0.13 score = 0.267

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
9. hfd_mean

z = -0.99 r = -0.24 score = 0.236

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. aridity

z = +0.81 r = -0.28 score = 0.223

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11141280: LOPEZ C NR ARROYO GRANDE CA

Dominant forcing: PRCP Peak TE: 0.0066 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +3.27 r = +0.23 score = 0.740

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.52 r = -0.24 score = 0.615

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.52 r = -0.27 score = 0.413

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. low_prec_freq

z = +1.82 r = -0.22 score = 0.408

many dry days indicate a drier climate where streamflow depends on stored water.
5. p_mean

z = -0.97 r = +0.34 score = 0.336

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. gvf_diff

z = -1.19 r = -0.24 score = 0.289

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. high_prec_freq

z = +1.39 r = -0.19 score = 0.262

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. soil_depth_statsgo

z = -1.83 r = +0.13 score = 0.244

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
9. hfd_mean

z = -1.00 r = -0.24 score = 0.239

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. q95

z = -0.69 r = +0.31 score = 0.212

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11143000: BIG SUR R NR BIG SUR CA

Dominant forcing: PRCP Peak TE: 0.0095 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gvf_diff.

Top 10 driver attributes

1. high_prec_dur

z = +3.88 r = +0.23 score = 0.878

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.50 r = -0.24 score = 0.609

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gvf_diff

z = -1.81 r = -0.24 score = 0.439

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. gauge_lon

z = -1.60 r = -0.27 score = 0.435

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. soil_depth_statsgo

z = -2.51 r = +0.13 score = 0.336

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
6. runoff_ratio

z = +1.40 r = +0.23 score = 0.322

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. low_prec_freq

z = +1.38 r = -0.22 score = 0.309

many dry days indicate a drier climate where streamflow depends on stored water.
8. high_prec_freq

z = +1.47 r = -0.19 score = 0.276

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
9. hfd_mean

z = -1.13 r = -0.24 score = 0.269

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. q95

z = +0.73 r = +0.31 score = 0.221

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11148900: NACIMIENTO R BL SAPAQUE C NR BRYSON CA

Dominant forcing: PRCP Peak TE: 0.0068 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +3.21 r = +0.23 score = 0.727

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.57 r = -0.24 score = 0.628

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.56 r = -0.27 score = 0.424

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. zero_q_freq

z = +2.74 r = -0.13 score = 0.366

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
5. low_prec_freq

z = +1.62 r = -0.22 score = 0.363

many dry days indicate a drier climate where streamflow depends on stored water.
6. gvf_diff

z = -1.38 r = -0.24 score = 0.336

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. hfd_mean

z = -1.34 r = -0.24 score = 0.318

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. soil_depth_statsgo

z = -2.09 r = +0.13 score = 0.280

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
9. slope_fdc

z = -1.38 r = +0.20 score = 0.278

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. high_prec_freq

z = +1.29 r = -0.19 score = 0.243

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11151300: SAN LORENZO C BL BITTERWATER C NR KING CITY CA

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_seasonality, high high_prec_dur.

Top 10 driver attributes

1. aridity

z = +2.73

r = -0.28

score = 0.753

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_seasonality

z = -2.56

r = -0.24

score = 0.625

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. high_prec_dur

z = +2.49

r = +0.23

score = 0.563

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. p_mean

z = -1.36

r = +0.34

score = 0.470

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. gauge_lon

z = -1.56

r = -0.27

score = 0.423

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. low_prec_freq

z = +1.78

r = -0.22

score = 0.398

many dry days indicate a drier climate where streamflow depends on stored water.
7. slope_fdc

z = -1.70

r = +0.20

score = 0.341

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. runoff_ratio

z = -1.45

r = +0.23

score = 0.334

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. hfd_mean

z = -1.35

r = -0.24

score = 0.321

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. q95

z = -0.98

r = +0.31

score = 0.299

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11162500: PESCADERO C NR PESCADERO CA

Dominant forcing: PRCP Peak TE: 0.0077 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +2.74 r = +0.23 score = 0.621

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.44 r = -0.24 score = 0.596

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.64 r = -0.27 score = 0.445

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gvf_diff

z = -1.56 r = -0.24 score = 0.380

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. hfd_mean

z = -1.36 r = -0.24 score = 0.323

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. low_prec_freq

z = +1.25 r = -0.22 score = 0.281

many dry days indicate a drier climate where streamflow depends on stored water.
7. high_prec_freq

z = +1.27 r = -0.19 score = 0.239

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. slope_fdc

z = -0.95 r = +0.20 score = 0.190

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. soil_depth_statsgo

z = -1.25 r = +0.13 score = 0.167

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. pet_mean

z = +1.19 r = -0.12 score = 0.145

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11176400: ARROYO VALLE BL LANG CN NR LIVERMORE CA

Dominant forcing: PRCP Peak TE: 0.0079 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high aridity, high zero_q_freq.

Top 10 driver attributes

1. p_seasonality

z = -2.31 r = -0.24 score = 0.563

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. aridity

z = +1.87 r = -0.28 score = 0.517

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. zero_q_freq

z = +3.37 r = -0.13 score = 0.451

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
4. p_mean

z = -1.26 r = +0.34 score = 0.435

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
5. gauge_lon

z = -1.60 r = -0.27 score = 0.434

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. soil_depth_statsgo

z = -2.53 r = +0.13 score = 0.338

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. slope_fdc

z = -1.60 r = +0.20 score = 0.321

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. gvf_diff

z = -1.32 r = -0.24 score = 0.320

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. low_prec_freq

z = +1.40 r = -0.22 score = 0.312

many dry days indicate a drier climate where streamflow depends on stored water.
10. hfd_mean

z = -1.28 r = -0.24 score = 0.304

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11180500: DRY C A UNION CITY CA

Dominant forcing: PRCP Peak TE: 0.0049 Peak lag: 2 day(s)
Summary: PRCP-dominant at lag 2 day(s); top drivers consistent with high zero_q_freq, low p_seasonality, high high_prec_dur.

Top 10 driver attributes

1. zero_q_freq

z = +5.04 r = -0.13 score = 0.674

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. p_seasonality

z = -2.38 r = -0.24 score = 0.580

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. high_prec_dur

z = +2.14 r = +0.23 score = 0.484

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. slope_fdc

z = -2.20 r = +0.20 score = 0.443

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
5. gauge_lon

z = -1.62 r = -0.27 score = 0.439

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. p_mean

z = -1.25 r = +0.34 score = 0.432

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
7. soil_depth_statsgo

z = -2.87 r = +0.13 score = 0.384

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. aridity

z = +1.24 r = -0.28 score = 0.343

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
9. low_prec_freq

z = +1.41 r = -0.22 score = 0.316

many dry days indicate a drier climate where streamflow depends on stored water.
10. hfd_mean

z = -1.32 r = -0.24 score = 0.313

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11180960: CULL C AB CULL C RES NR CASTRO VALLEY CA

Dominant forcing: PRCP Peak TE: 0.0064 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, low soil_depth_statsgo.

Top 10 driver attributes

1. p_seasonality

z = -2.35 r = -0.24 score = 0.574

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. high_prec_dur

z = +2.38 r = +0.23 score = 0.540

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. soil_depth_statsgo

z = -3.31 r = +0.13 score = 0.442

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
4. gauge_lon

z = -1.62 r = -0.27 score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. p_mean

z = -1.15 r = +0.34 score = 0.395

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. zero_q_freq

z = +2.45 r = -0.13 score = 0.328

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
7. aridity

z = +1.12 r = -0.28 score = 0.310

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
8. hfd_mean

z = -1.29 r = -0.24 score = 0.307

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. low_prec_freq

z = +1.36 r = -0.22 score = 0.303

many dry days indicate a drier climate where streamflow depends on stored water.
10. slope_fdc

z = -1.39 r = +0.20 score = 0.279

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11224500: LOS GATOS C AB NUNEZ CYN NR COALINGA CA

Dominant forcing: PRCP Peak TE: 0.0010 Peak lag: 4 day(s)
Summary: PRCP-dominant at lag 4 day(s); top drivers consistent with high aridity, low p_seasonality, high zero_q_freq.

Top 10 driver attributes

1. aridity

z = +2.56 r = -0.28 score = 0.706

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_seasonality

z = -2.57 r = -0.24 score = 0.628

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. zero_q_freq

z = +3.85 r = -0.13 score = 0.514

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
4. high_prec_dur

z = +2.18 r = +0.23 score = 0.494

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. p_mean

z = -1.38 r = +0.34 score = 0.476

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. low_prec_freq

z = +1.95 r = -0.22 score = 0.436

many dry days indicate a drier climate where streamflow depends on stored water.
7. gauge_lon

z = -1.52 r = -0.27 score = 0.413

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. slope_fdc

z = -1.94 r = +0.20 score = 0.389

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
9. stream_elas

z = +3.05 r = -0.12 score = 0.377

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
10. runoff_ratio

z = -1.45 r = +0.23 score = 0.334

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11230500: BEAR C NR LAKE THOMAS A EDISON CA

Dominant forcing: SRAD Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high frac_snow, high runoff_ratio.

Top 10 driver attributes

1. p_seasonality

z = -2.04 r = -0.46 score = 0.932

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. frac_snow

z = +3.57 r = +0.26 score = 0.911

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. runoff_ratio

z = +1.99 r = +0.43 score = 0.852

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. slope_mean

z = +2.23 r = +0.37 score = 0.819

steep terrain promotes fast surface runoff and a quick streamflow response.
5. elev_mean

z = +2.88 r = +0.24 score = 0.694

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. other_frac

z = +4.76 r = +0.14 score = 0.662

a large 'other' land fraction reflects mixed or atypical surface cover.
7. high_prec_dur

z = +2.28 r = +0.29 score = 0.658

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. gauge_lon

z = -1.43 r = -0.43 score = 0.618

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. gvff_diff

z = -1.73 r = -0.33 score = 0.565

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. low_prec_freq

z = +1.51 r = -0.31 score = 0.470

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11237500: PITMAN C BL TAMARACK C CA

Dominant forcing: SRAD Peak TE: 0.0040 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, high frac_snow.

Top 10 driver attributes

1. p_seasonality

z = -2.20 r = -0.46 score = 1.003

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. high_prec_dur

z = +2.98 r = +0.29 score = 0.859

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. frac_snow

z = +2.81 r = +0.26 score = 0.719

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. gauge_lon

z = -1.44 r = -0.43 score = 0.624

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. high_q_dur

z = +3.00 r = +0.16 score = 0.471

long high-flow events suggest sustained wet conditions and slow recession.
6. high_prec_freq

z = +1.43 r = -0.33 score = 0.468

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
7. runoff_ratio

z = +1.09 r = +0.43 score = 0.466

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. slope_mean

z = +1.24 r = +0.37 score = 0.456

steep terrain promotes fast surface runoff and a quick streamflow response.
9. elev_mean

z = +1.83 r = +0.24 score = 0.443

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. low_prec_freq

z = +1.37 r = -0.31 score = 0.429

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11253310: CANTUA C NR CANTUA CREEK CA

Dominant forcing: PRCP Peak TE: 0.0016 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high aridity, low p_seasonality, high high_prec_dur.

Top 10 driver attributes

1. aridity

z = +2.41 r = -0.28 score = 0.666

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
2. p_seasonality

z = -2.56 r = -0.24 score = 0.624

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. high_prec_dur

z = +2.48 r = +0.23 score = 0.561

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. zero_q_freq

z = +3.82 r = -0.13 score = 0.510

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
5. p_mean

z = -1.48 r = +0.34 score = 0.509

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. low_prec_freq

z = +1.97 r = -0.22 score = 0.440

many dry days indicate a drier climate where streamflow depends on stored water.
7. gauge_lon

z = -1.52 r = -0.27 score = 0.413

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. runoff_ratio

z = -1.42 r = +0.23 score = 0.327

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
9. slope_fdc

z = -1.52 r = +0.20 score = 0.306

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
10. q95

z = -0.97 r = +0.31 score = 0.296

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11264500: MERCED R A HAPPY ISLES BRIDGE NR YOSEMITE CA

Dominant forcing: SRAD Peak TE: 0.0080 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, high frac_snow.

Top 10 driver attributes

1. p_seasonality

z = -2.19 r = -0.46 score = 1.000

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. high_prec_dur

z = +3.40 r = +0.29 score = 0.980

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. frac_snow

z = +3.58 r = +0.26 score = 0.915

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. slope_mean

z = +1.80 r = +0.37 score = 0.662

steep terrain promotes fast surface runoff and a quick streamflow response.
5. runoff_ratio

z = +1.49 r = +0.43 score = 0.638

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. gauge_lon

z = -1.47 r = -0.43 score = 0.634

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. other_frac

z = +4.14 r = +0.14 score = 0.576

a large 'other' land fraction reflects mixed or atypical surface cover.
8. elev_mean

z = +2.38 r = +0.24 score = 0.575

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. gvfdiff

z = -1.57 r = -0.33 score = 0.513

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. high_prec_freq

z = +1.29 r = -0.33 score = 0.421

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11266500: MERCED R A POHONO BRIDGE NR YOSEMITE CA

Dominant forcing: SRAD Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, high frac_snow.

Top 10 driver attributes

1. p_seasonality

z = -2.19 r = -0.46 score = 0.999

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. high_prec_dur

z = +3.11 r = +0.29 score = 0.898

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. frac_snow

z = +3.48 r = +0.26 score = 0.889

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. slope_mean

z = +1.99 r = +0.37 score = 0.733

steep terrain promotes fast surface runoff and a quick streamflow response.
5. gauge_lon

z = -1.47 r = -0.43 score = 0.636

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. runoff_ratio

z = +1.40 r = +0.43 score = 0.597

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. other_frac

z = +4.17 r = +0.14 score = 0.581

a large 'other' land fraction reflects mixed or atypical surface cover.
8. elev_mean

z = +2.23 r = +0.24 score = 0.538

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. gvf_diff

z = -1.47 r = -0.33 score = 0.480

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. high_prec_freq

z = +1.28 r = -0.33 score = 0.418

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11274500: ORESTIMBA C NR NEWMAN CA

Dominant forcing: PRCP Peak TE: 0.0050 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high zero_q_freq, low p_seasonality, high aridity.

Top 10 driver attributes

1. zero_q_freq

z = +5.96 r = -0.13 score = 0.796

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
2. p_seasonality

z = -2.40 r = -0.24 score = 0.586

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. aridity

z = +1.99 r = -0.28 score = 0.548

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
4. high_prec_dur

z = +2.41 r = +0.23 score = 0.546

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. p_mean

z = -1.45 r = +0.34 score = 0.501

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
6. slope_fdc

z = -2.44 r = +0.20 score = 0.490

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
7. gauge_lon

z = -1.56 r = -0.27 score = 0.424

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. glim_2nd_class_frac

z = +2.31 r = +0.15 score = 0.355

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. low_prec_freq

z = +1.51 r = -0.22 score = 0.338

many dry days indicate a drier climate where streamflow depends on stored water.
10. stream_elas

z = +2.42 r = -0.12 score = 0.298

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11274630: DEL PUERTO C NR PATTERSON CA

Dominant forcing: PRCP Peak TE: 0.0042 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high aridity, low p_mean.

Top 10 driver attributes

1. **p_seasonality**

z = -2.35

r = -0.24

score = 0.574

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **aridity**

z = +1.83

r = -0.28

score = 0.506

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.
3. **p_mean**

z = -1.42

r = +0.34

score = 0.488

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
4. **zero_q_freq**

z = +3.45

r = -0.13

score = 0.461

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
5. **gauge_lon**

z = -1.57

r = -0.27

score = 0.426

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. **high_prec_dur**

z = +1.71

r = +0.23

score = 0.388

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. **stream_elas**

z = +3.06

r = -0.12

score = 0.378

high streamflow elasticity means small changes in rainfall produce large changes in streamflow.
8. **low_prec_freq**

z = +1.51

r = -0.22

score = 0.338

many dry days indicate a drier climate where streamflow depends on stored water.
9. **runoff_ratio**

z = -1.31

r = +0.23

score = 0.302

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
10. **q95**

z = -0.94

r = +0.31

score = 0.287

low peak-flow values reflect a damped streamflow response with limited peaks.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11284400: BIG C AB WHITES GULCH NR GROVELAND CA

Dominant forcing: PRCP Peak TE: 0.0057 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high high_prec_dur, low gauge_lon.

Top 10 driver attributes

1. **p_seasonality**

z = -2.27 r = -0.24 score = 0.554

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **high_prec_dur**

z = +2.29 r = +0.23 score = 0.518

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. **gauge_lon**

z = -1.50 r = -0.27 score = 0.409

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. **gvf_diff**

z = -1.66 r = -0.24 score = 0.403

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. **zero_q_freq**

z = +2.90 r = -0.13 score = 0.387

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
6. **low_prec_freq**

z = +1.36 r = -0.22 score = 0.305

many dry days indicate a drier climate where streamflow depends on stored water.
7. **slope_fdc**

z = -1.45 r = +0.20 score = 0.291

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
8. **hfd_mean**

z = -1.08 r = -0.24 score = 0.257

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. **soil_depth_statsgo**

z = -1.83 r = +0.13 score = 0.244

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. **high_prec_freq**

z = +1.12 r = -0.19 score = 0.210

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11299600: BLACK C NR COPPEROPOLIS CA

Dominant forcing: PRCP Peak TE: 0.0053 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high zero_q_freq, low gauge_lon.

Top 10 driver attributes

1. **p_seasonality**

z = -2.24

r = -0.24

score = 0.546

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **zero_q_freq**

z = +3.26

r = -0.13

score = 0.435

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
3. **gauge_lon**

z = -1.53

r = -0.27

score = 0.416

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. **soil_depth_statsgo**

z = -3.00

r = +0.13

score = 0.402

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
5. **slope_fdc**

z = -1.69

r = +0.20

score = 0.341

a flat flow-duration curve indicates a stable basin with slow streamflow variability.
6. **high_prec_dur**

z = +1.46

r = +0.23

score = 0.330

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. **hfd_mean**

z = -1.36

r = -0.24

score = 0.324

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. **low_prec_freq**

z = +1.36

r = -0.22

score = 0.303

many dry days indicate a drier climate where streamflow depends on stored water.
9. **p_mean**

z = -0.84

r = +0.34

score = 0.290

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
10. **aridity**

z = +1.03

r = -0.28

score = 0.283

the basin is arid, so streamflow is water-limited and sensitive to evaporative demand.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11381500: MILL C NR LOS MOLINOS CA

Dominant forcing: PRCP Peak TE: 0.0045 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, high q5.

Top 10 driver attributes

1. p_seasonality

z = -2.01 r = -0.24 score = 0.491

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.62 r = -0.27 score = 0.439

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. q5

z = +1.70 r = +0.18 score = 0.307

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. high_prec_dur

z = +1.30 r = +0.23 score = 0.294

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. glim_2nd_class_frac

z = +1.78 r = +0.15 score = 0.272

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. soil_depth_statsgo

z = -1.91 r = +0.13 score = 0.256

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. runoff_ratio

z = +1.09 r = +0.23 score = 0.252

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. gvf_diff

z = -1.02 r = -0.24 score = 0.248

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. high_prec_freq

z = +0.86 r = -0.19 score = 0.162

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. glim_1st_class_frac

z = -1.10 r = -0.15 score = 0.161

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11383500: DEER C NR VINA CA

Dominant forcing: PRCP Peak TE: 0.0050 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, high glim_2nd_class_frac.

Top 10 driver attributes

1. p_seasonality

z = -2.07 r = -0.24 score = 0.506

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.61 r = -0.27 score = 0.438

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. glim_2nd_class_frac

z = +2.32 r = +0.15 score = 0.357

a large secondary geological class adds heterogeneity to the subsurface flow response.
4. high_prec_dur

z = +1.29 r = +0.23 score = 0.291

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. gvf_diff

z = -1.00 r = -0.24 score = 0.243

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. glim_1st_class_frac

z = -1.43 r = -0.15 score = 0.208

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. pet_mean

z = +1.70 r = -0.12 score = 0.207

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
8. hfd_mean

z = -0.79 r = -0.24 score = 0.189

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. high_prec_freq

z = +0.99 r = -0.19 score = 0.187

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. low_prec_freq

z = +0.71 r = -0.22 score = 0.158

many dry days indicate a drier climate where streamflow depends on stored water.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11451100: NF CACHE C A HOUGH SPRING NR CLEARLAKE OAKS CA

Dominant forcing: PRCP Peak TE: 0.0065 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +3.21 r = +0.23 score = 0.727

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.34 r = -0.24 score = 0.571

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.65 r = -0.27 score = 0.449

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gvf_diff

z = -1.54 r = -0.24 score = 0.373

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. soil_depth_statsgo

z = -2.52 r = +0.13 score = 0.338

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
6. hfd_mean

z = -1.41 r = -0.24 score = 0.334

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. high_prec_freq

z = +1.34 r = -0.19 score = 0.251

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. low_prec_freq

z = +1.12 r = -0.22 score = 0.250

many dry days indicate a drier climate where streamflow depends on stored water.
9. runoff_ratio

z = +0.97 r = +0.23 score = 0.223

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. q95

z = +0.64 r = +0.31 score = 0.194

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11468500: NOYO R NR FORT BRAGG CA

Dominant forcing: PRCP Peak TE: 0.0095 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, low gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = -2.26 r = -0.24 score = 0.553

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.72 r = -0.27 score = 0.468

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. gvf_diff

z = -1.63 r = -0.24 score = 0.396

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. high_prec_dur

z = +1.58 r = +0.23 score = 0.358

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. hfd_mean

z = -1.41 r = -0.24 score = 0.335

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. q95

z = +0.76 r = +0.31 score = 0.231

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. high_prec_freq

z = +1.08 r = -0.19 score = 0.204

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
8. soil_depth_statsgo

z = -1.49 r = +0.13 score = 0.200

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
9. frac_forest

z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. glim_2nd_class_frac

z = -0.97 r = +0.15 score = 0.149

a small secondary geological class means the subsurface is more uniform.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11473900: MF EEL R NR DOS RIOS CA

Dominant forcing: PRCP Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. high_prec_dur

z = +2.79 r = +0.23 score = 0.632

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. p_seasonality

z = -2.25 r = -0.24 score = 0.550

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.70 r = -0.27 score = 0.461

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. slope_fdc

z = +1.68 r = +0.20 score = 0.338

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. gvf_diff

z = -1.23 r = -0.24 score = 0.299

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. hfd_mean

z = -1.12 r = -0.24 score = 0.267

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. soil_depth_statsgo

z = -1.89 r = +0.13 score = 0.253

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. glim_2nd_class_frac

z = +1.64 r = +0.15 score = 0.252

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. high_prec_freq

z = +1.26 r = -0.19 score = 0.237

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.
10. glim_1st_class_frac

z = -1.04 r = -0.15 score = 0.151

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11475560: ELDER C NR BRANSCOMB CA

Dominant forcing: PRCP Peak TE: 0.0063 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high high_prec_dur, low p_seasonality.

Top 10 driver attributes

1. q95

z = +2.16

r = +0.31

score = 0.658

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. high_prec_dur

z = +2.60

r = +0.23

score = 0.589

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_seasonality

z = -2.23

r = -0.24

score = 0.545

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. gauge_lon

z = -1.72

r = -0.27

score = 0.467

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. p_mean

z = +1.35

r = +0.34

score = 0.466

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. gvf_diff

z = -1.74

r = -0.24

score = 0.423

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
7. q_mean

z = +1.39

r = +0.28

score = 0.386

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
8. hfd_mean

z = -1.38

r = -0.24

score = 0.328

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. runoff_ratio

z = +1.37

r = +0.23

score = 0.315

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. high_prec_freq

z = +1.35

r = -0.19

score = 0.253

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11476600: BULL C NR WEOTT CA

Dominant forcing: PRCP Peak TE: 0.0117 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. q95

z = +2.33

r = +0.31

score = 0.710

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_seasonality

z = -2.16

r = -0.24

score = 0.528

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.74

r = -0.27

score = 0.473

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. p_mean

z = +1.12

r = +0.34

score = 0.385

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gvf_diff

z = -1.58

r = -0.24

score = 0.384

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. q_mean

z = +1.35

r = +0.28

score = 0.373

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
7. hfd_mean

z = -1.48

r = -0.24

score = 0.353

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. runoff_ratio

z = +1.51

r = +0.23

score = 0.349

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. high_prec_dur

z = +1.33

r = +0.23

score = 0.302

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. aridity

z = -0.93

r = -0.28

score = 0.258

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11478500: VAN DUZEN R NR BRIDGEVILLE CA

Dominant forcing: PRCP Peak TE: 0.0155 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. q95

z = +1.92

r = +0.31

score = 0.586

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_seasonality

z = -2.13

r = -0.24

score = 0.520

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.73

r = -0.27

score = 0.471

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. high_prec_dur

z = +1.60

r = +0.23

score = 0.363

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. runoff_ratio

z = +1.55

r = +0.23

score = 0.358

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. hfd_mean

z = -1.49

r = -0.24

score = 0.354

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. q_mean

z = +1.24

r = +0.28

score = 0.343

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
8. gvf_diff

z = -1.32

r = -0.24

score = 0.321

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. p_mean

z = +0.92

r = +0.34

score = 0.316

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. slope_fdc

z = +1.04

r = +0.20

score = 0.209

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11480390: MAD R AB RUTH RES NR FOREST GLEN CA

Dominant forcing: PRCP Peak TE: 0.0137 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, high high_prec_dur.

Top 10 driver attributes

1. p_seasonality

z = -2.19 r = -0.24 score = 0.533

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.70 r = -0.27 score = 0.461

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. high_prec_dur

z = +1.75 r = +0.23 score = 0.396

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. gvf_diff

z = -1.41 r = -0.24 score = 0.343

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. hfd_mean

z = -1.42 r = -0.24 score = 0.338

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. q95

z = +1.10 r = +0.31 score = 0.334

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. runoff_ratio

z = +0.80 r = +0.23 score = 0.184

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. p_mean

z = +0.53 r = +0.34 score = 0.181

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
9. zero_q_freq

z = +1.22 r = -0.13 score = 0.163

frequent zero-flow days indicate an ephemeral or highly water-limited basin.
10. high_prec_freq

z = +0.84 r = -0.19 score = 0.158

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11481200: LITTLE R NR TRINIDAD CA

Dominant forcing: PRCP Peak TE: 0.0115 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, high q95.

Top 10 driver attributes

1. **p_seasonality** **z = -2.03** **r = -0.24** **score = 0.494**

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **gauge_lon** **z = -1.74** **r = -0.27** **score = 0.474**

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. **q95** **z = +1.55** **r = +0.31** **score = 0.472**

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. **hfd_mean** **z = -1.48** **r = -0.24** **score = 0.352**

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. **gvf_diff** **z = -1.42** **r = -0.24** **score = 0.345**

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. **runoff_ratio** **z = +1.25** **r = +0.23** **score = 0.288**

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. **high_prec_dur** **z = +1.22** **r = +0.23** **score = 0.277**

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. **p_mean** **z = +0.78** **r = +0.34** **score = 0.270**

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
9. **q_mean** **z = +0.95** **r = +0.28** **score = 0.263**

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
10. **aridity** **z = -0.83** **r = -0.28** **score = 0.230**

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11482500: REDWOOD C A ORICK CA

Dominant forcing: PRCP Peak TE: 0.0157 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, high q95, low gauge_lon.

Top 10 driver attributes

1. p_seasonality

z = -2.02 r = -0.24 score = 0.493

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. q95

z = +1.60 r = +0.31 score = 0.490

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. gauge_lon

z = -1.74 r = -0.27 score = 0.473

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gvf_diff

z = -1.50 r = -0.24 score = 0.364

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. glim_2nd_class_frac

z = +2.13 r = +0.15 score = 0.327

a large secondary geological class adds heterogeneity to the subsurface flow response.
6. hfd_mean

z = -1.36 r = -0.24 score = 0.324

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. p_mean

z = +0.94 r = +0.34 score = 0.323

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. slope_fdc

z = +1.53 r = +0.20 score = 0.307

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. high_prec_dur

z = +1.32 r = +0.23 score = 0.300

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. q_mean

z = +1.06 r = +0.28 score = 0.292

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11522500: SALMON R A SOMES BAR CA

Dominant forcing: PRCP Peak TE: 0.0059 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high slope_fdc, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. slope_fdc

z = +2.48 r = +0.20 score = 0.499

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
2. p_seasonality

z = -1.92 r = -0.24 score = 0.468

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.71 r = -0.27 score = 0.464

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. runoff_ratio

z = +1.32 r = +0.23 score = 0.304

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. soil_depth_statsgo

z = -2.17 r = +0.13 score = 0.291

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
6. high_prec_dur

z = +1.16 r = +0.23 score = 0.262

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. gvfdiff

z = -0.92 r = -0.24 score = 0.224

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. glim1stclassfrac

z = -1.52 r = -0.15 score = 0.220

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. frac_forest

z = +0.94 r = +0.17 score = 0.162

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. elev_mean

z = +0.66 r = -0.20 score = 0.129

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11523200: TRINITY R AB COFFEE C NR TRINITY CTR CA

Dominant forcing: PRCP Peak TE: 0.0054 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high runoff_ratio, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. runoff_ratio

z = +2.22 r = +0.23 score = 0.512

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. p_seasonality

z = -1.92 r = -0.24 score = 0.469

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.66 r = -0.27 score = 0.451

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. slope_fdc

z = +1.95 r = +0.20 score = 0.392

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. high_prec_dur

z = +1.62 r = +0.23 score = 0.367

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. q95

z = +0.99 r = +0.31 score = 0.303

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. soil_depth_statsgo

z = -1.59 r = +0.13 score = 0.213

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. elev_mean

z = +1.09 r = -0.20 score = 0.212

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. q_mean

z = +0.75 r = +0.28 score = 0.208

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
10. gvf_diff

z = -0.85 r = -0.24 score = 0.207

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11528700: SF TRINITY R BL HYAMPOM CA

Dominant forcing: PRCP Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, low gvf_diff.

Top 10 driver attributes

1. p_seasonality

z = -2.11 r = -0.24 score = 0.514

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.71 r = -0.27 score = 0.464

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. gvf_diff

z = -1.32 r = -0.24 score = 0.320

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. high_prec_dur

z = +1.18 r = +0.23 score = 0.268

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. hfd_mean

z = -1.06 r = -0.24 score = 0.252

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. slope_fdc

z = +1.10 r = +0.20 score = 0.221

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. soil_depth_statsgo

z = -1.61 r = +0.13 score = 0.216

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. pet_mean

z = +1.44 r = -0.12 score = 0.175

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
9. frac_forest

z = +0.94 r = +0.17 score = 0.162

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
10. area_gages2

z = +0.70 r = +0.18 score = 0.126

a large drainage area integrates many sub-basin responses, producing a smoother streamflow signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 11532500: SMITH R NR CRESCENT CITY CA

Dominant forcing: SRAD Peak TE: 0.0169 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q95, high runoff_ratio, high q_mean.

Top 10 driver attributes

1. q95

z = +3.00

r = +0.43

score = 1.293

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. runoff_ratio

z = +2.48

r = +0.43

score = 1.061

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. q_mean

z = +2.53

r = +0.35

score = 0.893

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. p_seasonality

z = -1.86

r = -0.46

score = 0.848

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. gauge_lon

z = -1.74

r = -0.43

score = 0.754

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. slope_mean

z = +1.40

r = +0.37

score = 0.514

steep terrain promotes fast surface runoff and a quick streamflow response.
7. gvf_diff

z = -1.39

r = -0.33

score = 0.454

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. glim_1st_class_frac

z = -1.77

r = -0.21

score = 0.380

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. high_prec_dur

z = +1.15

r = +0.29

score = 0.332

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. p_mean

z = +1.65

r = +0.20

score = 0.331

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12010000: NASELLE RIVER NEAR NASELLE, WA

Dominant forcing: PRCP Peak TE: 0.0209 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high p_mean, high q_mean.

Top 10 driver attributes

1. q95

z = +4.41

r = +0.31

score = 1.346

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_mean

z = +3.31

r = +0.34

score = 1.141

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +3.70

r = +0.28

score = 1.023

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. low_prec_freq

z = -2.41

r = -0.22

score = 0.540

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. runoff_ratio

z = +2.23

r = +0.23

score = 0.515

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. hfd_mean

z = -2.07

r = -0.24

score = 0.492

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. high_prec_freq

z = -2.55

r = -0.19

score = 0.480

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. gauge_lon

z = -1.72

r = -0.27

score = 0.468

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. aridity

z = -1.31

r = -0.28

score = 0.363

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. glim_2nd_class_frac

z = +2.27

r = +0.15

score = 0.349

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12013500: WILLAPA RIVER NEAR WILLAPA, WA

Dominant forcing: PRCP Peak TE: 0.0303 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.56

r = +0.34

score = 0.883

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.60

r = +0.31

score = 0.793

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.05

r = +0.28

score = 0.566

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. low_prec_freq

z = -2.34

r = -0.22

score = 0.523

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. hfd_mean

z = -1.99

r = -0.24

score = 0.473

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. gauge_lon

z = -1.72

r = -0.27

score = 0.467

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_freq

z = -2.11

r = -0.19

score = 0.397

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. p_seasonality

z = -1.44

r = -0.24

score = 0.351

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. aridity

z = -1.22

r = -0.28

score = 0.336

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gvf_diff

z = -1.32

r = -0.24

score = 0.321

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12020000: CHEHALIS RIVER NEAR DOTY, WA

Dominant forcing: PRCP Peak TE: 0.0023 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +3.01

r = +0.34

score = 1.036

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.74

r = +0.31

score = 0.837

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.23

r = +0.28

score = 0.618

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. high_prec_freq

z = -2.87

r = -0.19

score = 0.540

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. low_prec_freq

z = -2.23

r = -0.22

score = 0.498

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. hfd_mean

z = -2.05

r = -0.24

score = 0.488

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. gauge_lon

z = -1.70

r = -0.27

score = 0.460

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. p_seasonality

z = -1.39

r = -0.24

score = 0.340

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. aridity

z = -1.12

r = -0.28

score = 0.310

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gvf_diff

z = -1.23

r = -0.24

score = 0.300

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12025000: NEWAUKUM RIVER NEAR CHEHALIS, WA

Dominant forcing: PRCP Peak TE: 0.0186 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low gauge_lon, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq z = -2.04 r = -0.22 score = 0.456

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gauge_lon z = -1.67 r = -0.27 score = 0.455

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. high_prec_freq z = -2.29 r = -0.19 score = 0.432

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. hfd_mean z = -1.80 r = -0.24 score = 0.427

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. q95 z = +1.30 r = +0.31 score = 0.397

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. p_mean z = +1.03 r = +0.34 score = 0.354

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
7. p_seasonality z = -1.40 r = -0.24 score = 0.342

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q_mean z = +1.10 r = +0.28 score = 0.305

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
9. slope_fdc z = +1.45 r = +0.20 score = 0.291

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. runoff_ratio z = +1.25 r = +0.23 score = 0.288

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12025700: SKOOKUMCHUCK RIVER NEAR VAIL, WA

Dominant forcing: PRCP Peak TE: 0.0118 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high runoff_ratio.

Top 10 driver attributes

1. q95 z = +2.44 r = +0.31 score = 0.744

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean z = +2.17 r = +0.28 score = 0.599

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. runoff_ratio z = +2.41 r = +0.23 score = 0.557

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. low_prec_freq z = -2.04 r = -0.22 score = 0.457

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. high_prec_freq z = -2.39 r = -0.19 score = 0.451

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. gauge_lon z = -1.65 r = -0.27 score = 0.449

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. p_mean z = +1.30 r = +0.34 score = 0.448

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
8. hfd_mean z = -1.69 r = -0.24 score = 0.401

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. p_seasonality z = -1.36 r = -0.24 score = 0.331

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. aridity z = -1.06 r = -0.28 score = 0.294

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12035000: SATSOP RIVER NEAR SATSOP, WA

Dominant forcing: PRCP Peak TE: 0.0275 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high p_mean.

Top 10 driver attributes

1. q95 z = +3.59 r = +0.31 score = 1.095

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean z = +3.41 r = +0.28 score = 0.944

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean z = +2.64 r = +0.34 score = 0.911

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio z = +2.48 r = +0.23 score = 0.573

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. hfd_mean z = -2.00 r = -0.24 score = 0.475

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. low_prec_freq z = -2.12 r = -0.22 score = 0.474

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lon z = -1.71 r = -0.27 score = 0.464

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. q5 z = +2.18 r = +0.18 score = 0.393

high low-flow values reflect strong baseflow and steady year-round streamflow.
9. p_seasonality z = -1.51 r = -0.24 score = 0.368

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. aridity z = -1.26 r = -0.28 score = 0.347

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12040500: QUEETS RIVER NEAR CLEARWATER, WA

Dominant forcing: PRCP Peak TE: 0.0258 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high p_mean.

Top 10 driver attributes

1. q95 z = +5.42 r = +0.31 score = 1.654

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean z = +5.31 r = +0.28 score = 1.470

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean z = +3.47 r = +0.34 score = 1.196

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio z = +3.43 r = +0.23 score = 0.793

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. q5 z = +3.30 r = +0.18 score = 0.595

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. low_prec_freq z = -2.40 r = -0.22 score = 0.537

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lon z = -1.76 r = -0.27 score = 0.478

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. hfd_mean z = -1.95 r = -0.24 score = 0.464

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. aridity z = -1.34 r = -0.28 score = 0.369

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. high_prec_freq z = -1.94 r = -0.19 score = 0.366

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12041200: HOH RIVER AT US HIGHWAY 101 NEAR FORKS, WA

Dominant forcing: PRCP Peak TE: 0.0290 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high q_mean, high q95.

Top 10 driver attributes

1. q5 z = +8.36 r = +0.18 score = 1.508

high low-flow values reflect strong baseflow and steady year-round streamflow.

2. q_mean z = +5.24 r = +0.28 score = 1.451

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.

3. q95 z = +4.14 r = +0.31 score = 1.264

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

4. p_mean z = +3.21 r = +0.34 score = 1.106

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

5. runoff_ratio z = +3.62 r = +0.23 score = 0.835

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

6. low_prec_freq z = -2.32 r = -0.22 score = 0.520

few dry days indicate steady moisture supply, which supports continuous streamflow.

7. gauge_lon z = -1.76 r = -0.27 score = 0.477

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

8. hfd_mean z = -1.59 r = -0.24 score = 0.378

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

9. high_prec_freq z = -1.94 r = -0.19 score = 0.366

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

10. p_seasonality z = -1.42 r = -0.24 score = 0.347

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12043000: CALAWAH RIVER NEAR FORKS, WA

Dominant forcing: PRCP Peak TE: 0.0237 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high p_mean, high q_mean.

Top 10 driver attributes

1. q95 z = +4.46 r = +0.31 score = 1.360

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_mean z = +3.56 r = +0.34 score = 1.228

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean z = +4.10 r = +0.28 score = 1.136

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio z = +2.39 r = +0.23 score = 0.552

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. low_prec_freq z = -2.35 r = -0.22 score = 0.526

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. hfd_mean z = -2.10 r = -0.24 score = 0.498

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. gauge_lon z = -1.76 r = -0.27 score = 0.479

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. high_prec_freq z = -1.94 r = -0.19 score = 0.366

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. aridity z = -1.28 r = -0.28 score = 0.352

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality z = -1.37 r = -0.24 score = 0.335

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12048000: DUNGENESS RIVER NEAR SEQUIM, WA

Dominant forcing: PRCP Peak TE: 0.0162 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high high_prec_dur, high p_mean.

Top 10 driver attributes

1. gauge_lon z = -1.69 r = -0.27 score = 0.458

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. high_prec_dur z = +1.83 r = +0.23 score = 0.414

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_mean z = +1.08 r = +0.34 score = 0.371

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. p_seasonality z = -1.50 r = -0.24 score = 0.366

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. q5 z = +1.69 r = +0.18 score = 0.306

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. low_prec_freq z = -1.11 r = -0.22 score = 0.248

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. aridity z = -0.86 r = -0.28 score = 0.237

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. frac_snow z = +1.58 r = -0.15 score = 0.231

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
9. gauge_lat z = +1.68 r = +0.13 score = 0.213

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. frac_forest z = +0.94 r = +0.17 score = 0.161

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12054000: DUCKABUSH RIVER NEAR BRINNON, WA

Dominant forcing: PRCP Peak TE: 0.0186 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high q95, high runoff_ratio.

Top 10 driver attributes

1. q_mean

z = +2.86 r = +0.28 score = 0.790

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. q95

z = +2.29 r = +0.31 score = 0.700

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. runoff_ratio

z = +2.77 r = +0.23 score = 0.639

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. p_mean

z = +1.74 r = +0.34 score = 0.600

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. q5

z = +2.85 r = +0.18 score = 0.515

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. high_prec_dur

z = +2.12 r = +0.23 score = 0.481

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. gauge_lon

z = -1.68 r = -0.27 score = 0.456

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. p_seasonality

z = -1.57 r = -0.24 score = 0.383

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. glim_2nd_class_frac

z = +2.00 r = +0.15 score = 0.307

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. low_prec_freq

z = -1.26 r = -0.22 score = 0.281

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12056500: NF SKOKOMISH R BL STAIRCASE RPDS NR HOODSPORT, WA

Dominant forcing: PRCP Peak TE: 0.0177 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high q95, high p_mean.

Top 10 driver attributes

1. q_mean

z = +4.67

r = +0.28

score = 1.293

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. q95

z = +4.17

r = +0.31

score = 1.272

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. p_mean

z = +2.65

r = +0.34

score = 0.913

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio

z = +3.68

r = +0.23

score = 0.848

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. q5

z = +3.01

r = +0.18

score = 0.542

high low-flow values reflect strong baseflow and steady year-round streamflow.
6. high_prec_dur

z = +2.17

r = +0.23

score = 0.493

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. gauge_lon

z = -1.70

r = -0.27

score = 0.461

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. low_prec_freq

z = -1.72

r = -0.22

score = 0.384

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. p_seasonality

z = -1.53

r = -0.24

score = 0.374

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. aridity

z = -1.25

r = -0.28

score = 0.346

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12073500: HUGE CREEK NEAR WAUNA, WA

Dominant forcing: PRCP Peak TE: 0.0061 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high high_prec_dur, low p_seasonality.

Top 10 driver attributes

1. gauge_lon

z = -1.66 r = -0.27 score = 0.451

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. high_prec_dur

z = +1.91 r = +0.23 score = 0.432

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_seasonality

z = -1.53 r = -0.24 score = 0.374

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. hfd_mean

z = -1.44 r = -0.24 score = 0.342

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
5. low_prec_freq

z = -1.34 r = -0.22 score = 0.300

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. q5

z = +1.40 r = +0.18 score = 0.253

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. gvf_diff

z = -0.96 r = -0.24 score = 0.234

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. gauge_lat

z = +1.56 r = +0.13 score = 0.197

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. high_prec_freq

z = -1.04 r = -0.19 score = 0.196

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. p_mean

z = +0.53 r = +0.34 score = 0.184

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12082500: NISQUALLY RIVER NEAR NATIONAL, WA

Dominant forcing: PRCP Peak TE: 0.0101 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high p_mean, high q_mean.

Top 10 driver attributes

1. q5

z = +5.98

r = +0.18

score = 1.078

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. p_mean

z = +2.70

r = +0.34

score = 0.929

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +2.59

r = +0.28

score = 0.716

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. low_prec_freq

z = -1.98

r = -0.22

score = 0.443

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. gauge_lon

z = -1.62

r = -0.27

score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. q95

z = +1.44

r = +0.31

score = 0.439

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. runoff_ratio

z = +1.67

r = +0.23

score = 0.386

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. high_prec_freq

z = -1.94

r = -0.19

score = 0.366

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. p_seasonality

z = -1.37

r = -0.24

score = 0.335

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. glim_1st_class_frac

z = -2.16

r = -0.15

score = 0.313

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12092000: PUYALLUP RIVER NEAR ELECTRON, WA

Dominant forcing: PRCP Peak TE: 0.0095 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high p_mean, high q_mean.

Top 10 driver attributes

1. q5

z = +6.41

r = +0.18

score = 1.156

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. p_mean

z = +2.85

r = +0.34

score = 0.981

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +2.54

r = +0.28

score = 0.704

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. low_prec_freq

z = -2.02

r = -0.22

score = 0.451

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. gauge_lon

z = -1.62

r = -0.27

score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. high_prec_freq

z = -2.23

r = -0.19

score = 0.420

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. q95

z = +1.28

r = +0.31

score = 0.390

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
8. runoff_ratio

z = +1.53

r = +0.23

score = 0.354

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. aridity

z = -1.25

r = -0.28

score = 0.346

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality

z = -1.33

r = -0.24

score = 0.324

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12095000: SOUTH PRAIRIE CREEK AT SOUTH PRAIRIE, WA

Dominant forcing: PRCP Peak TE: 0.0069 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low low_prec_freq, low gauge_lon, low high_prec_freq.

Top 10 driver attributes

1. low_prec_freq z = -2.00 r = -0.22 score = 0.446

few dry days indicate steady moisture supply, which supports continuous streamflow.
2. gauge_lon z = -1.62 r = -0.27 score = 0.441

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. high_prec_freq z = -2.25 r = -0.19 score = 0.424

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. p_mean z = +0.92 r = +0.34 score = 0.318

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. p_seasonality z = -1.12 r = -0.24 score = 0.274

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. hfd_mean z = -1.12 r = -0.24 score = 0.266

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. gvfdiff z = -1.00 r = -0.24 score = 0.243

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. aridity z = -0.83 r = -0.28 score = 0.230

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. q_mean z = +0.75 r = +0.28 score = 0.209

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
10. runoff_ratio z = +0.84 r = +0.23 score = 0.194

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12114500: CEDAR RIVER BELOW BEAR CREEK NEAR CEDAR FALLS, WA

Dominant forcing: PRCP Peak TE: 0.0133 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean , high q_mean , high $q95$.

Top 10 driver attributes

1. **p_mean**

z = +2.82

r = +0.34

score = 0.973

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **q_mean**

z = +2.77

r = +0.28

score = 0.765

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. **q95**

z = +2.35

r = +0.31

score = 0.716

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. **gauge_lon**

z = -1.59

r = -0.27

score = 0.431

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. **runoff_ratio**

z = +1.75

r = +0.23

score = 0.404

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. **q5**

z = +2.23

r = +0.18

score = 0.402

high low-flow values reflect strong baseflow and steady year-round streamflow.
7. **low_prec_freq**

z = -1.79

r = -0.22

score = 0.399

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. **aridity**

z = -1.27

r = -0.28

score = 0.351

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. **p_seasonality**

z = -1.34

r = -0.24

score = 0.328

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. **high_prec_freq**

z = -1.72

r = -0.19

score = 0.324

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12115000: CEDAR RIVER NEAR CEDAR FALLS, WA

Dominant forcing: PRCP Peak TE: 0.0143 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q_mean, high q95.

Top 10 driver attributes

1. p_mean

z = +2.80

r = +0.34

score = 0.965

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q_mean

z = +2.83

r = +0.28

score = 0.783

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. q95

z = +2.25

r = +0.31

score = 0.688

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. gauge_lon

z = -1.59

r = -0.27

score = 0.433

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. low_prec_freq

z = -1.89

r = -0.22

score = 0.423

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. runoff_ratio

z = +1.83

r = +0.23

score = 0.422

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. q5

z = +2.01

r = +0.18

score = 0.363

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. aridity

z = -1.27

r = -0.28

score = 0.351

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. high_prec_freq

z = -1.85

r = -0.19

score = 0.349

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. p_seasonality

z = -1.31

r = -0.24

score = 0.321

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12115500: REX RIVER NEAR CEDAR FALLS, WA

Dominant forcing: PRCP Peak TE: 0.0125 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q_mean, high q95.

Top 10 driver attributes

1. p_mean

z = +2.77

r = +0.34

score = 0.953

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q_mean

z = +3.44

r = +0.28

score = 0.951

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. q95

z = +3.03

r = +0.31

score = 0.924

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. runoff_ratio

z = +2.41

r = +0.23

score = 0.556

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. low_prec_freq

z = -1.98

r = -0.22

score = 0.443

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. gauge_lon

z = -1.60

r = -0.27

score = 0.433

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_freq

z = -2.01

r = -0.19

score = 0.378

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. aridity

z = -1.27

r = -0.28

score = 0.351

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. p_seasonality

z = -1.22

r = -0.24

score = 0.298

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. q5

z = +1.56

r = +0.18

score = 0.281

high low-flow values reflect strong baseflow and steady year-round streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12117000: TAYLOR CREEK NEAR SELLECK, WA

Dominant forcing: PRCP Peak TE: 0.0095 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q5, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.31

r = +0.34

score = 0.794

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q5

z = +3.61

r = +0.18

score = 0.652

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. q_mean

z = +2.34

r = +0.28

score = 0.648

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. low_prec_freq

z = -2.19

r = -0.22

score = 0.491

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. q95

z = +1.57

r = +0.31

score = 0.480

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. high_prec_freq

z = -2.38

r = -0.19

score = 0.449

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. gauge_lon

z = -1.61

r = -0.27

score = 0.436

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. runoff_ratio

z = +1.71

r = +0.23

score = 0.394

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. aridity

z = -1.22

r = -0.28

score = 0.337

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. gvf_diff

z = -1.23

r = -0.24

score = 0.299

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12141300: MIDDLE FORK SNOQUALMIE RIVER NEAR TANNER, WA

Dominant forcing: PRCP Peak TE: 0.0225 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean , high q_mean , high $q95$.

Top 10 driver attributes

1. **p_mean**

$z = +3.32$

$r = +0.34$

score = 1.143

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **q_mean**

$z = +3.89$

$r = +0.28$

score = 1.077

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. **q95**

$z = +3.06$

$r = +0.31$

score = 0.934

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. **q5**

$z = +3.23$

$r = +0.18$

score = 0.582

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. **runoff_ratio**

$z = +2.36$

$r = +0.23$

score = 0.545

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. **low_prec_freq**

$z = -2.25$

$r = -0.22$

score = 0.504

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. **gauge_lon**

$z = -1.59$

$r = -0.27$

score = 0.433

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. **high_prec_freq**

$z = -2.25$

$r = -0.19$

score = 0.424

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. **aridity**

$z = -1.30$

$r = -0.28$

score = 0.359

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. **p_seasonality**

$z = -1.28$

$r = -0.24$

score = 0.311

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12143600: SF SNOQUALMIE RIVER AT EDGEWICK, WA

Dominant forcing: PRCP Peak TE: 0.0176 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q_mean, high q95.

Top 10 driver attributes

1. p_mean

z = +3.10

r = +0.34

score = 1.066

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q_mean

z = +3.26

r = +0.28

score = 0.901

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. q95

z = +2.64

r = +0.31

score = 0.805

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. q5

z = +2.72

r = +0.18

score = 0.490

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. low_prec_freq

z = -2.09

r = -0.22

score = 0.468

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. runoff_ratio

z = +2.01

r = +0.23

score = 0.463

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gauge_lon

z = -1.60

r = -0.27

score = 0.434

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. high_prec_freq

z = -1.96

r = -0.19

score = 0.370

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. aridity

z = -1.27

r = -0.28

score = 0.351

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality

z = -1.29

r = -0.24

score = 0.316

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12144000: SF SNOQUALMIE RIVER AT NORTH BEND, WA

Dominant forcing: PRCP Peak TE: 0.0225 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q_mean, high q5.

Top 10 driver attributes

1. p_mean

z = +2.80 r = +0.34 score = 0.966

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q_mean

z = +3.21 r = +0.28 score = 0.888

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. q5

z = +4.46 r = +0.18 score = 0.805

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. q95

z = +2.21 r = +0.31 score = 0.673

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
5. low_prec_freq

z = -2.29 r = -0.22 score = 0.512

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. runoff_ratio

z = +2.17 r = +0.23 score = 0.501

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gauge_lon

z = -1.60 r = -0.27 score = 0.436

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. high_prec_freq

z = -2.00 r = -0.19 score = 0.376

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. aridity

z = -1.22 r = -0.28 score = 0.336

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality

z = -1.27 r = -0.24 score = 0.309

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12145500: RAGING RIVER NEAR FALL CITY, WA

Dominant forcing: PRCP Peak TE: 0.0166 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, low low_prec_freq.

Top 10 driver attributes

1. p_mean

z = +1.74

r = +0.34

score = 0.600

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +1.67

r = +0.31

score = 0.508

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. low_prec_freq

z = -2.12

r = -0.22

score = 0.473

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. high_prec_freq

z = -2.34

r = -0.19

score = 0.440

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. gauge_lon

z = -1.61

r = -0.27

score = 0.438

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. q_mean

z = +1.56

r = +0.28

score = 0.432

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
7. hfd_mean

z = -1.81

r = -0.24

score = 0.431

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. aridity

z = -1.11

r = -0.28

score = 0.306

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. runoff_ratio

z = +1.27

r = +0.23

score = 0.293

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. p_seasonality

z = -1.11

r = -0.24

score = 0.270

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12147500: NORTH FORK TOLT RIVER NEAR CARNATION, WA

Dominant forcing: PRCP Peak TE: 0.0114 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high p_mean, high q95.

Top 10 driver attributes

1. q_mean

z = +4.41

r = +0.28

score = 1.221

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. p_mean

z = +3.36

r = +0.34

score = 1.157

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q95

z = +3.17

r = +0.31

score = 0.967

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. q5

z = +4.48

r = +0.18

score = 0.808

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. runoff_ratio

z = +2.79

r = +0.23

score = 0.645

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. low_prec_freq

z = -2.23

r = -0.22

score = 0.498

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. high_prec_freq

z = -2.47

r = -0.19

score = 0.465

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. gauge_lon

z = -1.60

r = -0.27

score = 0.435

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. aridity

z = -1.32

r = -0.28

score = 0.364

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. hfd_mean

z = -1.25

r = -0.24

score = 0.296

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12147600: SOUTH FORK TOLT RIVER NEAR INDEX, WA

Dominant forcing: PRCP Peak TE: 0.0115 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high p_mean.

Top 10 driver attributes

1. q95 z = +4.94 r = +0.31 score = 1.506

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean z = +5.19 r = +0.28 score = 1.437

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean z = +3.92 r = +0.34 score = 1.352

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio z = +2.97 r = +0.23 score = 0.687

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. low_prec_freq z = -2.18 r = -0.22 score = 0.487

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. high_prec_freq z = -2.48 r = -0.19 score = 0.467

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. gauge_lon z = -1.59 r = -0.27 score = 0.432

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. q5 z = +2.32 r = +0.18 score = 0.418

high low-flow values reflect strong baseflow and steady year-round streamflow.
9. aridity z = -1.36 r = -0.28 score = 0.375

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality z = -1.12 r = -0.24 score = 0.274

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12167000: NF STILLAGUAMISH RIVER NEAR ARLINGTON, WA

Dominant forcing: PRCP Peak TE: 0.0166 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high p_mean, high q95.

Top 10 driver attributes

1. q_mean

z = +3.45

r = +0.28

score = 0.956

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. p_mean

z = +2.46

r = +0.34

score = 0.846

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q95

z = +2.63

r = +0.31

score = 0.804

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. runoff_ratio

z = +2.69

r = +0.23

score = 0.622

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. high_prec_freq

z = -2.78

r = -0.19

score = 0.523

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. low_prec_freq

z = -2.26

r = -0.22

score = 0.506

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. q5

z = +2.47

r = +0.18

score = 0.445

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. gauge_lon

z = -1.62

r = -0.27

score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. hfd_mean

z = -1.47

r = -0.24

score = 0.349

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
10. aridity

z = -1.24

r = -0.28

score = 0.343

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12175500: THUNDER CREEK NEAR NEWHALEM, WA

Dominant forcing: SRAD Peak TE: 0.0086 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high slope_mean, high runoff_ratio, high q5.

Top 10 driver attributes

1. slope_mean

z = +4.23 r = +0.37 score = 1.555

steep terrain promotes fast surface runoff and a quick streamflow response.
2. runoff_ratio

z = +2.83 r = +0.43 score = 1.210

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. q5

z = +3.54 r = +0.31 score = 1.105

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. q_mean

z = +2.74 r = +0.35 score = 0.968

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
5. q95

z = +2.03 r = +0.43 score = 0.877

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. gauge_lon

z = -1.56 r = -0.43 score = 0.674

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. p_seasonality

z = -1.45 r = -0.46 score = 0.664

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. frac_snow

z = +2.52 r = +0.26 score = 0.644

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
9. high_prec_dur

z = +1.93 r = +0.29 score = 0.556

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. gauge_lat

z = +1.81 r = +0.29 score = 0.531

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12178100: NEWHALEM CREEK NEAR NEWHALEM, WA

Dominant forcing: PRCP Peak TE: 0.0045 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high q5, high q95.

Top 10 driver attributes

1. q_mean

z = +3.09 r = +0.28 score = 0.855

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. q5

z = +4.57 r = +0.18 score = 0.825

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. q95

z = +2.28 r = +0.31 score = 0.694

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. p_mean

z = +2.01 r = +0.34 score = 0.693

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. runoff_ratio

z = +2.75 r = +0.23 score = 0.634

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. gauge_lon

z = -1.57 r = -0.27 score = 0.426

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. high_prec_dur

z = +1.77 r = +0.23 score = 0.401

long heavy-rainfall events allow more soil saturation and stronger runoff response.
8. low_prec_freq

z = -1.58 r = -0.22 score = 0.354

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. high_prec_freq

z = -1.74 r = -0.19 score = 0.328

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. aridity

z = -1.18 r = -0.28 score = 0.326

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12186000: SAUK RIVER AB WHITECHUCK RIVER NEAR DARRINGTON, WA

Dominant forcing: PRCP Peak TE: 0.0050 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q_mean, high p_mean, high q95.

Top 10 driver attributes

1. q_mean

z = +3.42 r = +0.28 score = 0.947

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
2. p_mean

z = +2.29 r = +0.34 score = 0.790

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q95

z = +2.57 r = +0.31 score = 0.783

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. q5

z = +4.21 r = +0.18 score = 0.760

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. runoff_ratio

z = +2.81 r = +0.23 score = 0.649

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. low_prec_freq

z = -2.16 r = -0.22 score = 0.484

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lon

z = -1.58 r = -0.27 score = 0.430

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. high_prec_freq

z = -2.23 r = -0.19 score = 0.420

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. aridity

z = -1.22 r = -0.28 score = 0.336

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality

z = -1.30 r = -0.24 score = 0.317

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12189500: SAUK RIVER NEAR SAUK, WA

Dominant forcing: PRCP Peak TE: 0.0165 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high q_mean, high p_mean.

Top 10 driver attributes

1. q5

z = +5.14

r = +0.18

score = 0.927

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. q_mean

z = +2.78

r = +0.28

score = 0.769

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean

z = +1.87

r = +0.34

score = 0.645

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio

z = +2.54

r = +0.23

score = 0.586

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. q95

z = +1.78

r = +0.31

score = 0.543

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. low_prec_freq

z = -2.25

r = -0.22

score = 0.502

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gauge_lon

z = -1.59

r = -0.27

score = 0.432

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. high_prec_freq

z = -2.10

r = -0.19

score = 0.395

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. p_seasonality

z = -1.38

r = -0.24

score = 0.336

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. aridity

z = -1.16

r = -0.28

score = 0.319

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12358500: Middle Fork Flathead River nr West Glacier MT

Dominant forcing: SRAD Peak TE: 0.0149 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, high runoff_ratio, low low_prec_freq.

Top 10 driver attributes

1. high_prec_freq

z = -2.43 r = -0.33 score = 0.792

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. runoff_ratio

z = +1.35 r = +0.43 score = 0.578

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. low_prec_freq

z = -1.76 r = -0.31 score = 0.550

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. gauge_lat

z = +1.78 r = +0.29 score = 0.521

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. glim_1st_class_frac

z = -2.30 r = -0.21 score = 0.493

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. gauge_lon

z = -1.12 r = -0.43 score = 0.486

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. frac_snow

z = +1.88 r = +0.26 score = 0.479

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. slope_mean

z = +1.28 r = +0.37 score = 0.470

steep terrain promotes fast surface runoff and a quick streamflow response.
9. q95

z = +0.85 r = +0.43 score = 0.366

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. other_frac

z = +2.28 r = +0.14 score = 0.318

a large 'other' land fraction reflects mixed or atypical surface cover.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12374250: Mill Cr ab Bassoo Cr nr Niarada MT

Dominant forcing: SRAD Peak TE: 0.0043 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, low gauge_lon, high gauge_lat.

Top 10 driver attributes

1. high_prec_freq

z = -1.80 r = -0.33 score = 0.587

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. gauge_lon

z = -1.17 r = -0.43 score = 0.504

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. gauge_lat

z = +1.65 r = +0.29 score = 0.483

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. slope_mean

z = +1.24 r = +0.37 score = 0.454

steep terrain promotes fast surface runoff and a quick streamflow response.
5. runoff_ratio

z = -0.91 r = +0.43 score = 0.388

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
6. frac_snow

z = +1.29 r = +0.26 score = 0.329

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
7. q95

z = -0.73 r = +0.43 score = 0.316

low peak-flow values reflect a damped streamflow response with limited peaks.
8. gvf_diff

z = -0.80 r = -0.33 score = 0.262

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. q_mean

z = -0.71 r = +0.35 score = 0.252

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. glim_1st_class_frac

z = +1.05 r = -0.21 score = 0.225

the basin is dominated by one main geological class, which gives a more uniform subsurface response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12375900: South Crow Creek near Ronan MT

Dominant forcing: SRAD Peak TE: 0.0085 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with high slope_mean, low high_prec_freq, high q5.

Top 10 driver attributes

1. slope_mean

z = +2.40 r = +0.37 score = 0.883

steep terrain promotes fast surface runoff and a quick streamflow response.
2. high_prec_freq

z = -2.01 r = -0.33 score = 0.655

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. q5

z = +1.89 r = +0.31 score = 0.591

high low-flow values reflect strong baseflow and steady year-round streamflow.
4. runoff_ratio

z = +1.35 r = +0.43 score = 0.578

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. frac_snow

z = +2.11 r = +0.26 score = 0.539

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. other_frac

z = +3.56 r = +0.14 score = 0.495

a large 'other' land fraction reflects mixed or atypical surface cover.
7. gauge_lon

z = -1.12 r = -0.43 score = 0.486

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. gauge_lat

z = +1.58 r = +0.29 score = 0.464

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. q95

z = +0.75 r = +0.43 score = 0.326

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. stream_elas

z = -1.19 r = -0.26 score = 0.308

low streamflow elasticity means streamflow is buffered against rainfall variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12377150: Mission Cr ab Reservoir nr ST Ignatius MT

Dominant forcing: SRAD Peak TE: 0.0168 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high runoff_ratio, low high_prec_freq, high q95.

Top 10 driver attributes

1. runoff_ratio

z = +2.24 r = +0.43 score = 0.959

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. high_prec_freq

z = -2.38 r = -0.33 score = 0.777

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. q95

z = +1.79 r = +0.43 score = 0.774

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. frac_snow

z = +2.67 r = +0.26 score = 0.681

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. slope_mean

z = +1.77 r = +0.37 score = 0.650

steep terrain promotes fast surface runoff and a quick streamflow response.
6. other_frac

z = +3.82 r = +0.14 score = 0.532

a large 'other' land fraction reflects mixed or atypical surface cover.
7. q5

z = +1.62 r = +0.31 score = 0.506

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. q_mean

z = +1.42 r = +0.35 score = 0.500

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
9. gauge_lon

z = -1.12 r = -0.43 score = 0.485

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
10. gauge_lat

z = +1.55 r = +0.29 score = 0.455

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12381400: South Fork Jocko River near Arlee MT

Dominant forcing: SRAD Peak TE: 0.0085 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, high frac_snow, low stream_elas.

Top 10 driver attributes

1. high_prec_freq

z = -1.83 r = -0.33 score = 0.598

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. frac_snow

z = +2.05 r = +0.26 score = 0.523

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. stream_elas

z = -1.93 r = -0.26 score = 0.500

low streamflow elasticity means streamflow is buffered against rainfall variability.
4. gauge_lon

z = -1.11 r = -0.43 score = 0.481

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. gauge_lat

z = +1.53 r = +0.29 score = 0.447

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. slope_mean

z = +1.19 r = +0.37 score = 0.436

steep terrain promotes fast surface runoff and a quick streamflow response.
7. p_seasonality

z = -0.77 r = -0.46 score = 0.353

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. elev_mean

z = +1.42 r = +0.24 score = 0.343

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. low_prec_freq

z = -0.80 r = -0.31 score = 0.249

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. frac_forest

z = +0.86 r = +0.23 score = 0.202

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12383500: Big Knife Creek near Arlee MT

Dominant forcing: SRAD Peak TE: 0.0102 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, high slope_mean, high frac_snow.

Top 10 driver attributes

1. high_prec_freq

z = -1.77 r = -0.33 score = 0.576

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. slope_mean

z = +1.49 r = +0.37 score = 0.548

steep terrain promotes fast surface runoff and a quick streamflow response.
3. frac_snow

z = +2.14 r = +0.26 score = 0.546

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. stream_elas

z = -1.94 r = -0.26 score = 0.502

low streamflow elasticity means streamflow is buffered against rainfall variability.
5. gauge_lon

z = -1.12 r = -0.43 score = 0.485

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. gauge_lat

z = +1.52 r = +0.29 score = 0.445

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. q5

z = +1.37 r = +0.31 score = 0.427

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. baseflow_index

z = +1.83 r = +0.22 score = 0.400

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
9. elev_mean

z = +1.63 r = +0.24 score = 0.393

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. glim_2nd_class_frac

z = +2.18 r = +0.18 score = 0.382

a large secondary geological class adds heterogeneity to the subsurface flow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12388400: Revais Cr bl West Fork nr Dixon MT

Dominant forcing: SRAD Peak TE: 0.0095 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high slope_mean, low gauge_lon, high gauge_lat.

Top 10 driver attributes

1. slope_mean

z = +1.50 r = +0.37 score = 0.553

steep terrain promotes fast surface runoff and a quick streamflow response.
2. gauge_lon

z = -1.15 r = -0.43 score = 0.496

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. gauge_lat

z = +1.54 r = +0.29 score = 0.451

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. high_prec_freq

z = -1.29 r = -0.33 score = 0.422

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. frac_snow

z = +1.43 r = +0.26 score = 0.365

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. glim_2nd_class_frac

z = +1.65 r = +0.18 score = 0.289

a large secondary geological class adds heterogeneity to the subsurface flow response.
7. stream_elas

z = -1.08 r = -0.26 score = 0.281

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. elev_mean

z = +1.12 r = +0.24 score = 0.269

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
9. p_seasonality

z = -0.50 r = -0.46 score = 0.228

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. gvf_diff

z = -0.64 r = -0.33 score = 0.210

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12390700: Prospect Creek at Thompson Falls MT

Dominant forcing: SRAD Peak TE: 0.0062 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high slope_mean, low gauge_lon, low high_prec_freq.

Top 10 driver attributes

1. slope_mean

z = +1.46 r = +0.37 score = 0.538

steep terrain promotes fast surface runoff and a quick streamflow response.
2. gauge_lon

z = -1.21 r = -0.43 score = 0.521

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. high_prec_freq

z = -1.57 r = -0.33 score = 0.512

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. gauge_lat

z = +1.60 r = +0.29 score = 0.469

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. p_seasonality

z = -1.00 r = -0.46 score = 0.459

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. frac_snow

z = +1.68 r = +0.26 score = 0.429

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
7. low_prec_freq

z = -1.19 r = -0.31 score = 0.371

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. frac_forest

z = +0.96 r = +0.23 score = 0.224

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. baseflow_index

z = +0.99 r = +0.22 score = 0.217

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. elev_mean

z = +0.83 r = +0.24 score = 0.201

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12411000: NF COEUR D ALENE R AB SHOSHONE CK NR PRICHARD ID

Dominant forcing: PRCP Peak TE: 0.0068 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, low high_prec_freq, low p_seasonality.

Top 10 driver attributes

1. gauge_lon z = -1.25 r = -0.27 score = 0.338

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. high_prec_freq z = -1.60 r = -0.19 score = 0.302

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. p_seasonality z = -1.12 r = -0.24 score = 0.272

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. low_prec_freq z = -1.08 r = -0.22 score = 0.242

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. frac_snow z = +1.41 r = -0.15 score = 0.207

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. gauge_lat z = +1.62 r = +0.13 score = 0.205

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. aridity z = -0.68 r = -0.28 score = 0.189

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. runoff_ratio z = +0.74 r = +0.23 score = 0.171

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. slope_fdc z = +0.81 r = +0.20 score = 0.164

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. frac_forest z = +0.95 r = +0.17 score = 0.163

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12414500: ST JOE RIVER AT CALDER ID

Dominant forcing: SRAD Peak TE: 0.0096 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with low p_seasonality, low gauge_lon, low high_prec_freq.

Top 10 driver attributes

1. p_seasonality

z = -1.22 r = -0.46 score = 0.560

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. gauge_lon

z = -1.26 r = -0.43 score = 0.544

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. high_prec_freq

z = -1.46 r = -0.33 score = 0.476

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. frac_snow

z = +1.83 r = +0.26 score = 0.467

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. slope_mean

z = +1.23 r = +0.37 score = 0.454

steep terrain promotes fast surface runoff and a quick streamflow response.
6. gauge_lat

z = +1.54 r = +0.29 score = 0.452

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
7. low_prec_freq

z = -1.43 r = -0.31 score = 0.446

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. runoff_ratio

z = +0.77 r = +0.43 score = 0.329

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. glim_2nd_class_frac

z = +1.84 r = +0.18 score = 0.323

a large secondary geological class adds heterogeneity to the subsurface flow response.
10. glim_1st_class_frac

z = -1.35 r = -0.21 score = 0.288

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12447390: ANDREWS CREEK NEAR MAZAMA, WA

Dominant forcing: SRAD Peak TE: 0.0082 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high frac_snow, low gauge_lon.

Top 10 driver attributes

1. slope_mean

z = +2.68 r = +0.37 score = 0.986

steep terrain promotes fast surface runoff and a quick streamflow response.
2. frac_snow

z = +2.95 r = +0.26 score = 0.753

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. gauge_lon

z = -1.50 r = -0.43 score = 0.649

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. gauge_lat

z = +1.84 r = +0.29 score = 0.539

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. p_seasonality

z = -1.06 r = -0.46 score = 0.483

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. high_prec_dur

z = +1.16 r = +0.29 score = 0.336

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. stream_elas

z = -1.24 r = -0.26 score = 0.321

low streamflow elasticity means streamflow is buffered against rainfall variability.
8. gvf_diff

z = -0.96 r = -0.33 score = 0.314

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. high_q_dur

z = +1.93 r = +0.16 score = 0.303

long high-flow events suggest sustained wet conditions and slow recession.
10. elev_mean

z = +1.20 r = +0.24 score = 0.289

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12451000: STEHEKIN RIVER AT STEHEKIN, WA

Dominant forcing: SRAD Peak TE: 0.0170 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, high runoff_ratio, high q95.

Top 10 driver attributes

1. slope_mean

z = +3.83 r = +0.37 score = 1.409

steep terrain promotes fast surface runoff and a quick streamflow response.
2. runoff_ratio

z = +2.36 r = +0.43 score = 1.009

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. q95

z = +1.77 r = +0.43 score = 0.766

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. p_seasonality

z = -1.60 r = -0.46 score = 0.730

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. frac_snow

z = +2.65 r = +0.26 score = 0.677

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. gauge_lon

z = -1.54 r = -0.43 score = 0.664

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. q_mean

z = +1.75 r = +0.35 score = 0.618

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
8. q5

z = +1.76 r = +0.31 score = 0.548

high low-flow values reflect strong baseflow and steady year-round streamflow.
9. gauge_lat

z = +1.74 r = +0.29 score = 0.511

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
10. high_prec_freq

z = -1.34 r = -0.33 score = 0.436

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 12488500: AMERICAN RIVER NEAR NILE, WA

Dominant forcing: SRAD Peak TE: 0.0093 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high slope_mean, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. slope_mean

z = +2.07 r = +0.37 score = 0.762

steep terrain promotes fast surface runoff and a quick streamflow response.
2. p_seasonality

z = -1.64 r = -0.46 score = 0.748

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon

z = -1.57 r = -0.43 score = 0.676

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. frac_snow

z = +2.28 r = +0.26 score = 0.583

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. gauge_lat

z = +1.49 r = +0.29 score = 0.435

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. glim_1st_class_frac

z = -1.95 r = -0.21 score = 0.418

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. low_prec_freq

z = -1.33 r = -0.31 score = 0.414

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. q95

z = +0.82 r = +0.43 score = 0.353

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. runoff_ratio

z = +0.80 r = +0.43 score = 0.342

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
10. high_prec_freq

z = -1.04 r = -0.33 score = 0.340

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13011500: PACIFIC CREEK AT MORAN WY

Dominant forcing: SRAD Peak TE: 0.0090 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high runoff_ratio, high elev_mean.

Top 10 driver attributes

1. frac_snow

z = +2.37 r = +0.26 score = 0.604

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. runoff_ratio

z = +1.21 r = +0.43 score = 0.516

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
3. elev_mean

z = +2.09 r = +0.24 score = 0.504

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. high_prec_freq

z = -1.23 r = -0.33 score = 0.401

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. gauge_lon

z = -0.91 r = -0.43 score = 0.393

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. glim_1st_class_frac

z = -1.82 r = -0.21 score = 0.391

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. p_seasonality

z = -0.81 r = -0.46 score = 0.368

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q95

z = +0.75 r = +0.43 score = 0.326

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. high_q_dur

z = +1.79 r = +0.16 score = 0.281

long high-flow events suggest sustained wet conditions and slow recession.
10. gauge_lat

z = +0.88 r = +0.29 score = 0.259

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13011900: BUFFALO FORK AB LAVA CREEK NR MORAN WY

Dominant forcing: SRAD Peak TE: 0.0113 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low high_prec_freq.

Top 10 driver attributes

1. frac_snow

z = +2.59 r = +0.26 score = 0.662

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.45 r = +0.24 score = 0.591

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = -1.57 r = -0.33 score = 0.512

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. runoff_ratio

z = +0.98 r = +0.43 score = 0.421

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -0.90 r = -0.43 score = 0.390

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. slope_mean

z = +1.01 r = +0.37 score = 0.372

steep terrain promotes fast surface runoff and a quick streamflow response.
7. glim_1st_class_frac

z = -1.56 r = -0.21 score = 0.335

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. p_seasonality

z = -0.67 r = -0.46 score = 0.304

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. low_prec_freq

z = -0.97 r = -0.31 score = 0.301

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. gauge_lat

z = +0.88 r = +0.29 score = 0.258

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13018300: CACHE CREEK NEAR JACKSON, WY

Dominant forcing: Tair Peak TE: 0.0051 Peak lag: 7 day(s)
Summary: Tair-dominant at lag 7 day(s); top drivers consistent with high frac_snow, high slope_mean, high elev_mean.

Top 10 driver attributes

1. frac_snow

z = +2.36 r = +0.46 score = 1.088

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. slope_mean

z = +1.34 r = +0.41 score = 0.548

steep terrain promotes fast surface runoff and a quick streamflow response.
3. elev_mean

z = +1.83 r = +0.30 score = 0.544

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
4. gauge_lat

z = +0.81 r = +0.51 score = 0.408

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. glim_1st_class_frac

z = -2.12 r = -0.16 score = 0.342

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. pet_mean

z = +0.98 r = -0.33 score = 0.325

high potential evapotranspiration reflects an energy-rich climate where evaporation strongly modifies the water balance.
7. p_seasonality

z = -0.82 r = -0.38 score = 0.313

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. high_prec_freq

z = -0.61 r = -0.49 score = 0.301

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. clay_frac

z = -0.98 r = -0.30 score = 0.290

low clay fraction allows easier infiltration and reduces surface runoff.
10. baseflow_index

z = +1.21 r = +0.22 score = 0.272

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13023000: GREYS RIVER AB RESERVOIR NR ALPINE WY

Dominant forcing: Tair Peak TE: 0.0101 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high elev_mean, low high_prec_freq.

Top 10 driver attributes

1. frac_snow

z = +2.31 r = +0.46 score = 1.064

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. elev_mean

z = +2.14 r = +0.30 score = 0.636

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
3. high_prec_freq

z = -0.95 r = -0.49 score = 0.469

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
4. slope_mean

z = +1.11 r = +0.41 score = 0.452

steep terrain promotes fast surface runoff and a quick streamflow response.
5. gauge_lat

z = +0.75 r = +0.51 score = 0.378

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
6. stream_elas

z = -0.84 r = -0.43 score = 0.362

low streamflow elasticity means streamflow is buffered against rainfall variability.
7. runoff_ratio

z = +0.60 r = +0.56 score = 0.334

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. p_seasonality

z = -0.80 r = -0.38 score = 0.306

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. baseflow_index

z = +1.33 r = +0.22 score = 0.299

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
10. glim_1st_class_frac

z = -1.85 r = -0.16 score = 0.298

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13083000: TRAPPER CREEK NR OAKLEY ID

Dominant forcing: SRAD Peak TE: 0.0030 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with low gauge_lon, high baseflow_index, low p_seasonality.

Top 10 driver attributes

1. gauge_lon

z = -1.12 r = -0.43 score = 0.485

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. baseflow_index

z = +2.13 r = +0.22 score = 0.466

a high baseflow index means streamflow is dominated by groundwater and slow drainage.
3. p_seasonality

z = -0.93 r = -0.46 score = 0.423

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. runoff_ratio

z = -0.97 r = +0.43 score = 0.415

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. frac_forest

z = -1.70 r = +0.23 score = 0.398

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
6. q95

z = -0.92 r = +0.43 score = 0.396

low peak-flow values reflect a damped streamflow response with limited peaks.
7. frac_snow

z = +1.52 r = +0.26 score = 0.389

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. stream_elas

z = -1.37 r = -0.26 score = 0.354

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. elev_mean

z = +1.40 r = +0.24 score = 0.339

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. high_prec_freq

z = +0.97 r = -0.33 score = 0.317

frequent heavy rainfall events drive a strong, fast rainfall-runoff signal.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13161500: BRUNEAU RIVER AT ROWLAND NV

Dominant forcing: SRAD Peak TE: 0.0033 Peak lag: 3 day(s)
Summary: SRAD-dominant at lag 3 day(s); top drivers consistent with low gauge_lon, high frac_snow, low p_seasonality.

Top 10 driver attributes

1. gauge_lon

z = -1.23 r = -0.43 score = 0.530

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. frac_snow

z = +1.68 r = +0.26 score = 0.428

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. p_seasonality

z = -0.93 r = -0.46 score = 0.427

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. runoff_ratio

z = -0.96 r = +0.43 score = 0.413

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
5. elev_mean

z = +1.65 r = +0.24 score = 0.399

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. frac_forest

z = -1.63 r = +0.23 score = 0.382

low forest cover reduces evapotranspiration and may produce stronger surface runoff.
7. glim_1st_class_frac

z = -1.67 r = -0.21 score = 0.357

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
8. q95

z = -0.82 r = +0.43 score = 0.355

low peak-flow values reflect a damped streamflow response with limited peaks.
9. q_mean

z = -0.82 r = +0.35 score = 0.291

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. p_mean

z = -1.34 r = +0.20 score = 0.269

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13235000: SF PAYETTE RIVER AT LOWMAN ID

Dominant forcing: SRAD Peak TE: 0.0114 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low p_seasonality, high frac_snow, high slope_mean.

Top 10 driver attributes

1. p_seasonality

z = -1.46 r = -0.46 score = 0.667

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. frac_snow

z = +2.46 r = +0.26 score = 0.628

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
3. slope_mean

z = +1.70 r = +0.37 score = 0.624

steep terrain promotes fast surface runoff and a quick streamflow response.
4. gauge_lon

z = -1.22 r = -0.43 score = 0.529

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. runoff_ratio

z = +1.11 r = +0.43 score = 0.476

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. elev_mean

z = +1.67 r = +0.24 score = 0.404

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
7. q5

z = +1.17 r = +0.31 score = 0.366

high low-flow values reflect strong baseflow and steady year-round streamflow.
8. clay_frac

z = -1.45 r = -0.22 score = 0.319

low clay fraction allows easier infiltration and reduces surface runoff.
9. high_prec_dur

z = +1.05 r = +0.29 score = 0.303

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. baseflow_index

z = +1.29 r = +0.22 score = 0.282

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13240000: LAKE FORK PAYETTE RIVER AB JUMBO CR NR MCCALL ID

Dominant forcing: SRAD Peak TE: 0.0064 Peak lag: 2 day(s)

Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high runoff_ratio, high q95, high frac_snow.

Top 10 driver attributes

1. runoff_ratio

z = +1.74 r = +0.43 score = 0.743

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
2. q95

z = +1.60 r = +0.43 score = 0.691

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. frac_snow

z = +2.70 r = +0.26 score = 0.689

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. p_seasonality

z = -1.35 r = -0.46 score = 0.617

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. gauge_lon

z = -1.25 r = -0.43 score = 0.539

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. slope_mean

z = +1.36 r = +0.37 score = 0.498

steep terrain promotes fast surface runoff and a quick streamflow response.
7. soil_conductivity

z = +1.90 r = +0.25 score = 0.481

high soil conductivity allows fast infiltration and subsurface flow.
8. high_prec_dur

z = +1.30 r = +0.29 score = 0.373

long heavy-rainfall events allow more soil saturation and stronger runoff response.
9. elev_mean

z = +1.53 r = +0.24 score = 0.370

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. soil_porosity

z = -1.88 r = -0.17 score = 0.323

low soil porosity reduces water storage and produces a flashier response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13310700: SF SALMON RIVER NR KRASSEL RANGER STATION ID

Dominant forcing: Tair Peak TE: 0.0100 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high gauge_lat, low p_seasonality.

Top 10 driver attributes

1. frac_snow

z = +2.37

r = +0.46

score = 1.093

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. gauge_lat

z = +1.10

r = +0.51

score = 0.557

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
3. p_seasonality

z = -1.39

r = -0.38

score = 0.532

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. slope_mean

z = +1.28

r = +0.41

score = 0.523

steep terrain promotes fast surface runoff and a quick streamflow response.
5. elev_mean

z = +1.55

r = +0.30

score = 0.462

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
6. clay_frac

z = -1.33

r = -0.30

score = 0.393

low clay fraction allows easier infiltration and reduces surface runoff.
7. gauge_lon

z = -1.23

r = -0.25

score = 0.313

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
8. frac_forest

z = +0.87

r = +0.35

score = 0.302

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
9. geol_porostiy

z = -1.45

r = -0.13

score = 0.185

low geological porosity reduces subsurface storage and produces a flashier response.
10. baseflow_index

z = +0.80

r = +0.22

score = 0.180

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13313000: JOHNSON CREEK AT YELLOW PINE ID

Dominant forcing: Tair Peak TE: 0.0083 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high frac_snow, high slope_mean, high gauge_lat.

Top 10 driver attributes

1. frac_snow

z = +2.79

r = +0.46

score = 1.286

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. slope_mean

z = +1.38

r = +0.41

score = 0.562

steep terrain promotes fast surface runoff and a quick streamflow response.
3. gauge_lat

z = +1.10

r = +0.51

score = 0.555

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. p_seasonality

z = -1.43

r = -0.38

score = 0.548

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. clay_frac

z = -1.79

r = -0.30

score = 0.530

low clay fraction allows easier infiltration and reduces surface runoff.
6. elev_mean

z = +1.78

r = +0.30

score = 0.528

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
7. frac_forest

z = +0.92

r = +0.35

score = 0.317

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.
8. gauge_lon

z = -1.22

r = -0.25

score = 0.310

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
9. silt_frac

z = -1.86

r = +0.13

score = 0.241

low silt fraction limits water retention in the soil column.
10. slope_fdc

z = -0.95

r = +0.24

score = 0.229

a flat flow-duration curve indicates a stable basin with slow streamflow variability.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13331500: MINAM RIVER NEAR MINAM, OR

Dominant forcing: SRAD Peak TE: 0.0113 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with high slope_mean, low gauge_lon, high frac_snow.

Top 10 driver attributes

1. slope_mean

z = +1.71 r = +0.37 score = 0.627

steep terrain promotes fast surface runoff and a quick streamflow response.
2. gauge_lon

z = -1.35 r = -0.43 score = 0.585

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. frac_snow

z = +1.87 r = +0.26 score = 0.478

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
4. high_prec_freq

z = -1.46 r = -0.33 score = 0.476

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. p_seasonality

z = -1.02 r = -0.46 score = 0.466

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. runoff_ratio

z = +1.05 r = +0.43 score = 0.451

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gauge_lat

z = +1.22 r = +0.29 score = 0.359

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
8. glim_2nd_class_frac

z = +2.02 r = +0.18 score = 0.354

a large secondary geological class adds heterogeneity to the subsurface flow response.
9. elev_mean

z = +1.14 r = +0.24 score = 0.275

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.
10. glim_1st_class_frac

z = -1.25 r = -0.21 score = 0.268

the basin's main geological class covers only a small share, so the subsurface response is more mixed.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13337000: LOCHSA RIVER NR LOWELL ID

Dominant forcing: SRAD Peak TE: 0.0117 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, high slope_mean, low low_prec_freq.

Top 10 driver attributes

1. high_prec_freq

z = -1.99 r = -0.33 score = 0.648

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. slope_mean

z = +1.58 r = +0.37 score = 0.582

steep terrain promotes fast surface runoff and a quick streamflow response.
3. low_prec_freq

z = -1.71 r = -0.31 score = 0.534

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. gauge_lon

z = -1.22 r = -0.43 score = 0.528

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. frac_snow

z = +1.90 r = +0.26 score = 0.486

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
6. p_seasonality

z = -1.02 r = -0.46 score = 0.468

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. runoff_ratio

z = +0.92 r = +0.43 score = 0.393

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. gauge_lat

z = +1.33 r = +0.29 score = 0.389

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. q95

z = +0.63 r = +0.43 score = 0.272

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. clay_frac

z = -1.21 r = -0.22 score = 0.266

low clay fraction allows easier infiltration and reduces surface runoff.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13338500: SF CLEARWATER RIVER AT STITES ID

Dominant forcing: SRAD Peak TE: 0.0081 Peak lag: 2 day(s)
Summary: SRAD-dominant at lag 2 day(s); top drivers consistent with low high_prec_freq, low gauge_lon, high slope_mean.

Top 10 driver attributes

1. high_prec_freq z = -2.04 r = -0.33 score = 0.666

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. gauge_lon z = -1.25 r = -0.43 score = 0.538

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. slope_mean z = +1.36 r = +0.37 score = 0.500

steep terrain promotes fast surface runoff and a quick streamflow response.
4. gauge_lat z = +1.31 r = +0.29 score = 0.385

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
5. low_prec_freq z = -1.16 r = -0.31 score = 0.362

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. glim_1st_class_frac z = -1.68 r = -0.21 score = 0.361

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
7. frac_snow z = +0.90 r = +0.26 score = 0.230

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
8. soil_porosity z = +1.34 r = -0.17 score = 0.230

high soil porosity increases water storage capacity and dampens streamflow response.
9. q95 z = -0.51 r = +0.43 score = 0.218

low peak-flow values reflect a damped streamflow response with limited peaks.
10. baseflow_index z = +0.99 r = +0.22 score = 0.217

a high baseflow index means streamflow is dominated by groundwater and slow drainage.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13340000: CLEARWATER RIVER AT OROFINO ID

Dominant forcing: Tair Peak TE: 0.0122 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with low high_prec_freq, low low_prec_freq, high gauge_lat.

Top 10 driver attributes

1. high_prec_freq

z = -2.47 r = -0.49 score = 1.216

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
2. low_prec_freq

z = -1.77 r = -0.50 score = 0.891

few dry days indicate steady moisture supply, which supports continuous streamflow.
3. gauge_lat

z = +1.39 r = +0.51 score = 0.702

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. frac_snow

z = +1.37 r = +0.46 score = 0.631

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
5. slope_mean

z = +1.44 r = +0.41 score = 0.585

steep terrain promotes fast surface runoff and a quick streamflow response.
6. gauge_lon

z = -1.26 r = -0.25 score = 0.321

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. p_seasonality

z = -0.80 r = -0.38 score = 0.307

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. stream_elas

z = -0.70 r = -0.43 score = 0.299

low streamflow elasticity means streamflow is buffered against rainfall variability.
9. pet_mean

z = -0.81 r = -0.33 score = 0.269

low potential evapotranspiration reflects a cool climate where evaporation losses are limited.
10. elev_mean

z = +0.89 r = +0.30 score = 0.264

high elevation is associated with cooler temperatures, more snow, and delayed streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 13340600: NF CLEARWATER RIVER NR CANYON RANGER STATION ID

Dominant forcing: Tair Peak TE: 0.0115 Peak lag: 1 day(s)
Summary: Tair-dominant at lag 1 day(s); top drivers consistent with high frac_snow, low high_prec_freq, high gauge_lat.

Top 10 driver attributes

1. frac_snow

z = +1.87 r = +0.46 score = 0.862

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
2. high_prec_freq

z = -1.58 r = -0.49 score = 0.777

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
3. gauge_lat

z = +1.46 r = +0.51 score = 0.737

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
4. low_prec_freq

z = -1.43 r = -0.50 score = 0.722

few dry days indicate steady moisture supply, which supports continuous streamflow.
5. runoff_ratio

z = +1.11 r = +0.56 score = 0.619

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. slope_mean

z = +1.34 r = +0.41 score = 0.544

steep terrain promotes fast surface runoff and a quick streamflow response.
7. p_seasonality

z = -1.17 r = -0.38 score = 0.449

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q5

z = +1.20 r = +0.36 score = 0.437

high low-flow values reflect strong baseflow and steady year-round streamflow.
9. stream_elas

z = -0.88 r = -0.43 score = 0.379

low streamflow elasticity means streamflow is buffered against rainfall variability.
10. frac_forest

z = +0.96 r = +0.35 score = 0.333

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14020000: UMATILLA RIVER ABOVE MEACHAM CREEK, NR GIBBON, OR

Dominant forcing: PRCP Peak TE: 0.0104 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high slope_fdc, low p_seasonality.

Top 10 driver attributes

1. gauge_lon

z = -1.39 r = -0.27 score = 0.377

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. slope_fdc

z = +1.78 r = +0.20 score = 0.357

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. p_seasonality

z = -1.30 r = -0.24 score = 0.317

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. runoff_ratio

z = +1.01 r = +0.23 score = 0.234

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. high_prec_dur

z = +0.87 r = +0.23 score = 0.197

long heavy-rainfall events allow more soil saturation and stronger runoff response.
6. soil_depth_statsgo

z = -1.25 r = +0.13 score = 0.168

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. glim_2nd_class_frac

z = -1.08 r = +0.15 score = 0.166

a small secondary geological class means the subsurface is more uniform.
8. gauge_lat

z = +1.24 r = +0.13 score = 0.157

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.
9. glim_1st_class_frac

z = +1.06 r = -0.15 score = 0.154

the basin is dominated by one main geological class, which gives a more uniform subsurface response.
10. p_mean

z = -0.44 r = +0.34 score = 0.152

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14092750: SHITIKE CR, AT PETERS PASTURE, NR WARM SPRINGS, OR

Dominant forcing: PRCP Peak TE: 0.0094 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, low gauge_lon, low p_seasonality.

Top 10 driver attributes

1. q5

z = +3.15

r = +0.18

score = 0.569

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. gauge_lon

z = -1.59

r = -0.27

score = 0.433

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. p_seasonality

z = -1.54

r = -0.24

score = 0.376

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. glim_2nd_class_frac

z = +2.23

r = +0.15

score = 0.343

a large secondary geological class adds heterogeneity to the subsurface flow response.
5. p_mean

z = +0.95

r = +0.34

score = 0.326

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. runoff_ratio

z = +1.39

r = +0.23

score = 0.322

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. q_mean

z = +1.15

r = +0.28

score = 0.319

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
8. frac_snow

z = +1.76

r = -0.15

score = 0.258

a large fraction of precipitation falls as snow, so streamflow is shaped by snow accumulation and melt rather than direct rainfall.
9. high_prec_dur

z = +1.06

r = +0.23

score = 0.241

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. high_prec_freq

z = -1.24

r = -0.19

score = 0.233

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14096850: BEAVER CREEK, BLW QUARTZ CR, NR SIMNASHO, OR.

Dominant forcing: PRCP Peak TE: 0.0051 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, low p_seasonality, low gvf_diff.

Top 10 driver attributes

1. gauge_lon

z = -1.58 r = -0.27 score = 0.429

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. p_seasonality

z = -1.58 r = -0.24 score = 0.386

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gvf_diff

z = -1.27 r = -0.24 score = 0.309

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
4. q95

z = -0.72 r = +0.31 score = 0.220

low peak-flow values reflect a damped streamflow response with limited peaks.
5. glim_1st_class_frac

z = -1.48 r = -0.15 score = 0.215

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
6. runoff_ratio

z = -0.75 r = +0.23 score = 0.174

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. p_mean

z = -0.50 r = +0.34 score = 0.173

low mean rainfall limits the water input, so rainfall-driven information flow to streamflow is weaker.
8. q_mean

z = -0.62 r = +0.28 score = 0.171

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
9. soil_depth_statsgo

z = -1.05 r = +0.13 score = 0.140

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
10. gauge_lat

z = +1.10 r = +0.13 score = 0.138

northern latitude is associated with colder climates, more snow, and snowmelt-driven streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14137000: SANDY RIVER NEAR MARMOT, OR

Dominant forcing: PRCP Peak TE: 0.0141 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q5, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.16

r = +0.34

score = 0.744

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q5

z = +3.23

r = +0.18

score = 0.582

high low-flow values reflect strong baseflow and steady year-round streamflow.
3. q_mean

z = +2.10

r = +0.28

score = 0.580

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. gauge_lon

z = -1.63

r = -0.27

score = 0.441

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. q95

z = +1.42

r = +0.31

score = 0.432

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
6. low_prec_freq

z = -1.75

r = -0.22

score = 0.390

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. runoff_ratio

z = +1.56

r = +0.23

score = 0.359

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. p_seasonality

z = -1.46

r = -0.24

score = 0.357

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. high_prec_freq

z = -1.85

r = -0.19

score = 0.349

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. aridity

z = -1.19

r = -0.28

score = 0.330

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14138800: BLAZED ALDER CREEK NEAR RHODODENDRON, OREG.

Dominant forcing: PRCP Peak TE: 0.0182 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high p_mean, high q_mean.

Top 10 driver attributes

1. q95

z = +3.75

r = +0.31

score = 1.143

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_mean

z = +3.17

r = +0.34

score = 1.092

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +3.25

r = +0.28

score = 0.900

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio

z = +1.95

r = +0.23

score = 0.450

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -1.61

r = -0.27

score = 0.437

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. p_seasonality

z = -1.47

r = -0.24

score = 0.358

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. aridity

z = -1.28

r = -0.28

score = 0.352

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. high_prec_freq

z = -1.84

r = -0.19

score = 0.347

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. low_prec_freq

z = -1.53

r = -0.22

score = 0.343

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. hfd_mean

z = -1.39

r = -0.24

score = 0.331

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14138870: FIR CREEK NEAR BRIGHTWOOD, OR

Dominant forcing: PRCP Peak TE: 0.0129 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +3.65 r = +0.34 score = 1.256

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.97 r = +0.31 score = 0.906

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.95 r = +0.28 score = 0.815

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. gauge_lon

z = -1.62 r = -0.27 score = 0.439

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. high_prec_freq

z = -2.21 r = -0.19 score = 0.415

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
6. low_prec_freq

z = -1.66 r = -0.22 score = 0.371

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. hfd_mean

z = -1.55 r = -0.24 score = 0.369

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. aridity

z = -1.31 r = -0.28 score = 0.361

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. p_seasonality

z = -1.42 r = -0.24 score = 0.346

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. runoff_ratio

z = +1.42 r = +0.23 score = 0.328

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14138900: NORTH FORK BULL RUN RIVER NEAR MULTNOMAH FALLS, OR

Dominant forcing: PRCP Peak TE: 0.0189 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean , high $q95$, high q_mean .

Top 10 driver attributes

1. **p_mean**

$z = +4.03$

$r = +0.34$

score = 1.387

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **q95**

$z = +4.20$

$r = +0.31$

score = 1.283

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. **q_mean**

$z = +4.25$

$r = +0.28$

score = 1.175

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. **q5**

$z = +3.97$

$r = +0.18$

score = 0.716

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. **runoff_ratio**

$z = +2.20$

$r = +0.23$

score = 0.508

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
6. **gauge_lon**

$z = -1.62$

$r = -0.27$

score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
7. **high_prec_freq**

$z = -2.29$

$r = -0.19$

score = 0.432

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. **low_prec_freq**

$z = -1.71$

$r = -0.22$

score = 0.383

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. **aridity**

$z = -1.33$

$r = -0.28$

score = 0.367

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. **hfd_mean**

$z = -1.47$

$r = -0.24$

score = 0.350

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14139800: SOUTH FORK BULL RUN RIVER NEAR BULL RUN, OR

Dominant forcing: PRCP Peak TE: 0.0220 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +3.31

r = +0.34

score = 1.141

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +3.27

r = +0.31

score = 0.998

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +3.44

r = +0.28

score = 0.952

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio

z = +2.02

r = +0.23

score = 0.466

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -1.62

r = -0.27

score = 0.441

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. high_prec_freq

z = -2.15

r = -0.19

score = 0.405

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. low_prec_freq

z = -1.76

r = -0.22

score = 0.394

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. hfd_mean

z = -1.52

r = -0.24

score = 0.361

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. aridity

z = -1.28

r = -0.28

score = 0.354

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. p_seasonality

z = -1.40

r = -0.24

score = 0.342

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14141500: LITTLE SANDY RIVER NEAR BULL RUN, OREG.

Dominant forcing: PRCP Peak TE: 0.0217 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.94

r = +0.34

score = 1.012

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.54

r = +0.31

score = 0.776

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.66

r = +0.28

score = 0.737

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. gauge_lon

z = -1.63

r = -0.27

score = 0.442

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. low_prec_freq

z = -1.89

r = -0.22

score = 0.422

few dry days indicate steady moisture supply, which supports continuous streamflow.
6. high_prec_freq

z = -2.10

r = -0.19

score = 0.395

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. runoff_ratio

z = +1.59

r = +0.23

score = 0.366

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
8. aridity

z = -1.24

r = -0.28

score = 0.343

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. p_seasonality

z = -1.38

r = -0.24

score = 0.337

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. hfd_mean

z = -1.41

r = -0.24

score = 0.335

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14154500: ROW RIVER ABOVE PITCHER CREEK NEAR, DORENA, OREG

Dominant forcing: PRCP Peak TE: 0.0157 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high slope_fdc, low p_seasonality.

Top 10 driver attributes

1. gauge_lon

z = -1.67 r = -0.27 score = 0.454

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. slope_fdc

z = +2.19 r = +0.20 score = 0.439

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
3. p_seasonality

z = -1.42 r = -0.24 score = 0.347

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. high_prec_freq

z = -1.78 r = -0.19 score = 0.335

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. p_mean

z = +0.97 r = +0.34 score = 0.334

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
6. hfd_mean

z = -1.39 r = -0.24 score = 0.332

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. low_prec_freq

z = -1.34 r = -0.22 score = 0.299

few dry days indicate steady moisture supply, which supports continuous streamflow.
8. glim_1st_class_frac

z = -1.98 r = -0.15 score = 0.288

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
9. gvf_diff

z = -1.14 r = -0.24 score = 0.277

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. aridity

z = -0.92 r = -0.28 score = 0.255

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14158500: MCKENZIE RIVER AT OUTLET OF CLEAR LAKE, OR

Dominant forcing: PRCP Peak TE: 0.0187 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q5, high p_mean, high q_mean.

Top 10 driver attributes

1. q5

z = +5.95

r = +0.18

score = 1.073

high low-flow values reflect strong baseflow and steady year-round streamflow.
2. p_mean

z = +1.80

r = +0.34

score = 0.620

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +1.93

r = +0.28

score = 0.534

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. gauge_lon

z = -1.62

r = -0.27

score = 0.439

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. p_seasonality

z = -1.57

r = -0.24

score = 0.383

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. runoff_ratio

z = +1.65

r = +0.23

score = 0.380

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. aridity

z = -1.09

r = -0.28

score = 0.300

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
8. q95

z = +0.91

r = +0.31

score = 0.278

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
9. high_prec_freq

z = -1.40

r = -0.19

score = 0.264

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. low_prec_freq

z = -1.12

r = -0.22

score = 0.250

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14158790: SMITH R AB SMITH R RES NR BELKNAP SPRGS,OREG.

Dominant forcing: PRCP Peak TE: 0.0182 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high p_mean, high q_mean.

Top 10 driver attributes

1. q95

z = +2.52

r = +0.31

score = 0.768

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_mean

z = +2.00

r = +0.34

score = 0.689

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +2.39

r = +0.28

score = 0.662

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio

z = +2.00

r = +0.23

score = 0.461

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -1.62

r = -0.27

score = 0.440

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. slope_fdc

z = +2.08

r = +0.20

score = 0.417

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
7. p_seasonality

z = -1.53

r = -0.24

score = 0.373

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. aridity

z = -1.13

r = -0.28

score = 0.312

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
9. high_prec_freq

z = -1.49

r = -0.19

score = 0.281

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. low_prec_freq

z = -1.13

r = -0.22

score = 0.253

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14166500: LONG TOM RIVER NEAR NOTI, OREG.

Dominant forcing: PRCP Peak TE: 0.0287 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low gauge_lon, high p_mean.

Top 10 driver attributes

1. high_prec_dur

z = +2.20 r = +0.23 score = 0.499

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. gauge_lon

z = -1.70 r = -0.27 score = 0.463

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. p_mean

z = +1.27 r = +0.34 score = 0.438

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. p_seasonality

z = -1.70 r = -0.24 score = 0.415

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
5. hfd_mean

z = -1.54 r = -0.24 score = 0.366

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. low_prec_freq

z = -1.44 r = -0.22 score = 0.322

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. gvf_diff

z = -1.26 r = -0.24 score = 0.306

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. high_prec_freq

z = -1.55 r = -0.19 score = 0.291

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. q95

z = +0.93 r = +0.31 score = 0.284

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
10. slope_fdc

z = +1.20 r = +0.20 score = 0.241

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14182500: LITTLE NORTH SANTIAM RIVER NEAR MEHAMA, OR

Dominant forcing: PRCP Peak TE: 0.0065 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high p_mean.

Top 10 driver attributes

1. q95 z = +2.76 r = +0.31 score = 0.842

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

2. q_mean z = +2.86 r = +0.28 score = 0.792

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.

3. p_mean z = +2.10 r = +0.34 score = 0.723

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

4. runoff_ratio z = +2.41 r = +0.23 score = 0.557

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

5. gauge_lon z = -1.65 r = -0.27 score = 0.449

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

6. slope_fdc z = +1.96 r = +0.20 score = 0.394

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

7. low_prec_freq z = -1.75 r = -0.22 score = 0.392

few dry days indicate steady moisture supply, which supports continuous streamflow.

8. high_prec_freq z = -2.02 r = -0.19 score = 0.380

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

9. hfd_mean z = -1.55 r = -0.24 score = 0.368

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

10. p_seasonality z = -1.47 r = -0.24 score = 0.359

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14185000: SOUTH SANTIAM RIVER BELOW CASCADIA, OR

Dominant forcing: PRCP Peak TE: 0.0168 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +1.74

r = +0.34

score = 0.599

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +1.73

r = +0.31

score = 0.529

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +1.80

r = +0.28

score = 0.499

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. gauge_lon

z = -1.65

r = -0.27

score = 0.447

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. slope_fdc

z = +2.17

r = +0.20

score = 0.435

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
6. low_prec_freq

z = -1.62

r = -0.22

score = 0.362

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. p_seasonality

z = -1.47

r = -0.24

score = 0.359

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. runoff_ratio

z = +1.55

r = +0.23

score = 0.358

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. high_prec_freq

z = -1.85

r = -0.19

score = 0.349

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. hfd_mean

z = -1.33

r = -0.24

score = 0.315

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14185900: QUARTZVILLE CREEK NEAR CASCADIA, OREG.

Dominant forcing: PRCP Peak TE: 0.0247 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high p_mean, high q_mean.

Top 10 driver attributes

1. q95

z = +3.25

r = +0.31

score = 0.992

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_mean

z = +2.38

r = +0.34

score = 0.818

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
3. q_mean

z = +2.92

r = +0.28

score = 0.808

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. runoff_ratio

z = +2.23

r = +0.23

score = 0.516

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -1.64

r = -0.27

score = 0.446

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. high_prec_dur

z = +1.73

r = +0.23

score = 0.393

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. slope_fdc

z = +1.86

r = +0.20

score = 0.374

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. p_seasonality

z = -1.50

r = -0.24

score = 0.366

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. low_prec_freq

z = -1.62

r = -0.22

score = 0.362

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. hfd_mean

z = -1.47

r = -0.24

score = 0.350

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14187000: WILEY CREEK NEAR FOSTER, OR

Dominant forcing: PRCP Peak TE: 0.0117 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, low gauge_lon.

Top 10 driver attributes

1. p_mean

z = +1.49

r = +0.34

score = 0.513

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +1.54

r = +0.31

score = 0.470

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. gauge_lon

z = -1.65

r = -0.27

score = 0.449

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. slope_fdc

z = +2.05

r = +0.20

score = 0.411

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
5. q_mean

z = +1.29

r = +0.28

score = 0.356

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. high_prec_freq

z = -1.89

r = -0.19

score = 0.355

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
7. p_seasonality

z = -1.44

r = -0.24

score = 0.352

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. hfd_mean

z = -1.42

r = -0.24

score = 0.337

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. low_prec_freq

z = -1.43

r = -0.22

score = 0.320

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. gvf_diff

z = -1.26

r = -0.24

score = 0.306

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14216500: MUDDY CREEK BELOW CLEAR CREEK NEAR COUGAR, WA

Dominant forcing: PRCP Peak TE: 0.0142 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean , high q_mean , high $q95$.

Top 10 driver attributes

1. **p_mean**

z = +2.87

r = +0.34

score = 0.990

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. **q_mean**

z = +2.95

r = +0.28

score = 0.817

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. **q95**

z = +2.42

r = +0.31

score = 0.739

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
4. **q5**

z = +2.56

r = +0.18

score = 0.461

high low-flow values reflect strong baseflow and steady year-round streamflow.
5. **gauge_lon**

z = -1.62

r = -0.27

score = 0.439

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. **runoff_ratio**

z = +1.89

r = +0.23

score = 0.435

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. **slope_fdc**

z = +1.93

r = +0.20

score = 0.388

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
8. **p_seasonality**

z = -1.56

r = -0.24

score = 0.380

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
9. **aridity**

z = -1.26

r = -0.28

score = 0.348

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.
10. **low_prec_freq**

z = -1.51

r = -0.22

score = 0.337

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14222500: EAST FORK LEWIS RIVER NEAR HEISSON, WA

Dominant forcing: PRCP Peak TE: 0.0112 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.92 r = +0.34 score = 1.005

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.48 r = +0.31 score = 0.756

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.46 r = +0.28 score = 0.682

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. high_prec_freq

z = -2.53 r = -0.19 score = 0.475

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
5. gauge_lon

z = -1.65 r = -0.27 score = 0.447

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. low_prec_freq

z = -1.91 r = -0.22 score = 0.428

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. hfd_mean

z = -1.65 r = -0.24 score = 0.393

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. slope_fdc

z = +1.77 r = +0.20 score = 0.356

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. p_seasonality

z = -1.46 r = -0.24 score = 0.356

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. aridity

z = -1.24 r = -0.28 score = 0.343

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14236200: TILTON RIVER AB BEAR CANYON CREEK NEAR CINEBAR, WA

Dominant forcing: PRCP Peak TE: 0.0116 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high runoff_ratio.

Top 10 driver attributes

1. q95

z = +2.50

r = +0.31

score = 0.764

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean

z = +2.58

r = +0.28

score = 0.713

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. runoff_ratio

z = +2.54

r = +0.23

score = 0.585

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. p_mean

z = +1.65

r = +0.34

score = 0.570

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gauge_lon

z = -1.64

r = -0.27

score = 0.447

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. low_prec_freq

z = -1.90

r = -0.22

score = 0.424

few dry days indicate steady moisture supply, which supports continuous streamflow.
7. high_prec_freq

z = -2.20

r = -0.19

score = 0.413

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. hfd_mean

z = -1.61

r = -0.24

score = 0.383

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
9. p_seasonality

z = -1.35

r = -0.24

score = 0.330

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
10. slope_fdc

z = +1.61

r = +0.20

score = 0.324

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14301000: NEHALEM RIVER NEAR FOSS, OR

Dominant forcing: PRCP Peak TE: 0.0249 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, low low_prec_freq.

Top 10 driver attributes

1. p_mean

z = +1.82 r = +0.34 score = 0.626

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +1.85 r = +0.31 score = 0.563

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. low_prec_freq

z = -2.14 r = -0.22 score = 0.479

few dry days indicate steady moisture supply, which supports continuous streamflow.
4. gauge_lon

z = -1.72 r = -0.27 score = 0.468

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. hfd_mean

z = -1.88 r = -0.24 score = 0.448

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. high_prec_dur

z = +1.75 r = +0.23 score = 0.398

long heavy-rainfall events allow more soil saturation and stronger runoff response.
7. p_seasonality

z = -1.58 r = -0.24 score = 0.385

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. q_mean

z = +1.38 r = +0.28 score = 0.381

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
9. high_prec_freq

z = -1.95 r = -0.19 score = 0.368

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
10. slope_fdc

z = +1.63 r = +0.20 score = 0.327

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14303200: TUCCA CREEK NEAR BLAINE, OR.

Dominant forcing: PRCP Peak TE: 0.0215 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high q_mean.

Top 10 driver attributes

1. p_mean

z = +2.44

r = +0.34

score = 0.839

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.66

r = +0.31

score = 0.813

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. q_mean

z = +2.50

r = +0.28

score = 0.693

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
4. high_prec_dur

z = +2.53

r = +0.23

score = 0.574

long heavy-rainfall events allow more soil saturation and stronger runoff response.
5. gauge_lon

z = -1.71

r = -0.27

score = 0.465

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. hfd_mean

z = -1.82

r = -0.24

score = 0.432

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. p_seasonality

z = -1.70

r = -0.24

score = 0.414

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. runoff_ratio

z = +1.77

r = +0.23

score = 0.409

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
9. gvf_diff

z = -1.65

r = -0.24

score = 0.401

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. aridity

z = -1.19

r = -0.28

score = 0.329

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14305500: SILETZ RIVER AT SILETZ, OR

Dominant forcing: PRCP Peak TE: 0.0200 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high p_mean.

Top 10 driver attributes

1. q95

z = +3.90

r = +0.31

score = 1.189

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean

z = +3.31

r = +0.28

score = 0.917

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. p_mean

z = +2.42

r = +0.34

score = 0.833

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
4. runoff_ratio

z = +2.59

r = +0.23

score = 0.597

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
5. gauge_lon

z = -1.73

r = -0.27

score = 0.471

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. hfd_mean

z = -1.87

r = -0.24

score = 0.446

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
7. p_seasonality

z = -1.71

r = -0.24

score = 0.416

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
8. low_prec_freq

z = -1.62

r = -0.22

score = 0.362

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. high_prec_dur

z = +1.53

r = +0.23

score = 0.347

long heavy-rainfall events allow more soil saturation and stronger runoff response.
10. aridity

z = -1.19

r = -0.28

score = 0.328

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14306340: EAST FORK LOBSTER CREEK NEAR ALSEA, OR.

Dominant forcing: PRCP Peak TE: 0.0210 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high high_prec_dur.

Top 10 driver attributes

1. p_mean

z = +2.29 r = +0.34 score = 0.789

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.51 r = +0.31 score = 0.765

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. high_prec_dur

z = +2.63 r = +0.23 score = 0.597

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. q_mean

z = +1.73 r = +0.28 score = 0.478

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
5. gauge_lon

z = -1.72 r = -0.27 score = 0.466

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. p_seasonality

z = -1.74 r = -0.24 score = 0.424

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. hfd_mean

z = -1.77 r = -0.24 score = 0.420

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. gvf_diff

z = -1.53 r = -0.24 score = 0.372

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. low_prec_freq

z = -1.54 r = -0.22 score = 0.345

few dry days indicate steady moisture supply, which supports continuous streamflow.
10. slope_fdc

z = +1.59 r = +0.20 score = 0.319

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14306500: ALSEA RIVER NEAR TIDEWATER, OR

Dominant forcing: PRCP Peak TE: 0.0341 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high p_mean, high q95, high high_prec_dur.

Top 10 driver attributes

1. p_mean

z = +2.15

r = +0.34

score = 0.740

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
2. q95

z = +2.00

r = +0.31

score = 0.612

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
3. high_prec_dur

z = +2.35

r = +0.23

score = 0.534

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. gauge_lon

z = -1.73

r = -0.27

score = 0.470

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
5. p_seasonality

z = -1.73

r = -0.24

score = 0.423

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
6. q_mean

z = +1.52

r = +0.28

score = 0.420

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
7. hfd_mean

z = -1.68

r = -0.24

score = 0.400

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. low_prec_freq

z = -1.76

r = -0.22

score = 0.393

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. gvf_diff

z = -1.28

r = -0.24

score = 0.310

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
10. slope_fdc

z = +1.53

r = +0.20

score = 0.308

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14308990: COW CREEK ABV GALESVILLE RES, NR AZALEA, OR.

Dominant forcing: PRCP Peak TE: 0.0099 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high high_prec_dur, low p_seasonality.

Top 10 driver attributes

1. gauge_lon z = -1.69 r = -0.27 score = 0.458

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
2. high_prec_dur z = +1.95 r = +0.23 score = 0.441

long heavy-rainfall events allow more soil saturation and stronger runoff response.
3. p_seasonality z = -1.70 r = -0.24 score = 0.416

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. gvf_diff z = -1.43 r = -0.24 score = 0.346

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. hfd_mean z = -1.23 r = -0.24 score = 0.293

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. soil_depth_statsgo z = -1.64 r = +0.13 score = 0.220

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
7. high_prec_freq z = -1.06 r = -0.19 score = 0.200

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. slope_fdc z = +0.96 r = +0.20 score = 0.193

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
9. glim_1st_class_frac z = -1.16 r = -0.15 score = 0.169

the basin's main geological class covers only a small share, so the subsurface response is more mixed.
10. frac_forest z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14309500: WEST FORK COW CREEK NEAR GLENDALE, OR

Dominant forcing: SRAD Peak TE: 0.0140 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with low p_seasonality, low gauge_lon, high high_prec_dur.

Top 10 driver attributes

1. **p_seasonality**

z = -1.81

r = -0.46

score = 0.827

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
2. **gauge_lon**

z = -1.72

r = -0.43

score = 0.742

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. **high_prec_dur**

z = +2.23

r = +0.29

score = 0.643

long heavy-rainfall events allow more soil saturation and stronger runoff response.
4. **slope_mean**

z = +1.36

r = +0.37

score = 0.501

steep terrain promotes fast surface runoff and a quick streamflow response.
5. **gvf_diff**

z = -1.39

r = -0.33

score = 0.456

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
6. **q95**

z = +1.04

r = +0.43

score = 0.450

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
7. **high_prec_freq**

z = -1.28

r = -0.33

score = 0.418

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
8. **low_prec_freq**

z = -0.91

r = -0.31

score = 0.284

few dry days indicate steady moisture supply, which supports continuous streamflow.
9. **p_mean**

z = +1.22

r = +0.20

score = 0.245

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. **frac_forest**

z = +0.97

r = +0.23

score = 0.227

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14316700: STEAMBOAT CREEK NEAR GLIDE, OR

Dominant forcing: PRCP Peak TE: 0.0190 Peak lag: 1 day(s)

Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with low gauge_lon, high slope_fdc, low p_seasonality.

Top 10 driver attributes

1. gauge_lon z = -1.66 r = -0.27 score = 0.451

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.

2. slope_fdc z = +1.96 r = +0.20 score = 0.394

a steep flow-duration curve indicates a flashy basin with quick rainfall response.

3. p_seasonality z = -1.48 r = -0.24 score = 0.361

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.

4. q95 z = +1.13 r = +0.31 score = 0.346

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.

5. hfd_mean z = -1.40 r = -0.24 score = 0.333

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.

6. p_mean z = +0.91 r = +0.34 score = 0.314

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.

7. q_mean z = +0.93 r = +0.28 score = 0.257

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.

8. high_prec_freq z = -1.36 r = -0.19 score = 0.256

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.

9. runoff_ratio z = +1.10 r = +0.23 score = 0.254

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.

10. low_prec_freq z = -1.13 r = -0.22 score = 0.253

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14325000: SOUTH FORK COQUILLE RIVER AT POWERS, OR

Dominant forcing: SRAD Peak TE: 0.0163 Peak lag: 1 day(s)
Summary: SRAD-dominant at lag 1 day(s); top drivers consistent with high q95, low p_seasonality, low gauge_lon.

Top 10 driver attributes

1. q95 z = +2.19 r = +0.43 score = 0.947

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. p_seasonality z = -1.73 r = -0.46 score = 0.789

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
3. gauge_lon z = -1.74 r = -0.43 score = 0.754

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
4. slope_mean z = +1.67 r = +0.37 score = 0.614

steep terrain promotes fast surface runoff and a quick streamflow response.
5. q_mean z = +1.67 r = +0.35 score = 0.591

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
6. runoff_ratio z = +1.11 r = +0.43 score = 0.476

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
7. gvf_diff z = -1.39 r = -0.33 score = 0.456

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
8. high_prec_freq z = -1.39 r = -0.33 score = 0.454

few heavy rainfall events mean streamflow is shaped more by slow processes such as snowmelt or baseflow.
9. p_mean z = +2.16 r = +0.20 score = 0.433

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
10. low_prec_freq z = -1.01 r = -0.31 score = 0.316

few dry days indicate steady moisture supply, which supports continuous streamflow.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14362250: STAR GULCH NEAR RUCH, OR.

Dominant forcing: PRCP Peak TE: 0.0025 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high high_prec_dur, low gauge_lon, low p_seasonality.

Top 10 driver attributes

1. high_prec_dur

z = +2.28 r = +0.23 score = 0.517

long heavy-rainfall events allow more soil saturation and stronger runoff response.
2. gauge_lon

z = -1.68 r = -0.27 score = 0.457

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
3. p_seasonality

z = -1.79 r = -0.24 score = 0.437

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
4. gvf_diff

z = -1.38 r = -0.24 score = 0.336

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
5. hfd_mean

z = -1.33 r = -0.24 score = 0.315

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
6. runoff_ratio

z = -1.15 r = +0.23 score = 0.265

a low runoff ratio means most rainfall is lost to evaporation or storage, weakening the forcing-to-Q link.
7. soil_depth_statsgo

z = -1.83 r = +0.13 score = 0.244

shallow soils (STATSGO) reduce storage and lead to faster streamflow response.
8. q95

z = -0.75 r = +0.31 score = 0.229

low peak-flow values reflect a damped streamflow response with limited peaks.
9. q_mean

z = -0.75 r = +0.28 score = 0.208

low mean streamflow limits the dynamic range of Q and weakens information flow from forcings.
10. frac_forest

z = +0.97 r = +0.17 score = 0.167

high forest cover increases evapotranspiration and modifies the rainfall-runoff response.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.

Basin 14400000: CHETCO RIVER NEAR BROOKINGS, OR

Dominant forcing: PRCP Peak TE: 0.0199 Peak lag: 1 day(s)
Summary: PRCP-dominant at lag 1 day(s); top drivers consistent with high q95, high q_mean, high runoff_ratio.

Top 10 driver attributes

1. q95

z = +5.30

r = +0.31

score = 1.617

high peak-flow values reflect strong rainfall-runoff response and a wide streamflow range.
2. q_mean

z = +3.94

r = +0.28

score = 1.089

high mean streamflow gives a more dynamic Q signal, which generally raises information flow from forcings.
3. runoff_ratio

z = +4.18

r = +0.23

score = 0.965

a high runoff ratio means most rainfall becomes streamflow, so the forcing-to-Q link is direct.
4. p_mean

z = +1.63

r = +0.34

score = 0.561

high mean rainfall provides more water input, so rainfall information transfers more strongly to streamflow.
5. gauge_lon

z = -1.75

r = -0.27

score = 0.476

western longitude (more negative) is generally associated with drier conditions and energy-limited streamflow.
6. p_seasonality

z = -1.85

r = -0.24

score = 0.451

weak precipitation seasonality means rainfall is spread evenly through the year, which smooths the streamflow response.
7. hfd_mean

z = -1.75

r = -0.24

score = 0.416

an early half-flow date indicates streamflow peaks early, typical of rainfall-dominated basins.
8. gvf_diff

z = -1.52

r = -0.24

score = 0.369

weak seasonal vegetation cycle is associated with evergreen or sparse vegetation.
9. slope_fdc

z = +1.54

r = +0.20

score = 0.309

a steep flow-duration curve indicates a flashy basin with quick rainfall response.
10. aridity

z = -1.05

r = -0.28

score = 0.290

the basin is humid, so streamflow is energy-limited and rainfall is rarely a limiting factor.

Note: TE shows lagged information flow, not direct causation. Driver scores combine population-level relevance with how unusual the basin is on each attribute.